## CHAPTER I

# SPHERICAL TRIGONOMETRY

### 1. *Introduction.*

When we look at the stars on a clear night we have the familiar impression that they are all sparkling points of light, apparently situated on the surface of a vast sphere of which the individual observer is the centre. The eye, of course, fails to give any indication of the distances of the stars from us; however, it allows us to make some estimate of the angles subtended at the observer by any pairs of stars and, with suitable instruments, these angles can be measured with great precision. Spherical Astronomy is concerned essentially with the *directions* in which the stars are viewed, and it is convenient to define these directions in terms of the positions on the surface of a sphere—the *celestial sphere*—in which the straight lines, joining the observer to the stars, intersect this surface. It is in this sense that the usual expression "the position of a star on the celestial sphere" is to be interpreted. The radius of the sphere is entirely arbitrary. The foundation of Spherical Astronomy is the geometry of the sphere.

### 2. *The spherical triangle.*

Any plane passing through the centre of a sphere cuts the surface in a circle which is called a *great circle*. Any other plane intersecting the sphere but not passing through the centre will also cut the surface in a circle which, in this case, is called a *small circle*. In Fig. 1, $EAB$ is a great circle, for its plane passes through $O$, the centre of the sphere. Let $QOP$ be the diameter of the sphere perpendicular to the plane of the great circle $EAB$. Let $R$ be any point in $OP$ and suppose a plane drawn through $R$ parallel to the plane of $EAB$; the surface of the sphere is then intersected in the small circle $FCD$. It follows from the construction that $OP$ is also perpendicular to the plane of $FCD$. The extremities $P$ and $Q$ of the common perpendicular diameter $QOP$ are called the *poles* of the great circle and of the parallel small circle. Now let $PCAQ$ be any great circle passing through the

poles $P$ and $Q$ and intersecting the small circle $FCD$ and the great circle $EAB$ in $C$ and $A$ respectively. Similarly, $PDB$ is part of another great circle passing through $P$ and $Q$. We shall find it convenient to refer to a particular great circle by specifying simply any portion of its circumference. When two great circles intersect at a point they are said to include a *spherical angle* which is defined as follows. Consider the two great circles $PA$ and $PB$ intersecting at $P$. Draw $PS$ and $PT$, the tangents to the

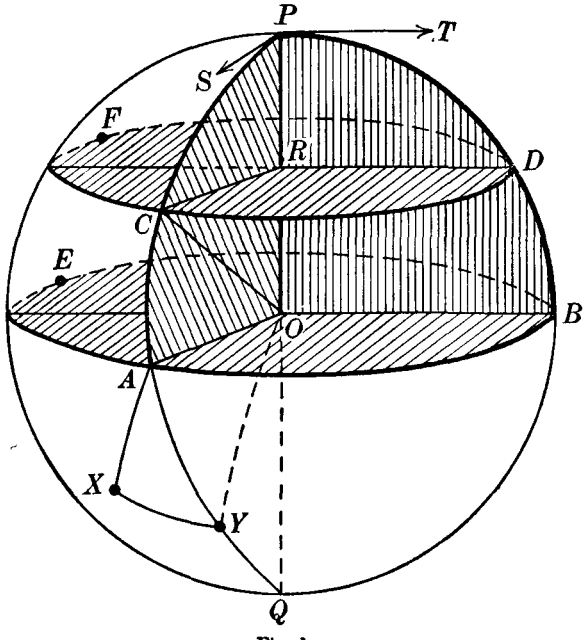



Fig. 1.

circumferences of $PA$ and $PB$ respectively. $PT$ is, by construction, perpendicular to the radius $OP$ of the great circle $PB$ and, being in the plane $PBO$, is therefore parallel to the radius $OB$. Similarly $PS$ is parallel to the radius $OA$. The angle $SPT$ defines the spherical angle at $P$ between the two great circles $PA$ and $PB$, and it is equal to the angle $AOB$, $AB$ being the arc intercepted on the great circle, of which $P$ is the pole, between the two great circles $PA$ and $PB$. It is to be emphasised that a spherical angle is defined only with reference to two intersecting *great circles.*

If we are given any three points on the surface of a sphere, then the sphere can be bisected so that all three points lie in the same hemisphere. If the points are joined by great circle arcs all lying on this hemisphere the figure obtained is called a *spherical triangle*. Thus, in Fig. 1, the three points $A$, $X$ and $Y$ on the spherical surface are joined by great circle arcs to form the spherical triangle $AXY$. $AX$, $AY$ and $XY$ are the *sides* and the spherical angles at $A$, $X$ and $Y$ are the angles of the spherical triangle. Actually, if $R$ is the radius of the sphere, the length of the great circle arc $AY$ is given by

$$AY = R \times \text{angle } AOY,$$

the angle $AOY$ being expressed in circular measure, i.e. in radians. Now for all great circle arcs on the sphere the radius $R$ is constant and it is convenient to consider its length as unity. The arc $AY$ is then simply the angle which it subtends at the centre of the sphere. If $AY$ is, let us say, one-eighth of the circumference of the complete great circle through $A$ and $Y$, the side $AY$ is then $\frac{\pi}{4}$ in circular measure and there is no ambiguity if it is expressed as 45°; similarly, for the remaining sides of the triangle. It follows from the definition of a spherical triangle that no side can be equal to or greater than 180°. As another example, $PAB$ is a spherical triangle two of whose sides $PA$ and $PB$ each subtend $\frac{\pi}{2}$ radians or 90° at $O$; in this instance we say that $PA$ and $PB$ are each equal to $\frac{\pi}{2}$ radians or 90°. But $PCD$ is *not* a spherical triangle, for the arc $CD$ is not a part of a great circle. Accordingly, the formulae which will be derived for spherical triangles will not be applicable to such a figure as $PCD$.

### 3. *Length of a small circle arc.*

Consider, in Fig. 1, the small circle arc $CD$. Its length is given by

$$CD = RC \times \text{angle } CRD.$$

Also, the length of the great circle arc $AB$ is given by

$$AB = OA \times \text{angle } AOB.$$

But since the plane of $FCD$ is parallel to the plane of $EAB$, then $C\hat{R}D = A\hat{O}B$, for $RC$, $RD$ are respectively parallel to $OA$, $OB$.

Therefore $$CD = \frac{RC}{OA} . AB.$$

But, since $OA = OC$ (radii of the sphere), we have

$$CD = \frac{RC}{OC} . AB.$$

Now $RC$ is perpendicular to $OR$; $\therefore$ $RC = OC \cos R\hat{C}O$. From the parallelism of $RC$ and $OA$, $R\hat{C}O = A\hat{O}C$. Hence

$$CD = AB \cos A\hat{O}C.$$

Now $AOC$ is the angle subtended at the centre of the sphere by the great circle arc $AC$. The formula can then be written as

$$CD = AB \cos AC,$$

or, since $PA = 90°$,

$$CD = AB \sin PC \qquad \text{.....(1).}$$

**4.** *Terrestrial latitude and longitude.*

The concepts introduced so far will now be illustrated with reference to the earth. For many practical problems, the earth can be regarded as a spherical body spinning about a diameter $PQ$ (Fig. 2). $P$ is the *north pole* and $Q$ is the *south pole*. The great circle whose plane is perpendicular to $PQ$ is called the *equator*. Any semi-great circle terminated by $P$ and $Q$ is a

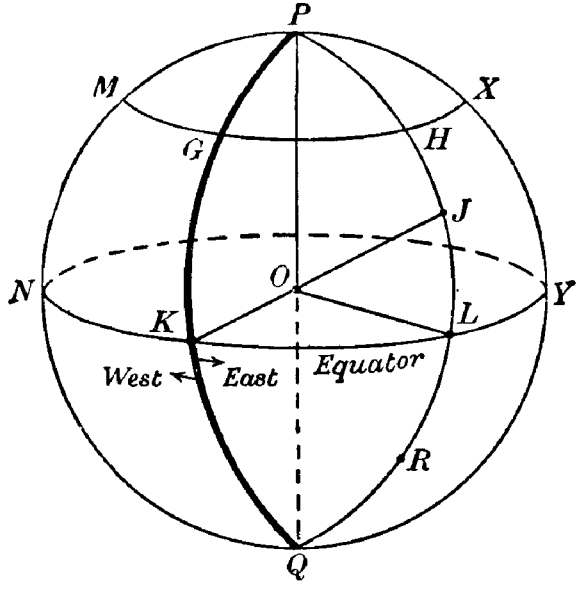



Fig. 2.

*meridian*. In particular, the meridian which passed through the fundamental instrument (the transit circle) of Greenwich Observatory is, by universal agreement, regarded as the principal or standard meridian; let it be $PGKQ$ in Fig. 2, intersecting the equator in $K$. Let $PHLQ$ be any other meridian cutting the equator in $L$. The angle $KOL$ is defined to be the *longitude* of the meridian $PHQ$ and it can be described equally well as the equatorial arc $KL$ or the spherical angle $KPL$. Longitudes are measured from 0° to 180° *east* of the Greenwich meridian and from 0° to 180° *west*, following the directions of the arrows near $K$ in Fig. 2. Thus, from the figure, the longitude of the meridian $PHQ$ is about 100° east (E) and that of the meridian $PMQ$ is about 60° west (W). All places on the same meridian have the same longitude and the meridian on which a particular place is situated is specified with reference to the principal meridian $PGQ$. To specify completely the position of a place on the surface of the earth, we require to describe its position on its meridian of longitude. This is done with reference to the equator. Consider a place $J$ on the meridian $PHQ$. The meridian through $J$ cuts the equator in $L$ and the angle $LOJ$, or the great circle arc $LJ$, is called the *latitude* of $J$. If $J$ is between the equator and the north pole $P$, as in Fig. 2, the latitude is said to be north (N); a place such as $R$, between the equator and the south pole $Q$, is said to have south latitude (S). In this way the position of any point on the surface of the earth is referred to the two fundamental great circles, the equator and the meridian of Greenwich.

Let $\phi$ denote the latitude of $J$; then $L\hat{O}J$ or $LJ = \phi$. Since $OP$ is perpendicular to the plane of the equator, $P\hat{O}L = 90°$ and therefore $POJ = 90° - \phi$. The angle $POJ$ or the spherical arc $PJ$ is the *colatitude* of $J$. We have thus

$$\text{Colat.} = 90° - \text{Lat.}$$

All places which have the same latitude lie on a small circle parallel to the equator, called a *parallel of latitude*. Thus all places with the same latitude as Greenwich lie on the small circle $MGHX$. If $\theta$ denotes the latitude of Greenwich, then by formula (1) the length of the small circle arc $HX$, for example, is given in terms of the length of the corresponding equatorial arc $LY$ by

$$HX = LY \cos\theta \qquad \text{.....(2).}$$

To give greater precision to the meaning of this formula, we consider the units in which distances on the surface of the earth are expressed. The simplest is that defined as the great circle distance between two points subtending an angle of one minute of arc at the centre of the earth—this unit is known as the *nautical mile* and is equivalent to 6080 feet (we neglect the small variations in this value due to the fact that the earth is not quite a sphere). If the difference in longitude between any two places on the same parallel of latitude is known, e.g. $LY$, then $LY$ can be expressed as so many minutes of arc and this number is the number of nautical miles between the two points $L$ and $Y$ on the equator. The formula (2) then provides the means of calculating the distance between $H$ and $X$ expressed in nautical miles (or minutes of arc) and *measured along the parallel of latitude.*

**5.** *The cosine-formula.*

Let $ABC$ be a spherical triangle (Fig. 3). Denote the sides $BC$, $CA$, $AB$ by $a$, $b$ and $c$ respectively. Then, by our definition, the

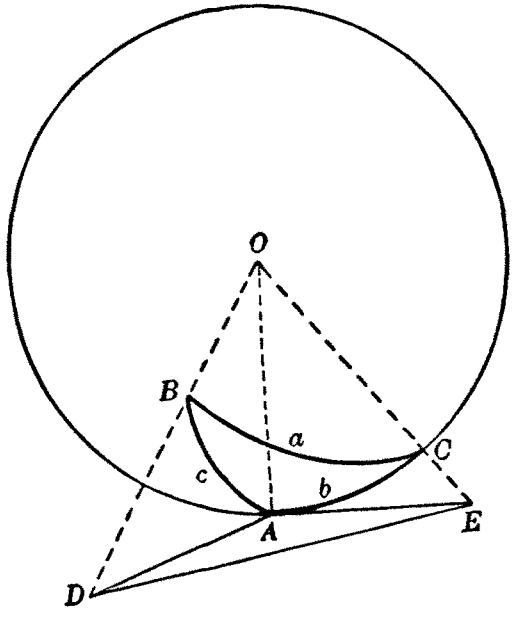



Fig. 3.

side $a$ is measured by the angle $BOC$ subtended at the centre $O$ of the sphere by the great circle arc $BC$. Similarly, $b$ and $c$ are measured respectively by the angles $AOC$ and $AOB$. Let $AD$ be the tangent at $A$ to the great circle $AB$ and $AE$ the tangent at $A$ to the great circle $AC$. Then the radius $OA$ is perpendicular to $AD$ and $AE$. By construction, $AD$ lies in the plane of the great circle $AB$; hence, if the radius $OB$ is produced, it will intersect the tangent $AD$ at a point $D$. Similarly, the radius $OC$ when produced will meet the tangent $AE$ in $E$. Now the spherical angle $BAC$ is defined to be the angle between the tangents at $A$ to the great circles $AB$ and $AC$, so that the spherical angle $BAC = D\hat{A}E$. The spherical angle $BAC$ will be denoted simply by $A$, so that $D\hat{A}E = A$.

Now, in the plane triangle $OAD$, $O\hat{A}D$ is 90° and $A\hat{O}D$, identical with $A\hat{O}B$, is $c$. We have then

$$AD = OA \tan c; \quad OD = OA \sec c \qquad \text{.....(3).}$$

From the plane triangle $OAE$ we have, similarly,

$$AE = OA \tan b; \quad OE = OA \sec b \qquad \text{.....(4).}$$

From the plane triangle $DAE$ we have

$$DE^2 = AD^2 + AE^2 - 2AD.AE \cos D\hat{A}E,$$

or $$DE^2 = OA^2 [\tan^2 c + \tan^2 b - 2 \tan b \tan c \cos A] \qquad \text{.....(5).}$$

From the plane triangle $DOE$,

$$DE^2 = OD^2 + OE^2 - 2OD.OE \cos D\hat{O}E.$$

But $D\hat{O}E = B\hat{O}C = a$;

$$\therefore \quad DE^2 = OA^2 [\sec^2 c + \sec^2 b - 2 \sec b \sec c \cos a] \qquad \text{.....(6).}$$

Hence, from (5) and (6),

$$\sec^2 c + \sec^2 b - 2 \sec b \sec c \cos a = \tan^2 c + \tan^2 b - 2 \tan b \tan c \cos A.$$

Now $\sec^2 c = 1 + \tan^2 c$, $\sec^2 b = 1 + \tan^2 b$,

and after some simplification we obtain

$$\textbf{cos a = cos b cos c + sin b sin c cos A} \qquad \text{.....(A).}$$

This is the fundamental formula of spherical trigonometry and it will be referred to in the following pages as the *cosine-formula*

or formula **A**. There are clearly two companion formulae; they are

$$\cos b = \cos c \cos a + \sin c \sin a \cos B \quad \ldots\ldots(7),$$

$$\cos c = \cos a \cos b + \sin a \sin b \cos C \quad \ldots\ldots(8).$$

From the three formulae—**A**, (7) and (8)—all the other formulae of spherical trigonometry in use can be derived. The fundamental formula has two direct practical applications:

(1) *If two sides, e.g.* b *and* c, *and the included angle* A *of a spherical triangle* ABC *are known,* formula **A** enables the calculation of the third side $a$ to be made.

(2) *If all three sides are known,* the angles of the triangle can be found successively by means of **A**, (7) and (8).

For suppose the value of $A$ is required; then by **A**

$$\cos A = \operatorname{cosec} b \operatorname{cosec} c \,[\cos a - \cos b \cos c] \quad \ldots\ldots(9).$$

Formula (9) can be replaced by one more suitable for logarithmic calculations as follows. Since $\cos A = 1 - 2\sin^2 \frac{A}{2}$, we have, from **A**,

$$\cos a = \cos b \cos c + \sin b \sin c \left(1 - 2\sin^2 \frac{A}{2}\right)$$

$$= \cos (b - c) - 2 \sin b \sin c \sin^2 \frac{A}{2},$$

or $$\cos (b - c) - \cos a = 2 \sin b \sin c \sin^2 \frac{A}{2};$$

$$\therefore\ 2 \sin \frac{a + (b - c)}{2} \sin \frac{a - (b - c)}{2} = 2 \sin b \sin c \sin^2 \frac{A}{2}.$$

Let $s$ be defined by $$2s = a + b + c \quad \ldots\ldots(10).$$

Then $a + b - c = 2(s - c)$ and $a - b + c = 2(s - b)$.

Hence $$\sin (s - b) \sin (s - c) = \sin b \sin c \sin^2 \frac{A}{2};$$

$$\therefore\ \sin \frac{A}{2} = \sqrt{\frac{\sin (s - b) \sin (s - c)}{\sin b \sin c}} \quad \ldots\ldots(11).$$

This form is useful in numerical work. There are two similar equations giving $\sin \frac{B}{2}$ and $\sin \frac{C}{2}$.

If we write $\cos A = 2\cos^2 \frac{A}{2} - 1$ in the formula **A** and proceed

as before, we shall obtain

$$\cos\frac{A}{2} = \sqrt{\frac{\sin s \sin(s-a)}{\sin b \sin c}} \qquad \ldots\ldots(12)$$

with two similar equations giving $\cos\frac{B}{2}$ and $\cos\frac{C}{2}$.

From (11) and (12) by division we have

$$\tan\frac{A}{2} = \sqrt{\frac{\sin(s-b)\sin(s-c)}{\sin s \sin(s-a)}} \qquad \ldots\ldots(13).$$

There are two similar equations, giving $\tan\frac{B}{2}$ and $\tan\frac{C}{2}$. Any one of (11), (12) and (13) can be used to calculate $A$, the three sides being known.

6. *The sine-formula.*

We shall now derive what is known as the sine-formula. From the cosine-formula A, we have

$$\sin b \sin c \cos A = \cos a - \cos b \cos c.$$

By squaring, we obtain

$$\sin^2 b \sin^2 c \cos^2 A = \cos^2 a - 2\cos a \cos b \cos c + \cos^2 b \cos^2 c.$$

The left-hand side can be written

$$\sin^2 b \sin^2 c - \sin^2 b \sin^2 c \sin^2 A,$$

or $$1 - \cos^2 b - \cos^2 c + \cos^2 b \cos^2 c - \sin^2 b \sin^2 c \sin^2 A.$$

Hence

$$\sin^2 b \sin^2 c \sin^2 A = 1 - \cos^2 a - \cos^2 b - \cos^2 c + 2\cos a \cos b \cos c.$$

Let a positive quantity $X$ be defined by

$$X^2 \sin^2 a \sin^2 b \sin^2 c = 1 - \cos^2 a - \cos^2 b - \cos^2 c + 2\cos a \cos b \cos c.$$

Then, from the previous equation,

$$\frac{\sin^2 A}{\sin^2 a} = X^2,$$

so that

$$X = \pm\frac{\sin A}{\sin a}.$$

But in a spherical triangle the sides are each less than 180°, and this applies also to the angles. As $\sin\theta$ is positive for all

values of $\theta$ between 0° and 180°, the minus sign in the above equation is inadmissible, and we have

$$X = \frac{\sin A}{\sin a}.$$

By treating (7) and (8) in a similar way, we shall obtain

$$X = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c}.$$

Hence

$$\frac{\sin \mathbf{A}}{\sin \mathbf{a}} = \frac{\sin \mathbf{B}}{\sin \mathbf{b}} = \frac{\sin \mathbf{C}}{\sin \mathbf{c}} \qquad \text{.....(B).}$$

This result we shall refer to as the *sine-formula* or formula **B**.

Formula **B** gives a relation between any two sides of a triangle and the two angles *opposite* these sides. It has to be used, however, with circumspection in numerical calculations; for, suppose that the two sides $a$ and $b$ and the angle $B$ are given, then by **B**

$$\sin A = \frac{\sin a \sin B}{\sin b},$$

from which the value of $\sin A$ can be calculated. But $\sin (180° - A) = \sin A$, and without further information it is not possible to decide which of the two angles $A$ or $180° - A$ represents the correct solution. The analogous ambiguity in plane trigonometry may be recalled to the reader's attention.

7. *The analogue formula.*

Write equation (7) in the form

$$\begin{aligned}\sin c \sin a \cos B &= \cos b - \cos c \cos a \\ &= \cos b - \cos c (\cos b \cos c + \sin b \sin c \cos A) \\ &= \sin^2 c \cos b - \sin b \sin c \cos c \cos A.\end{aligned}$$

Hence, dividing by $\sin c$, we have

$$\sin \mathbf{a} \cos \underline{\mathbf{B}} = \cos \underline{\mathbf{b}} \sin \mathbf{c} - \sin \underline{\mathbf{b}} \cos \mathbf{c} \cos \mathbf{A} \quad \text{...(C),}$$

a relation involving all three sides and two angles.

We can easily prove in a similar manner, beginning with equation (8), that

$$\sin a \cos \underline{C} = \cos \underline{c} \sin b - \sin \underline{c} \cos b \cos A \quad \text{...(14).}$$

As we have seen, the cosine-formula **A** gives $\cos a$ in terms of $b, c$ and the included angle $A$. Formulae **C** and (14) are, in some ways, analogous to **A** as they give $\sin a \times$ cosine of one of the

two angles $B$ and $C$, adjacent to the side $a$, in terms of $b$, $c$ and $A$. We shall, therefore, refer to formula **C** or (14) as the analogue formula.

The formula **C** can also be proved as follows. Suppose the side $c$ of the triangle $ABC$ to be less than 90° (the case when $c$ is between 90° and 180° is left as an exercise to the student). Produce the great circle arc $BA$ to $D$ so that $BD$ is 90° (Fig. 4).

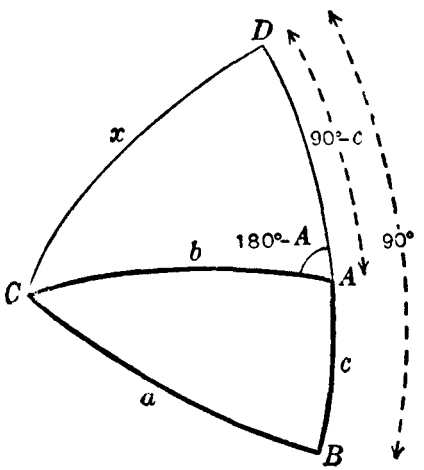



Fig. 4.

Then $AD = 90° - c$ and $C\hat{A}D = 180° - A$. Join $C$ and $D$ by a great circle arc and denote it by $x$. From the triangle $DAC$, by **A**,

$$\cos x = \cos(90° - c)\cos b + \sin(90° - c)\sin b \cos(180° - A),$$

or

$$\cos x = \sin c \cos b - \cos c \sin b \cos A \qquad \ldots\ldots(15).$$

From the triangle $DBC$, by **A**,

$$\cos x = \cos 90° \cos a + \sin 90° \sin a \cos B,$$

or

$$\cos x = \sin a \cos B \qquad \ldots\ldots(16),$$

and therefore from (15) and (16)

$$\sin a \cos B = \cos b \sin c - \sin b \cos c \cos A,$$

which is formula **C**.

**8.** *The four-parts formula.*

Another useful formula, known as the four-parts formula, will now be derived. In the spherical triangle $ABC$ (Fig. 5) consider the four consecutive parts $B$, $a$, $C$, $b$. The angle $C$ is contained by the two sides $a$ and $b$ and is called the "inner angle". The side $a$ is flanked by the two angles $B$ and $C$ and is called the "inner side". Introduce $B$ and $C$ by means of the cosine-formula; then we have

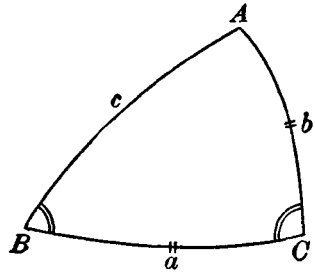


Fig. 5.

$$\cos b = \cos a \cos c + \sin a \sin c \cos B \quad \ldots\ldots(17),$$

$$\cos c = \cos b \cos a + \sin b \sin a \cos C \quad \ldots\ldots(18).$$

Substitute the value of $\cos c$ given by (18) on the right-hand side of (17); then

$$\cos b = \cos a (\cos b \cos a + \sin b \sin a \cos C) + \sin a \sin c \cos B;$$

$$\therefore \cos b \sin^2 a = \cos a \sin b \sin a \cos C + \sin a \sin c \cos B.$$

Divide throughout by $\sin a \sin b$; then

$$\cot b \sin a = \cos a \cos C + \frac{\sin c}{\sin b} \cos B.$$

But by the sine-formula **B**,

$$\frac{\sin c}{\sin b} = \frac{\sin C}{\sin B}.$$

Hence $$\mathbf{\cos a \cos C = \sin a \cot b - \sin C \cot B} \quad \ldots\ldots(D),$$

which may be put into words, as an aid to the memory, as follows:

cos (*inner side*).cos (*inner angle*)
= sin (*inner side*).cot (*other side*)
− sin (*inner angle*).cot (*other angle*).

**9.** *Alternative proofs of the formulae* **A**, **B** *and* **C**.

The formulae **B**, **C** and **D** have been derived by algebraic transformations of the fundamental formula. Another proof of each of **A**, **B** and **C** will now be briefly obtained from a simple and instructive geometrical construction. Let $ABC$ (Fig. 6) be a spherical triangle and $O$ the centre of the sphere. Join $O$ to the

vertices and take any point $P$ in $OC$. From $P$ draw $PQ$ perpendicular to $OA$ and $PR$ perpendicular to $OB$. In the plane $OAB$, draw $QS$ perpendicular to $OA$ and $RS$ perpendicular to $OB$. These perpendiculars meet in $S$. Join $PS$ and $OS$. If we draw tangents at $A$ to the great circle arcs $AB$ and $AC$, these tangents, by definition, include the spherical angle $A$. But $QS$ and $QP$ are by construction parallel to these tangents. Hence $P\hat{Q}S = A$. Similarly $P\hat{R}S = B$. Also $C\hat{O}B = a$, $C\hat{O}A = b$ and $A\hat{O}B = c$.

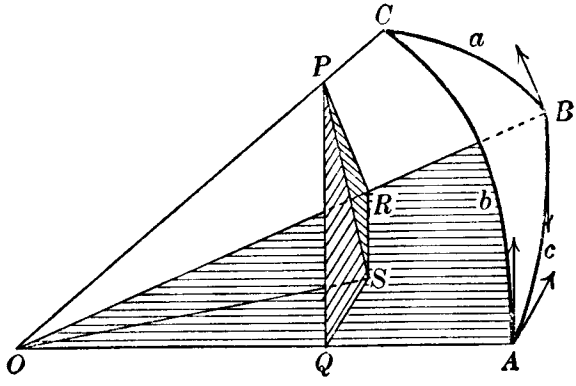



Fig. 6.

The first step is to prove that $PS$ is perpendicular to the plane $AOB$. By the construction, $OQ$ is perpendicular to both $PQ$ and $QS$; hence $OQ$ is perpendicular to the plane $PQS$; therefore $OQ$ is perpendicular to $PS$ which is a line lying in the plane $PQS$. Similarly, $OR$ is perpendicular to $PS$. Thus $PS$ is perpendicular to both $OQ$ and $OR$ and is therefore perpendicular to every line in the plane of $OQ$ and $OR$, that is, $PS$ is perpendicular to the plane $OAB$ and, in particular, to $OS$, $SQ$ and $SR$. Thus $PQS$ and $PRS$ are right-angled triangles.

(1) We have, from the right-angled triangles $OQP$ and $ORP$,

$$PQ = OP \sin b; \quad PR = OP \sin a \qquad \ldots\ldots(19).$$

$$OQ = OP \cos b; \quad OR = OP \cos a \qquad \ldots\ldots(20).$$

Let $x$ denote the angle $SOQ$; then $R\hat{O}S = c - x$.

Now $OS = OQ \sec x$ and $OS = OR \sec (c - x)$.

Hence $OR \cos x = OQ \cos (c - x)$;

$\therefore$ by (20), $OP \cos a \cos x = OP \cos b \cos (c - x)$;

$\therefore \cos a = \cos b \cos c + \cos b \sin c \tan x.$

But $$\tan x = \frac{QS}{OQ} = \frac{PQ \cos A}{OQ} = \tan b \cos A,$$

and hence $$\cos a = \cos b \cos c + \sin b \sin c \cos A,$$

which is formula **A**.

(2) Again, from the right-angled triangles $PQS$ and $PRS$,

$$PS = PQ \sin PQS = PQ \sin A,$$

and $$PS = PR \sin PRS = PR \sin B.$$

Hence $$PQ \sin A = PR \sin B,$$

and $\therefore$ by (19),

$$OP \sin b \sin A = OP \sin a \sin B,$$

from which formula **B** follows.

(3) We have, from the right-angled triangles $OSQ$ and $OSR$,

$$QS = OS \sin x \quad \text{and} \quad RS = OS \sin (c - x);$$

$$\therefore \quad RS \sin x = QS (\sin c \cos x - \cos c \sin x),$$

or $$RS = QS (\sin c \cot x - \cos c).$$

Now $$RS = PR \cos B = OP \sin a \cos B,$$

and $$QS = PQ \cos A = OP \sin b \cos A,$$

and $$QS \cot x = OQ = OP \cos b.$$

Hence $$\sin a \cos B = \cos b \sin c - \sin b \cos c \cos A,$$

which is formula **C**.

**10.** *Right-angled and quadrantal triangles.*

When one of the spherical angles is 90°, the formulae **A**, **B**, **C** and **D** assume simple forms. This is also the case when one side of a spherical triangle is 90°—the triangle is then said to be *quadrantal*. Rules have been given by Napier according to which the various simple formulae can be written down. The rules, however, impose an additional charge on the memory and it is much simpler to apply one of the main formulae **A** to **D** to the particular right-

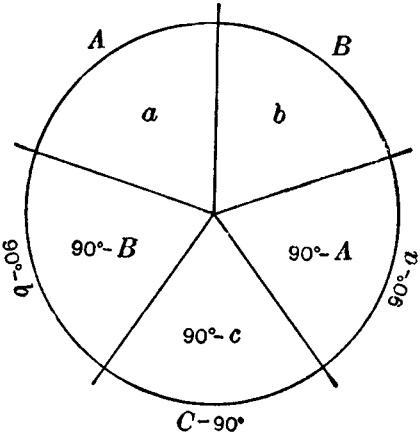


Fig. 7.

angled or quadrantal triangle concerned. The rules are as follows:

(1) Right-angled triangle in which $C = 90°$. Arrange *inside* a circle the five "circular parts" $a$, $b$, $90° - A$, $90° - c$, $90° - B$, as in Fig. 7. If any one circular part is chosen as a "middle", the two flanking parts are called "adjacents" and the two others the "opposites". The rules then are:

**sin** (*middle*) = product of tangents of *adjacents*;

**sin** (*middle*) = product of cosines of *opposites*.

(2) Quadrantal triangle in which $c = 90°$. Arrange *outside* the circle (Fig. 7) the five "circular parts" $A$, $B$, $90° - a$, $C - 90°$, $90° - b$. The two rules are then the same as for right-angled triangles.

**11.** *Polar formulae.*

Certain useful formulae can be obtained by means of the polar triangle which is constructed as follows (Fig. 8). Let $ABC$ be a spherical triangle. The great circle of which $BC$ is an arc has two poles, one in each of the hemispheres into which the sphere is divided by the great circle. Let $A'$ be the pole in the hemisphere in which $A$ lies. Similarly $B'$ and $C'$ are the appropriate poles of $CA$ and $AB$. Produce $BC$ both ways to meet $A'B'$ and $A'C'$ in $L$ and $M$ respectively. Then, since $A'$ is the pole of the great circle $LBCM$, the spherical angle $B'A'C'$ (or simply $A'$) is equal to the arc $LM$. Again, $B'$ is the pole of $AC$, that is, the angular distance of $B'$ from any point on $AC$ is 90°; similarly the angular distance of $A'$ from any point on $BC$ is 90°. Hence the angular distance of $C$ from $B'$ and from $A'$ is in each instance 90°; in other words, $C$ is the pole of $A'B'$. Hence $CL = 90°$, and similarly $BM = 90°$. Now $LM = LB + BM = LB + 90°$. Also $BC = a$; $\therefore$ $LB = 90° - a$. Hence $A' = 180° - a$. Similarly $B' = 180° - b$ and $C' = 180° - c$. We obtain in a similar manner

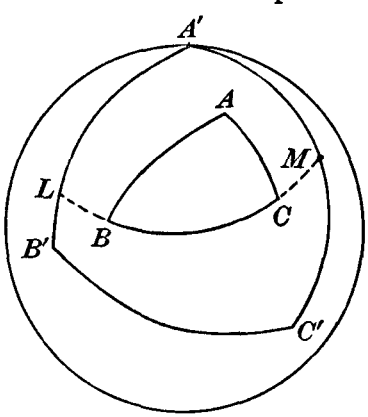


Fig. 8.

$$a' = 180° - A; \quad b' = 180° - B; \quad c' = 180° - C.$$

Now apply formula **A** to the triangle $A'B'C'$ and we have, for example,

$$\cos a' = \cos b' \cos c' + \sin b' \sin c' \cos A'.$$

Using the relations just found, we obtain from this equation

$$-\cos A = \cos B \cos C - \sin B \sin C \cos a,$$

*which is a formula for the triangle* $ABC$, giving the angle $A$ in terms of the two remaining angles and the included side. The procedure in this instance can be extended to any of the principal formulae which we have already derived, by writing $180° - a$ for $A$, $180° - b$ for $B$, etc., in the formulae **A** to **D**.

**12.** *Numerical example.*

To illustrate the numerical solution of a spherical triangle, we shall consider the following problem. In Fig. 9 let $A$ and $B$ represent two places, in north latitude, on the surface of the earth; their latitudes are respectively 24° 18′ N and 36° 47′ N, and their longitudes 133° 39′ E and 125° 24′ W respectively; it is required to find (i) the length of the great circle arc $AB$, (ii) the angle $PAB$, $P$ being the north pole, and (iii) the most northerly point on the great circle $AB$.

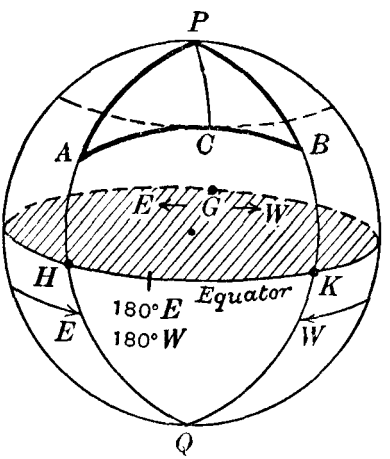


Fig. 9.

$PAHQ$ is the meridian through $A$ cutting the equator in $H$. $HA$ measures the latitude of $A$, i.e. $HA = 24° 18'$. $PA$ is the colatitude of $A$; $\therefore PA = 90° - 24° 18' = 65° 42'$. Similarly $PB = 53° 13'$. Let the Greenwich meridian intersect the equator in $G$. Then, following the arrows,

$$GH = \text{long. (E) of } A = 133° 39',$$

and

$$GK = \text{long. (W) of } B = 125° 24'.$$

Hence the arc

$$HGK = 259° 3',$$

and therefore $HK$ (the shorter of the two great circle arcs joining $H$ and $K$) is 100° 57′; that is $A\hat{P}B = 100° 57'$. In the

triangle $APB$ we now are given the two sides $PA$ and $PB$ and the contained angle $APB$.

(i) *Calculation of* $AB$. By formula **A**, we have

$$\cos AB = \cos PA \cos PB + \sin PA \sin PB \cos APB,$$

which becomes, on inserting the data,

$$\cos AB = \cos 65° 42' \cos 53° 13' - \sin 65° 42' \sin 53° 13' \cos 79° 3'$$
$$\equiv \quad M \quad - N.$$

We shall use five-figure logarithms.

| | | | |
|---|---|---|---|
| $\log \cos 65° 42'{\cdot}0$ | $\bar{1}{\cdot}61\,438$ | $\log \sin 65° 42'{\cdot}0$ | $\bar{1}{\cdot}95\,971$ |
| $\log \cos 53° 13'{\cdot}0$ | $\bar{1}{\cdot}77\,728$ | $\log \sin 53° 13'{\cdot}0$ | $\bar{1}{\cdot}90\,358$ |
| $\therefore \log M =$ | $\overline{\bar{1}{\cdot}39\,166}$ | $\log \cos 79° 3'{\cdot}0$ | $\bar{1}{\cdot}27\,864$ |
| | | $\therefore \log N =$ | $\overline{\bar{1}{\cdot}14\,193}$ |
| $\therefore M =$ | $0{\cdot}24\,641;$ | $\therefore N =$ | $0{\cdot}13\,865.$ |

Hence $\cos AB \equiv M - N = 0{\cdot}10\,776;$

$$\therefore AB = 83° 48'{\cdot}8 \equiv 5028'{\cdot}8.$$

Thus the great circle distance between $A$ and $B$ is $83° 48'{\cdot}8$ or $5028{\cdot}8$ nautical miles. To the nearest minute of arc, $AB = 83° 49'$.

(ii) *Calculation of* $PAB$. By formula **A**,

$$\cos PB = \cos AB \cos PA + \sin AB \sin PA \cos PAB.$$

In this equation, all three sides $PB$, $AB$, $PA$ are now known and hence we can derive $P\hat{A}B$. In this instance simple geometrical considerations show that $P\hat{A}B$ is less than $90°$ and consequently the sine-formula **B** can be used without ambiguity; the appropriate equation is

$$\sin PAB = \frac{\sin APB . \sin PB}{\sin AB},$$

all the quantities on the right-hand side being now known. However, for purposes of illustration, we shall calculate $P\hat{A}B$ by means of formula (11). Denote $AB$ by $p$, $PB$ by $a$ and $PA$ by $b$; then

$$2s = p + a + b = 83° 49' + 53° 13' + 65° 42' = 202° 44'.$$

Hence $s = 101° 22'$; $s - p = 17° 33'$; $s - b = 35° 40'$.

In this instance, formula (11) is written

$$\sin \frac{A}{2} = \sqrt{\frac{\sin (s-b) \sin (s-p)}{\sin b \sin p}}.$$

$$\begin{aligned}
\log\sin(s-b) &\equiv \log\sin\ 35°\,40' \quad \bar{1}\cdot76\,572\\
\log\sin(s-p) &\equiv \log\sin\ 17°\,33' \quad \bar{1}\cdot47\,934\\
\log\operatorname{cosec} b &\equiv \log\operatorname{cosec}\ 65°\,42' \quad 0\cdot04\,029\\
\log\operatorname{cosec} p &\equiv \log\operatorname{cosec}\ 83°\,49' \quad 0\cdot00\,253\\
\therefore\ \log\sin^2\frac{A}{2} &= \bar{1}\cdot28\,788\\
\therefore\ \log\sin\frac{A}{2} &= \bar{1}\cdot64\,394\\
\therefore\ \frac{A}{2} &= 26°\,8'\\
\therefore\ A &= 52°\,16'.
\end{aligned}$$

(iii) *Calculation of the most northerly latitude reached by the great circle AB.* Let $C$ be the most northerly point on $AB$ (Fig. 9). Then it is evident that the parallel of latitude through $C$ will touch the great circle at $C$ and that the meridian $PC$ will be perpendicular to the great circle $AB$ at $C$. Thus $P\hat{C}A$ and $P\hat{C}B$ are each 90°. In the triangle $PAC$, we now know $PA$, $P\hat{A}C$ and $P\hat{C}A$ and it is required to find $PC$. Clearly, formula B can be used; it is

$$\frac{\sin PC}{\sin PAC} = \frac{\sin PA}{\sin PCA},$$

and, since $P\hat{C}A = 90°$, we obtain

$$\sin PC = \sin PA \sin PAC$$

$$\begin{aligned}
\log\sin PA &\equiv \log\sin\ 65°\,42' \quad \bar{1}\cdot95\,971\\
\log\sin PAC &\equiv \log\sin\ 52°\,16' \quad \bar{1}\cdot89\,810\\
\therefore\ \log\sin PC &= \bar{1}\cdot85\,781\\
\therefore\ PC &= 46°\,7'.
\end{aligned}$$

Thus the latitude of $C$ is 43° 53′.

The calculation of the longitude of $C$ is left as an exercise to the reader.

**13.** *The haversine formula.*

Many calculations are appreciably shortened by the use of "haversines". The *haversine* of an angle $\theta$ (written hav $\theta$) is defined by

$$\text{hav}\,\theta = \tfrac{1}{2}(1-\cos\theta) = \sin^2\frac{\theta}{2} \qquad \text{.....(21).}$$

Since $\cos\theta = 1 - 2\sin^2\frac{\theta}{2}$, we have

$$\cos\theta = 1 - 2\,\text{hav}\,\theta \qquad \text{.....(22).}$$

We can now modify formula **A**, which is

$$\cos a = \cos b \cos c + \sin b \sin c \cos A.$$

According to (22) write $(1 - 2 \text{ hav } a)$ for $\cos a$, and $(1 - 2 \text{ hav } A)$ for $\cos A$. Then

$$1 - 2 \text{ hav } a = \cos (b - c) - 2 \sin b \sin c \text{ hav } A.$$

Write $1 - 2 \text{ hav } (b - c)$ for $\cos (b - c)$. Then we obtain

$$\textbf{hav a = hav (b - c) + sin b sin c hav A} \quad \ldots\ldots(23),$$

which is the form of the fundamental formula expressed in terms of haversines.

From the definition in (21), hav $\theta$ is always positive and hav $(-\theta) =$ hav $\theta$.

The haversines and log haversines of angles from 0° to 180° are found in some collections of mathematical tables among which may be mentioned *Inman's Nautical Tables* (J. D. Potter, 156 Minories, London, E. 1), which, in addition to the usual logarithmic and trigonometrical tables (to five figures), contain several other tables of astronomical value.

The calculation of the side $AB$ (Fig. 9) by means of haversines will now be given in order to show the convenience of the method. We write (23) as follows for the triangle $PAB$:

$$\begin{aligned}
\text{hav } AB &= \text{hav } (PA - PB) + \sin PA \sin PB \text{ hav } APB \\
&\equiv \text{hav } (PA - PB) + X \\
\log \text{hav } APB &\equiv \log \text{hav } 100° 57' \quad \bar{1}\cdot 77\,450 \\
\log \sin PA &\equiv \log \sin \ 65° 42' \quad \bar{1}\cdot 95\,971 \\
\log \sin PB &\equiv \log \sin \ 53° 13' \quad \bar{1}\cdot 90\,358 \\
&\therefore \log X = \overline{\bar{1}\cdot 63\,779} \\
&\therefore X = 0\cdot 43\,430 \\
\text{hav } (PA - PB) &\equiv \text{hav } 12° 29' = 0\cdot 01\,182 \\
&\therefore \text{hav } AB = \overline{0\cdot 44\,612} \\
&\therefore AB = 83° 49',
\end{aligned}$$

which agrees with our result on p. 17.

**14.** *Another method.*

When two sides and the contained angle of a triangle are given, the following method is sometimes used when it is required to find the third side *and* one of the remaining angles.

To illustrate the method we shall find $AB$ and $P\hat{A}B$ (Fig. 9). Denote $AB$ by $p$, $PB$ by $a$, $PA$ by $b$ and $A\hat{P}B$ by $P$. Then $a = 53° 13'$, $b = 65° 42'$ and $P = 100° 57'$.

By formulae **A**, **C** and **B**, we have

$$\cos p = \cos a \cos b + \sin a \sin b \cos P \quad \ldots\ldots(24),$$

$$\sin p \cos A = \cos a \sin b - \sin a \cos b \cos P \quad \ldots\ldots(25),$$

$$\sin p \sin A = \sin a \sin P \quad \ldots\ldots(26).$$

Define $d$ (a positive quantity) and $D$ by

$$\cos a = d \cos D \quad \ldots\ldots(27),$$

$$\sin a \cos P = d \sin D \quad \ldots\ldots(28).$$

Hence we can write (24)–(26) as follows:

$$\cos p = d \cos (b - D) \quad \ldots\ldots(29),$$

$$\sin p \cos A = d \sin (b - D) \quad \ldots\ldots(30),$$

$$\sin p \sin A = \sin a \sin P \quad \ldots\ldots(31).$$

(i) From (27) and (28), by division,

$$\tan D = \tan a \cos P \quad \ldots\ldots(32),$$

from which $D$ can be calculated.

(ii) From (30) and (31),

$$\tan A = \frac{\sin a \sin P}{d \sin (b - D)},$$

which, by inserting the value of $d$ given by (28), becomes

$$\tan A = \tan P \sin D \operatorname{cosec} (b - D) \quad \ldots\ldots(33),$$

from which $A$ can be calculated.

(iii) From (29) and (30),

$$\tan p = \tan (b - D) \sec A \quad \ldots\ldots(34),$$

from which $p$ can be calculated.

*The calculations.*

(i)
$$\log \tan a \equiv \log \tan \ 53° 13' \quad 0{\cdot}12\,631$$
$$\log \cos P \equiv \log \cos \ 100° 57' \quad \bar{1}{\cdot}27\,864\,n$$
$$\therefore \ \log \tan D = \overline{\bar{1}{\cdot}40\,495\,n}$$

$\cos P$ is *negative* and we attach the letter $n$ to its logarithm to remind us of this fact. It follows that $\tan D$ is *negative*. We have assumed in formulae (27) and (28) that $d$ is a positive quantity. Then, from the given values of $a$ and $P$, it follows that

cos $D$ is positive and sin $D$ is negative; thus $D$ is in the fourth quadrant, and from the value of log tan $D$ which we have found we obtain

$$D = 360° - 14° 15'·6 = 345° 44'·4.$$

Hence

$$b - D \equiv 65° 42' - 345° 44'·4 = -280° 2'·4 = 79° 57'·6.$$

(ii)

| | | |
|---|---|---|
| log tan $P$ | $\equiv$ log tan 100° 57′ | 0·71 338 $n$ |
| log sin $D$ | $\equiv$ log sin 345° 44′·4 | $\bar{1}$·39 151 $n$ |
| log cosec $(b - D)$ | $\equiv$ log cosec 79° 57′·6 | 0·00 670 |
| | $\therefore$ log tan $A$ = | 0·11 159 |

and, as $A$ is less than 180°, we have

$$P\hat{A}B \equiv A = 52° 16'·9.$$

(iii)

| | | |
|---|---|---|
| log tan $(b - D)$ | $\equiv$ log tan 79° 57′·6 | 0·75 192 |
| log sec $A$ | $\equiv$ log sec 52° 16′·9 | 0·21 340 |
| | $\therefore$ log tan $p$ = | 0·96 532 |

$$\therefore AB \equiv p = 83° 49',$$

agreeing with the previous calculations of $AB$.

15. *The trigonometrical ratios for small angles.*

If $\theta$ is a small angle and *expressed in circular measure*, we have the well-known approximate formulae:

$$\sin\theta = \theta \text{ radians}; \quad \cos\theta = 1; \quad \tan\theta = \theta \text{ radians} \qquad \ldots\ldots(35).$$

Now

$$1 \text{ radian} = 57° 17' 45''$$
$$= 3437\tfrac{3}{4}'$$
$$= 206265'',$$

so that $1'' = \dfrac{1}{206265}$ radian,

and $1' = \dfrac{1}{3438}$ radian, approximately.

Hence, by the first equation of (35), when $\theta$ is successively 1″ and 1′,

$$\sin 1'' = \frac{1}{206265} \qquad \ldots\ldots(36),$$

and

$$\sin 1' = \frac{1}{3438} \qquad \ldots\ldots(37).$$

If $\theta''$ denotes the *number* of seconds of arc in $\theta$ radians, then $\theta = \frac{\theta''}{206265}$ and consequently

$$\sin\theta = \frac{\theta''}{206265},$$

which may be written

$$\sin\theta'' = \theta'' \sin 1'' \qquad \ldots\ldots(38).$$

Similarly,
$$\sin\theta' = \theta' \sin 1' \qquad \ldots\ldots(39),$$

where $\theta'$ is expressed in minutes of arc.

In a similar way, we find

$$\tan\theta'' = \theta'' \sin 1''.$$

In spherical astronomy, certain angles are frequently expressed in terms of hours, minutes and seconds of time, according to the following relations:

$$24 \text{ hours} = 360^\circ;\ 1^h = 15^\circ;\ 1^m = 15' \text{ and } 1^s = 15'' \qquad \ldots\ldots(40).$$

Thus we obtain the approximate formulae

$$\sin 1^m = \sin 15' = 15 \sin 1' \qquad \ldots\ldots(41),$$

$$\sin 1^s = \sin 15'' = 15 \sin 1'' \qquad \ldots\ldots(42).$$

If $H$ is a small angle, which, when expressed in minutes of time, will be denoted by $H^m$, then

$$\sin H = H^m \sin 1^m = 15H^m \sin 1' \qquad \ldots\ldots(43).$$

Similarly, if we express $H$ in terms of seconds of time, we have

$$\sin H = H^s \sin 1^s = 15H^s \sin 1'' \qquad \ldots\ldots(44).$$

These results will be of use in subsequent chapters.

**16.** *Delambre's and Napier's analogies.*

For reference, we give the following formulae, originally due to Delambre, and known as Delambre's analogies:

$$\sin\tfrac{1}{2}c \sin\tfrac{1}{2}(A-B) = \cos\tfrac{1}{2}C \sin\tfrac{1}{2}(a-b) \ldots\ldots(45),$$

$$\sin\tfrac{1}{2}c \cos\tfrac{1}{2}(A-B) = \sin\tfrac{1}{2}C \sin\tfrac{1}{2}(a+b) \ldots\ldots(46),$$

$$\cos\tfrac{1}{2}c \sin\tfrac{1}{2}(A+B) = \cos\tfrac{1}{2}C \cos\tfrac{1}{2}(a-b) \ldots\ldots(47),$$

$$\cos\tfrac{1}{2}c \cos\tfrac{1}{2}(A+B) = \sin\tfrac{1}{2}C \cos\tfrac{1}{2}(a+b) \ldots\ldots(48).$$

These formulae are easily derived from the principal formulae already discussed in the previous pages.

Taking these equations in pairs, we obtain Napier's analogies:

$$\tan \tfrac{1}{2}(a+b) = \frac{\cos \tfrac{1}{2}(A-B)}{\cos \tfrac{1}{2}(A+B)} \tan \tfrac{1}{2}c \quad \ldots\ldots(49),$$

$$\tan \tfrac{1}{2}(a-b) = \frac{\sin \tfrac{1}{2}(A-B)}{\sin \tfrac{1}{2}(A+B)} \tan \tfrac{1}{2}c \quad \ldots\ldots(50),$$

$$\tan \tfrac{1}{2}(A+B) = \frac{\cos \tfrac{1}{2}(a-b)}{\cos \tfrac{1}{2}(a+b)} \cot \tfrac{1}{2}C \quad \ldots\ldots(51),$$

$$\tan \tfrac{1}{2}(A-B) = \frac{\sin \tfrac{1}{2}(a-b)}{\sin \tfrac{1}{2}(a+b)} \cot \tfrac{1}{2}C \quad \ldots\ldots(52).$$

## EXERCISES

**1.** In the spherical triangle $ABC$, $C = 90°$, $a = 119° 46' 36''$ and $B = 52° 25' 38''$. Calculate the values of $b$, $c$ and $A$.

[Ans. $48° 26' 49''$, $109° 14' 0''$ and $113° 10' 46''$.]

**2.** In the triangle $ABC$, $a = 57° 22' 11''$, $b = 72° 12' 19''$ and $C = 94° 1' 49''$. Calculate the values of $c$, $A$ and $B$.

[Ans. $83° 46' 32''$, $57° 40' 45''$ and $72° 49' 50''$.]

**3.** In the triangle $ABC$, $c = 90°$, $B = 62° 20' 42$ and $a = 136° 19' 0''$. Calculate the values of $A$, $C$ and $b$.

[Ans. $139° 46' 13''$, $69° 14' 45''$ and $71° 18' 9''$.]

**4.** Two ships $X$ and $Y$ are steaming along the parallels of latitude 48° N and 15° S respectively, in such a way that at any given moment the two ships are on the same meridian of longitude. If the speed of $X$ is 15 knots,* find the speed of $Y$.

**5.** $A$ and $B$ are two places on the earth's surface with the same latitude $\phi$; the difference of longitude between $A$ and $B$ is $2l$. Prove that (i) the highest latitude reached by the great circle $AB$ is $\tan^{-1}(\tan \phi \sec l)$, and (ii) the distance measured along the parallel of latitude between $A$ and $B$ exceeds the great circle distance $AB$ by

$$2 \operatorname{cosec} 1' [l \cos \phi - \sin^{-1}(\sin l \cos \phi)] \text{ nautical miles.}$$

**6.** The most southerly latitude reached by the great circle joining a place $A$ on the equator to a place $B$ in south latitude $\phi$ is $\phi_1$. Prove that the difference of longitude between $A$ and $B$ is $90° + \cos^{-1}(\tan \phi \cot \phi_1)$.

**7.** The positions of $A$ and $B$ are respectively: Lat. 39° 20′ S, Long. 110° 10′ E and Lat. 44° 30′ S, Long. 46° 20′ W. Show that, if a ship steams from $A$ to $B$ by the shortest possible route without crossing the parallel of 62° S, the distance steamed is 5847·6 nautical miles.

* The *knot* is the unit of speed in use at sea; it is 1 nautical mile per hour.

**8.** If the elements $a, b, c, A, B, C$ of a spherical triangle receive increments $da, \ldots dC$, show that, if

$$K = \frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c},$$

$$da = \cos C.db + \cos B.dc + K \sin b \sin c.dA,$$

$$db = \cos A.dc + \cos C.da + K \sin c \sin a.dB,$$

$$dc = \cos B.da + \cos A.db + K \sin a \sin b.dC,$$

$$dA = -\cos c.dB - \cos b.dC + \frac{1}{K} \sin B \sin C.da,$$

$$dB = -\cos a.dC - \cos c.dA + \frac{1}{K} \sin C \sin A.db,$$

$$dC = -\cos b.dA - \cos a.dB + \frac{1}{K} \sin A \sin B.dc.$$

**9.** Prove that two sides of a spherical triangle are equal if and only if their opposite angles are equal.

$ABC$ is an equilateral spherical triangle in which small displacements are made, in the sides and angles, of such a nature that the triangle remains equilateral. Prove that

$$\frac{da}{dA} = \cos \frac{A}{2} \cot \frac{a}{2}.$$

[*Glas.* 1967.]

CHAPTER II

# THE CELESTIAL SPHERE

## 17. *Introduction.*

In Chapter I we have seen that positions on the surface of the earth are completely specified by reference to two principal great circles, the Greenwich meridian and the equator. The principle of specifying positions on the celestial sphere is fundamentally similar and there are several methods depending on the particular great circles chosen as the principal circles. These methods will now be described.

## 18. *Altitude and azimuth.*

Let *O*—the observer on the surface of the earth (supposed spherical)—be the centre of the celestial sphere (Fig. 10). Let *Z*

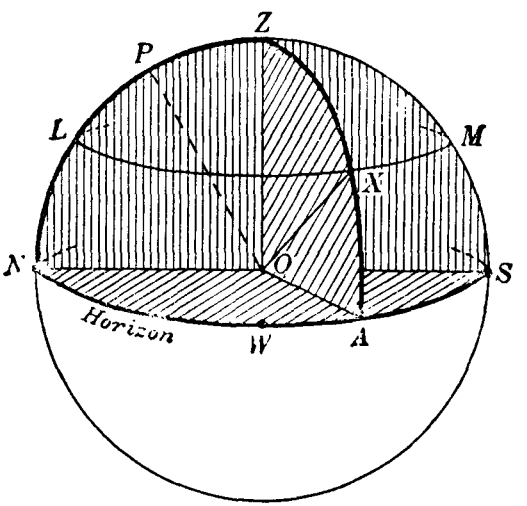



Fig. 10.

(the *zenith*) be the point on the celestial sphere vertically overhead—its direction can be defined by means of a plumb-line. *OZ* is thus the continuation of the straight line joining the earth's centre to *O*. The plane through *O* at right angles to *OZ* is the

plane of the horizon, cutting the celestial sphere in the great circle $NAS$, called the *celestial horizon* or simply the horizon. Thus, in Fig. 10, the horizon divides the celestial sphere into two hemispheres, of which the upper is the visible hemisphere, the lower being hidden from the observer by the earth. Let $X$ be the position of a star on the celestial sphere at a given moment. Any great circle drawn through $Z$ is called a *vertical circle*; in particular, the vertical circle in Fig. 10 passing through $X$ is $ZXA$. In the plane of $ZXA$, the angle $AOX$ or the great circle arc $AX$ is called the *altitude*, which will be denoted by $a$. Since $OZ$ is perpendicular to the plane of the horizon, the great circle arc $ZA$ is 90°; hence $ZX = 90° - a$. $ZX$ is called the *zenith distance* (Z.D.) of the star $X$ and will be denoted by $z$. Thus

$$z = 90° - a \qquad \text{.....(1).}$$

Let $LXM$ be a small circle through $X$ parallel to the horizon; it is called a *parallel of altitude* and is such that all heavenly bodies, whose positions at a given instant lie on this small circle, have the same altitude and also, by (1), the same zenith distance as $X$. Thus if the altitude or zenith distance of a star is given, the parallel of altitude on which it must lie can be definitely specified. To define its position completely on the celestial sphere, the particular vertical circle on which it lies must also be specified. This is done as follows.

Let $OP$ be parallel to the axis about which the earth spins. If the latitude of the observer is north (as in Fig. 10), the position $P$ is called the *north celestial pole*, or simply the north pole. We are not directly conscious of the earth's rotation, but the effect is shown in the apparent rotation of the celestial sphere. The stars thus appear to travel across the sky and their altitudes and directions are continually changing. In the northern hemisphere there is, however, one star, visible to the naked eye, which appears to change very little. This is Polaris, or the north pole star, whose direction in the sky is very nearly that given by $OP$. If there happened to be a star exactly situated at $P$ on the celestial sphere, its altitude and direction would be invariable throughout a night. We define the vertical circle through $P$, that is $ZPN$ (which cuts the horizon in $N$), as the principal vertical circle and the point $N$ as the *north point of the horizon*

The point $S$ on the horizon exactly opposite to $N$ is the south point; the west ($W$) and east ($E$) points* have directions at right angles to the directions of $N$ and $S$ ($E$ is not shown in Fig. 10). The points $N$, $E$, $S$ and $W$ are called the *cardinal points.*

We now specify the position of a star $X$ on the celestial sphere at a given moment by reference to the horizon and the principal vertical circle $ZPN$. If the star is in the western part of the celestial sphere (as in Fig. 10), the spherical angle $PZX$ (which is formed by the principal vertical circle and the vertical circle through $X$) or the great circle arc $NA$ is called the *azimuth* ($W$). If the star is in the eastern part of the celestial sphere, as in Fig. 11, the angle $PZX$ or the arc $NB$ is the azimuth ($E$). Thus at any instant the position of a heavenly body on the celestial sphere can be described completely by reference to the horizon and the north point of the horizon in terms of altitude and azimuth ($E$ or $W$) or, alternatively, in terms of zenith distance and azimuth. When the azimuth is 90° E or 90° W, the star is said to be on the *prime vertical*, which is thus the vertical circle through the east point $E$ or the west point $W$.

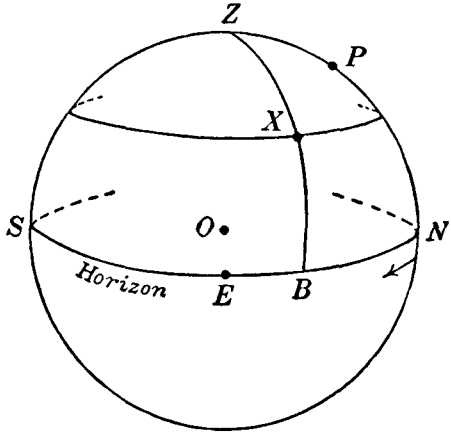


Fig. 11.

Since in Figs. 10 and 11 the angle $POZ$ (or the great circle arc $PZ$) is equivalent to the angle between the radius of the earth which passes through the observer's position and the earth's axis, then $P\hat{O}Z$ (or $PZ$) is equal to the colatitude of the observer or

$$PZ = 90° - \phi \qquad \ldots\ldots(2),$$

where $\phi$ is the observer's latitude. Also $PN = 90° - PZ = \phi$; hence the altitude of the pole is equal to the observer's latitude.

* The positions of $W$ and $E$ relatively to $N$ and $S$ are obtained from the consideration that, if the observer faces north, the west point is towards his left hand and the east point towards his right hand.

**19.** *Declination and hour angle.*

As in the preceding section, suppose that the celestial sphere is drawn for an observer $O$ in latitude $\phi$, showing the horizon, the zenith $Z$ and the north pole $P$ (Fig. 12). The great circle $RWT$ whose plane is perpendicular to $OP$ is the *celestial equator* and its plane, clearly, is parallel to that of the earth's equator. The celestial equator and the horizon intersect in two points $W$ and $E$. Now $Z$ is the pole of the great circle $NWS$ and $P$ is the pole of the great circle $RWT$; hence $W$ is 90° from both $Z$ and $P$

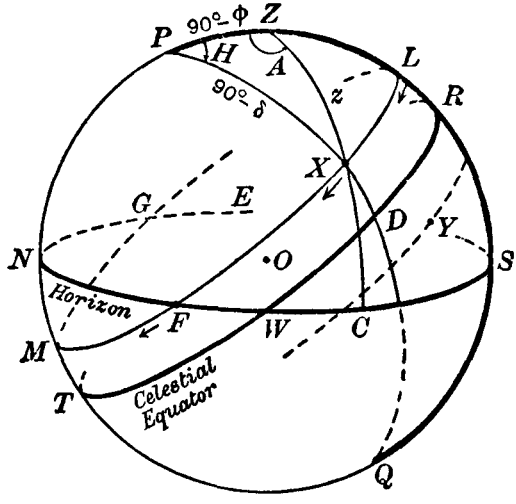



Fig. 12.

and therefore is 90° from all points on the great circle through $Z$ and $P$. In other words, $W$ is the pole of the great circle $NPZSQ$; hence $NW = 90°$ and $WS = 90°$. Similarly $EN = 90°$ and $ES = 90°$. Hence $W$ and $E$ are the two remaining cardinal points, $N$ and $S$ having been previously defined explicitly.

As already stated, the rotation of the earth results in an apparent rotation of the celestial sphere from east to west about $OP$. It follows that, as the stars are at such great distances from the earth, the angle between the straight line joining the observer at $O$ to any particular star and the straight line $OP$ (parallel to the earth's axis) remains unaltered. If we consider a star $X$, the earth's rotation makes it appear to describe a small circle $LXM$,

parallel to the celestial equator, in the direction indicated by the arrows in Fig. 12. Let $PXDQ$ be the semi-great circle through $X$ and the poles of the celestial sphere. Then the arc $DX$ is called the *declination* of the star and is north declination if the star is between the celestial equator and the north pole $P$ (as for the star $X$). The star's declination is south (as for $Y$) when it is between the celestial equator and the south pole $Q$. Declination is thus analogous to latitude as defined for points on the earth's surface. Denote the declination of $X$ by $\delta$; then $DX = \delta$ and $PX = 90° - \delta$. $PX$ is called the *north polar distance* (N.P.D.) of the star. It is convenient to treat declination as an algebraic quantity, so that the various formulae to be derived will hold equally for north and south declinations. North declinations carry the positive sign (+) and south declinations the negative sign (−). Thus the formula for north polar distance, viz. N.P.D. $= 90° - \delta$, is applicable to all stars, whatever their declinations may be.

The declination of a star being known, we can thus specify a small circle, called the *parallel of declination*, on which it must lie. To fix completely its position on the celestial sphere at any moment we require another great circle of reference. This is the semi-great circle $PZRSQ$, called the *observer's meridian*. When the star is at $L$ on the observer's meridian, it is said to *transit* or *culminate*, and it is clear from Fig. 12 that its altitude (that is $SL$) is then greatest and its zenith distance $ZL$ is least. Thereafter, owing to the earth's rotation, it moves along the small circle $LFM$ crossing the horizon at $F$ where it is said to *set*; its altitude at $F$ is of course 0° and its zenith distance is 90°. During an interval of time depending on its declination, the star is below the horizon, reaching its maximum depression below the horizon at $M$; eventually it reaches the horizon at $G$ where it is said to *rise*. Its altitude gradually increasing, it returns after an interval, equivalent to that in which the earth makes a complete rotation about its axis, to the observer's meridian at $L$. At any moment the star's position on the parallel of declination is specified by the angle at $P$ between the observer's meridian and the meridian ($PXQ$) through the star at this time; this angle is $RPX$ or $ZPX$ or the arc $RD$ on the equator. This angle, denoted by $H$, is called the *hour angle* and is measured from the

observer's meridian *westwards* from 0° (at $L$) to 360° (when the star again returns to the observer's meridian) or, as is more usual, from $0^h$ to $24^h$. We can express this in a slightly different way. When the star is in transit, its meridian coincides with the observer's meridian; thereafter, the star's meridian moves steadily westwards and, when it has made a complete circuit of the celestial sphere, it has described an angle of 360° or $24^h$ with reference to the observer's meridian. From Fig. 12 it is seen that if the star is *west* of the observer's meridian, that is, if the

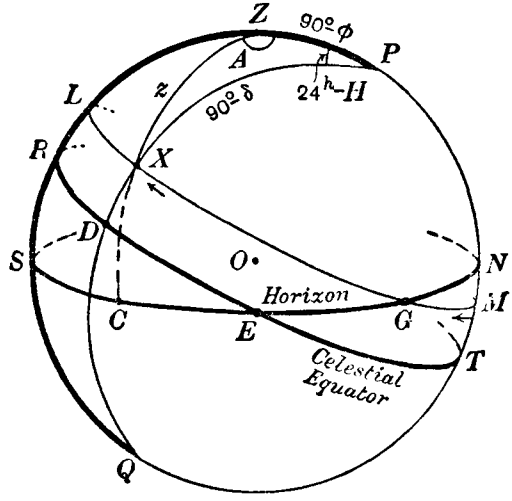



Fig. 13.

azimuth is west, its hour angle is between 0° and 180°, that is, between $0^h$ and $12^h$. Similarly, if the star is *east* of the meridian (azimuth east)—as in Fig. 13—the hour angle is between $12^h$ and $24^h$. We thus have the rules:

*If the star's azimuth is west, the hour angle is between* $0^h$ *and* $12^h$ (*and* vice versa); *if the star's azimuth is east, the hour angle is between* $12^h$ *and* $24^h$.

**20.** *Diagram for the southern hemisphere.*

The diagrams described so far in this chapter refer to the celestial sphere for an observer in north latitude. We shall now describe the corresponding diagrams for an observer in the

southern hemisphere. In Fig. 14, we shall place the observer's zenith as in the previous diagrams. The celestial horizon is then as indicated. In the southern hemisphere, the south celestial pole $Q$ is above the horizon. Then, if $\phi$ denotes the southern latitude of the observer, $QZ = 90° - \phi$. The principal vertical circle is now $ZQS$, intersecting the horizon in the *south* point $S$. The north point $N$ can then be placed in the diagram. The celestial equator and the horizon intersect in the west and east points $W$ and $E$ (the latter is not shown in Fig. 14) according to

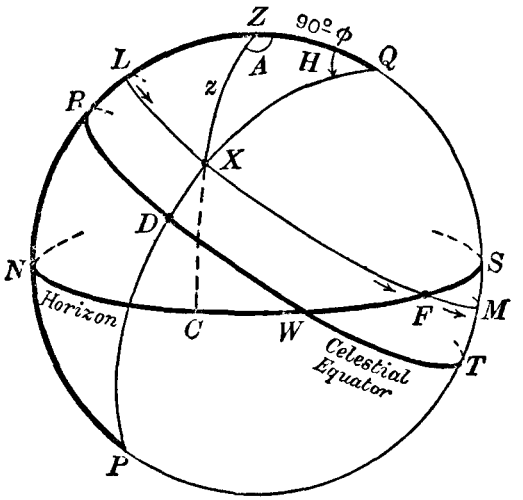



Fig. 14.

the rule in the footnote to page 27. Consider a star $X$ with *south* declination. Owing to the earth's rotation it will describe a small circle $LXM$, parallel to the celestial equator and lying between the celestial equator and the south pole $Q$. At $L$, the star will have its greatest altitude—it is *then* on the observer's meridian, which is the semi-circle $QZRNP$. In consequence of the earth's rotation the star will move from the observer's meridian *westwards*, that is, in the direction $LXM$, as indicated by the arrows in the diagram. The angle $ZQX$ is the hour angle measured, as before, from $0^h$ to $24^h$ westwards from the observer's meridian. $QZX$ is the azimuth; in this instance it is west. If $\delta$ is the (negative) declination of the star, then $DX = -\delta$ and

$QX = 90° + \delta$. The other parts of the spherical triangle $QZX$ are: $QZ = 90° - \phi$, $ZX = z$ (the zenith distance), $QZX = A$ (the azimuth) and $ZQX = H$ (the hour angle). When the star's azimuth is *west*, the hour angle is between $0^h$ and $12^h$. When the star's azimuth is east, the appropriate diagram can be similarly drawn; this is left as an exercise to the student; it will then be found that the hour angle is between $12^h$ and $24^h$. The rules stated at the end of section 19 are seen to hold for southern as well as northern latitudes.

21. *Circumpolar stars.*

Consider the celestial sphere for an observer in northern latitude $\phi$ (Fig. 15). The parallels of declination are drawn for two stars $X$ and $Y$, both of which are always above the horizon and consequently do not set. Such stars are called circumpolar

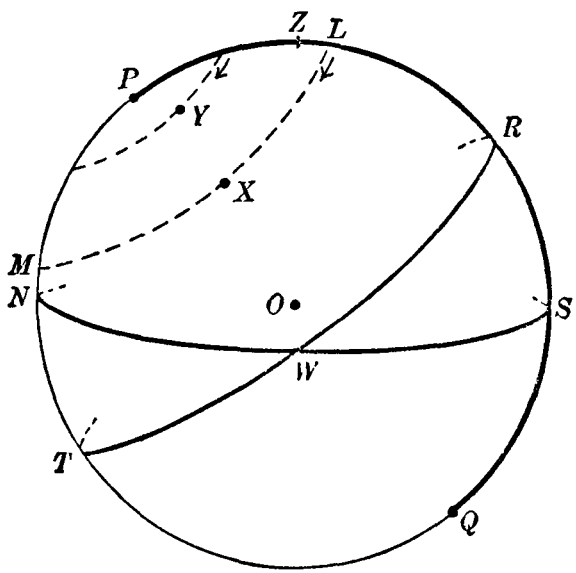



Fig. 15.

stars. It is readily seen from the figure that the condition that a star should not set is: $PM$ must be less than $PN$; that is, the north polar distance must be less than the latitude, or, in other words, the declination must be greater than the colatitude.

When the star $X$ is on the observer's meridian at $L$, it is at *upper culmination* or in *transit*; when the star reaches $M$, it is at *lower culmination*. The expressions "culmination above pole" and "culmination below pole" are frequently used. At upper culmination, the star's zenith distance is $ZL$ or $(PL - PZ)$, that is, $\phi - \delta$. At lower culmination, the star's zenith distance is $ZM$ or $(ZP + PM)$, that is, $180° - (\phi + \delta)$. When $\delta = \phi$, the star's upper culmination occurs in the zenith. When $\delta > \phi$, the upper culmination occurs between $P$ and $Z$, as for the star $Y$; then the azimuth does not exceed 90°, as can be readily inferred from the diagram. Southern circumpolar stars can be considered in the same way.

22. *The standard or geocentric celestial sphere.*

In the previous sections, the declination of a star on the celestial sphere whose centre is the observer has been defined. As the stars are at distances almost infinitely great compared with the dimensions of the earth, the star's declination or polar distance so defined is independent of the observer's position on the surface of the earth, as may be readily seen from Fig. 16. (It is more convenient for our present purpose to deal with the star's north polar distance than with its declination.) In Fig. 16, $P_1CQ_1$ is the earth's axis of rotation, $C$ being the centre of the earth; $O$ is the observer and $COZ$ the direction of the zenith at $O$; $OP$ is parallel to $CP_1$ and the direction of a star that is transiting at $O$ is $OX$. By definition, the north polar distance of the star for an observer at $O$ is $P\hat{O}X$. If $CY$ is drawn parallel to $OX$, then $CY$ is the direction of the star with reference to $C$, the earth's centre. Thus $P_1\hat{C}Y = P\hat{O}X$; in other words the north polar

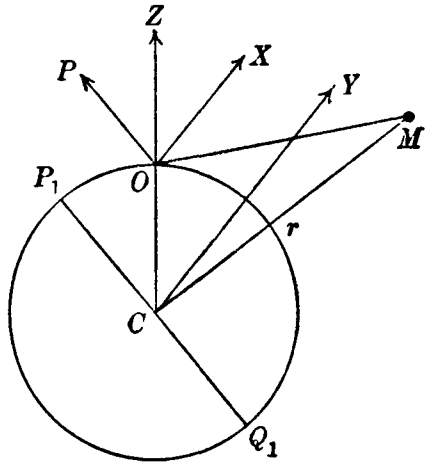


Fig. 16.

distance of the star (and consequently its declination) is the same on the celestial sphere centred at $O$ (or any other position on the earth's surface) as it is on the celestial sphere centred at $C$. But when a comparatively near body such as the moon, or sun, or a planet is observed, the definition of north polar distance (and therefore of declination) previously given is dependent on the particular position of the observer on the earth. Thus if $M$ is the moon (Fig. 16) at the distance $r$ from the centre of the earth, it is evident that $P\hat{O}M = P_1\hat{C}M + O\hat{M}C$; also $O\hat{M}C$ clearly depends on the position of $O$, whereas $P_1CM$ is entirely independent

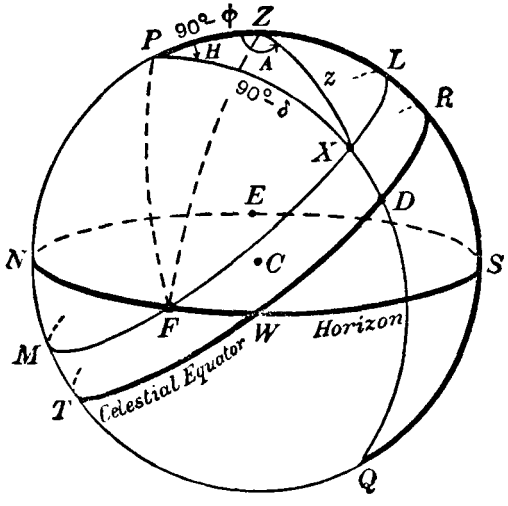



Fig. 17.

of $O$. $P_1\hat{C}M$ is defined as the north polar distance of $M$ which is thus the angle between the earth's axis and the straight line joining the earth's centre to the heavenly body. This definition is entirely general and is applicable to every heavenly body. Accordingly, the centre of the standard celestial sphere (or the geocentric celestial sphere, as it may be called) is taken to be at $C$, the earth's centre (Fig. 17). $CZ$ is the direction of the observer's zenith, the diameter $QCP$ is coincident with the earth's axis, $NWSE$ is the celestial horizon (the great circle whose plane is perpendicular to $CZ$), and $RWTE$ is the celestial equator (its plane is coincident with the plane of the earth's equator). The

arc $PX$ is the north polar distance of the heavenly body according to the definition just given and $DX$ is the declination $\delta$ (N.P.D. $= 90° - \delta$). The observer's meridian is $PZRSQ$, the zenith distance of the heavenly body is $ZX$ (denoted by $z$) and the azimuth $A$ ($P\hat{Z}X$) and the hour angle $H$ ($Z\hat{P}X$) are as described previously. The declinations of the principal heavenly bodies (moon, sun, planets and the brightest stars) are tabulated in the *Astronomical Ephemeris* (the American and British publication) and in the other national ephemerides.

Hereafter, the celestial sphere will be assumed to be as in Fig. 17, that is, centred at $C$, the earth's centre.

**23.** *Solution of the spherical triangle PZX.*

We shall consider two common problems associated with the triangle $PZX$.

(i) Given the observer's latitude $\phi$, the declination $\delta$ and hour angle $H$ of the heavenly body, to calculate its zenith distance and azimuth. By formula **A** (the cosine formula), we have, since two sides $PZ$ and $PX$ and the contained angle $ZPX$ are given (Fig. 17),

$$\cos ZX = \cos PZ \cos PX + \sin PZ \sin PX \cos ZPX,$$

or $$\cos z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H \quad \ldots\ldots(3).$$

Thus $z$ can be calculated directly from (3) or by means of the haversine formula (section 13), which in this instance can be written $$\text{hav } z = \text{hav } (\phi - \delta) + \cos \phi \cos \delta \text{ hav } H \ldots\ldots(4).$$

Again, by **A**,

$$\cos PX = \cos PZ \cos ZX + \sin PZ \sin ZX \cos PZX,$$

or $$\sin \delta = \sin \phi \cos z + \cos \phi \sin z \cos A \quad \ldots\ldots(5),$$

from which the azimuth $A$ can be calculated. In the haversine form (5) may be written

$$\cos \phi \cos a \text{ hav } A = \text{hav } (90° - \delta) - \text{hav } (\phi - a) \quad \ldots(6),$$

where $a$ is the altitude.

(ii) Given the observer's latitude $\phi$, the star's zenith distance and azimuth, to calculate the star's declination and hour angle. We are given $\phi$, $z$ and $A$; hence, by (5), we can calculate the declination. Either equation (3) or (4) is available for calculating the hour angle $H$. Thus from (3)

$$\cos H = \cos z \sec \phi \sec \delta - \tan \phi \tan \delta \quad \ldots\ldots(7).$$

Consider now the spherical triangle $PZX$ in Fig. 13. The angle $PZX$ is the azimuth (east). Remembering that the hour angle is measured at the pole from the observer's meridian *westwards*, we see that $Z\hat{P}X = 24^h - H$. The solution of the triangle proceeds as before.

**24.** *Right ascension and declination.*

In the hour angle and declination method of specifying a star's position on the celestial sphere only one co-ordinate, namely declination, remains constant as the star travels across the sky, whereas the hour angle increases uniformly from $0^h$ to $24^h$. But

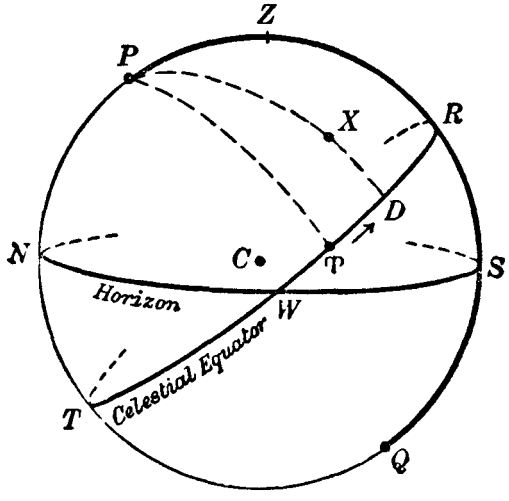



Fig. 18.

the positions of the stars on the celestial sphere may be likened to the positions of fixed points on the surface of the earth and can therefore be specified with reference to the celestial equator and any particular star on the equator. For example, in Fig. 18, let $\Upsilon$ be an equatorial star and $X$ any other star; let the meridian through $X$ cut the celestial equator in $D$. As the stars pass across the sky we know in particular that the declination of $X$, that is, $DX$, remains constant and that the relative configuration of the stars also remains constant. It follows that $\Upsilon D$ is constant; in other words, that the angle between the meridians of $\Upsilon$ and $D$ remains constant. We may regard $\Upsilon$ as a reference

point on the celestial equator; with respect to $\Upsilon$ and the celestial equator, we can clearly specify the position of the star $X$ by means of the great circle arc $\Upsilon D$ and the declination $DX$. The reference point chosen in practice is called the *vernal equinox* or the *first point of Aries*, and it is convenient to regard the position of $\Upsilon$ as specified by a particular star in the sky. Later we shall define $\Upsilon$ more precisely. The arc $\Upsilon D$ or $\Upsilon\hat{P}X$ is called the *right ascension* (R.A.) of the star $X$ (denoted by $\alpha$) and is measured *eastwards* from $\Upsilon$ from $0^h$ to $24^h$ (in the direction of the arrow near $\Upsilon$). This direction is opposite to that in which hour angle is measured. From Fig. 18, we see that $R\Upsilon = RD + \Upsilon D$. Now $RD$ (or $R\hat{P}X$) is the hour angle $H$ of $X$ and $R\Upsilon$ is the hour angle of $\Upsilon$. The hour angle of $\Upsilon$ is called the *sidereal time* (S.T.). We have, accordingly,

$$\text{Sid. time} = \text{H.A. } X + \text{R.A. } X \qquad \ldots\ldots(8),$$

or

$$\text{S.T.} = H + \alpha \qquad \ldots\ldots(9).$$

When $\Upsilon$ is on the observer's meridian, the hour angle of $\Upsilon$ is $0^h$, that is to say, the sidereal time is $0^h$. When $\Upsilon$ is next on the observer's meridian, an interval of $24^h$ of *sidereal time* has elapsed. This interval is, of course, the same as that required for the complete rotation of the earth about its axis and it is called a *sidereal day*. The rotating earth is, in fact, the standard time-keeper.

**25.** *The earth's orbit.*

The earth is a planet revolving around the sun in an elliptical path or orbit, the sun being situated at a focus $S$ of the ellipse (Fig. 19). This is Kepler's first law of planetary motion. The time required for the earth to make a complete revolution of its orbit is a *year*. As the earth progresses in its orbit, the direction of the earth, as viewed from the sun, is continually altering; the angular velocity is, however, not uniform. Since our observations are made from the earth, then relative to the earth the sun appears to describe an elliptical orbit around the earth. In Fig. 20, $C$ is the centre of the earth and the ellipse represents the *apparent* orbit of the sun relative to the earth. The sequence of positions of the sun, namely $a, e, f, b, g$ in this orbit, corresponds to the sequence of positions $A, E, F, B, G$ of the earth in its orbit round the sun (Fig. 19). In the course of the year, the sun thus

appears to make a complete circuit of the heavens against the background of the stars. The plane of the orbit is called the plane of the ecliptic, and the great circle in which this plane intersects the celestial sphere, whose centre is the earth's centre $C$, is called the *ecliptic*. In Fig. 21, let $C$ be the centre of the celestial sphere on which the celestial equator $TTR$ and the

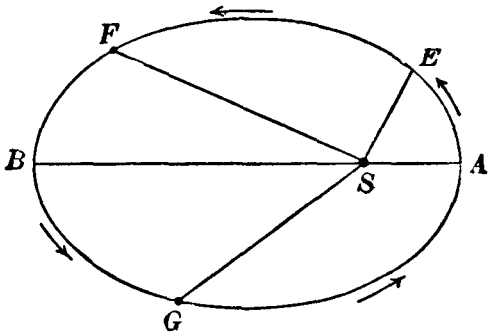



Fig. 19.

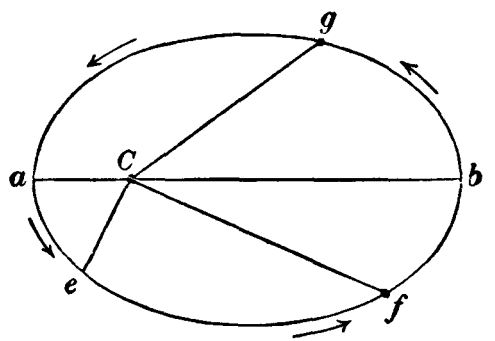



Fig. 20.

north pole $P$ are drawn. We may imagine that the stars can be viewed from the centre of the earth, that is, from $C$, and accordingly they will occupy definite positions on the celestial sphere in Fig. 21. With reference to the stars, the plane of the ecliptic will have a definite position and, consequently, the ecliptic will be a particular great circle, which is found from observations to be inclined at an angle of about $23\frac{1}{2}°$ to the celestial equator. In Fig. 21, $\Upsilon TMU$ represents the ecliptic and

its inclination to the celestial equator is $M\hat{\Upsilon}R$, which is known as the *obliquity of the ecliptic*. Relative to the earth, the sun appears to move on the celestial sphere along the ecliptic—in the direction $Y\Upsilon M$—and twice yearly, at $\Upsilon$ and at $U$, its position on the celestial sphere coincides with the intersections of the ecliptic with the celestial equator. Between $\Upsilon$ and $M$ and between $M$ and $U$ the sun is on the north pole side of the equator; its declination is then north. Similarly between $U$ and $Y$ and

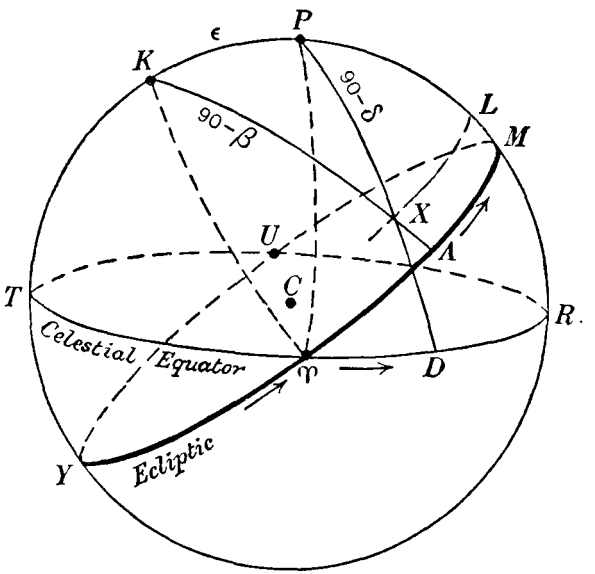



Fig. 21.

between $Y$ and $\Upsilon$ its declination is south. The position $\Upsilon$, at which the sun's declination changes from south to north, is the *vernal equinox*. It is in this way that the reference point $\Upsilon$, from which are measured the right ascensions of the stars, is obtained. Thus if $X$ is a star, its right ascension is $\Upsilon D$ or $\alpha$ measured along the equator from $\Upsilon$ eastwards, and its declination $\delta$ is $DX$. From the diagram it is seen that the right ascension and declination of the sun are both changing continually. When the sun is at $\Upsilon$, its right ascension and declination are both zero (this occurs about March 21—the *vernal equinox*); at $M$ the right ascension is $6^h$ and declination about $23\frac{1}{2}°$ N (this occurs about June 21—

the *summer solstice*); at $U$ the right ascension is $12^h$ and declination $0°$ (this occurs about September 21—the *autumnal equinox*) and at $Y$ the right ascension is $18^h$ and the declination about $23\frac{1}{2}°$ S (this occurs about December 21—the *winter solstice*).

**26.** *Celestial latitude and longitude.*

The position of a heavenly body can be referred to the ecliptic as fundamental great circle and the vernal equinox $\Upsilon$ as principal reference point. In Fig. 21 $K$ is the north pole of the ecliptic and $KXA$ is a great circle arc passing through $X$ and meeting the ecliptic in $A$. The arc $\Upsilon A$, measured from $\Upsilon$ to $A$ along the ecliptic in the *direction of the sun's annual motion*, i.e. eastwards, is called the *longitude* of the heavenly body $X$ and is measured from $0°$ to $360°$ round the ecliptic. The arc $AX$ is the *latitude* and north latitude is considered positive and south negative. If we know the star's right ascension and declination we can obtain its latitude ($\beta$) and its longitude ($\lambda$) from the triangle $KPX$; and *vice versa*. Now $\Upsilon$ is the pole of the great circle $KPMR$; hence $K\hat{P}\Upsilon = 90°$, and since $\Upsilon D = \Upsilon\hat{P}X = \alpha$, then $K\hat{P}X = 90° + \alpha$. Also $P\hat{K}\Upsilon = 90°$, and since $\Upsilon A = \Upsilon\hat{K}X = \lambda$, then $P\hat{K}X = 90° - \lambda$. Also $PX = 90° - \delta$ and $KX = 90° - \beta$. Let $\epsilon$ denote the obliquity of the ecliptic; it is the angle between the radii $CM$ and $CR$; thus the arc $RM = \epsilon$. But $KM = 90°$ and $PR = 90°$; hence $KP = \epsilon$. Applying the formulae **A**, **B** and **C**, we have

$$\cos KX = \cos PX \cos KP + \sin PX \sin KP \cos KPX,$$

$$\sin KX \sin PKX = \sin PX \sin KPX,$$

$$\sin KX \cos PKX = \cos PX \sin KP - \sin PX \cos KP \cos KPX,$$

or

$$\sin \beta = \sin \delta \cos \epsilon - \cos \delta \sin \epsilon \sin \alpha \quad \ldots\ldots(10),$$

$$\cos \beta \cos \lambda = \cos \delta \cos \alpha \quad \ldots\ldots(11),$$

$$\cos \beta \sin \lambda = \sin \delta \sin \epsilon + \cos \delta \cos \epsilon \sin \alpha \quad \ldots\ldots(12).$$

By a similar process, the right ascension $\alpha$ and the declination $\delta$ can be expressed in terms of $\beta$, $\lambda$ and $\epsilon$. The formulae are

$$\sin \delta = \sin \beta \cos \epsilon + \cos \beta \sin \epsilon \sin \lambda,$$

$$\cos \delta \cos \alpha = \cos \beta \cos \lambda,$$

$$\cos \delta \sin \alpha = -\sin \beta \sin \epsilon + \cos \beta \cos \epsilon \sin \lambda.$$

**27.** *Sidereal time.*

Let the earth and the celestial sphere (centred at $C$) be drawn as in Fig. 22; let $g$ denote the position of Greenwich on the earth's surface and $l$ that of any other place. The angle between the meridians $plq$ and $pgq$ is, of course, the longitude (terrestrial) of $l$; in this instance $l$ is west of Greenwich. Produce $Cg$, $Cl$ to meet the celestial sphere in $G$ and $L$. Then $G$ and $L$ are the zeniths of Greenwich and $l$ respectively. If $X$ is the position of a heavenly body on the celestial sphere at a given moment, $G\hat{P}X$ is the hour angle of $X$ for an observer on the Greenwich meridian and $L\hat{P}X$ is the hour angle for an observer on the meridian of $l$. But $G\hat{P}X = L\hat{P}X + G\hat{P}L$ and $G\hat{P}L = g\hat{p}l$; hence

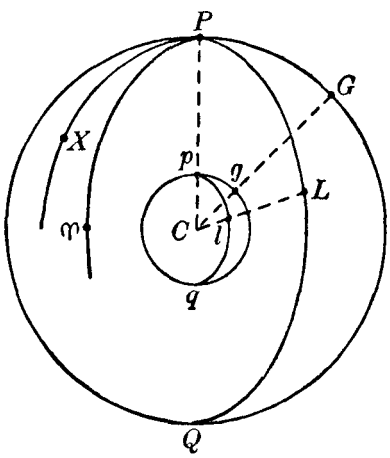


Fig. 22.

H.A. of $X$ at Greenwich = H.A. of $X$ at $l$ + long. (W) of $l$ .....(13).

In this formula we suppose that the longitude of $l$ is expressed in time-measure ($15° = 1^h$; $15' = 1^m$; $15'' = 1^s$). The formula (13) is a general one and it clearly holds for the vernal equinox $\Upsilon$. We thus obtain—since sidereal time is the hour angle of $\Upsilon$—

Sid. time at Greenwich = Sid. time at $l \pm$ long. of $l$ ...(14),

the + sign being taken when $l$ is *west* of Greenwich and the − sign when $l$ is *east* of Greenwich. The sidereal time at $l$ is called the *local sidereal time* (L.S.T.).

**28.** *Mean solar time.*

The sidereal day is an observatory unit of time and is obviously unsuited to the regulation of everyday affairs which in the main are governed according to the position of the sun in the sky. When the sun is on the meridian of a place, it is *apparent noon* there; when the sun is next on the meridian, an *apparent solar day* is said to have elapsed. This interval can be

measured, for example, by means of a clock keeping accurate sidereal time and it is found that an apparent solar day is not constant. We have seen that, relative to the earth, the sun appears to describe an elliptic orbit around the earth and the rate at which its direction in the orbit changes is not constant. It follows that the sun appears to describe the ecliptic at a non-uniform rate; in other words, the sun appears to move somewhat irregularly against the background of the stars. Due to this and also to the fact that it is moving in the ecliptic and not along the celestial equator (the fundamental great circle with which the measurement of hour angle or time is associated) its right ascension does not increase uniformly. The average apparent solar day throughout the year is called a *mean solar day* and it is convenient to define the mean solar day as the interval between two successive transits across the observer's meridian of a fictitious body called the *mean sun*. The mean sun is assumed to move in the *celestial equator* at a uniform rate around the earth. This rate is such that the mean sun completes a revolution in the same time as that required by the sun for a complete circuit of the ecliptic. According to this definition, the right ascension of the mean sun (denoted by R.A.M.S.) increases at a uniform rate.

Now if we regard the mean sun as an ordinary celestial body, then at any given moment, we can assume that it has a particular hour angle (H.A.M.S.) at a given place on the earth's surface. At this moment we shall assume that its right ascension is known; hence by (8) or (9),

$$\text{Sid. time} = \text{H.A.M.S.} + \text{R.A.M.S.} \qquad \ldots\ldots(15).$$

The time shown by a mean time clock, say, at Greenwich at any moment is simply related to the value of H.A.M.S. there, and if the R.A.M.S. is known, (15) forms the basis of comparison between the sidereal and the mean time clocks. The mean sun is related to the true sun according to certain principles which will be discussed in a later chapter. Meanwhile it will be sufficient to state that the difference at any moment between the right ascension of the mean sun and of the true sun can be calculated; this difference is called the *equation of time** (denoted by $E$). We thus have

$$E = \text{R.A.M.S.} - \text{R.A.}\ \odot \qquad \ldots\ldots(16),$$

* In older text-books the equation of time is defined by $E = \text{R.A.}\ \odot - \text{R.A.M.S.}$, but the convention of (16) is generally adopted.

in which R.A. ⊙ denotes the right ascension of the true sun. $E$ can be positive or negative and varies in a complicated manner. The detailed computation of $E$ is discussed in section 91. In Fig. 23, let us suppose that at a given instant the right ascension and declination of the sun (⊙) are known. Let $\Upsilon$ be the vernal equinox at this instant so that $R\hat{P}\Upsilon$ or $R\Upsilon$ is the hour angle of $\Upsilon$, that is, the local sidereal time. If this is known, the position of $\Upsilon$ on the celestial sphere can be definitely specified. The position of the sun can then be indicated on the celestial sphere. $\Upsilon K$ = R.A. ⊙ and $K$ ⊙ is the sun's declination and both of these

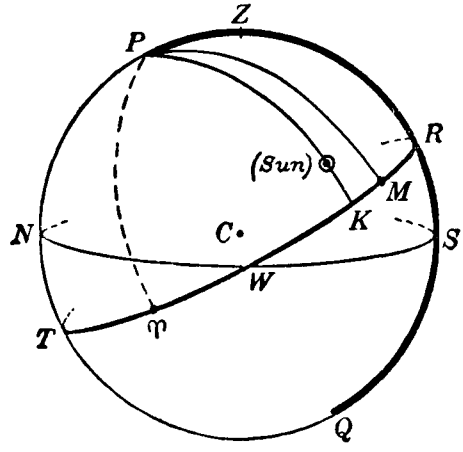



Fig. 23.

are supposed known. Suppose the value of $E$ is positive; then by (16), R.A.M.S. is greater than R.A. ⊙, and if $E$ is known the position of the mean sun $M$ at this instant can be indicated in the diagram. $R\hat{P}M$ or $RM$ is the hour angle of $M$ (H.A.M.S.). It is clear from Fig. 23 that, as $RK = RM + MK$, then

$$\text{H.A. } ⊙ = \text{H.A.M.S.} + E \qquad \ldots\ldots(17),$$

which is an important relation connecting H.A.M.S. and H.A. ⊙, enabling us to calculate the hour angle of the sun (H.A. ⊙) when the other quantities are known. When the mean sun is on the observer's meridian, it is *local mean noon* there. When the mean sun is on the meridian of Greenwich, it is *Greenwich mean noon*. The hour angle of the mean sun at Greenwich will be denoted in this book by G.M.A.T. (*Greenwich mean astronomical time*). When

the mean sun is at $T$—the H.A.M.S. being then $12^h$—it is said to be *mean midnight*. When G.M.A.T. $= 12^h$, it is mean midnight at Greenwich and this is the moment when a new civil day at Greenwich begins. Mean time reckoned from midnight at Greenwich is called *Greenwich Mean Time* (G.M.T.)*, now designated Universal Time (U.T.). It is clear that

$$\text{U.T.} \equiv \text{G.M.T.} = \text{G.M.A.T.} + 12^h \qquad \ldots\ldots(18).$$

Similarly, for any place keeping the mean time appropriate to its meridian, we shall have

$$\text{Local M.T.} = \text{Local M.A.T.} + 12^h \qquad \ldots\ldots(19)$$

$$= \text{H.A.M.S.} \pm 12^h \qquad \ldots\ldots(20).$$

Formula (14) gives the relation between sidereal time at Greenwich and the sidereal time at any place $l$, and it is clear from Fig. 22 and from (18) and (19) that we shall have a similar relation between mean time at Greenwich and the mean time at the place; it is

$$\text{U.T.} \equiv \text{G.M.T.} = \text{Local M.T.} \pm \text{long. of } l \qquad \ldots\ldots(21),$$

the + sign being taken when the longitude of $l$ is west and the − sign when the longitude is east.

Confusion would be inevitable if every place kept the local mean time appropriate to its meridian, and so in small countries a standard mean time is chosen, corresponding to a particular meridian of longitude (the standard meridian), which is in use uniformly throughout the country. In Great Britain, the standard mean time is G.M.T. In extensive countries such as Russia and the United States of America, two or more standard times are in use in zones of longitude; within each zone, a standard time appropriate to a definite meridian within the zone is kept. The standard time, based on a particular meridian, we shall designate *zone time* (Z.T.). This system is, in effect, kept by ships at sea which are generally less troubled by geographical complications. We have, as in (21),

$$\text{U.T.} \equiv \text{G.M.T.} = \text{Z.T.} \pm \text{long. of standard meridian} \qquad \ldots(22).$$

* Before 1925, G.M.T. was used in the almanacs to signify Greenwich mean astronomical time (G.M.A.T.). Beginning with 1925, the time used was G.M.T. ($\equiv$G.C.T.) later superseded, as stated above, by U.T. Recently, for reasons stated in Appendix E (p. 424), U.T. has been replaced in the almanacs by Ephemeris Time (E.T.). The difference between U.T. and E.T. is so slight that we shall use the former generally, unless otherwise stated.

29. *Example.*

We shall use as an illustration the following problem of a common and important type. At a place in longitude 163° 14′ E, it is required to calculate the hour angle of the sun (H.A. ⊙) corresponding to an observation made at zone time $8^h\ 46^m\ 22^s$ on 1975 March 10; the zone time is that of the standard meridian of 165° E ($11^h$ E).

The first step is to derive the U.T. at which the observation was made. We have

| | | |
|---|---|---|
| Zone time | $8^h\ 46^m\ 22^s$ | March 10 |
| Long. of standard meridian | $-11^h$ | |
| U.T. = | $21^h\ 46^m\ 22^s$ | *March 9* |

We subtract $11^h$ from the zone time in accordance with formula (22). (Clearly, we can write the zone time as $32^h\ 46^m\ 22^s$ *March 9.*)

We next find the local mean time (that is, the mean time corresponding to the longitude of the place) by means of (21).

| | | |
|---|---|---|
| U.T. | $21^h\ 46^m\ 22^s$ | March 9 |
| Long. of place (E) | $+10^h\ 52^m\ 56^s$ | |
| Local M.T. = | $32^h\ 39^m\ 18^s$ | March 9 |
| = | $8^h\ 39^m\ 18^s$ | March 10 |

Formula (20) enables us to write down the H.A.M.S. (the hour angle of the mean sun *at the place*); it is

$$\text{H.A.M.S.} = 20^h\ 39^m\ 18^s.$$

The next step is to apply the equation of time to H.A.M.S. From the *Astronomical Ephemeris* it is found, by interpolation, that at U.T. $21^h\ 46^m\ 22^s$, *March 9*, $E = -10^m\ 36^s$.

Thus, by (17), $\text{H.A.}\odot = 20^h\ 39^m\ 18^s - 10^m\ 36^s$,

or $\text{H.A.}\odot = 20^h\ 28^m\ 42^s$.

30. *Hour angle of a heavenly body.*

To calculate the hour angle of any heavenly body ($X$) other than the sun, we proceed as follows. By (8) and (14) we have

$$\text{L.S.T.} = \text{H.A. } X + \text{R.A. } X$$

and $$\text{G.S.T.} = \text{L.S.T.} \pm l,$$

whence $$\textbf{H.A. } X + \textbf{R.A. } X = \text{G.S.T.} \pm l \qquad \ldots(23).$$

In the *Astronomical Ephemeris*, the Greenwich sidereal time is tabulated at $0^h$ U.T. for each day throughout the year. Since, by (15),

$$\text{Sid. time} = \text{H.A.M.S.} + \text{R.A.M.S.},$$

we have

**R.A.M.S. at U.T. $0^h$ for any day = the tabulated Greenwich sidereal time at U.T. $0^h$ for that day $- 12^h$.**

*The* R.A.M.S. *increases uniformly at the rate of* $3^m\ 56^s{\cdot}56$ *per mean solar day or at the rate of* $9^s{\cdot}856$ *per mean solar hour;* by means of this we can calculate the R.A.M.S. for any given U.T.

Tables are given in the almanacs for facilitating this calculation.

The use of the formula (23) is best illustrated by means of an example. It is required to calculate the hour angle of Betelgeuse ($\alpha$ Orionis) at zone time $18^h\ 35^m\ 46^s$ on 1975 January 26, in a place whose longitude is 64° 28′ 49″ W. (Zone $+4^h$: this means that the standard meridian of the zone is $4^h$ W or 60° W.)

| | | |
|---|---|---|
| Zone time | $18^h\ 35^m\ 46^s$ | January 26 |
| Zone | $+\ 4^h$ | |
| U.T. | $22^h\ 35^m\ 46^s$ | January 26 |
| Sid. time correction | $3^m\ 43^s$ | ($3^m\ 56^s{\cdot}56$ per day) |
| | $22^h\ 39^m\ 29^s$ | |
| G.S.T. at $0^h$ U.T. | $8^h\ 18^m\ 39^s$ | *From* A.E. |
| G.S.T. | $30^h\ 58^m\ 08^s$ | |
| Longitude of place (W) | $-\ 4^h\ 17^m\ 55^s$ | |
| L.S.T. | $26^h\ 40^m\ 13^s$ | |
| *Subtract* R.A. of Betelgeuse | $5^h\ 53^m\ 49^s$ | *From* A.E. |
| H.A. of Betelgeuse | $20^h\ 46^m\ 24^s$ | |

**31.** *Rising and setting.*

Consider Fig. 24. The heavenly body $X$ is said to set at $F$, the point where it reaches the horizon. Then the zenith distance is 90°, that is, $ZF = 90°$. Let $H$ be the hour angle of $X$ at setting, so that $ZPF = H$. Also $PF = 90° - \delta$. Let $A$ be the azimuth at setting ($PZF$) and $\phi$ the latitude.

From formula **A**,

$$\cos ZF = \cos PZ \cos PF + \sin PZ \sin PF \cos ZPF,$$

or $$\cos 90° = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H,$$

so that, as $\cos 90° = 0$,

$$\cos H = - \tan \phi \tan \delta \qquad \text{.....(24)},$$

from which the hour angle at setting can be calculated.

Also from **A**,

$$\cos PF = \cos PZ \cos ZF + \sin PZ \sin ZF \cos PZF,$$

or $$\sin \delta = 0 + \cos \phi \cos A,$$

whence $$\cos A = \sin \delta \sec \phi \qquad \text{.....(25)},$$

from which the azimuth at setting can be calculated.

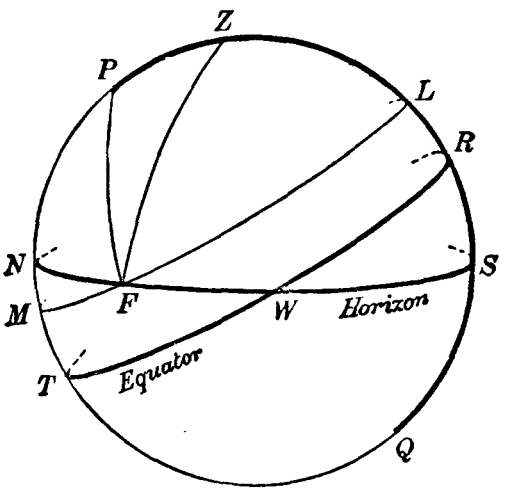



Fig. 24.

In north latitudes, it is seen either from the equations (24) and (25), or from Fig. 24, that if the declination is north the hour angle at setting is between $6^h$ and $12^h$ and that the azimuth is less than 90° (that is to say, the body sets between west and north); and that if the declination is south, the hour angle at setting is between $0^h$ and $6^h$ and the body sets between south and west. The problem as it concerns the rising of a heavenly body can be treated in a similar way. When the observer's latitude is south, the procedure is similar.

If the heavenly body concerned is a star, the hour angle at setting gives the interval between meridian transit and setting expressed in sidereal time. If the heavenly body is the sun, the interval between meridian transit and setting is expressed in apparent solar time. But during this interval the relative positions of the sun and the mean sun will alter but little (in other words, the change in the equation of time can be usually disregarded unless extreme accuracy is desired), and so the interval can be described, for all practical purposes, in terms of mean time. Thus, if from formula (24) the hour angle $H$ at setting is found to be $7^h\ 30^m$, then the interval between the sun's meridian transit and setting is $7^h\ 30^m$ mean solar time. Leaving out of consideration any change in the sun's declination, we infer that this is also the interval between sunrise and meridian passage. Thus the sun is above the horizon for $15^h$ and below the horizon for $9^h$. Actually, of course, the sun's declination is generally slightly different at sunrise from that at sunset owing to its motion along the ecliptic and the effect can be calculated.

Formula (24) shows that if $\phi > 90° - \delta$, $\cos H$ is, numerically, greater than unity, so that the equation fails to give a value of $H$. In this instance, the sun does not set in latitudes and on days such that $\phi > 90° - \delta$, as may also be verified from a diagram. On midsummer day, the sun's north declination is greatest; it is then $23\frac{1}{2}°$ N approximately, so that in latitudes north of $66\frac{1}{2}°$ N, the sun is above the horizon on that day without setting.* At the north pole, since $\phi > 90° - \delta$, provided $\delta$ is north, the sun is above the horizon continuously between March 21 and September 21; for the remaining six months it is below the horizon. The parallel of $66\frac{1}{2}°$ N is called the *Arctic Circle* and the corresponding parallel in the southern hemisphere ($66\frac{1}{2}°$ S) is the *Antarctic Circle*.

32. *Rate of change of zenith distance and azimuth.*

Let $X$ in Fig. 25 be the position of a heavenly body on the celestial sphere at a certain instant and $Y$ its position a little later. Assume the declination to be constant so that $X$ and $Y$ lie on the small circle $LM$ (the parallel of declination), of which $P$ is the pole. Draw the great circle arcs $PX$, $PY$, $ZX$, $ZY$. Let

* Hence the expression, *the midnight sun.*

$UX$ be an arc of a small circle of which $Z$ is the pole; then $ZX = ZU$. Let $Z\hat{P}X = H$ and $Z\hat{P}Y = H + \Delta H$, so that $X\hat{P}Y = \Delta H$. Let $P\hat{Z}X = A$ and $X\hat{Z}Y = \Delta A$; $ZX = z$ and $ZY = z + \Delta z$. Then $UY = \Delta z$. Since $XY$ is supposed to be a small arc, we may assume that $UXY$ is a plane triangle, right-angled at $U$.

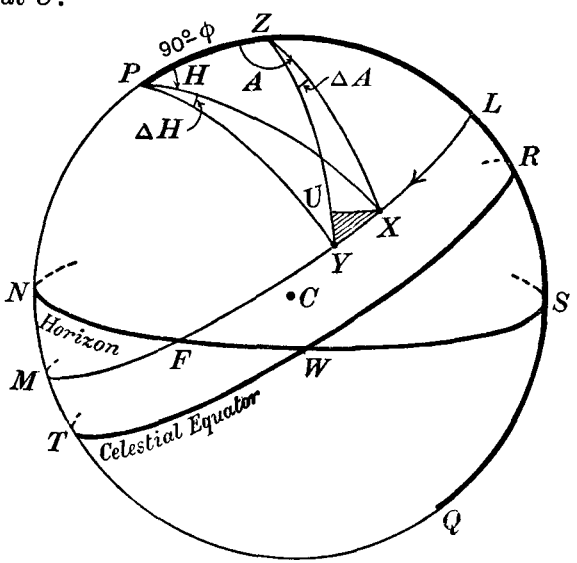



Fig. 25.

As the heavenly body moves, owing to the diurnal motion, from $X$ to $Y$ its zenith distance increases by $\Delta z$, its hour angle by $\Delta H$ and its azimuth decreases by $\Delta A$.

By formula (1) of section 3 (p. 4),

$$XY = X\hat{P}Y \sin PX = \Delta H \cos \delta,$$

and

$$UX = X\hat{Z}Y \sin ZX = \Delta A \sin z.$$

Let $\eta$ denote the angle $PXZ$; $\eta$ is called the parallactic angle. Then, since $Y$ is very close to $X$, we may take $P\hat{Y}Z$ to be $\eta$. Then

$$UY = XY \cos UYX,$$

and

$$UX = XY \sin UYX.$$

Now $P\hat{Y}Z = \eta$ and $P\hat{Y}X = 90°$; hence

$$UY \equiv \Delta z = \Delta H \cos \delta \sin \eta,$$

and

$$UX \equiv \Delta A \sin z = \Delta H \cos \delta \cos \eta.$$

Now in the spherical triangle $PXZ$, by formula **B**,

$$\cos\delta \sin\eta = \sin A \cos\phi,$$

and, by formula **C**,

$$\cos\delta \cos\eta = \sin\phi \sin z - \cos\phi \cos z \cos A.$$

Hence $$\Delta z = \Delta H \sin A \cos\phi \quad \text{.....(26),}$$

and $$\Delta A = \Delta H (\sin\phi - \cos\phi \cot z \cos A) \quad \text{.....(27).}$$

In these formulae, $\Delta H$, $\Delta z$ and $\Delta A$ are supposed expressed in circular measure. Let $\Delta H^s$ denote the number of seconds of time in $\Delta H$ radians; let $\Delta z''$, $\Delta A''$ denote the number of seconds of arc in $\Delta z$, $\Delta A$ radians respectively. Then, by the principles of section 15, p. 22,

$$\Delta z = \Delta z'' \sin 1''; \quad \Delta A = \Delta A'' \sin 1''; \quad \Delta H = \Delta H^s \sin 1^s,$$

and, since $\sin 1^s = 15 \sin 1''$, we have

$$\Delta z'' = 15 \Delta H^s . \sin A \cos\phi,$$

$$\Delta A'' = 15 \Delta H^s (\sin\phi - \cos\phi \cot z \cos A).$$

If $\Delta H^s = 1$ second, these equations express respectively that the zenith distance increases at the rate of $15 \sin A \cos\phi$ seconds of arc per second of time and that the azimuth is decreasing at the rate of $15 [\sin\phi - \cos\phi \cot z \cos A]$ seconds of arc per second of time.

If the heavenly body is a star, the rates of change of zenith distance and of azimuth are expressed in terms of seconds of arc per second of sidereal time; in the case of the sun, the rates are in terms of seconds of arc per second of apparent solar time or with sufficient accuracy, of mean solar time.

The results just obtained can be easily derived by calculus methods, as follows. From the triangle $PZX$, by formula **A**,

$$\cos z = \sin\delta \sin\phi + \cos\delta \cos\phi \cos H,$$

in which $\delta$ and $\phi$ are supposed constant. By differentiation

$$\sin z \frac{dz}{dH} = \cos\delta \cos\phi \sin H.$$

By **B**, $$\sin z \sin A = \sin H \cos\delta \quad \text{.....(28);}$$

$$\therefore \frac{dz}{dH} = \sin A \cos\phi \quad \text{.....(29),}$$

which is essentially the same as (26). If $z$ and $H$ are expressed in terms of seconds of arc and seconds of time respectively,

$$\frac{dz}{dH} = 15 \sin A \cos \phi.$$

Differentiate (28)—in which $z$, $A$ and $H$ are variables—with respect to $H$. Then

$$\sin z \cos A \frac{dA}{dH} = \cos H \cos \delta - \sin A \cos z \frac{dz}{dH}$$
$$= \cos H \cos \delta - \sin^2 A \cos z \cos \phi,$$

by means of (29).

Also, by C,

$$\cos \delta \cos H = \cos z \cos \phi - \sin z \sin \phi \cos A;$$
$$\therefore \sin z \cos A \frac{dA}{dH} = \cos^2 A \cos z \cos \phi - \sin z \sin \phi \cos A;$$
$$\therefore \frac{dA}{dH} = -(\sin \phi - \cot z \cos A \cos \phi),$$

or, if $A$ and $H$ are expressed in seconds of arc and seconds of time respectively, this last formula becomes

$$\frac{dA}{dH} = -15(\sin \phi - \cot z \cos A \cos \phi),$$

which is that already derived.

**33.** *Twilight.*

After the sun has set, indirect sunlight, reflected and scattered by the upper atmosphere, still continues to illumine the earth, diminishing however as the sun sinks farther below the horizon. When the sun is 18° below the horizon (its zenith distance is then 108°) this indirect illumination has become quite negligible. The interval between sunset and the time when the sun's zenith distance has increased to 108° is called the duration of evening twilight. In a similar way, we define the duration of morning twilight. The duration of evening twilight, for example, can be calculated as follows. In Fig. 26, $LFM$ is the sun's parallel of declination (as no great accuracy is required in this particular calculation, we ignore changes in the sun's declination during the particular day concerned) and $JGK$ is a small circle, parallel to the horizon, every point of which is 108° from $Z$. This small

circle intersects the parallel of declination in $G$. Then the interval of time required for the sun to travel from $F$ to $G$, that is $F\hat{P}G$, is the duration of evening twilight. Now $F\hat{P}G = Z\hat{P}G - Z\hat{P}F$; as $Z\hat{P}F$ is the hour angle of sunset it can be calculated by formula (24). Now in the triangle $ZPG$, we have: $ZG = 108°$, $PZ = 90° - \phi$ and $PG = 90° - \delta$; hence, by **A**,

$$\cos 108° = \sin \phi \sin \delta + \cos \phi \cos \delta \cos Z\hat{P}G,$$

which enables the calculation of $Z\hat{P}G$ to be made. The value of $\delta$, used in this formula, depends of course on the particular day of

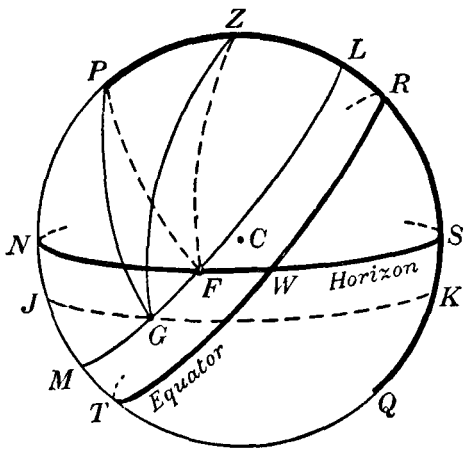



Fig. 26.

the year concerned. Thus the duration of evening twilight is found.

It is clear from Fig. 26 that evening twilight will come to an end if $NM$ is greater than $NJ$, in other words, if at apparent midnight the sun is more than 18° below the horizon. Now $NT = 90° - \phi$ and $MT = \delta$; $\therefore NM = 90° - \phi - \delta$. Hence evening twilight ends if $90° - \phi - \delta > 18°$, or if $\delta < 72° - \phi$. For example, in latitude 60° N, twilight will end if $\delta < 12°$. When $\delta$ is greater than 12°, the sun's zenith distance is less than 108° between sunset and apparent midnight, and also between apparent midnight and sunrise; therefore, in 60° N it is never completely dark on those days of the year when the sun's declination exceeds 12° N. These days are between April 23 and August 22.

## EXERCISES

[Symbols used:

$\phi$ = latitude of observer, $z$ = zenith distance,
$A$ = azimuth of heavenly body, $\epsilon$ = obliquity of the ecliptic.]
$H$ = hour angle,

**1.** If $z_1$ and $z_2$ are the zenith distances of a star on the meridian and on the prime vertical respectively, prove that

$$\text{(i)}\quad \cot\delta = \operatorname{cosec} z_1 \sec z_2 - \cot z_1,$$
$$\text{(ii)}\quad \cot\phi = \cot z_1 - \operatorname{cosec} z_1 \cos z_2,$$

where $\delta$ is the star's declination. [*Lond.* 1929.]

**2.** If $\psi$ is the angle which a star's path at rising makes with the horizon, prove that

$$\cos\psi = \sin\phi \sec\delta.$$

**3.** If $h$, $H$ are the hour angles of a star, of declination $+\delta$, on the prime vertical (west) and at setting respectively, for a place in north latitude, show that

$$\cos h \cos H + \tan^2\delta = 0.$$

Calculate the interval (correct to 0·1 minute of mean solar time) for a place in latitude 36° N between the passage of Aldebaran (declination $+16°\,22'$) over the prime vertical (west) and its setting. [*Lond.* 1926.]

**4.** A boat travelling at 5 knots is steered continually towards a star. Prove that the distance travelled towards the west is approximately $\frac{1}{3}(z_2° - z_1°)\sec\phi$ miles, where $z_1°$ and $z_2°$ are the initial and final zenith distances, in degrees, of the star and $\phi$ the mean latitude. [*M.T.* 1917.]

**5.** If the colatitude is $C$, prove that

$$C = x + \cos^{-1}(\cos z \sec y),$$

where

$$\tan x = \cot\delta \cos H,$$
$$\sin y = \cos\delta \sin H,$$

$H$ being the hour angle.

**6.** Find to the nearest second of mean solar time the interval between the passages across the meridian of two stars whose declinations are 60° N and 60° S, and whose distance apart is $\cos^{-1}(-\frac{5}{8})$. (Assume that 1 year is $365\frac{1}{4}$ days.) [*M.T.* 1923.]

**7.** If the declination $\delta$ of a star is greater than the latitude $\phi$, prove that the star's greatest azimuth east or west is

$$\sin^{-1}(\cos\delta \sec\phi).$$

**8.** At U.T. $21^h\,56^m$ on 1927 March 28 a bright star was observed through a break in the clouds as follows: altitude (approximate) 37° 10′; azimuth 136° west. The observer's position was: lat. 50° N, long. 7° 15′ W. Identify the star. (The R.A.M.S. is $0^h\,21^m$ approximately.) [*Lond.* 1927.]

**9.** The R.A. and declination of Capella at upper transit at Greenwich on 1930 May 30 were $5^h\ 11^m$ and $+\ 45°\ 55'$. Find the altitude and azimuth of the star at the same instant at New York, Columbia University Observatory, Latitude 40° 49′ N, Longitude $4^h\ 56^m$ W.

**10.** In north latitude 45° the greatest azimuth of a circumpolar star is 45° (east or west). Prove that the star's declination is + 60°.

**11.** If the latitude $\phi$ and the declination of a star be known, show that the error in the deduced value of the hour angle caused by an error of size $\Delta z$ in the zenith distance is $\Delta z \operatorname{cosec} A \sec \phi$, where $A$ is the star's azimuth.

**12.** If the observer increased his latitude by an amount $\Delta\phi$ while the hour angle of a star increased by $\Delta H$, show that the change in altitude is

$$\Delta\phi \cos A - \Delta H \sin A \cos \phi.$$

**13.** $a$ and $a + \Delta a$ are the altitudes of the sun observed simultaneously at two neighbouring places on the same meridian. If $\phi$ is the latitude of one of the places and $\delta$ is the sun's declination, prove that the difference of latitude between the places is approximately

$$\Delta a \cos a \cos \phi/(\sin \delta - \sin a \sin \phi). \qquad [\textit{Ball.}]$$

**14.** Two stars $(\alpha, \delta)$ and $(\alpha', \delta')$ are observed at the same moment on the same vertical circle. If $H$ is the hour angle of the first star, prove that

$$\cos(\chi + H) = \tan \phi \cos \chi \cot \delta,$$

where $\chi$ is given by

$$\tan \tfrac{1}{2}(\alpha - \alpha' - 2\chi) = \frac{\sin(\delta' - \delta)}{\sin(\delta' + \delta)} \cot \tfrac{1}{2}(\alpha' - \alpha).$$

**15.** If $x$ is the length of the shadow cast on level ground by a vertical pole at apparent noon at an equinox, and if $y$ is the length of shadow cast by the same pole at the summer solstice when the sun is on the prime vertical, show that

$$x = y \tan \psi \tan \phi,$$

where

$$\sin \psi = \sin \epsilon \operatorname{cosec} \phi. \qquad [\textit{Lond.}\ 1928.]$$

**16.** A straight wall of height $h$ runs in the direction $\theta$ degrees west of south. Prove that at an equinox the wall casts no shadow when the sun's hour angle $H$ is given by

$$\tan H = \sin \phi \tan \theta,$$

and that at apparent noon the breadth of the shadow is $h \tan \phi \sin \theta$.

**17.** An observer in latitude 50° sees a star set due west behind a low ridge a mile away, which slopes down to the north at an inclination of 30° to the horizontal. Prove that by stepping a yard to his right he will see the star for about 22 seconds longer. [*M.T.* 1913.

**18.** Two places are in the same latitude and the polar distance of the great circle through them is equal to the sun's declination. Prove that at these places the length of the night is equal to their difference of longitude.

**19.** Let $\alpha$, $\delta$ be the co-ordinates of a star with respect to a great circle $S$, and $\alpha'$, $\delta'$ the co-ordinates of the same star with respect to another great circle $S'$. If $i$ be the inclination of $S'$ to $S$ and if the ascending node of $S'$ on $S$ has co-ordinates $(\theta, 0)$ in the first system and $(\theta', 0)$ in the second, show that

$$\cos\delta'\cos(\alpha'-\theta') = \cos\delta\cos(\alpha-\theta),$$
$$\cos\delta'\sin(\alpha'-\theta') = \sin\delta\sin i + \cos\delta\cos i\sin(\alpha-\theta),$$
$$\sin\delta' = \sin\delta\cos i - \cos\delta\sin i\sin(\alpha-\theta).$$

If $\alpha = 75°$, $\delta = 15°$, $\theta = 215°$, $\theta' = 115°$, $i = 23°\,30'$, show that from the last equations $\alpha' = 327°\,12'$, $\delta' = 29°\,0'$.

**20.** Show that if $a$ is the altitude of the pole star, $H$ the hour angle and $p$ (in seconds of arc) the polar distance, the latitude is approximately given by

$$\phi = a - p\cos H + \tfrac{1}{2}p^2\sin^2 H\tan a\sin 1''.$$

**21.** A heavenly body (declination $\delta$) is at a small angle $H$ from the meridian. Prove that the zenith distance $z$ is given approximately by

$$z = \phi - \delta + \alpha_1 - \alpha_2,$$

where $\alpha_1$ (expressed in minutes of arc) is given by

$$\alpha_1 = \frac{2\cos\phi\cos\delta}{\sin(\phi-\delta)}\sin^2\frac{H}{2}\operatorname{cosec} 1',$$

and

$$\alpha_2 = \tfrac{1}{2}\alpha_1^2\cot(\phi-\delta)\sin 1'.$$

**22.** If $a$ is the sun's altitude in the prime vertical at a place in latitude $\phi$ and $L$ is its longitude, prove that

$$\phi = \sin^{-1}(\sin L\sin\epsilon\operatorname{cosec} a). \qquad [\textit{Ball.}]$$

**23.** Prove that, in latitude 45°, the interval between the moment at which a star's azimuth is 90° east and the moment of setting is constant.

**24.** If $\delta$ be a star's declination and $A$ its maximum azimuth, show that in $t$ seconds of time from the moment when the azimuth is $A$ the azimuth has changed by $\tfrac{1}{2}\,15^2t^2\sin 1''\sin^2\delta\tan A$ seconds of arc.

**25.** If $\eta$ is the parallactic angle and $\phi$ and $\delta$ are constant, prove that

(i) $\dfrac{d\eta}{dH} = -\cos\phi\cos A\operatorname{cosec} z;$

(ii) $\dfrac{d^2z}{dH^2} = \dfrac{d\eta}{dH}\cos\delta\cos\eta;$

(iii) $\dfrac{d^2A}{dH^2} = -\dfrac{\cos\delta}{\sin^2 z}\left(\cos z\cos\eta\dfrac{dz}{dH} + \sin z\sin\eta\dfrac{d\eta}{dH}\right).$

**26.** If $H$ is the hour angle of a star at rising, show that

$$\tan^2\frac{H}{2} = \frac{\cos(\phi-\delta)}{\cos(\phi+\delta)}.$$

**27.** At a place in north latitude $\phi$, two stars $A$ and $B$ (declinations $\delta$ and $\delta_1$ respectively) rise at the same moment and $A$ transits when $B$ is setting. Prove that

$$\tan\phi\tan\delta = 1 - 2\tan^2\phi\tan^2\delta_1.$$

**28.** If two stars $(\alpha, \delta)$ and $(\alpha_1, \delta_1)$ rise at the same moment at a place in latitude $\phi$, show that

$$\cot^2\phi \sin^2(\alpha_1 - \alpha) = \tan^2\delta + \tan^2\delta_1 - 2\tan\delta\tan\delta_1\cos(\alpha_1 - \alpha).$$

[*Ball.*]

**29.** At a place in latitude $\phi$ the sun is observed to rise $h$ hours before apparent noon, and the next day it rises $m$ minutes later. Its declination on the first day is $\delta$. Show that the distance in minutes of arc between the two points of rising is

$$15m\cos^2\delta\operatorname{cosec}\phi.$$

[*Coll. Exam.*]

**30.** If evening twilight ends when the sun's centre is 18° below the horizon, show that at the equator the duration of evening twilight is given in hours by

$$\frac{12}{\pi}\sin^{-1}(\sin 18°\sec\delta).$$

Use this formula to calculate the duration of evening twilight at the summer solstice. [*Lond.* 1930.]

**31.** Show that at a place in latitude $\phi$ the shortest duration of twilight, expressed in hours, is

$$\tfrac{2}{15}\sin^{-1}(\sin 9°\sec\phi),$$

where $\sin^{-1}(\sin 9°\sec\phi)$ is expressed in degrees. [*Ball.*]

**32.** If twilight begins or ends when the sun is 18° below the horizon, show that all places have a day of more than twelve hours, including twilight, so long as the declination of the sun is numerically less than 18°.

**33.** If the day is considered to begin and end when the sun is at an angle $\theta$ below the horizon, show that the shortest day will not occur at the winter solstice if the latitude is less than $\phi$, where

$$\sin\phi = \sin\epsilon\sin\theta,$$

and $\epsilon$ is the obliquity of the ecliptic. [*M.T.* 1917.]

**34.** Assuming that the sun travels uniformly in the ecliptic, completing a revolution in 365 days, show that the number of nights in which there is twilight even at midnight at a place in latitude $\phi$ is the integer next greater than

$$\tfrac{7}{8}\tfrac{3}{6}\cos^{-1}\{\cos(\phi + 18°)/\sin\epsilon\},$$

twilight beginning or ending when the sun is 18° below the horizon.

[*Coll. Exam.*]

**35.** If $\theta$ denotes the sun's depression below the horizon at the end of evening twilight, and $\eta$, $\eta'$ the parallactic angles at end of twilight and at sunset respectively, prove that the duration $(T)$ of twilight is given by

$$2\sin^2\frac{T}{2}\cos^2\phi = 1 - \cos\theta\cos(\eta' - \eta).$$

**36.** The right ascension of a star is $5^h\,49^m$ and its declination is $+\,7°\,23'$, and the obliquity of the ecliptic is $23°\,27'$. Show that the longitude and latitude of the star are respectively $87°\,10'$, $-\,16°\,2'$.

**37.** Two stars $(\alpha_1, \delta_1)$ and $(\alpha_2, \delta_2)$ have the same longitude; prove that

$$\sin(\alpha_1 - \alpha_2) = \tan\epsilon(\cos\alpha_1\tan\delta_2 - \cos\alpha_2\tan\delta_1).$$

**38.** A star of right ascension $\alpha$ and declination $\delta$ has a small latitude $\beta$. Prove that the longitude of the sun, when its R.A. is $\alpha$, differs from the longitude of the star by $\beta \sin \delta \cot \alpha$ approximately.

**39.** Show that the obliquity of the ecliptic can be determined by making observation of the sun's declination $\delta$ at a noon near the summer solstice by means of the formula $\epsilon = \delta + q^2 \sin 2\delta$, where $q$ is one-half the defect from a right angle of the sun's right ascension. [*M.T.* 1924.]

**40.** The pole of the Milky Way is at R.A. $12^h\,48^m$, Dec. $+\,27°$. About what dates will the sun pass through the Milky Way? (Obliquity of the ecliptic $= 23° 27'$.) [*M.T.* 1925.]

**41.** A star is displaced a small amount $dr$ towards a point $O$ on the celestial sphere with equatorial co-ordinates $(\alpha_0, \delta_0)$. Show that the resulting changes in the star's equatorial co-ordinates $(\alpha, \delta)$ are given by

$$\cos\delta\, d\alpha = \cos\delta_0 \sin(\alpha - \alpha_0) \operatorname{cosec} r\, dr,$$
$$d\delta = (\cos\delta_0 \sin\delta \cos(\alpha - \alpha_0) - \sin\delta_0 \cos\delta) \operatorname{cosec} r\, dr,$$

where $r$ is the arc length on the celestial sphere from the star to the point $O$. [*Glas.* 1974.]

**42.** Prove that the zenith distance $z$ of the north pole of the ecliptic is given by

$$z = \cos^{-1}(\cos\epsilon \sin\phi - \sin\epsilon \cos\phi \sin T).$$

Here $\epsilon$ is the obliquity of the ecliptic, $\phi$ is the latitude of the observer, and $T$ is the local sidereal time.

CHAPTER III

# REFRACTION

**34.** *The laws of refraction.*

In astronomical observations, the light from the particular heavenly body observed has to pass through the earth's atmosphere before reaching the observer, and during its passage a ray of light suffers a change in direction, owing to refraction, the amount of which depends on the physical characteristics of the atmosphere and on the altitude (or zenith distance) of the body concerned. It is thus necessary, at the outset, to eliminate from the observations the effects of our terrestrial atmosphere on them. From the study of meteors the deduction is made that the atmosphere extends to a height of at least 100 miles, for even at that height the friction of the air on a rapidly moving meteor is sufficient to render it luminescent. But beyond a height of about 40 miles the air is so tenuous that it has an inappreciable effect on the course of a ray of light.

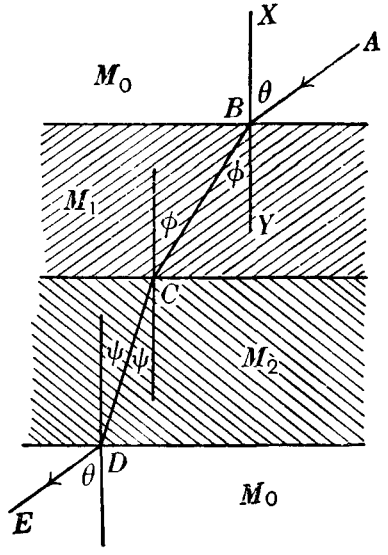


Fig. 27.

We shall first state the laws of refraction.

Consider (Fig. 27) a ray of light $AB$ passing through a transparent medium $M_0$ (such as air) and falling at $B$ on a slab, with parallel plane faces, of another transparent medium $M_1$ (such as glass). In the medium $M_1$ the path of the ray will be along some such line as $BC$, different in direction from $AB$; the ray is said to be *refracted* at $B$. Let $YBX$ be perpendicular to the slab at $B$. The angle $ABX$ (denoted by $\theta$) is called the angle of incidence and the angle $YBC$ (denoted by $\phi$)

the angle of emergence. The laws of refraction are: (i), the incident ray $AB$, the refracted ray $BC$ and the perpendicular at $B$ to the surface separating the two media $M_0$ and $M_1$ lie in the same plane; (ii), the relation between $\phi$ and $\theta$ is

$$\frac{\sin\theta}{\sin\phi} = \mu_1 \qquad \text{.....(1),}$$

$\mu_1$ being a constant depending on the optical properties of the two media concerned. In this instance, $\mu_1$ is called the *index of refraction* for the two media $M_0$ and $M_1$; its value can be determined by laboratory experiment.

Let the ray $BC$ now pass from the medium $M_1$ into the medium $M_2$. The angle of incidence is now $\phi$ and the angle of emergence is $\psi$. Then

$$\frac{\sin\phi}{\sin\psi} = \mu \qquad \text{.....(2),}$$

where $\mu$ is the refractive index for the two media $M_1$ and $M_2$. At $D$, we shall suppose that the ray emerges into the medium $M_0$. Its path $DE$ in $M_0$ is parallel to its original direction $AB$ in the same medium. The path of the ray is reversible, that is to say, a ray in the direction $ED$ will be refracted, at the surface between the media $M_0$ and $M_2$, along $DC$. Hence

$$\frac{\sin\theta}{\sin\psi} = \mu_2 \qquad \text{.....(3),}$$

where $\mu_2$ is the index of refraction between the media $M_0$ and $M_2$. From (1) and (3), we have by division

$$\frac{\sin\phi}{\sin\psi} = \frac{\mu_2}{\mu_1},$$

or

$$\mu_1 \sin\phi = \mu_2 \sin\psi \qquad \text{.....(4).}$$

It follows from (2) that $\mu = \mu_2/\mu_1$.

Regarding $M_0$ as a standard medium, we can define $\mu_1$ simply as the refractive index for medium $M_1$ and $\mu_2$ for medium $M_2$, and the values of $\mu_1$ and $\mu_2$ may be supposed known. Consider now only the two media $M_1$ and $M_2$ in Fig. 27, $BC$ being a ray in $M_1$ incident at $C$ on the bounding surface between $M_1$ and $M_2$ and $CD$ the ray in $M_2$. $\phi$ is the angle of incidence, $\psi$ the angle of emergence, and the relation between $\phi$ and $\psi$ is given by formula (4).

**35.** *Refraction for small zenith distances.*

As the density of the air diminishes with increasing height above the earth's surface, it is convenient to regard the atmosphere as made up of a large number of thin spherical layers, concentric with the earth's surface regarded as spherical, throughout each of which the density and other physical characteristics are uniform. The simplest case in the investigation of astronomical refraction occurs when the heavenly body observed—for example, a star—is nearly overhead; in this

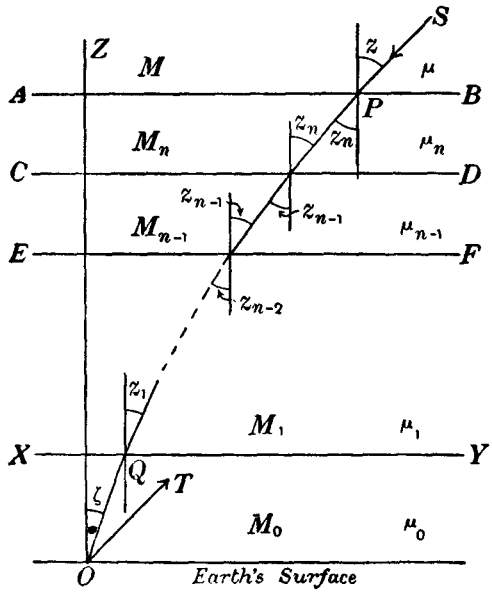



Fig. 28.

instance we can ignore the curvature of the atmospheric strata through which the rays from the star pass and thus we can regard their bounding surfaces simply as a series of parallel planes. Let there be $n + 1$ parallel layers (of which only a few are shown in Fig. 28) and let $AB$ be the upper effective limit of the atmosphere beyond which the air, owing to its extreme tenuity, is ineffective in causing refraction. Each layer has its own optical properties and, in particular, its own index of refraction. Let the medium above $AB$ be denoted by $M$, the layer between $CD$ and $AB$ by $M_n$, and so on, with the corresponding refractive

indices $\mu$, $\mu_n$, $\mu_{n-1}$, ... $\mu_0$, the last being the refractive index for the lowest layer $M_0$. We can regard $M$ as the standard medium —effectively, it corresponds to a vacuum—and so we can put $\mu = 1$. Let $z$ denote the angle of incidence, at the surface $AB$, of a ray from the star that finally reaches the observer at $O$. Then $z$ is called the true zenith distance of the star; if there were no atmospherical refraction the star would be seen by the observer in the direction $OT$, which is parallel to $PS$. Applying formula (4) we have for the successive pairs of layers the following series of equations (the notation is indicated in Fig. 28):

$$\mu \sin z = \mu_n \sin z_n,$$

or, since $\mu = 1$,

$$\begin{aligned} \sin z &= \mu_n \sin z_n, \\ \mu_n \sin z_n &= \mu_{n-1} \sin z_{n-1}, \\ \mu_{n-1} \sin z_{n-1} &= \mu_{n-2} \sin z_{n-2}, \\ &\vdots \qquad\qquad \vdots \\ \mu_1 \sin z_1 &= \mu_0 \sin \zeta, \end{aligned}$$

in the last of which $\zeta$ is the angle between the direction of the zenith $OZ$ and the final element $OQ$ in the path of the ray. $\zeta$ is thus the *observed zenith distance* of the star. From these equations we have, clearly,

$$\sin z = \mu_0 \sin \zeta \qquad \ldots\ldots(5);$$

$\mu_0$ is the index of refraction of the air at the earth's surface. From the first law of refraction it is evident that the path of the ray through the various strata lies in a vertical plane. The values of the refractive index increase from $\mu$ in $M$ continuously to $\mu_0$ in $M_0$, corresponding to the increase in the density of the atmospherical layers from $M$ downwards; accordingly, it follows from (4) that the angles $z$, $z_n$, $z_{n-1}$, ... $z_1$, $\zeta$ form a decreasing sequence and the path of the ray is thus bent in the way indicated in the diagram. In particular $\zeta$ is less than $z$, that is to say, the star is observed *nearer* the zenith than it would be if the atmosphere were non-effective as regards refraction. The angle $z - \zeta$ is called the *angle of refraction*; denote it by $R$. Then (5) becomes

$$\sin (\zeta + R) = \mu_0 \sin \zeta,$$

or

$$\sin \zeta \cos R + \cos \zeta \sin R = \mu_0 \sin \zeta.$$

Now $R$ is a small angle and we can write $\cos R = 1$ and $\sin R = R$ ($R$ being supposed expressed in circular measure). Thus

$$\sin \zeta + R \cos \zeta = \mu_0 \sin \zeta,$$

or

$$R = (\mu_0 - 1) \tan \zeta \qquad \ldots\ldots(6).$$

We thus have the result that, *at small zenith distances*, the angle of refraction is proportional to the tangent of the observed zenith distance. Now $\mu_0$ is the index of refraction of the air at the surface of the earth, and its value will be dependent on the density and temperature at any given time. The standard conditions are taken in practice to be: barometric pressure = 760 mm and temperature = 10° C.; the refraction is then called *mean refraction*. For these conditions, $\mu_0 - 1$ is approximately 0·00029, so that $R$ is approximately $0{\cdot}00029 \tan \zeta$ or, in seconds of arc, $206265 \times 0{\cdot}00029 \tan \zeta$. The coefficient of $\tan \zeta$ is more accurately determined by means of astronomical observations and the value usually adopted is $58''{\cdot}2$; we then have

$$R = 58''{\cdot}2 \tan \zeta.$$

The coefficient of $\tan \zeta$ is called the *constant of mean refraction*; denote it by $k$. Then

$$R = k \tan \zeta \qquad \ldots\ldots(7).$$

At any barometric pressure $P$ (in mm Hg) and temperature $T$ (in degrees Centigrade), the corresponding refraction $R'$ in terms of the mean refraction $R$ is given by

$$\frac{R'}{R} = \frac{0{\cdot}372P}{273+T} \qquad \ldots\ldots(8).$$

For many purposes, the formula (7) is sufficiently accurate for zenith distances not exceeding 45°.

**36.** *General formula for refraction.*

When the zenith distance of the body observed is considerable, the atmosphere through which the rays pass can no longer be regarded as stratified in plane layers. Assume that the earth is spherical and that the atmosphere is arranged in spherical layers. In Fig. 29, let $C$ be the centre of the earth, $O$ the observer and $COZ$ the direction of his zenith. Let $\mu'$, $\mu$ be the indices of refraction in two adjacent thin layers $M'$ and $M$. Let $LP$ be the section of a ray in $M'$ which finally reaches the observer at $O$. At $P$ it is refracted along $PQ$. Similarly, it is refracted at the surfaces between successive layers and the final element of its path is $TO$. If the layers are thin, the path of the ray is curved and the direction in which the observer sees the object is along $OT$, the tangent to this curve at $O$. The observed zenith distance

is thus $Z\hat{O}T$, denoted by $\zeta$. Draw the radii $CP$ and $CE$. Let $GPL = \phi'$, $QPF = \psi$, $EQP = \phi$. Then, since the radius $CP$ is perpendicular at $P$ to the bounding surface between the layers $M'$ and $M$, by the laws of refraction we have

$$\mu' \sin \phi' = \mu \sin \psi \qquad \text{.....(9).}$$

Now from the triangle $CQP$, in which $CP = r'$ and $CQ = r$ and $CQP = 180° - \phi$, we have

$$r \sin \phi = r' \sin \psi \qquad \text{.....(10).}$$

Eliminate $\sin \psi$ from (9) and (10); then

$$r'\mu' \sin \phi' = r\mu \sin \phi \qquad \text{.....(11).}$$

This is a general relation which holds for any two contiguous layers and consequently for any two layers whatever their heights above the earth's surface may be. If $r_0$, $\mu_0$, $\phi_0$ denote the values of $r$, $\mu$ and $\phi$ for the lowest layer—next the earth's surface—we have, from (11),

$$r\mu \sin \phi = r_0 \mu_0 \sin \phi_0.$$

But $r_0 = a$, the earth's radius, and $\phi_0$ is simply the angle $ZOT$ or $\zeta$, the observed zenith distance. Hence

$$r\mu \sin \phi = \mu_0 a \sin \zeta \qquad \text{.....(12).}$$

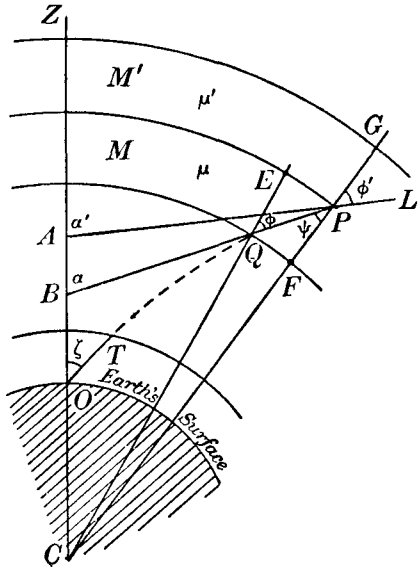



Fig. 29.

Consider now the angle through which the ray is deviated in its passage from one layer through the next. Produce $LP$ and $PQ$ to meet $OZ$ in $A$ and $B$ respectively, and let $Z\hat{A}P$ and $Z\hat{B}P$ be denoted by $\alpha'$ and $\alpha$. Then the angle through which the ray is refracted at $P$ is $A\hat{P}B$ or $\alpha' - \alpha$; let it be denoted by $\Delta R$, so that

$$\Delta R = \alpha' - \alpha \qquad \text{.....(13).}$$

Let $A\hat{C}P = \theta'$ and $B\hat{C}Q = \theta$, and let

$$\Delta\theta = \theta' - \theta \qquad \text{.....(14).}$$

Then, assuming that the layers are thin, we can write

$$QF = EP = r\Delta\theta.$$

Also if $r' = r + \Delta r$, then $QE = \Delta r$. In the infinitesimal triangle $EQP$ right-angled at $E$, we have

$$\tan\phi = \frac{EP}{QE} = \frac{r\Delta\theta}{\Delta r} \qquad \text{.....(15).}$$

Now, $\alpha' = \theta' + \phi'$ and $\alpha = \theta + \phi$; hence

$$\Delta R \equiv \alpha' - \alpha = (\phi' - \phi) + (\theta' - \theta),$$

or, if $\Delta\phi$ denotes $\phi' - \phi$,

$$\Delta R = \Delta\phi + \Delta\theta \qquad \text{.....(16).}$$

Now from (11), writing $\mu' = \mu - \Delta\mu$ ($\mu$ decreases as $r$ increases), we obtain

$$(r + \Delta r)(\mu - \Delta\mu)\sin(\phi + \Delta\phi) = r\mu\sin\phi,$$

or $$(r + \Delta r)(\mu - \Delta\mu)(\sin\phi + \Delta\phi\cos\phi) = r\mu\sin\phi,$$

since $\Delta\phi$ is a small angle.

Omitting products of the infinitesimal quantities $\Delta r$, $\Delta\mu$, $\Delta\phi$ we obtain, after dividing throughout by $r\mu\sin\phi$,

$$\frac{\Delta r}{r} - \frac{\Delta\mu}{\mu} + \Delta\phi\cot\phi = 0 \qquad \text{.....(17).}$$

But, by (15),

$$\frac{\Delta r}{r} = \Delta\theta\cot\phi.$$

Hence (17) becomes

$$(\Delta\theta + \Delta\phi)\cot\phi - \frac{\Delta\mu}{\mu} = 0$$

and, by (16),

$$\Delta R = \frac{\Delta\mu}{\mu}\tan\phi \qquad \text{.....(18).}$$

Now (12) enables us to express $\tan\phi$ in terms of the variables $\mu$ and $r$ and the constants $a$, $\mu_0$, $\zeta$. When this is done, (18) becomes

$$\Delta R = \frac{\Delta\mu}{\mu} \cdot \frac{a\mu_0\sin\zeta}{(r^2\mu^2 - a^2\mu_0^2\sin^2\zeta)^{\frac{1}{2}}} \qquad \text{.....(19).}$$

This equation expresses the amount of refraction suffered by a ray in passing from one spherical layer with index of refraction $\mu - \Delta\mu$ to the next lower layer with index of refraction $\mu$. The total refraction $R$ due to the whole atmosphere is given by

$$R = a\mu_0\sin\zeta\int_1^{\mu_0}\frac{d\mu}{\mu(r^2\mu^2 - a^2\mu_0^2\sin^2\zeta)^{\frac{1}{2}}} \qquad \text{.....(20),}$$

the limits of integration being $\mu_0$ at the earth's surface and unity at the highest layer.

Formula (20) is the general expression for refraction. It is to be noted that the integral involves two variables, $r$ and $\mu$. Now the index of refraction of any layer is dependent on the physical characteristics of that layer which, in turn, will depend on the height of the layer above the earth's surface, that is, in effect, on $r$. Before the integral in (20) can be rigorously calculated the relation between $\mu$ and $r$ must be specified. This involves the application of physical laws embracing the pressure, density and temperature of the air. Unfortunately, our knowledge of the physical state of the upper atmosphere is insufficient to indicate the precise dependence of $\mu$ on $r$, and we are thus forced to treat equation (20) by approximate methods.

**37.** *Development of the general formula for refraction.*

The height of the atmosphere is small in comparison with the radius of the earth, and if we write

$$\frac{r}{a} = 1 + s \qquad \text{.....(21)}$$

we can regard $s$ as a small quantity; it varies from zero at the earth's surface to about 0·01 at a height of 40 miles, which may be regarded as the limit beyond which the air is ineffective in producing any appreciable refraction.* Using (21) and neglecting terms in $s^2$, $s^3$, etc., we can write

$$(r^2\mu^2 - a^2\mu_0^2 \sin^2\zeta)^{-\frac{1}{2}} = \frac{1}{a}(\mu^2 - \mu_0^2 \sin^2\zeta + 2s\mu^2)^{-\frac{1}{2}}$$

$$= \frac{1}{a}(\mu^2 - \mu_0^2 \sin^2\zeta)^{-\frac{1}{2}}\left(1 - \frac{s\mu^2}{\mu^2 - \mu_0^2 \sin^2\zeta}\right).$$

Hence (20) becomes

$$R = \mu_0 \sin\zeta \int_1^{\mu_s} \frac{d\mu}{\mu(\mu^2 - \mu_0^2 \sin^2\zeta)^{\frac{1}{2}}} - \mu_0 \sin\zeta \int_1^{\mu_s} \frac{s\mu\, d\mu}{(\mu^2 - \mu_0^2 \sin^2\zeta)^{\frac{3}{2}}}$$

$$\equiv \qquad R_1 \qquad - \qquad R_2 \qquad \text{.....(22).}$$

The expansion by the binomial theorem is only valid if $2s\mu^2$ is small compared with $(\mu^2 - \mu_0^2 \sin^2\zeta)$, and since $\mu_0$ and $\mu$ are approximately unity it is invalid when the observed heavenly

* The earth's mean radius is 3960 miles.

body is on or near the horizon (then $\zeta$ is equal to or close to 90° and sin $\zeta$ is equal to or close to unity). Consider first the term $R_1$ in (22). It is of the form

$$\alpha \int \frac{d\mu}{\mu (\mu^2 - \alpha^2)^{\frac{1}{2}}},$$

which is a well-known integral whose value is $-\sin^{-1}\frac{\alpha}{\mu}$. Hence, inserting the limits $\mu_0$ and 1, we obtain

$$R_1 = \sin^{-1}(\mu_0 \sin\zeta) - \zeta.$$

Now $\mu_0$—the index of refraction of air at the earth's surface—is a little greater than unity; denote it by $(1 + x)$, where $x$ is a small quantity. Then

$$R_1 \equiv f(x) = \sin^{-1}[(1 + x)\sin\zeta] - \zeta,$$

and by Maclaurin's theorem, terms in $x^2$, $x^3$, etc. being neglected,

$$R_1 \equiv f(x) = f(0) + x\left(\frac{df}{dx}\right)_{x=0}.$$

Now $$f(0) = \sin^{-1}(\sin\zeta) - \zeta = 0,$$

and $$\frac{df}{dx} = \frac{\sin\zeta}{\{1 - (1+x)^2\sin^2\zeta\}^{\frac{1}{2}}},$$

so that $$\left(\frac{df}{dx}\right)_{x=0} = \tan\zeta.$$

Hence to the degree of approximation indicated

$$R_1 = x\tan\zeta,$$

or $$R_1 = (\mu_0 - 1)\tan\zeta \qquad \ldots\ldots(23),$$

which, on reference to section 35, is seen to be the result obtained for small zenith distances.

Consider now the second integral in (22), namely,

$$R_2 = \mu_0 \sin\zeta \int_1^{\mu_0} \frac{s\mu\, d\mu}{(\mu^2 - \mu_0^2\sin^2\zeta)^{\frac{1}{2}}}.$$

In this expression, we regard $s$ as a small quantity varying, as we have seen, from 0 to about 0·01. Now, by Gladstone and Dale's law, we can express $\mu$ in terms of the atmospheric density $\rho$ by

$$\mu = 1 + c\rho \qquad \ldots\ldots(24),$$

where $c$ is a constant with a numerical value 0·226. For the air at the earth's surface (density $\rho_0$)

$$c\rho_0 = 0{\cdot}00029.$$

From (24), $$d\mu = c\,d\rho,$$

an equation which expresses the change in refractive index with atmospheric density. We then have

$$R_2 = c\mu_0 \sin\zeta \int_0^{\rho_0} \frac{s\mu\,d\rho}{(\mu^2 - \mu_0^2 \sin^2\zeta)^{\frac{3}{2}}} \qquad \ldots\ldots(25).$$

In this integral, since $\mu$ and $\mu_0$ are very nearly unity, and $s$ is small, the value of $R_2$ will be little affected if we write $\mu = \mu_0 = 1$ in (25). Then

$$R_2 = c \sin\zeta \int_0^{\rho_0} \frac{s\,d\rho}{(1 - \sin^2\zeta)^{\frac{3}{2}}}$$

$$= c \tan\zeta \sec^2\zeta \int_0^{\rho_0} s\,d\rho \qquad \ldots\ldots(26).$$

Now $$\int_0^{\rho_0} s\,d\rho = [s\rho]_0^{\rho_0} - \int_{s'}^{0} \rho\,ds$$

$$= [s\rho]_0^{\rho_0} + \int_0^{s'} \rho\,ds \qquad \ldots\ldots(27),$$

in which $s'$ denotes the value of $s$ at the highest atmospheric level where, of course, $\rho = 0$. Now $s = 0$ when $\rho = \rho_0$, and it follows that $[s\rho]_0^{\rho_0} = 0$.

Now $$a \int_0^{s'} \rho\,ds = \int_0^{s'} \rho\,d\,(as),$$

and the second integral is the expression for the mass of a column of air, of unit cross section, extending from the earth's surface to the effective limit of the atmosphere; it is therefore independent of the actual law according to which atmospheric density changes with height. The mass of a column of air is, however, related to temperature and barometric pressure, and if we write

$$R_2 = -B \tan\zeta \sec^2\zeta \qquad \ldots\ldots(28),$$

the quantity $-B$ (which is $\frac{c}{a} \times$ mass of the column considered) must be regarded as dependent on temperature and pressure. Combining (23) and (28) we obtain the expression for the refraction in the form

$$R = (\mu_0 - 1)\tan\zeta + B \tan\zeta\,(1 + \tan^2\zeta),$$

or $$R = A \tan\zeta + B \tan^3\zeta \qquad \ldots\ldots(29),$$

in which $A$ has been written for $(\mu_0 - 1) + B$.

Instead of calculating the values of $A$ and $B$ from the physical data concerned, it is preferable to assume that the refraction can be expressed by a formula of the type (29), and to derive the values of the coefficients $A$ and $B$ from observations of stars. The numerical expression for the mean refraction (for barometric pressure 760 mm and temperature 10° C.) is

$$R = 58''{\cdot}16 \tan \zeta - 0''{\cdot}067 \tan^3 \zeta \quad \ldots\ldots(30).$$

The approximations which we introduced in deriving formula (29) are insufficient when the zenith distance exceeds 75° approximately. For observations made near the horizon, special tables of refraction have been prepared (based mainly on observational data) and are in use in observatories where an accurate knowledge of the refraction is essential. Among these may be mentioned the Greenwich Tables (1898) and the Pulkova Tables (4th edition, 1956).

**38.** *The determination of the constants* A *and* B *in the formula* (29).

In the next chapter, we shall consider in some detail the instrument by which the zenith distances of heavenly bodies at upper or at lower culmination can be measured; meanwhile we shall take for granted the practical results of the method. Consider a star at upper and lower culmination. At upper culmination it is observed at $X_1$; the displacement due to refraction is $XX_1$ (Fig. 30). $ZX_1 = \zeta$ which is obtained from the observation, and $ZX$ is the true zenith distance $z$, so that

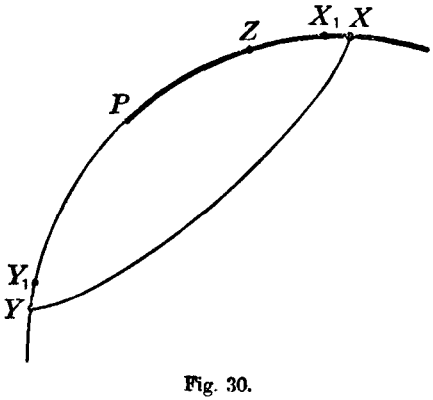


Fig. 30.

$$z = (90° - \delta) - (90° - \phi),$$

or $z = \phi - \delta$. Now $z = \zeta + R$, where $R$ is the refraction corresponding to the observed zenith distance $\zeta$; hence by (29),

$$\phi - \delta = \zeta + A \tan \zeta + B \tan^3 \zeta \quad \ldots\ldots(31).$$

Let $\zeta'$ be the observed zenith distance at lower culmination. Then $\zeta' = ZY_1$ and the true zenith distance $ZY$ or $z'$ is given by

$z' = (90° - \phi) + (90° - \delta)$ or $z' = 180° - \phi - \delta$. Also $z' = \zeta' + R'$; hence we have

$$180° - \phi - \delta = \zeta' + A \tan \zeta' + B \tan^3 \zeta' \quad \text{.....(32).}$$

If the values of $\phi$ and $\delta$ are known accurately, we have two equations (31) and (32) in which $A$ and $B$ are the only unknowns, for $\zeta$ and $\zeta'$ have been determined by observation. If we suppose that $\delta$ is known, we can eliminate $\phi$ from (31) and (32) and so obtain

$$180° - 2\delta = \zeta + \zeta' + R + R' \quad \text{.....(33).}$$

Observations of another star will lead to an equation similar in form to (33), and from this equation and (33) the values of $A$ and $B$ can be determined. As no observation is free from error, such errors are incorporated on the right-hand side of (33), thus vitiating slightly the deduced values of $A$ and $B$. To reduce the errors in $A$ and $B$ to a minimum, a large number ($n$) of stars are observed and the $n$ equations of the type of (33) are solved for $A$ and $B$ by the method of "least squares".

**39.** *The effect of refraction on the time of sunset.*

When a heavenly body is on the horizon at rising or setting —the zenith distance is then 90°—the numerical value of the refraction is 34′ (this is called the *horizontal refraction*). Now the effect of refraction is to make the body appear nearer the zenith than it would be seen if the atmosphere were non-existent or ineffective in deviating rays of light. Hence it follows that at setting, for example, when the observed zenith distance $\zeta$ is 90°, the true zenith distance $z$ of the heavenly body is 90° + horizontal refraction, or 90° 34′. If we take the case of the sun, it is clear that the time of visible sunset is *later* than the time of theoretical sunset, which is discussed in section 31 of Chapter II. The interval between theoretical and visible sunset is easily found. Let $H$ be the hour angle when the true zenith distance of the sun's centre is 90° and let $H + \Delta H$ be the hour angle when the sun's centre is seen on the horizon. Then $\zeta = 90°$ and $z = 90°\ 34'$. Then we have, as in section 31,

$$\cos H = - \tan \phi \tan \delta \quad \text{.....(34),}$$

and also, by the cosine formula **A**,

$$\cos (90°\ 34') = \sin \phi \sin \delta + \cos \phi \cos \delta \cos (H + \Delta H),$$

which may be written

$$- \sin 34' = \sin \phi \sin \delta + \cos \phi \cos \delta (\cos H \cos \Delta H - \sin H \sin \Delta H).$$

As $34'$ and $\Delta H$ are small angles, we can reduce the last equation to

$$-34 \sin 1' = \sin\phi \sin\delta + \cos\phi \cos\delta \cos H - 15\Delta H \sin 1' . \cos\phi \cos\delta \sin H,$$

in which $\Delta H$ is now supposed to be expressed in minutes of time. Using (34), we find that

$$\Delta H = \tfrac{34}{15} \sec\phi \sec\delta \operatorname{cosec} H \quad \text{minutes} \quad \ldots\ldots(35).$$

Let us take a simple example. Suppose the latitude is $60°$ and $\delta = 0°$ (about March 21 or about September 21). $H$ being the hour angle of theoretical sunset is easily seen to be $6^h$. Hence from (35), by calculation, $\Delta H = 4^m.5$.

The sun's declination $\delta$ which we have introduced into the formulae is the declination of the sun's centre, and the zenith distances concerned are the zenith distances of the sun's centre. To find the hour angle when the sun's upper limb just disappears below the horizon, we notice that the true zenith distance of the sun's centre then is $90° + 34' +$ the angle subtended by a radius of the sun. The last quantity is the sun's *semi-diameter*, which is tabulated in the almanacs for each day of the year and, for the purposes under consideration, may be taken to be $16'$; thus the true zenith distance of the sun's centre under the circumstances now contemplated is $90° 50'$. If $\Delta H$ now denotes the interval between the time of theoretical sunset and the time of the disappearance of the sun's upper limb below the horizon, $\Delta H$ is given by $\Delta H = \frac{50}{15} \sec\phi \sec\delta \operatorname{cosec} H$ minutes.

Hence in latitude $60°$ when the sun's declination is zero, $\Delta H = 6^m.7$. There is a similar interval between visible sunrise and theoretical sunrise. The effect of refraction is thus to increase the length of the "day" (by the "day" is here meant the interval during which some part of the sun is above the horizon) by about $13\frac{1}{3}$ minutes at the latitude and at the dates indicated.

**40.** *Effect of refraction on the right ascension and declination of a star.*

Consider the position $X$ of a star on the celestial sphere (Fig. 31). It is displaced towards the zenith $Z$ by refraction to the position $X'$. Through $X'$ draw a small circle of which $P$ is the pole to cut $PX$ in $Y$. Since $XX'$ is small, we can consider $XX'Y$ to be a

plane triangle right-angled at $Y$. Denote the hour angle $ZPX$ and the north polar distance $PX$ of $X$ by $H$ and $90° - \delta$ respectively. $X'$ is the observed position of the star and let $ZPX'$ and $PX'$ be denoted by $H'$ and $90° - \delta'$. It can be supposed that an observation of the star enables the values of $H'$ and $\delta'$

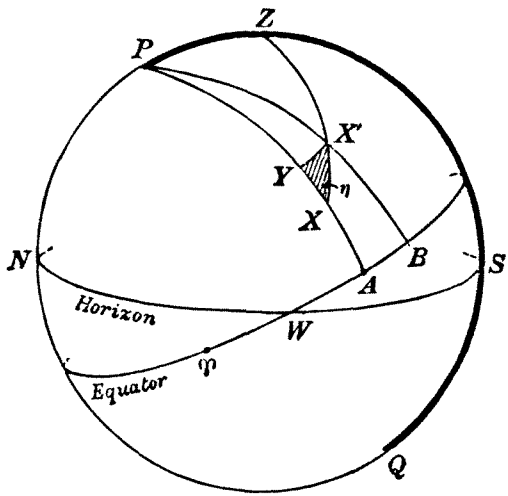



Fig. 31.

to be derived; it is required to find the values of $H$ and $\delta$. From the plane triangle $XX'Y$ in which $\eta$ denotes the angle $X'XY$, we have

$$X'Y = XX' \sin \eta \qquad \text{.....(36),}$$

and

$$XY = XX' \cos \eta \qquad \text{.....(37).}$$

Now $$X'Y = X'\hat{P}Y \sin PX' = (H - H') \cos \delta'.$$

Also $$XY = \delta' - \delta.$$

Using the simple expression for the refraction in formula (7), we have $XX' = k \tan \zeta$, where $ZX' = \zeta$. Hence (36) and (37) become

$$H - H' = k \tan \zeta \sec \delta' \sin \eta \qquad \text{.....(38),}$$

$$\delta - \delta' = -k \tan \zeta \cos \eta \qquad \text{.....(39).}$$

In these equations $\eta$ is the parallactic angle $PXZ$; as $XX'$ is small, $PX'Z$ will differ little from $PXZ$ and so $\eta$ may be regarded as defined by $PX'Z$. Thus $\eta$ can be calculated from the data by means of the cosine formula **A** (we assume $PZ \equiv 90° - \phi$ to be

known). Since we have supposed that $H'$ and $\delta'$ can be derived from the observation of the star, the right-hand sides of (38) and (39) can be calculated; consequently $H$ and $\delta$ can be found. Now if $\Upsilon$ is the vernal equinox the right ascension of the point $X$ is $\Upsilon A$ and of $X'$, $\Upsilon B$. If $\alpha$, $\alpha'$ denote the right ascensions of $X$ and of $X'$, $\alpha' - \alpha = AB = H - H'$. Hence

$$\alpha - \alpha' = -k \tan \zeta \sec \delta' \sin \eta \qquad \ldots\ldots(40).$$

(40) and (39) thus enable us to calculate the true right ascension and declination of the observed body.

## EXERCISES

**1.** Find the approximate north latitude where the effect of refraction, at a time when the sun's declination is 10° S, is to lengthen the day by 15 minutes. (The horizontal refraction is 34′.)

**2.** Assuming that the atmosphere is homogeneous and of height $h$ (*Cassini's hypothesis*), prove that

$$\sin R = \frac{a \sin \zeta}{a+h} (\mu^2 - 2\mu \cos R + 1)^{\frac{1}{2}},$$

where $a$ is the radius of the earth.

**3.** If the relation between $r$ and $\mu$ is $r\mu^{n+1} = \text{constant}$ (*Simpson's hypothesis*), prove that

$$R = \frac{1}{n} \left\{ \zeta - \sin^{-1} \left( \frac{\sin \zeta}{\mu_0{}^n} \right) \right\},$$

and deduce *Bradley's formula*

$$R = \frac{2}{n} \frac{\mu_0{}^n - 1}{\mu_0{}^n + 1} \tan (\zeta - \tfrac{1}{2} nR).$$

**4.** Assuming that the formula for refraction is $R = k \tan \zeta$, prove that the circular disc of the sun appears, due to refraction, as an ellipse whose semi-major and semi-minor axes are $a(1-k)$ and $a(1 - k \sec^2 z)$, where $k$ is expressed in circular measure, $\zeta$ is the observed zenith distance of the sun's centre, $z$ is the true zenith distance and $a$ is the sun's semi-diameter.

**5.** The mean of any two perpendicular diameters of the sun is observed to be $D$. If $z$ is the true zenith distance of the sun's centre, show that the true angular diameter of the sun is

$$D \left\{ 1 + \frac{k}{2} (1 + \sec^2 z) \right\},$$

where $k$ is expressed in circular measure.

**6.** If the declination of a star is unaffected by refraction at a given moment, prove that the azimuth is then a maximum.

**7.** $X_1$ and $X_2$ are two neighbouring stars, the true angular distance between them being $D$ (in seconds of arc). If $Z$ is the zenith, $Z\hat{X}_1X_2 = \psi$ and the true zenith distance of $X_1$ is $z$, prove that the observed angular distance is

$$D - kD\,(1 + \cos^2\psi \tan^2 z),$$

in which $k$ is expressed in circular measure.

**8.** If $\phi$ is the latitude, $H$ the hour angle and $\delta$ the declination of a star, show that refraction diminishes the apparent rate of change of hour angle at the rate of

$$0^s{\cdot}51 \frac{\sin 2\theta}{\sin^2(\delta + \theta)} (\tan\delta + \cot\phi \sec H) \text{ per hour,}$$

where $\tan\theta = \cot\phi \cos H$. (The constant of refraction $= 58''{\cdot}2$.)

Show also that the rate of change of refraction in declination is

$$+ 15''{\cdot}2 \cot\phi \sin H \cos^2\theta \operatorname{cosec}^2(\delta + \theta) \text{ per hour.}$$

**9.** If the formula for refraction is given *exactly* by an equation of the form $R = k\tan\zeta$, how will the refractive index vary with height? Determine $k$ and the height of the atmosphere in terms of the refractive index at ground level.

[*Glas.* 1965.]

**10.** Use the result of exercise 41 of Chapter II to show that the effects of refraction on the right ascension and declination of a star are given by

$$\Delta\alpha = \frac{k\sec^2\delta \sin H}{\tan\delta\tan\phi + \cos H},$$

$$\Delta\delta = \frac{k(\tan\phi - \tan\delta\cos H)}{\tan\delta\tan\phi + \cos H}.$$

## CHAPTER IV

# THE MERIDIAN CIRCLE

**41.** *General description.*

In this chapter we shall consider some of the principal features of the fundamental instrument of astronomy—the meridian circle—by which the right ascensions and declinations of the principal heavenly bodies can be determined with great precision. Meridian circle observations also provide the information by which sidereal clocks can be regulated (in practice it is sufficient to derive the error of a sidereal clock), and after a simple step the true mean time at any instant can be deduced so that the error of a clock keeping mean solar time can be easily found. The instrument consists primarily of a refracting telescope which can be rotated about a fixed horizontal axis (the rotation axis) oriented east and west. The telescope itself can thus move only in the plane of the meridian. Fig. 32 shows the main features of the instrument. In the focal plane of the object-glass are two systems of spider threads, or "wires"—as they are generally called—(*a*) one or sometimes two horizontal wires (a single horizontal wire is shown in Fig. 33), and (*b*) several wires, at right angles to the horizontal wire, arranged symmetrically about a central wire $AB$; these wires will be referred to as the vertical wires. In some instruments, attached to the plate carrying the wires are two micrometers, one capable of moving the horizontal wire (or wires) parallel to $HK$, the other

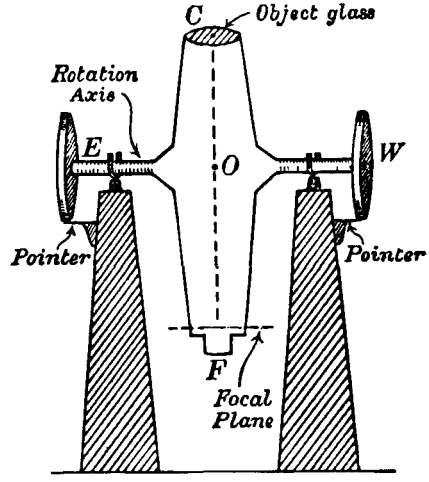



Fig. 32.

capable of moving the system of vertical wires parallel to $AB$. In other instruments the system of vertical wires is fixed, but there is an additional vertical wire which can be moved by a micrometer across the system of fixed wires.

The plane through the centre $C$ of the object-glass and perpendicular to the rotation axis $EW$ is called the *collimation plane*, and the straight line through $C$ lying on this plane and intersecting the rotation axis is called the *collimation axis* ($CO$ in Fig. 32). As the telescope rotates about the axis $EW$, the collimation axis will clearly sweep out the collimation plane. Assume, for a moment, that the instrument is mechanically perfect and that it has been set up accurately, with the central wire $AB$ in the collimation plane; then, as viewed through the eye-piece at $F$, any star which, at a given instant, is observed on the central wire will, at that moment, be on the meridian. If this instant is noted by means of a sidereal clock keeping accurate sidereal time, this sidereal time (otherwise described as the hour angle of the vernal equinox) is clearly equivalent to the star's right ascension. The function of the additional vertical wires in Fig. 33 is to give additional accuracy; for, as the star moves across the field of view owing to the diurnal motion, it coincides with the wires in succession and, if the coincidences with each of the vertical wires are noted, the mean will presumably give a much more accurate determination of the sidereal time of transit—and hence the star's right ascension—than if the observation is restricted to the single coincidence with the central wire.

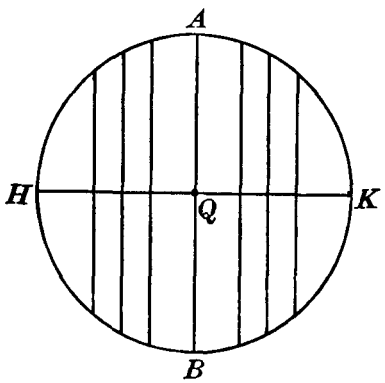


Fig. 33.

Attached to the axis of rotation are two finely graduated circles (shown at $E$ and $W$ in Fig. 32), with auxiliary optical arrangements for enabling an accurate measurement of a star's altitude to be made; as the latitude of the meridian circle may be presumed known, the star's declination can then be simply

deduced. Such then, in brief, are the main principles according to which the positions of the stars are obtained from meridian circle observations.

**42.** *Instrumental errors.*

It is impossible, however, to set up an instrument with the precision necessary for the accurate measurement of stellar positions, as just indicated, and it is necessary in actual practice to take into consideration the inherent errors of the instrument. These errors are (i) *azimuth error*—the axis of rotation is not accurately oriented east and west and its angular deviation from the true east and west direction is the azimuth error (sometimes called the deviation error), which will be denoted by $a$; (ii) *level error*—the axis of rotation is not accurately horizontal and the angular deviation from the horizontal is called the level error, which will be denoted by $b$; (iii) *collimation error*—the central wire is not quite in the collimation plane, and the angle between the collimation axis and the line joining the middle point of the vertical wire to the centre of the object-glass is called the collimation error (denoted by $c$). We shall examine in turn the effect of these errors on the observed time of transit of a star.

**43.** *Azimuth error.*

Assume that this is the only error. Fig. 34 shows the celestial sphere with the horizon and equator and the cardinal points; the centre of the sphere is taken to be the point of intersection of the collimation axis with the rotation axis. If the instrument were perfectly set up, one end of the axis would point to the west point $W$ and the other to the east point $E$. We shall suppose, with azimuth error presumed, that the west end of the rotation axis points towards the point $A$ of the horizon, $WA$ being the azimuth error; the convention is that the azimuth error is positive when $A$ is between $W$ and $S$ as in Fig. 34. As the telescope is rotated about its axis, the collimation axis will describe a plane perpendicular to the rotation axis and cutting the celestial sphere in a great circle of which $A$ is the pole: this great circle we shall designate the "fictitious meridian". Since $A$ is a point on the horizon, the fictitious meridian will pass through $Z$. Let $X$ be the position of a star at the moment of its coincidence

with the central wire (which is in the collimation plane); then $ZX$ is an arc of the fictitious meridian. Let $\tau_1$ denote the angle $ZPX$; it is seen from the figure that since $X$ is the position of the star when it is observed on the central wire, the true time of transit will occur later, after an interval $\tau_1$, when the star reaches the true meridian at $Y$. $\tau_1$ is thus the error in the time of transit of the star.

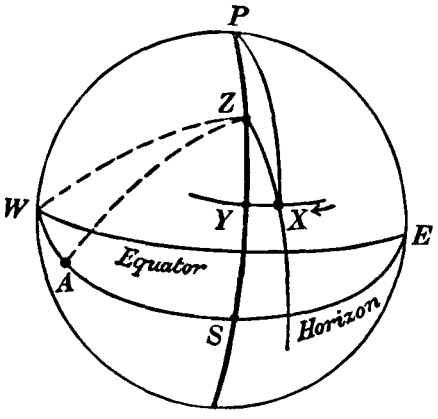


Fig. 34.

Now $W\hat{Z}A \equiv WA = a$, since $Z$ is the pole of $WASE$. Also $W\hat{Z}S = 90°$ and $A\hat{Z}X = 90°$, since $A$ is the pole of the great circle $ZX$ (the fictitious meridian). Hence

$$Y\hat{Z}X = a,$$

and therefore $P\hat{Z}X = 180° - a$. From the triangle $PZX$ in which $PZ = 90° - \phi$, $PX = 90° - \delta$, $Z\hat{P}X = \tau_1$ and $P\hat{Z}X = 180° - a$, we have by formula **B** (the sine-formula),

$$\sin PZX \sin ZX = \sin ZPX \sin PX,$$

or

$$\sin a \sin ZX = \sin \tau_1 \cos \delta.$$

Also, $a$ and $\tau_1$ are small angles so that, in general, $ZX$ is approximately equal to $ZY$, that is, to $(\phi - \delta)$; also $\sin a = a$, $\sin \tau_1 = \tau_1$, and consequently,

$$\tau_1 = a \sin (\phi - \delta) \sec \delta \qquad \ldots\ldots(1),$$

in which both $\tau_1$ and $a$ may be supposed expressed in seconds of time. This formula gives the correction to the observed time of transit due to azimuth error alone.

### 44. *Level error.*

Assume now that the only instrumental error is the level error and that the west end of the axis is tilted upwards by the angle $b$. In this case the sign of $b$ is, by convention, positive. On the celestial sphere the west end of the axis will be represented by $B$ (Fig. 35) so that $WB = b$; since there is no other error, $B$ will lie

in the prime vertical $ZW$. As the telescope is rotated about its axis, the collimation axis will describe a plane cutting the celestial sphere in a great circle of which $B$ is the pole. In Fig. 35 this great circle is $SX$ (it passes through $S$, since $S$ is the pole of $WZ$ and consequently $SB = 90°$). $SX$ is now the fictitious meridian. Now $WB \equiv W\hat{S}B = b$; also $W\hat{S}Z = 90°$ and $B\hat{S}X = 90°$; hence $Z\hat{S}X = b$. Consider a star at the observed moment of transit over the central wire; it is then at $X$, we shall suppose, on the fictitious meridian, and its observed time of transit occurs before it reaches the true meridian at $Y$. Let $X\hat{P}Y = \tau_2$; then $\tau_2$ is the correction to be applied to the observed time of transit to give the true time of transit at $Y$. In the triangle $PXS$, $PX = 90° - \delta$, $PS = PZ + ZS = 180° - \phi$, $S\hat{P}X = \tau_2$ and $P\hat{S}X = b$. By formula **B**,

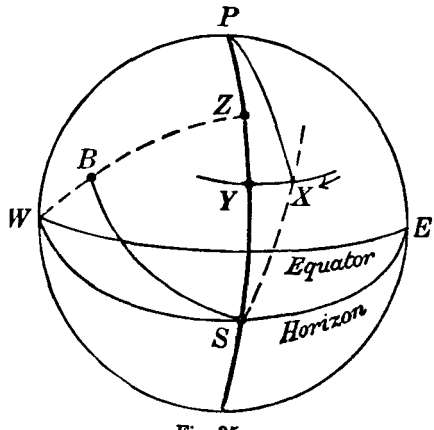


Fig. 35.

$$\sin \tau_2 \sin PX = \sin b \sin SX.$$

Since $b$ is always very small, $SX$ is approximately equal to $SY$. Also

$$SY = PS - PY = 180° - \phi - (90° - \delta) = 90° - (\phi - \delta).$$

Hence $$\sin \tau_2 \cos \delta = \sin b \cos (\phi - \delta).$$

Express the small angles $\tau_2$ and $b$ in seconds of time; then

$$\tau_2 = b \cos (\phi - \delta) \sec \delta \qquad \ldots\ldots(2).$$

This formula gives the correction to the observed time of transit due to level error alone.

### 45. *Collimation error.*

Assume now that the only error is collimation error. In this case the central wire lies outside the collimation plane, and as the telescope is rotated the central wire will describe a plane parallel to the plane of the true meridian; this plane will inter-

sect the celestial sphere in a small circle $XU$ (Fig. 36) which is now the fictitious meridian. The straight line joining the centre of the object-glass to the middle point of the central wire will make a small angle $c$ with the collimation axis and as the latter is perpendicular to the rotation axis, in this case $EW$, the angular distance of $W$ from any point on the small circle $XU$ is $90° + c$. (The sign of $c$ is taken to be positive when the fictitious meridian $XU$ cuts the horizon to the east of $S$, as in Fig. 36.) Consider now a star $X$ at the moment of transit over the central wire; when $c$ is positive, it is seen from the figure that the star is on the central wire before it reaches the true meridian at $Y$. Let $Z\hat{P}X = \tau_3$. In the spherical triangle $WPX$, $WX = 90° + c$, $WPX = 90° + \tau_3$, $PX = 90° - \delta$ and $WP = 90°$. Then, by formula **A**,

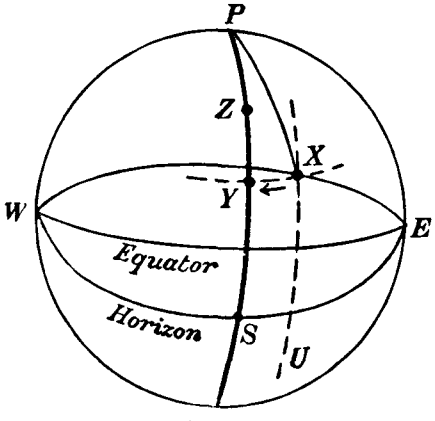


Fig. 36.

$$\cos WX = \cos WP \cos PX + \sin WP \sin PX \cos WPX,$$

from which $$-\sin c = -\cos\delta \sin\tau_3,$$

so that, $c$ and $\tau_3$ being small and both expressed in seconds of time,

$$\tau_3 = c \sec\delta \qquad \text{.....(3).}$$

**46.** *The total correction to the observed time of transit.*

Let $T$ be the observed time of transit over the central wire according to the sidereal clock—in practice, $T$ would be taken to be the mean of the observed times of transit over, say, seven wires, three on each side of the central wire. Suppose that the clock is in error by $\Delta T$, $\Delta T$ being regarded as positive when the clock is slow. Then the true time of transit over the central wire is $T + \Delta T$. The sidereal time of transit over the true meridian is then $T + \Delta T + \tau_1 + \tau_2 + \tau_3$. This is the right ascension ($\alpha$) of

the star. Thus, $\alpha = T + \Delta T + \tau_1 + \tau_2 + \tau_3$, or, from (1), (2) and (3)

$$\alpha = T + \Delta T + \sec\delta\{a \sin(\phi - \delta) + b\cos(\phi - \delta) + c\} \quad \ldots\ldots(4),$$

which is sometimes written in the form

$$\alpha = T + \Delta T + aA + bB + cC \quad \ldots\ldots(5),$$

where $A = \sin(\phi - \delta)\sec\delta$, $B = \cos(\phi - \delta)\sec\delta$, and $C = \sec\delta$. Actually, the error $\Delta T$ of the sidereal clock is not constant from day to day, but we shall assume, however, in determining the right ascensions of the stars according to (5), that its value is known for each observation. Also the formula (5) is incomplete in the form just given, as a small term due to aberration (see section 111) has not yet been included; the value of this term is $-0^s{\cdot}021\cos\phi\sec\delta$.

The formulae given so far refer to the upper culmination of a star. The corresponding formulae for lower culmination can be derived in a similar manner. The formula corresponding to (4) is found to be of similar character and is derivable from (4) by writing $(180° - \delta)$ for $\delta$ in that equation.

### 47. *Bessel's formula.*

The correction $\tau$ to the observed time of transit of a star due to all three errors $a$, $b$ and $c$, considered together, can be in-

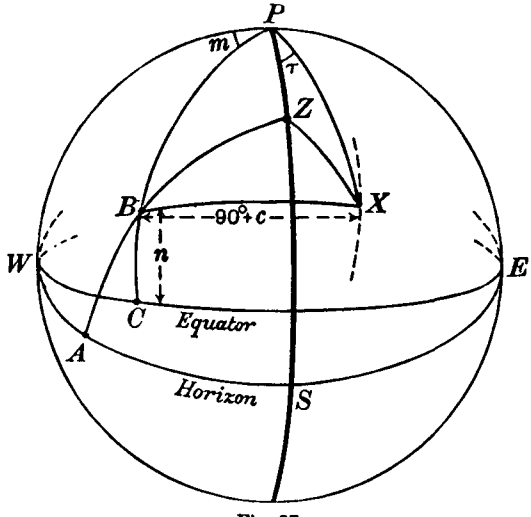



Fig 37.

vestigated as follows. Suppose that the west end of the rotation axis points to the position $B$ on the celestial sphere. Draw the vertical circle $ZBA$ to meet the horizon in $A$. Then $WA = a$ (azimuth error) and $AB = b$ (level error). Also let $X$ be the position of a star at the moment of coincidence with the central wire. Then $BX = 90° + c$. Draw the meridian $PBC$ to meet the equator in $C$ and let $WC \equiv W\hat{P}C = m$ and $BC = n$. The quantities $m$ and $n$ are simply related to $a$ and $b$. In the triangle $PBZ$, $PZ = 90° - \phi$, $PB = 90° - n$, $BZ = 90° - b$, $P\hat{Z}A = 90° + a$ and $Z\hat{P}B = 90° - m$. By **A**, we have

$$\cos PB = \cos PZ \cos ZB + \sin PZ \sin ZB \cos PZB,$$

or $$\sin n = \sin \phi \sin b - \cos \phi \cos b \sin a,$$

which can be written with sufficient accuracy, since $n$, $a$ and $b$ are small,

$$n = b \sin \phi - a \cos \phi \qquad \ldots\ldots(6).$$

Also, by formula **D**,

$$\cos PZ \cos PZB = \sin PZ \cot ZB - \sin PZB \cot ZPB,$$

or $$-\sin \phi \sin a = \cos \phi \tan b - \cos a \tan m,$$

from which we have, since $m$, $a$ and $b$ are small,

$$m = a \sin \phi + b \cos \phi \qquad \ldots\ldots(7).$$

Now in triangle $BPX$, $BX = 90° + c$, $PB = 90° - n$, $PX = 90° - \delta$ and $BPX = 90° + \tau - m$, where $\tau$ is the angle $ZPX$ (the correction to the observed time of transit at $X$ over the central wire to give the time of transit over the true meridian $PZS$). By **A**

$$\cos BX = \cos PB \cos PX + \sin PB \sin PX \cos BPX,$$

whence

$$-\sin c = \sin n \sin \delta - \cos n \cos \delta \sin (\tau - m),$$

or, since $c$, $n$ and $(\tau - m)$ are small angles,

$$-c = n \sin \delta - (\tau - m) \cos \delta,$$

from which we obtain

$$\tau = m + n \tan \delta + c \sec \delta \qquad \ldots\ldots(8).$$

This is Bessel's formula and is the form found most convenient in practice. It is easily verified that when the values of $m$ and $n$, given by (6) and (7) in terms of $a$ and $b$, are substituted in (8), formula (4) is obtained. Inserting the clock error $\Delta T$ and the aberration term we obtain, for the right ascension of the star under observation,

$$\alpha = T + \Delta T + m + n \tan \delta + (c - 0^{s}{\cdot}021 \cos \phi) \sec \delta \ldots(9).$$

48. *Wire intervals.*

As already stated, some meridian circles are fitted with a micrometer by means of which the entire system of vertical wires can be moved across the field of view; in other instruments the system of vertical wires is fixed and the micrometer actuates a moving wire, which can be placed successively on any two of the fixed wires. In both types it is necessary to determine

(*a*) the interval between two consecutive wires in terms of the micrometer scale, and

(*b*) the value of one revolution of the micrometer in terms of angular measure.

To determine (*a*) in the first type of instrument, suppose that a distant object is visible in the field of view when the telescope is approximately horizontal. By moving the micrometer so that two wires are successively coincident with the image of the distant object, the interval between the wires is expressed as the difference between the micrometer readings. The observation can then be repeated for any other pair of wires. It is obvious that, in the second type of instrument, the wire intervals can be easily found in terms of the micrometer scale.

To determine (*b*), the interval of time required by a star to pass from one wire $W_1$ to another wire $W_2$ is observed. A star near the pole is selected for this purpose as it moves comparatively slowly across the field of view. As the telescope is rotated about its axis, the wire $W_1$ will sweep out a plane which will intersect the celestial sphere in a small circle of which the west end $B$ (as in Fig. 37) is the pole. Let the angular distance of $B$ from any point on this small circle be $90° + c_1$; then $c_1$ is the collimation error of $W_1$. Then if $\tau_1$ is the interval between the transit of the star $X$ (Fig. 37) over wire $W_1$ and over the true meridian, we have by (8), taking into account that $c_1$ and $n$ are small,

$$\tau_1 = m + c_1 \sec \delta + n \tan \delta \qquad \text{.....(10).}$$

Now consider wire $W_2$ and let $\tau_2$ and $c_2$ be the corresponding quantities. Then we have, as before,

$$\tau_2 = m + c_2 \sec \delta + n \tan \delta \qquad \text{.....(11).}$$

If $T_1$, $T_2$ are the times by the sidereal clock when the star is on wires $W_1$ and $W_2$ respectively, and $T_0$ is the time by the clock

when the star is on the true meridian, we have

$$T_0 = T_1 + \Delta T + \tau_1 \quad \text{and} \quad T_0 = T_2 + \Delta T + \tau_2.$$

Also if $t = T_2 - T_1$ ($t$ being the time in seconds required by the star to pass from $W_1$ to $W_2$) then, eliminating $T_0$ and $\Delta T$,

$$t = \tau_1 - \tau_2,$$

so that, by (10) and (11),

$$t = (c_1 - c_2) \sec \delta,$$

or

$$c_1 - c_2 = t \cos \delta \qquad \text{.....(12).}$$

Now $c_1 - c_2$ is the angular separation of the wires $W_1$ and $W_2$ (expressed in seconds of time), and it can consequently be calculated from (12), for $t$ is observed and $\delta$ is known. Formula (12) is sufficiently accurate in practice unless the star's declination is very near to 90°.

As we have explained, the interval between the two wires $W_1$ and $W_2$ can be found in terms of the micrometer scale, and thus we are enabled to express one revolution of the micrometer head in terms of seconds of time.

From (12), it is seen that $c_1 - c_2$ (expressed in seconds of time) is the interval required for an equatorial star to pass from wire $W_1$ to wire $W_2$.

*Example.* On 1931 January 3, the interval between the transits of δ Ursae Minoris over two of the wires of the Cambridge Meridian Circle was $3^m\ 4^s$; to find the wire interval.

From the *Nautical Almanac*, $\delta = +\ 86°\ 36'\ 34''$. Also $t \equiv 3^m\ 4^s = 184^s$. Thus

$$c_1 - c_2 = 184^s \cos 86°\ 36'\ 34''$$
$$= 10^s{\cdot}88.$$

49. *Determination of the collimation error.*

The methods of determining the instrumental errors vary according to the size or elaborateness of the meridian circle concerned. It is not our purpose to describe all the different methods in use and for details the student is recommended to study a book dealing specifically with practical astronomy.* There are, however, certain general principles involved in the

* For example, *The Elements of Practical Astronomy*, by W. W. Campbell (Macmillan, 1899).

determination of the various errors and we shall consider these in turn, taking first the collimation error.

As regards large instruments the collimation error is determined by means of two *collimators*, one north of the telescope and the other south. Consider the south collimator. It consists of an object-glass mounted vertically, with its axis in or near the meridian and at the same height above the floor as the rotation axis of the telescope. In the focal plane ($AB$) of the object-glass ($C$) of the collimator are placed one or two vertical wires and one or two horizontal wires (Fig. 38). (For simplicity, we shall suppose that there is only one vertical wire and one horizontal wire.) These wires can be viewed by means of an eye-piece just outside the focal plane $AB$. When the meridian circle telescope is pointed horizontally and south, the observer can see in the field of view at $E$ the collimator wire $W$ when it is suitably illuminated. Since the collimator wire is in the focal plane of $C$, rays passing from $W$ through the collimator will emerge at $C$ as a parallel beam and after entering the object-glass $D$ of the telescope the wire will be seen sharply at $E$. In this optical arrangement the observation of the collimator wire is equivalent to the observation of an object at an infinite distance. The north collimator is arranged in a similar way, but we shall suppose that the vertical wire in this collimator is movable by means of a micrometer and that the south collimator wire is fixed. When the telescope is rotated to the vertical position, shutters in the framework of the axis can be opened, allowing an unimpeded view from one collimator to the other. Thus the observer, if stationed at the eye-end of the north collimator, can see the wire of the south collimator and by actuating the micrometer can make the north collimator wire coincide with the image of the south collimator wire. When this is done, the combined optical arrangement is such that the south



Fig. 38.

collimator wire, when viewed in the telescope, appears as if it is at an infinite distance in a particular direction and the north collimator wire, when viewed in the telescope, appears as if it is at an infinite distance in exactly the opposite direction. It is unlikely that the direction of the south collimator wire is exactly south; suppose that the direction is $\theta$ degrees west of south; then the north collimator wire will be in the direction $\theta$ degrees east of north. The method of determining the collimation error $c$ must be such that the unknown quantity $\theta$ is eliminated in the observations.

Suppose that the telescope is first pointed to the south collimator so that its vertical wire is visible. This wire corresponds to an object whose position on the celestial sphere is indicated by the point $F$ (Fig. 39), $FS$ being the angle $\theta$. Assume that the collimation error $c$ is positive; then on the celestial sphere the central wire will sweep out the small circle $CD$ (as the telescope is rotated about its axis) of which the west axis $B$ of the instrument is the pole, and the great circle arc joining $B$ to $C$ is $90° + c$. Since the level error is always small, this great circle arc will be practically coincident with the horizon in the neighbourhood of the south point $S$; for the present purpose we can thus ignore level error and assume that the west part of the axis of the telescope points to $A$, $A$ being on the horizon such that $WA$ is the azimuth error $a$. Then $AC = 90° + c$. Now $AS = 90° - a$ and $AC = 90° + c$; hence $SC = a + c$ and, since $FS = \theta$,

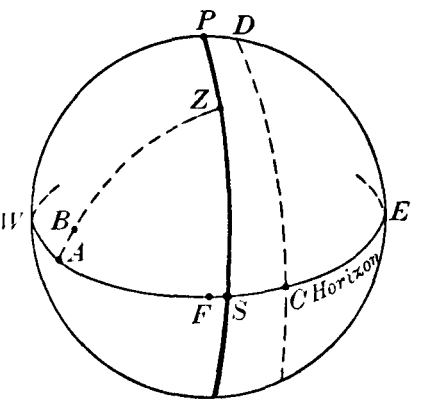


Fig. 39.

$$FC = \theta + a + c \qquad \text{.....(13).}$$

We shall consider a meridian circle fitted with a fixed system of vertical wires and a movable wire connected with a micrometer whose scale has been determined, for example, by the method of section 48. Place the movable wire over the central wire (corre-

sponding to $C$ in Fig. 39) and let the reading be $M_1$; then place it over the image of the south collimator wire $F$ and let the reading be $M_2$. Then, in terms of the micrometer scale, $M_2 - M_1 = CF$, and writing $D = M_2 - M_1$ we have, from (13),

$$D = \theta + a + c \qquad \text{.....(14).}$$

It is necessary to make some convention with regard to the sign of $D$; the sign will be assumed to be positive when, as in Fig. 39, $C$ is east of $F$; as viewed in the telescope (the optical combination is inverting) the central wire $C$ will thus appear to be west of the south collimator wire $F$.

Now suppose the telescope pointed to the north collimator. Neglecting the level error, the east end of the axis will point to $A_1$ (Fig. 40) such that $EA_1 = a$. The small circle on the celestial sphere, of which $A_1$ is the pole, associated with the central wire will intersect the horizon, between the north and east points, at $C_1$. Since $A_1$ in Fig. 40 and $A$ in Fig. 39 are diametrically opposite points and since $AC = 90° + c$, then we have $A_1C_1 = 90° - c$. Also $A_1N = 90° - a$; hence $C_1N = c - a$. Again, the vertical wire of the north collimator is in the direction $\theta$ degrees east of north; on the celestial sphere it will define the position $G$ such that $GN = \theta$. Hence

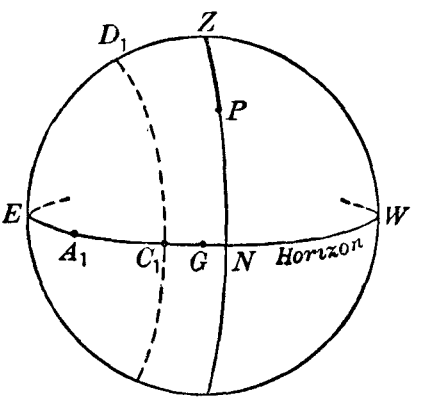


Fig. 40.

$$GC_1 = c - a - \theta \qquad \text{.....(15).}$$

Now the micrometer reading when the movable wire coincides with the central wire is $M_1$; let the reading be $M_3$ when the movable wire coincides with the north collimator wire $G$. Write $D' = M_3 - M_1$. Then according to Fig. 40, since $C_1$ is east of $G$ so that in the telescope the central wire appears west of the collimator wire, the sign of $D'$ is positive by our convention. By (15) we have

$$D' = c - a - \theta \qquad \text{.....(16).}$$

From (14) and (16),

$$c = \tfrac{1}{2}(D + D') \qquad \text{.....(17).}$$

This is the formula for the collimation error $c$. $D$ and $D'$ are at first expressed in terms of the micrometer scale, and, from the previously ascertained value of one revolution of the micrometer in terms of seconds of time, we are thus enabled to express $c$ in terms of the latter unit.

It is to be noted that for instruments in which the system of vertical wires is movable the effect of collimation error and the aberration correction (to which reference has been made in section 46) on the time of transit of a star can be removed. To each position of the central wire corresponds a definite value of $c$, and, if the wires are moved to such a position that

$$c - 0^{s}\cdot021 \cos \phi = 0 \qquad \text{.....(18)}$$

(in which $\phi$ is the latitude), the only corrections to be applied to the time of transit over the central wire to give the time of transit over the true meridian are the corrections due to the azimuth error $a$ and the level error $b$.

**50.** *Determination of the level error.*

As regards the smaller instruments, this error can generally be derived with sufficient accuracy by means of a striding level, placed on the axis of rotation. For large instruments, a more precise method is employed, which we shall now describe. The telescope is pointed vertically downwards towards a bowl of mercury which is generally placed below the floor level to avoid atmospherical and other disturbances. The surface of the mercury acts as a reflecting horizontal plane, the normal to which defines the direction of the zenith. A special eye-piece—the Bohnenberger eye-piece—enables a beam of light to pass into the telescope; after traversing the object-glass, the beam will be reflected at the mercury surface and will then re-enter the object-glass. The central wire and its image formed as a result of reflection at the mercury surface will both be seen in the field of

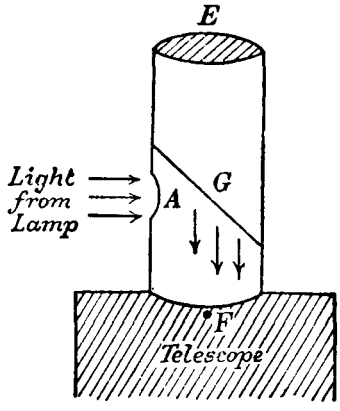



Fig. 41.

view. The special eye-piece (Fig. 41) consists of a tube in the side of which is an aperture $A$ through which the horizontal rays from a lamp, placed some distance away, enter the tube. In the tube is placed opposite $A$ a piece of clear glass $G$ inclined at 45° to the axis of the tube. The glass surface acts as a reflector and so part of the beam entering at $A$ is reflected by $G$ into the telescope. The glass being unsilvered, the observer with his eye at $E$ can see the vertical wires and their images due to the reflection at the mercury surface. In Fig. 42, let $EW$ be the rotation axis of the telescope inclined at the angle $b$ to the horizontal, $O$ the centre of the object-glass, $C$ the central wire and $D$ its image, and $AB$ the mercury surface. The ray $CO$ from $C$ to the centre of the object-glass will be undeviated by the latter (supposed thin) and will strike the mercury surface at $B$. Since $C$ is in the focal plane of the object-glass any other ray $CG$ will emerge from the object-glass along $GA$ parallel to $OB$. Thus a cone of rays from $C$ will emerge as a parallel beam. This beam will be reflected at the mercury surface into another parallel beam which will be brought by the object-glass to a focus at $D$. Consider the ray $COB$; at $B$ it is reflected along $BH$ in such a way that $O\hat{B}S = S\hat{B}H$, $BS$ being perpendicular to the mercury surface at $B$; after passing through the object-glass at $H$, it passes through $D$. Produce $DO$ to meet the mercury surface at $A$. Let $CG$ be the ray from $C$ which is incident on the mercury surface at $A$. Then this ray is reflected along $AO$ and will pass through $D$; thus $GA$ is parallel to $OB$ and $AO$ is parallel to $BH$.

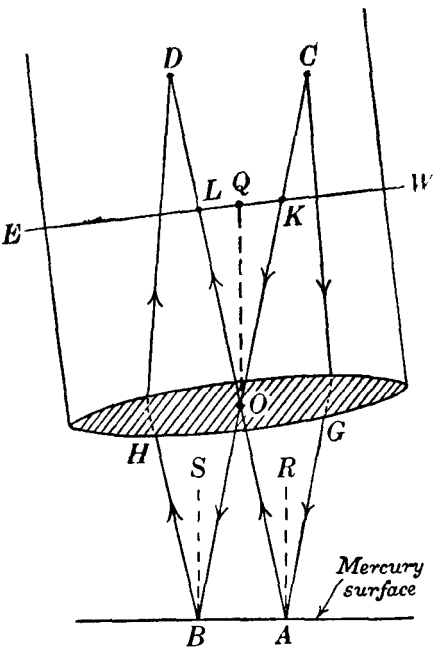


Fig. 42.

It follows that $C\hat{O}D = G\hat{A}O = O\hat{B}H$ and each is twice $O\hat{B}S$. Let $OQ$ be parallel to $BS$ and $AR$. Then, since $OQ$ is perpendicular to the mercury surface, $K\hat{Q}O = 90° + b$. Also $W\hat{K}O = 90° + c$, where $c$ is the collimation error. Now

$$K\hat{Q}O = W\hat{K}O - Q\hat{O}K,$$

that is

$$b = c - \tfrac{1}{2}C\hat{O}D.$$

The angle $COD$ can be measured. Place the movable wire over the central wire $C$ and then over its image at $D$; the difference in readings is, say, $R$, which can be converted into seconds from the known value of the micrometer scale. We thus have

$$b = c - \tfrac{1}{2}R \qquad \text{.....(19),}$$

from which $b$ can be derived, the value of $c$ being supposed known. When the system of vertical wires is movable, the value of $\frac{1}{2}R$ is obtained simply by moving the wires until the central wire coincides with its image. The convention as to the sign of $R$ is as follows: when the central wire is seen in the telescope to the west of its image, $R$ is defined to be positive.

**51.** *Determination of the azimuth error.*

The azimuth error $a$ is determined from observations of stars, the values of $b$ and $c$ having previously been derived. Let $(\alpha_1, \delta_1)$ and $(\alpha_2, \delta_2)$ be the equatorial co-ordinates of two stars, $T_1$ and $T_2$ the observed times of transit by the sidereal clock. By (5), we have

$$\alpha_1 = T_1 + \Delta T + aA_1 + bB_1 + cC_1,$$
$$\alpha_2 = T_2 + \Delta T + aA_2 + bB_2 + cC_2.$$

In these equations, $\Delta T$ is the unknown error of the clock, and if, by a suitable choice of stars, the observations are made within a few minutes of each other, the value of $\Delta T$ can be assumed to be the same in the two equations; also $A_1 = \sin(\phi - \delta_1)\sec\delta_1$, etc. By subtraction, $\Delta T$ is eliminated and we have

$$a = \frac{(\alpha_2 - \alpha_1) - (T_2 - T_1) - b(B_2 - B_1) - c(C_2 - C_1)}{A_2 - A_1} \qquad \text{.....(20).}$$

In this equation,

$$A_2 - A_1 = \sin(\phi - \delta_2)\sec\delta_2 - \sin(\phi - \delta_1)\sec\delta_1.$$

The right-hand side of (20) can be calculated from the known values of $b$, $c$, $\alpha_1$, $\alpha_2$, $\delta_1$, $\delta_2$ and the observed times $T_1$, $T_2$. It remains to indicate how best the choice of stars can be made to ensure an accurate determination of $a$. It is clear from (20) that $a$ can be most accurately determined if the denominator $(A_2 - A_1)$ is large. This condition is fulfilled if (i) a star of high declination ($\delta_2$) and a star near the equator (declination $\delta_1$) are selected, for then $A_2$ is large, since sec $\delta_2$ is large, or (ii) a star of high declination ($\delta_2$) observed at upper culmination and a star of high declination ($\delta_1$), differing from the first by about 12 hours in right ascension, observed at lower culmination. In (ii), the values of $A_1$ and $A_2$ will both be large but of opposite sign, so that $(A_2 - A_1)$ will be numerically large.

**52.** *The chronograph (mainly of historical interest).*

In modern instruments the times of transit of a star over the system of vertical wires are recorded electrically by means of a chronograph. A clockwork mechanism causes a paper tape to be drawn out at a uniform rate (or a cylinder covered by a sheet of paper to revolve at a uniform rate). The sidereal clock by which the times of transit are to be recorded is connected electrically with an electro-magnet belonging to the chronograph and to which a pen is attached. At the bottom of its swing, the pendulum of the clock closes an electrical circuit for an instant and the action of the momentary current passing through the electro-magnet is to cause the pen to "kick". When there is no current, the pen traces out a straight line on the moving tape (or a uniform line on the rotating cylindrical paper), but when contact is made by the pendulum, the "kick" of the pen results in a distinctive mark being made on the paper. In this way, the seconds of the clock are mechanically registered. Usually, there is an automatic arrangement by which the pen misses the 60th second of every minute, and thus it is easy, by simple reference to the clock, to ascertain the hour and minute of any "blank"; this time can be written down on the paper and any particular second of the following minute deduced by counting the "kicks" of the pen from the "blank". This is illustrated in the upper trace of Fig. 43. The chronograph carries a second pen, attached to an electro-magnet, which makes a trace parallel to that of the

clock pen, and this pen can be actuated by the observer closing a circuit by means of a tapping-key. When he sees, in the telescope, the star on a vertical wire he gives the key a sharp tap and the second pen makes a "kick" on the moving paper. The time of transit over this wire can be inferred later from the chronograph record (this is illustrated in the lower trace of Fig. 43).

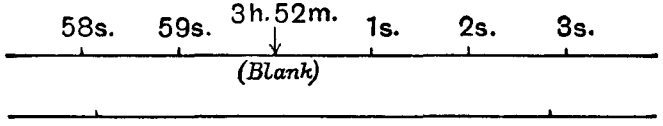



Fig. 43.

There are in use several modifications of the procedure just outlined. In one, the observer moves a vertical wire, by means of the attached micrometer, in such a way that the wire appears to bisect continuously the image of the star as it moves across the field of view; at definite points, corresponding to the positions of the system of vertical wires, the frame supporting the wire closes an electrical circuit and the chronograph pen makes the appropriate records. In other instruments, the moving wire is driven across by a small motor, at a rate adjusted according to the declination of the star, and the observer has only to make the necessary small corrections to the rate to ensure the accurate bisection of the stellar image by the moving wire.

**53.** *The measurement of declination.*

We have already mentioned that the rotation axis of the telescope carries two finely graduated circles, the graduations being generally at 5′ intervals from 0° to 360°. In the focal plane of the object-glass there is a horizontal wire and when observations of a star's declination are being made the telescope is adjusted so that, in the neighbourhood of the collimation axis, the star appears to travel along the horizontal wire. In this position, the circles indicate a particular reading $R_1$, the accurate determination of which is facilitated by the use of four microscopes (90° apart) on each circle. To obtain the zenith distance of the star we require to know the circle reading $R$ when the telescope is pointed accurately to the zenith. We shall suppose

for the moment that the horizontal wire is fixed in the focal plane of the object-glass. The reading $R$ can then be determined by pointing the telescope downwards towards the basin of mercury and moving the telescope gradually (by slow-motion screws) until the horizontal wire and its image coincide when viewed in the special eye-piece of Fig. 41; the addition of 180° to the reading of the circles in this position gives the reading $R$, corresponding to the position of the telescope when pointing to the zenith. The difference between the readings $R$ and $R_1$ is the meridian zenith distance of the star. This zenith distance of course contains the refraction and when the latter is removed we obtain the true zenith distance of the star. The latitude being presumed known, we finally deduce the declination of the star. But in practice it is inadvisable to attempt the delicate adjustment of the instrument just contemplated. As we invariably know the approximate declination of the star to be observed, the telescope can be pointed with sufficient accuracy to ensure that in due course the star will appear in the field of view. Generally, it will appear to move (disregarding a slight curvature in its path) parallel to and at some distance from the horizontal wire. In this position of the telescope the circles can be read either before or after the star's appearance. Let the reading be $R_2$. Now assume that the horizontal wire can be moved parallel to itself by means of a micrometer to such a position that the star appears to travel along the wire. Let the two micrometer readings be $M_0$ and $M_1$, the former the original reading (corresponding, say, to some definite reading on the micrometer head) and the latter the reading when the star travels along the wire. The value of one revolution of the micrometer can be found in terms of seconds of arc by observations of two stars of known declination which are close together in the sky; with the telescope fixed, the difference in declination can be measured in terms of the micrometer scale and hence the value of this scale (in seconds of arc) can be deduced. Now, corresponding to $M_0$, there is the particular reading $R$ of the circles when the telescope is pointing accurately to the zenith; the zenith distance of the star is not quite the difference between $R$ and $R_2$, as in the case of the fixed horizontal wire; we have to apply to this difference a correction given by $(M_1 - M_0)$ expressed in seconds of arc.

Thus the zenith distance of the star is determined and its declination can be deduced as before.

**54.** *The measurement of right ascension.*

The point on the celestial equator from which right ascensions are measured is the vernal equinox, whose position is given by the centre of the sun about March 21 when the sun passes from south declination to north declination. At the summer solstice (about June 21) the sun has reached its maximum northerly declination and then its declination is simply the obliquity of the ecliptic. Observations of its meridian zenith distances on several days before and after the solstice will lead to an accurate value of its declination exactly at the solstice; in other words the obliquity of the ecliptic can be found. It is to be noticed that in these considerations it is the sun's centre that is specified; in the actual observations the meridian zenith distance of the upper (or lower) limb is measured; to obtain the zenith distance of the centre, the sun's *semi-diameter* (the angle subtended at the observer by a radius of the sun) requires to be added (or subtracted).

Since the vernal equinox is so closely associated with the sun, it follows that the right ascension of any star is fundamentally related to the position of the sun on any day. Let us suppose that the complete observation of the sun on a particular day gives

(i) the clock time $T$ (corrected for the instrumental errors $a$, $b$ and $c$) of the transit of its centre, and

(ii) the zenith distance of its centre at transit.

Assuming that we know the latitude accurately we deduce, as in the previous section, the declination of the sun's centre [we omit in this discussion the correction due to "parallax" (see Chapter IX)]. From the declination and the value of the obliquity $\epsilon$ (which we shall assume known), we can calculate the right ascension $\alpha$ of the sun at the time of transit on the day in question. In the triangle $\Upsilon SA$ (Fig. 44), $PSA$ is the meridian through the sun $S$, and $SA\Upsilon = 90°$, $\Upsilon A = \alpha$, $AS = \delta$, $S\Upsilon A = \epsilon$. By formula D,

$$\cos\alpha\cos 90° = \sin\alpha\cot\delta - \sin 90°\cot\epsilon,$$

or

$$\sin\alpha = \tan\delta\cot\epsilon \qquad \text{.....(21).}$$

From (21) we obtain $\alpha$, the sun's right ascension at the moment of transit on the day concerned. Let $E$ be the error of the sidereal clock at the time $T$ of transit. Then the true sidereal time of transit is $T + E$. But this is precisely the sun's right ascension $\alpha$. Hence

$$\alpha = T + E \qquad \text{.....(22).}$$

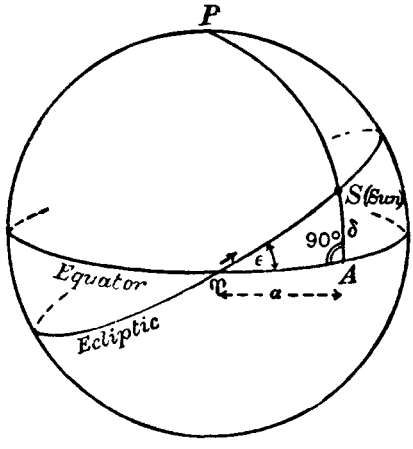


Fig. 44.

Suppose that the transit of a star is observed afterwards, as soon as practicable. Let the clock time of transit be $T_1$ (we suppose that both $T_1$ and $T$ have been corrected for the instrumental errors $a$, $b$ and $c$). The error of the clock may now be presumed to be slightly different; let it be $E_1$. Then, if $\alpha_1$ is the star's right ascension,

$$\alpha_1 = T_1 + E_1 \qquad \text{.....(23),}$$

so that, from (22) and (23),

$$\alpha_1 - \alpha = (T_1 - T) + (E_1 - E) \qquad \text{.....(24).}$$

$(T_1 - T)$ is the interval by the clock between the times of transit of the sun and star and is therefore known. If the clock neither gains nor loses, $(E_1 - E) = 0$, so that $\alpha_1$ can be found from (24). Generally, however, the mechanical perfection of the clock cannot be presumed and as a convenient hypothesis the clock may be supposed to gain (or lose)—over an interval of a few hours—at a uniform rate, so that $(E_1 - E)$ is proportional to the interval between observations; we can write

$$E_1 - E = r\,(T_1 - T),$$

so that (24) becomes

$$\alpha_1 - \alpha = (1 + r)\,(T_1 - T) \qquad \text{.....(25).}$$

Suppose that the observations are repeated next day. Then we shall have

$$\alpha_1 - \alpha' = (1 + r)\,(T_1' - T') \qquad \text{.....(26),}$$

in which $\alpha'$ is the sun's right ascension calculated from (21) and $T'$, $T_1'$ are the sidereal times of transit of the sun and star respectively. From (25) and (26) we have by division

$$\frac{\alpha_1 - \alpha'}{\alpha_1 - \alpha} = \frac{T_1' - T'}{T_1 - T} \qquad \ldots\ldots(27),$$

from which the right ascension $\alpha_1$ of the star can be calculated.

This, in brief, is the fundamental principle involved in the measurement of the right ascensions of the stars. In practice, we may assume that the right ascensions of a sufficiently large number of stars have been accurately measured; such stars—"fundamental stars"—form a series of reference points on the celestial sphere and by means of them (or a selection of them) we are enabled to determine the right ascension of any other star by observing the interval between the transits of one or more fundamental stars and of the star in question, as in the case of the sun.

**55.** *The measurement of time.*

Stars which are selected for the determination of clock error and hence of the accurate sidereal time—and, by a further step, of the accurate U.T.—are called "clock stars". If $\alpha_1$, $\alpha_2$ are the known right ascensions of two clock stars, $T_1$, $T_2$ the observed times of transit (both corrected for the instrumental errors of azimuth, etc.), $E_1$, $E_2$ the clock errors at the respective times of transit, then

$$\alpha_1 = T_1 + E_1,$$
$$\alpha_2 = T_2 + E_2.$$

$\alpha_1$, $\alpha_2$ are both known, $T_1$ and $T_2$ are observed, hence $E_1$ and $E_2$ are found. The clock rate $r$ is obtained from the equation

$$E_2 - E_1 = r\,(T_2 - T_1).$$

In practice several stars are observed to ensure greater accuracy in the values of the clock error and rate.

The modern clocks now installed in several important observatories are the most accurate time measurers ever devised. The time which they record is so nearly uniform that it may be considered as conforming to the Newtonian description of time as "equably flowing"—as for E.T.—and the normal behaviour of a modern clock is such that its error on any day can be almost

exactly predicted several months in advance. In the past the performance of a clock was checked by meridian observations of the stars. Today the situation is reversed and it is variations in the earth's period of rotation that are determined from these observations.

## EXERCISES

[Symbols used:

$\phi$ = latitude, $b$ = level error,

$a$ = azimuth error, $c$ = collimation error.]

**1.** If $a$ and $b$ are the only errors of a meridian circle, show that the time of transit of a star will be unaffected if its declination $\delta$ is given by

$$\delta = \phi + \tan^{-1}(b/a).$$

**2.** Prove that the error in the time of transit of a star due to the three instrumental errors is a minimum for a star whose declination is

$$\sin^{-1}\{(a\cos\phi - b\sin\phi)/c\}. \qquad [\textit{Coll. Exam.}]$$

**3.** If two stars of declinations $\delta_1$ and $\delta_2$ can be found for which the three errors of adjustment produce no error in the time of transit, show that the correction to be added to the observed time of transit of a star of declination $\delta$ is

$$2c\sin\tfrac{1}{2}(\delta - \delta_1)\sin\tfrac{1}{2}(\delta - \delta_2)\sec\delta\sec\tfrac{1}{2}(\delta_1 - \delta_2). \qquad [M.T.]$$

**4.** If the observed time of transit of a star whose declination is 30° is found to be correct, while the observed times for stars of declinations 15° and 60° are found to be $-\ 7^s{\cdot}4$ and $+\ 31^s{\cdot}5$ in error, show that the error to be expected for a star in declination 45° is approximately 11 seconds. [$M.T.$]

**5.** In a transit instrument of 10 feet focal length, which is in correct adjustment except for collimation error, a star of declination 60° is observed to cross the meridian $2^s$ too soon. Show that to adjust the instrument the cross-wires must be moved a distance of 0·0087 inch. [$M.T.$ 1900.]

**6.** The level constant $b$ and the collimation constant $c$ of a transit instrument are determined, in the usual way, with a possibility of errors $\Delta b$, $\Delta c$, respectively. The azimuth constant $a$ is determined by observations of a polar star and of an equatorial star. Show that the resulting possible error in $a$ is given by

$$\Delta a = \Delta b\tan\phi + \Delta c\sec\phi,$$

where $\phi$ is the latitude.

Prove also: $\Delta m = \Delta b\sec\phi + \Delta c\tan\phi,$

$$\Delta n = -\Delta c. \qquad [\textit{Lond.}\ 1926.]$$

**7.** On the same day, the observed sidereal times of transits of two stars, corrected for the level and collimation errors, in latitude 51° 30′ N are $18^h\ 51^m\ 27^s{\cdot}2$ and $19^h\ 7^m\ 38^s{\cdot}9$. Calculate the azimuth error and the clock

error, given the co-ordinates of the two stars as follows:

$$\alpha_1 = 18^h\,51^m\,36^s{\cdot}5; \quad \delta_1 = +89^\circ\,2'.$$
$$\alpha_2 = 7^h\,7^m\,11^s{\cdot}2; \quad \delta_2 = +87^\circ\,10'. \qquad [\textit{Lond.}\ 1926.]$$

**8.** Prove that a displacement of the pole of the earth's rotation by $s$ feet in longitude $L$ produces a change in the azimuth error of a meridian circle in latitude $\phi$, longitude $l$, of $s/a \sin(l - L) \sec\phi \operatorname{cosec} 1''$, where $a$ is the radius of the earth in feet. Estimate roughly the maximum value of this displacement for an observatory in latitude 60°, and discuss the possibility of detecting it in the routine of meridian circle observations. [*Lond.* 1922.]

**9.** Assuming that the instrumental correction for a transit instrument is of the form

$$r = m + n\tan\delta + c\sec\delta,$$

prove that the correction will be positive for stars of all declinations if

$$c > |n|,$$

and

$$m + (c^2 - n^2)^{\frac{1}{2}} > 0.$$

**10.** A meridian circle of unknown instrumental errors has been set up at a site of inexactly known latitude and longitude. Outline a method for the determination of the following quantities $a$, $b$, $c$, $\Delta T$, $m$, $n$, $\phi$, and $\lambda$, where these symbols have their usual meanings.

CHAPTER V

# PLANETARY MOTIONS

**56.** *Introduction.**

The nine major planets in the order of increasing distance from the sun are Mercury, Venus, the Earth, Mars, Jupiter, Saturn, Uranus, Neptune and Pluto. The laws according to which the planets move with reference to the sun were discovered by John Kepler (1571–1630), and half a century later Kepler's three laws were shown by Sir Isaac Newton (1642–1727) to be deducible from the law of universal gravitation which he stated in the *Principia* in 1687. A complete investigation into the motions of the planets is the province of Dynamical Astronomy and as such is outside the scope of this book. But certain principles and results relating to the planetary motions and, in particular, to the earth's motion are necessary if we are to understand clearly some of the problems with which Spherical Astronomy is more closely associated.

**57.** *Kepler's first law.*

Kepler's first law states that the path, or orbit, of a planet around the sun is an ellipse, the position of the sun being at a focus of the ellipse. Fig. 45 shows an ellipse of which $S$ and $F$

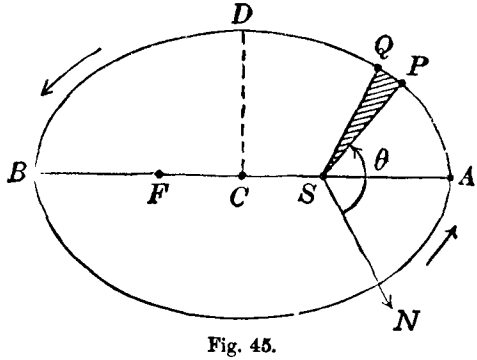



Fig. 45.

* In this chapter the time is assumed throughout to be expressed in terms of Ephemeris Time (E.T.); v. Appendix E (p. 424).

are the two foci, $C$ is the centre (midway between $S$ and $F$) and $AB$ is the *major axis*. The sun will be supposed to be at $S$ and the planet to move round the ellipse in the direction of the arrows. At $A$, the planet is nearest the sun; it is then said to be in *perihelion*. At $B$, it is furthest from the sun and it is then said to be in *aphelion*. $CA$ is called the *semi-major axis*; its length is denoted by $a$. $CD$, drawn perpendicularly to $CA$, is the *semi-minor axis*, denoted by $b$. The ratio $CS : CA$ is called the *eccentricity* which we denote by $e$. The semi-minor axis $b$ can be expressed in terms of $a$ and $e$ by the formula

$$b^2 = a^2 (1 - e^2) \qquad \ldots\ldots(1).$$

The perihelion distance $SA$ is $a(1 - e)$ and the aphelion distance $SB$ is $a(1 + e)$.

If $P$ is any position of the planet in its orbit, $SP$ is called the *radius vector* (denoted by $r$) and the distance $SP$ is the *heliocentric distance*, that is, the distance of the planet from the sun. Let $SN$ be a reference direction in the orbital plane. Then the position of $P$ is specified by the radius vector $r$ and the angle $\theta$ which $SP$ makes with $SN$, $\theta$ being measured in the direction of the planet's motion. Let $\omega$ be the value of $\theta$ when the planet is in perihelion, that is, at $A$ so that $N\hat{S}A = \omega$. Then $P\hat{S}A = \theta - \omega$. The equation of the ellipse is known to be

$$r = \frac{p}{1 + e \cos(\theta - \omega)} \qquad \ldots\ldots(2),$$

in which

$$p = b^2/a = a(1 - e^2) \qquad \ldots\ldots(3).$$

The time required for the planet to describe its orbit is called the *period* which we denote by $T$. The earth's orbital period is the *year* which, for the present, may be taken to be $365\frac{1}{4}$ mean solar days.

**58.** *Kepler's second law.*

Kepler's second law states that the radius vector $SP$ (Fig. 45) sweeps out equal areas in equal times. Let $P$ correspond to the planet's position at time $t$ and $Q$ its position at time $t + \Delta t$. Let $r + \Delta r$ denote the radius vector $SQ$ and $\theta + \Delta\theta$ the angle $QSN$. Then $Q\hat{S}P = \Delta\theta$. If $\Delta\theta$ is sufficiently small, the arc $PQ$ may be regarded as a straight line and the area swept out in the infinitesimal interval $\Delta t$ is simply the area of the triangle $QSP$,

that is, $\frac{1}{2}r(r+\Delta r)\sin\Delta\theta$, or, with sufficient accuracy, $\frac{1}{2}r^2\Delta\theta$. The rate of description of area is this last expression divided by $\Delta t$; as this rate is constant according to Kepler's second law, we can write

$$r^2\frac{d\theta}{dt}=h \qquad \text{.....(4),}$$

where $h$, a constant, is twice the rate of description of area by the radius vector.

Now the whole area of the ellipse is $\pi ab$ and this is described in the interval defined by the period $T$. Hence

$$\frac{2\pi ab}{T}=h \qquad \text{.....(5),}$$

or, using (1),

$$\frac{2\pi a^2(1-e^2)^{\frac{1}{2}}}{T}=h \qquad \text{.....(6).}$$

In time $T$, the radius vector sweeps out an angle of 360° or $2\pi$. Let $n$ denote the average rate of description of angle by the radius vector. Then

$$n=2\pi/T \qquad \text{.....(7).}$$

$n$ is called the *mean angular motion* of the planet. In moving from $SP$ to $SQ$ the radius vector sweeps out the angle $\Delta\theta$ in the interval $\Delta t$. At $P$ the angular velocity is thus $\frac{d\theta}{dt}$; thus $n$ is the mean value of $\frac{d\theta}{dt}$ for all points in the orbit. By the help of (7), formula (6) can be written

$$na^2(1-e^2)^{\frac{1}{2}}=h \qquad \text{.....(8).}$$

**59.** *Kepler's third law.*

Kepler's third law, expressed mathematically, is as follows. Let $a$, $a_1$ be the semi-major axes of two planetary orbits and $T$, $T_1$ the corresponding orbital periods. Then, by the third law,

$$\frac{a^3}{T^2}=\frac{a_1^3}{T_1^2} \qquad \text{.....(9),}$$

or, using (7),

$$n^2a^3=n_1^2a_1^3 \qquad \text{.....(10),}$$

in which $n$ and $n_1$ are the mean angular motions in the two orbits.

We have from (9)

$$\frac{a_1}{a}=\left(\frac{T_1}{T}\right)^{\frac{2}{3}} \qquad \text{.....(11).}$$

If $a$, $T$ refer to the earth's orbit around the sun, the ratio of the semi-major axis of the orbit of any planet $P_1$ to the semi-major axis of the earth's orbit is given by (11) when the orbital period $T_1$ for the planet $P_1$ is known in years, since for the earth $T$ is one year. The orbital periods of the planets are known from observations (see section 80) and hence their semi-major axes can be derived in terms of $a$ as the unit of distance. The semi-major axis $a$ of the earth's orbit is known as the *astronomical unit* of distance. Expressing $T_1$ in years, we have from (11), putting $T = 1$ year,

$$a_1 = (T_1)^{\frac{2}{3}} \text{ astronomical units.}$$

60. *Newton's law of gravitation.*

The statement of this law is as follows. Every particle of matter attracts every other particle of matter with a force proportional to the product of the masses of the two particles concerned and inversely proportional to the square of the distance between them. Stated mathematically, the law is:

$$F = G\frac{mm_1}{r^2} \qquad \text{.....(12),}$$

where $m$, $m_1$ are the masses of the particles, $r$ the distance between them, $F$ the gravitational force of attraction and $G$ a constant—the *constant of gravitation*. In the c.g.s. system of units, the value of $G$ is $6{\cdot}670 \times 10^{-8}$, which means that the force of attraction between two particles each of 1 gramme in mass and separated by a distance of 1 centimetre is $6{\cdot}670 \times 10^{-8}$ dynes. For our present purpose we can regard the sun and planets as "particles".

The application of the law of gravitation (12) to the motion of a planet around the sun leads to the three laws of Kepler. Let $M$, $m$ denote the masses of the sun and a planet respectively and let $\mu$ be defined by

$$\mu = G(M + m) \qquad \text{.....(13).}$$

Then it is found that the constant $h$ in formula (4) is given by

$$h^2 = \mu p = \mu a(1 - e^2) \qquad \text{.....(14).}$$

But, by (8), $h^2 = n^2a^4(1 - e^2)$.

Hence $$n^2a^3 = \mu \equiv G(M + m) \qquad \text{.....(15).}$$

For another planet,

$$n_1^2 a_1^3 = \mu_1 \equiv G(M + m_1) \qquad \ldots\ldots(16).$$

Hence from (15) and (16),

$$\frac{n^2 a^3}{n_1^2 a_1^3} = \frac{M + m}{M + m_1},$$

or, using (7),

$$\frac{a^3}{a_1^3} = \frac{M + m}{M + m_1} \cdot \frac{T^2}{T_1^2} \qquad \ldots\ldots(17).$$

The equation (17) gives the correct form of Kepler's third law which in our notation had been expressed by (9). Actually the mass of a planet is very small compared with that of the sun and so the quantity $\frac{M + m}{M + m_1}$ in (17) is very nearly equal to unity; consequently, Kepler's third law, although not strictly accurate, is nevertheless a very good approximation.

**61.** *The masses of the planets.*

The formulae of the preceding section enable us to calculate the mass of any planet, which is accompanied by one or more satellites, as a fraction of the sun's mass. If $m$, $a$, $T$ refer to the earth, we have, by (7) and (15),

$$G(M + m) = 4\pi^2 \frac{a^3}{T^2} \qquad \ldots\ldots(18).$$

Now the motion of a satellite around a planet is given by the same laws as the motion of a planet around the sun. In the case of the satellite, the planet is the controlling body and if $m_1$, $m'$ are the masses of the planet and the satellite respectively, $a_1$ the semi-major axis of the satellite's orbit around the planet, and $T_1$ the period of its orbital revolution, Newton's law of gravitation leads to an equation analogous to (18); it is

$$G(m_1 + m') = 4\pi^2 \frac{a_1^3}{T_1^2} \qquad \ldots\ldots(19).$$

Hence, by (18) and (19),

$$\frac{m_1 + m'}{M + m} = \left(\frac{a_1}{a}\right)^3 \left(\frac{T}{T_1}\right)^2 \qquad \ldots\ldots(20).$$

Now the mass $m'$ of the satellite is small compared with the mass of the planet $m_1$; hence in (19) and (20) we can neglect $m'$.

Similarly we can neglect the earth's mass $m$ in comparison with the sun's mass $M$, and (20) can be written

$$\frac{m_1}{M} = \left(\frac{a_1}{a}\right)^3 \left(\frac{T}{T_1}\right)^2 \qquad \ldots\ldots(21).$$

We shall suppose that the semi-major axis $a_1$ of the satellite's orbit is known in astronomical units and that its period of revolution $T_1$ is also known in terms of years. Then in (21), $a = 1$ astronomical unit, $T = 1$ year, and therefore the ratio of $m_1$ (the planet's mass) to $M$ (the sun's mass) is determined.

As an example we shall find the mass of Mars (in terms of the sun's mass) from the relevant orbital elements of the satellite Deimos. The semi-major axis of the orbit of Deimos around Mars is 0·00015695 astronomical unit; its period of revolution is 1·26244 day or $\frac{1{\cdot}26244}{365\frac{1}{4}}$ year.

Hence, by (21),

$$\frac{m_1}{M} = (0{\cdot}00015695)^3 \times \frac{(365\frac{1}{4})^2}{(1{\cdot}26244)^2}$$

$$= \frac{1}{3{\cdot}09 \times 10^6},$$

that is to say, the sun's mass is rather more than three million times the mass of Mars.

**62.** *Perturbations of the elements.*

We have assumed so far that the path of a planet around the sun is determined by the mutual gravitational attraction of the planet and the sun only. But every other planet and body in the solar system exerts a gravitational attraction on the planet concerned and the effects are shown in small changes in its orbital elements such as the semi-major axis $a$ and the eccentricity $e$. Such changes are known as *perturbations of the elements*; in general their values are small. The magnitude of any perturbation due to a particular disturbing planet depends, among other things, on the mass of the latter and it is thus possible to deduce from observations, in combination with the formulae of Dynamical Astronomy, the masses of planets which are unaccompanied by satellites. In this way, the masses of Mercury and Venus (which have no satellites) are derived. These two planets have, however, been visited by space probes, and while

the space probe is passing close to the planet it is effectively an artificial satellite of that planet. A slight variant of the method of section 61 can then be applied to give a much more accurate value for the planet's mass.

**63.** *The dynamical principles of orbital motion.*

Let $S$ and $P$ be the positions of the sun and a planet at any time $t$ and let their co-ordinates referred to unaccelerated rectangular axes $OA, OB, OC$ in space be $(X_1, Y_1, Z_1)$ and $(X, Y, Z)$ respectively (Fig. 46). If the masses of the sun and planet are

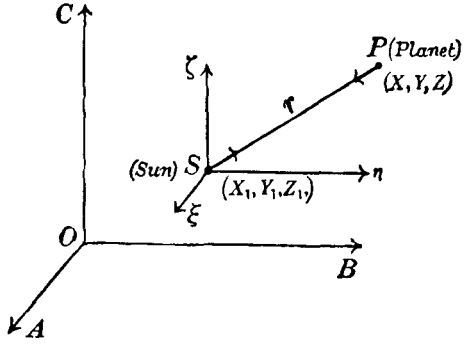



Fig. 46.

$M$ and $m$, then by Newton's law of gravitation, the planet $P$ is attracted towards $S$ with the force $GMm/r^2$, where $r$ is the distance $SP$. The component of this force in the positive direction of the axis $OA$ is

$$\frac{GMm}{r^2} \cdot \frac{X_1 - X}{r} \quad \text{or} \quad -\frac{GMm\,(X - X_1)}{r^3}.$$

If $\ddot{X}$ $\left(\equiv \dfrac{d^2X}{dt^2}\right)$ denotes the acceleration of $P$ parallel to $OA$, we have, by Newton's second law of motion,

$$m\ddot{X} = -\frac{GMm\,(X - X_1)}{r^3} \qquad \text{.....(22).}$$

Now gravitational attraction is mutual and the sun $S$ will be attracted towards $P$ with the force $GMm/r^2$ and the component of this force parallel to $OA$ is

$$\frac{GMm}{r^2}\,\frac{(X - X_1)}{r}.$$

If $\ddot{X}_1$ is the acceleration of the sun parallel to $OA$, we have as before

$$M\ddot{X}_1 = \frac{GMm\,(X - X_1)}{r^3} \qquad \text{.....(23).}$$

Dividing (22) by $m$ and (23) by $M$ and subtracting the resulting equations, we obtain

$$\ddot{X} - \ddot{X}_1 = -\,G\,(M + m)\,\frac{(X - X_1)}{r^3} \qquad \text{.....(24).}$$

There are two similar equations in $Y$ and $Z$.

Write $\quad \xi = X - X_1,\ \eta = Y - Y_1,\ \zeta = Z - Z_1.$

Then $(\xi, \eta, \zeta)$ are the co-ordinates of the planet $P$ referred to rectangular axes passing through the sun, and we have from (24), writing $\mu$ for $G\,(M + m)$ as in (13),

$$\ddot{\xi} + \frac{\mu\xi}{r^3} = 0 \qquad \text{.....(25).}$$

Similarly

$$\ddot{\eta} + \frac{\mu\eta}{r^3} = 0 \qquad \text{.....(26),}$$

$$\ddot{\zeta} + \frac{\mu\zeta}{r^3} = 0 \qquad \text{.....(27).}$$

These are the equations of motion of the planet $P$ with reference to the sun.

Multiply (26) by $\zeta$ and (27) by $\eta$ and subtract. We obtain

$$\zeta\ddot{\eta} - \eta\ddot{\zeta} = 0,$$

that is,

$$\frac{d}{dt}\,(\zeta\dot{\eta} - \eta\dot{\zeta}) = 0.$$

Integrating, we have $\quad \zeta\dot{\eta} - \eta\dot{\zeta} = A \qquad \text{.....(28),}$

in which $A$ is a constant of integration.

In a similar way, we derive from (25) and (27) and then from (25) and (26),

$$\xi\dot{\zeta} - \zeta\dot{\xi} = B \qquad \text{.....(29),}$$

and

$$\eta\dot{\xi} - \xi\dot{\eta} = C \qquad \text{.....(30),}$$

in which $B$ and $C$ are constants of integration.

Multiply (28), (29) and (30) by $\xi$, $\eta$, $\zeta$ in order and add. Then

$$A\xi + B\eta + C\zeta = 0 \qquad \text{.....(31),}$$

which is the equation of a plane passing through the origin of the $\xi$, $\eta$, $\zeta$ co-ordinates, that is, passing through the sun $S$. The interpretation is that, as the co-ordinates $(\xi, \eta, \zeta)$ of the planet

$P$ satisfy the relation given by (31), its motion with respect to the sun takes place in a plane. This plane is the orbital plane.

64. *The equation of the orbit.*

We can now refer the motion of the planet to two axes passing through the sun and lying in the orbital plane: let $(x, y)$ be the co-ordinates of the planet referred to these axes. In Fig. 45, we shall suppose that $SN$ is the $x$-axis, the $y$-axis being at right angles to $SN$ and of course in the plane of the orbit. The equations of motion of the planet are then

$$\ddot{x} + \mu \frac{x}{r^3} = 0 \qquad \text{.....(32),}$$

$$\ddot{y} + \mu \frac{y}{r^3} = 0 \qquad \text{.....(33).}$$

These equations, in rectangular co-ordinates, will now be transformed in terms of the polar co-ordinates $r$ and $\theta$. (In Fig. 45, $SP = r$ and $N\hat{S}P = \theta$.) We have

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta \qquad \text{.....(34).}$$

Let $\alpha$ and $\beta$ denote the components of the acceleration of the planet $P$ along $SP$ and at right angles to $SP$ respectively. Then

$$\alpha = \ddot{x}\cos\theta + \ddot{y}\sin\theta \qquad \text{.....(35),}$$

and

$$\beta = \ddot{y}\cos\theta - \ddot{x}\sin\theta \qquad \text{.....(36).}$$

We have, from the first of (34),

$$\dot{x} = \dot{r}\cos\theta - r\sin\theta\,.\dot{\theta},$$

and

$$\ddot{x} = \ddot{r}\cos\theta - 2\dot{r}\sin\theta\,.\dot{\theta} - r\cos\theta\,.\dot{\theta}^2 - r\sin\theta\,.\ddot{\theta} \qquad \text{...(37).}$$

The expression for $\ddot{y}$ can be derived in a similar way; it is

$$\ddot{y} = \ddot{r}\sin\theta + 2\dot{r}\cos\theta\,.\dot{\theta} - r\sin\theta\,.\dot{\theta}^2 + r\cos\theta\,.\ddot{\theta} \qquad \text{...(38).}$$

Inserting in (35) the values of $\ddot{x}$ and $\ddot{y}$ given by (37) and (38), we obtain, after some reduction,

$$\alpha = \ddot{r} - r\dot{\theta}^2 \qquad \text{.....(39).}$$

But by (32) and (33),

$$\alpha \equiv \ddot{x}\cos\theta + \ddot{y}\sin\theta = -\frac{\mu}{r^3}(x\cos\theta + y\sin\theta)$$

$$= -\frac{\mu}{r^2}.$$

Hence

$$\ddot{r} - r\dot{\theta}^2 = -\frac{\mu}{r^2} \qquad \text{.....(40).}$$

By a similar procedure, we find that

$$\beta = 2\dot{r}\dot{\theta} + r\ddot{\theta} = 0.$$

But $$2\dot{r}\dot{\theta} + r\ddot{\theta} = \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta}).$$

Hence $$\frac{d}{dt}(r^2\dot{\theta}) = 0,$$

and by integration $$r^2\dot{\theta} = h \qquad \text{.....(41)},$$

where $h$ is the constant of integration. Thus (41) is the mathematical expression of Kepler's second law which we considered previously in section 58.

The equation of the path of the planet around the sun can now be derived from (40) and (41); it is, of course, a relation between $r$ and $\theta$. The process consists in eliminating the time $t$ from (40) by means of (41); in these equations the time occurs only in the differential coefficients. We shall write for convenience

$$u = \frac{1}{r} \qquad \text{.....(42)},$$

so that, by (41), $$\dot{\theta} = hu^2 \qquad \text{.....(43)}.$$

Now $$\dot{r} = \frac{dr}{dt} = -\frac{1}{u^2}\frac{du}{dt} = -\frac{1}{u^2}\frac{du}{d\theta}\cdot\frac{d\theta}{dt}.$$

Hence, by (43), $$\dot{r} = -h\frac{du}{d\theta}.$$

Again, $$\ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{d}{d\theta}(\dot{r})\cdot\frac{d\theta}{dt}$$

$$= -hu^2\frac{d}{d\theta}\left(h\frac{du}{d\theta}\right),$$

so that $$\ddot{r} = -h^2u^2\frac{d^2u}{d\theta^2} \qquad \text{.....(44)}.$$

Also, using (43), $$r\dot{\theta}^2 = h^2u^3 \qquad \text{.....(45)}.$$

Hence (40) becomes, by means of (44) and (45),

$$-h^2u^2\frac{d^2u}{d\theta^2} - h^2u^3 = -\mu u^2,$$

or $$\frac{d^2u}{d\theta^2} + u = \frac{\mu}{h^2} \qquad \text{.....(46)}.$$

The general solution of (46) is given by

$$u = \frac{\mu}{h^2}[1 + e\cos(\theta - \omega)] \qquad \text{.....(47)},$$

in which $e$ and $\omega$ are the two essential constants of integration, or by (42),

$$r = \frac{h^2/\mu}{1 + e\cos(\theta - \omega)} \qquad \ldots\ldots(48).$$

This last equation is identical with (2)—the equation of an ellipse—if $p = h^2/\mu$, that is, making use of (3), if

$$h^2 = \mu a\,(1 - e^2) \qquad \ldots\ldots(49),$$

which is equation (14) previously considered. The constant of integration $e$ is seen to be identified with the eccentricity.

It is to be remarked that (48) is the general equation of a conic section which may be

(i) an ellipse, if $e < 1$,
(ii) a parabola, if $e = 1$,
(iii) a hyperbola, if $e > 1$.

Although case (i) is that with which we are closely concerned here, the extension of the possibilities concerning the motion of a body under the gravitational attraction of the sun should be noted.

The equation (41) is simply the mathematical expression of Kepler's second law. Also, by defining the mean angular motion $n$ as in (7), we are led—using (5), (7) and (49)—to the formula

$$n^2a^3 = \mu \equiv G\,(M + m)$$

already mentioned. Thus from the single law of gravitation we have derived the mathematical equivalents of Kepler's three laws.

**65.** *Velocity of a planet in its orbit.*

Let $V$ denote the velocity of the planet at the point $P$ in its orbit (Fig. 47). $V$ will be directed along the tangent $PT$. The components of $V$ are (i) $\dot{r}$ along the radius vector in the direction $PR$, and (ii) $r\dot{\theta}$ along $PL$ at right angles to the radius vector. We thus have

$$V^2 = \dot{r}^2 + r^2\dot{\theta}^2 \qquad \ldots\ldots(50).$$

Now

$$\dot{r} = -h\frac{du}{d\theta},$$

and from (47)

$$\frac{du}{d\theta} = -\frac{\mu}{h^2}e\sin(\theta - \omega),$$

so that

$$\dot{r} = \frac{\mu}{h}e\sin(\theta - \omega) \qquad \ldots\ldots(51).$$

Also, since $r^2\dot{\theta} = h$, we have

$$r\dot{\theta} = hu = \frac{\mu}{h}[1 + e\cos(\theta - \omega)] \qquad \ldots\ldots(52).$$

Hence, squaring (51) and (52), we have from (50)

$$V^2 = \frac{\mu^2}{h^2}[1 + 2e\cos(\theta - \omega) + e^2],$$

which can be written

$$V^2 = \frac{\mu^2}{h^2}[2 + 2e\cos(\theta - \omega) - (1 - e^2)],$$

or, using (47), $\qquad V^2 = 2\mu u - \frac{\mu^2}{h^2}(1 - e^2).$

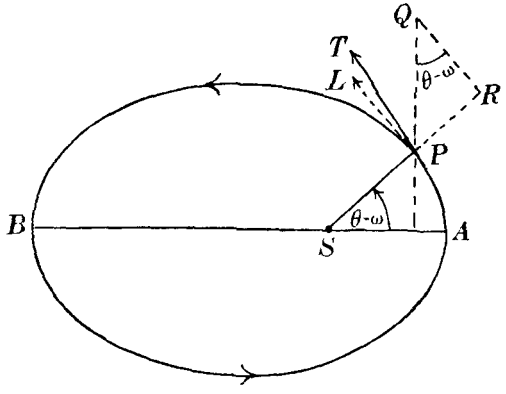



Fig. 47.

But from (49) $\qquad h^2 = \mu a(1 - e^2).$

Hence we obtain, writing $1/r$ for $u$,

$$V^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right) \qquad \ldots\ldots(53).$$

This formula gives the velocity $V$ as a function of the radius vector $r$.

It is seen from (53) that $V$ is greatest when $r$ is least, that is, when the planet is in perihelion. Then $r = a(1 - e)$, and if $V_1$ denotes the velocity at perihelion we have

$$V_1^2 = \frac{\mu}{a} \cdot \frac{1 + e}{1 - e} \qquad \ldots\ldots(54).$$

Similarly, the velocity is a minimum when $r$ is a maximum, that is, when the planet is in aphelion. If $V_2$ denotes the velocity at aphelion, we obtain

$$V_2^2 = \frac{\mu}{a} \cdot \frac{1-e}{1+e} \qquad \text{.....(55).}$$

From (54) and (55), $\qquad V_1 V_2 = \frac{\mu}{a}$.

Thus the product of the linear velocities at perihelion and aphelion is independent of the eccentricity of the orbit.

**66.** *Components of the linear velocity perpendicular to the radius vector and to the major axis.*

We now derive a theorem which will be used later in the investigation of certain problems. In Fig. 47 let $PR$ represent the velocity $\dot{r}$. Draw $PQ$ perpendicular to the major axis $AB$ and draw $RQ$ perpendicular to $PR$. Then the velocity $\dot{r}$ is equivalent to (i) a velocity along $PQ$ and represented in magnitude by $PQ$, and (ii) a velocity parallel to $\vec{QR}$ and represented in magnitude by $QR$. Now $P\hat{S}A = \theta - \omega$ and by the construction, $P\hat{Q}R = \theta - \omega$. Hence $PQ = PR \operatorname{cosec}(\theta - \omega)$ and $QR = PR \cot(\theta - \omega)$. We thus find that the velocity $\dot{r}$ is equivalent to

(i) $\dot{r} \operatorname{cosec}(\theta - \omega)$ along $PQ$,

(ii) $\dot{r} \cot(\theta - \omega)$ parallel to $\vec{QR}$.

Now the velocity $V$ is equivalent to $\dot{r}$ along $PR$ and $r\dot{\theta}$ along $PL$, $PL$ being perpendicular to $SP$. Hence $V$ is equivalent to

(i) $\dot{r} \operatorname{cosec}(\theta - \omega)$ along $PQ$, that is, perpendicular to the major axis,

and (ii) $r\dot{\theta} - \dot{r} \cot(\theta - \omega)$ along $PL$, that is, perpendicular to the radius vector.

From (51), we obtain for (i),

$$\dot{r} \operatorname{cosec}(\theta - \omega) = \frac{e\mu}{h} \qquad \text{.....(56),}$$

and, similarly, from (51) and (52) for (ii),

$$r\dot{\theta} - \dot{r} \cot(\theta - \omega) = \frac{\mu}{h} \qquad \text{.....(57).}$$

Hence (57) and (56) express the result that the velocity $V$ of a planet at any point of its orbit can be resolved into a constant

velocity $\mu/h$ perpendicular to the radius vector and a constant velocity $e\mu/h$ perpendicular to the major axis.

**67.** *The true and eccentric anomalies.*

Theoretically, Kepler's second law enables us to calculate the position of a planet in its orbit at any time provided we know the semi-major axis $a$, the eccentricity $e$, the time $\tau$ at which the planet passed through perihelion and the orbital period $T$. In Fig. 48 let $P$ be the position of the planet at time $t$. In the

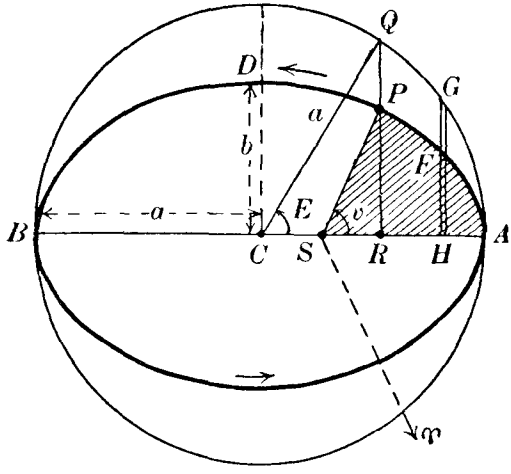



Fig. 48.

interval $(t - \tau)$ the radius vector moving from $SA$ to $SP$ sweeps out the shaded area $SPA$. Now by the second law

$$\text{Area } SPA : \text{Area of ellipse} = t - \tau : T,$$

that is,
$$\text{Area } SPA = \frac{\pi ab\,(t-\tau)}{T},$$

or, introducing the mean angular motion $n$ $(n = 2\pi/T)$, we can write

$$\text{Area } SPA = \tfrac{1}{2}\, nab\,(t-\tau) \qquad \ldots\ldots(58),$$

in which $b$ is given by $b^2 = a^2\,(1 - e^2)$. Thus, at a given instant $t$, the right-hand side of (58) can be calculated when $a$, $e$, $\tau$ and $n$ are known. Hence the area $SPA$ is found and the position of the planet in its orbit determined.

However simple this method appears theoretically, in practice

it is inconvenient and we shall now derive the formulae generally applied for determining the position of a planet in its orbit.

Let the radius vector $SP$ make the angle $v$ with $SA$; $v$ is called the *true anomaly* and it is clearly related to $\theta$ and $\omega$ by the formula

$$v = \theta - \omega \qquad \text{.....(59).}$$

Let a circle be described on the major axis $AB$ as diameter; its radius is thus $a$. Let $RP$, the perpendicular from $P$ to $AB$, be produced to meet this circle at $Q$. Then the angle $QCA$ is called the *eccentric anomaly* which we denote by $E$.

By a well-known property of the ellipse

$$PR : QR = b : a \qquad \text{.....(60),}$$

where $b$ is the semi-minor axis $CD$.

Now $PR = r \sin v$ and $QR = CQ \sin E = a \sin E$. Hence, from (60),

$$r \sin v = b \sin E \qquad \text{.....(61).}$$

Again, $SR = r \cos v$. Also $SR = CR - CS = a \cos E - ae$. Hence

$$r \cos v = a (\cos E - e) \qquad \text{.....(62).}$$

Square (61) and (62) and add. Then, putting $b^2 = a^2 (1 - e^2)$, we obtain, after a little reduction,

$$r = a (1 - e \cos E) \qquad \text{.....(63).}$$

Again,
$$2r \sin^2 \frac{v}{2} = r (1 - \cos v)$$
$$= a (1 - e \cos E) - a (\cos E - e),$$

applying (63) and (62); hence

$$2r \sin^2 \frac{v}{2} = a (1 + e) (1 - \cos E) \qquad \text{.....(64).}$$

Similarly,
$$2r \cos^2 \frac{v}{2} = a (1 - e) (1 + \cos E) \qquad \text{.....(65).}$$

Divide (64) by (65). Then

$$\tan^2 \frac{v}{2} = \frac{1 + e}{1 - e} \cdot \frac{1 - \cos E}{1 + \cos E},$$

from which
$$\tan \frac{v}{2} = \left(\frac{1 + e}{1 - e}\right)^{\frac{1}{2}} \tan \frac{E}{2} \qquad \text{.....(66).}$$

Thus, by means of (63) and (66), we can express the radius vector $r$ and the true anomaly $v$ in terms of the eccentric

anomaly $E$. We now derive Kepler's equation by means of which the eccentric anomaly $E$ can be expressed in terms of known quantities. The eccentric anomaly thus enters into our theoretical discussion as an intermediary angle.

**68.** *Kepler's equation.*

From our definition of the mean angular motion $n$ we see that the product $n(t-\tau)$, which occurs in formula (58), represents the angle described in an interval $(t-\tau)$ by a radius vector rotating about $S$ with constant angular velocity $n$. We define $n(t-\tau)$ as the *mean anomaly*, which we denote by $M$, so that

$$M = n(t-\tau) \qquad \text{.....(67).}$$

Hence, from (58), the area $SPA$ in Fig. 48 is given by

$$\text{Area } SPA = \tfrac{1}{2}abM \qquad \text{.....(68).}$$

We now express this area in terms of the eccentric anomaly $E$.

The shaded area $SPA$ is equal to the area of the triangle $PSR$ together with the area $RPA$. Take first the triangle $PSR$. Its area is $\frac{1}{2}SR.PR$. But $SR = CR - CS$ or $SR = a\cos E - ae$. Also $PR = \frac{b}{a}QR$ by (60), and therefore $PR = b\sin E$. Hence the area of the triangle $PSR$ is $\frac{1}{2}ab\sin E(\cos E - e)$. Consider now the area $RPA$. Divide it up into strips perpendicular to $AB$ and extend the strips to meet the circle on $AB$ as diameter. Consider the strip $FH$. Since $FH = \frac{b}{a}GH$, then the sum of all the strips forming the area $RPA$ is equal to $\frac{b}{a}$ times the sum of the strips forming the area $QRA$; hence

$$\text{Area } RPA = \frac{b}{a}.\text{Area } QRA \qquad \text{.....(69).}$$

But the area $QRA$ is the area of the sector $CQA$ minus the area of the triangle $QCR$. But since $Q\hat{C}A = E$, the area of $CQA$ is $\frac{1}{2}a^2E$ and the area of the triangle $QCR$ is $\frac{1}{2}a^2\sin E\cos E$. Hence (69) becomes

$$\text{Area } RPA = \frac{b}{a}[\tfrac{1}{2}a^2E - \tfrac{1}{2}a^2\sin E\cos E]$$

$$= \tfrac{1}{2}ab(E - \sin E\cos E).$$

Add to $RPA$ the area of the triangle $PSR$ previously found and we obtain

$$\text{Area } SPA = \tfrac{1}{2}ab\,(E - e\sin E) \qquad \ldots\ldots(70).$$

From (68) and (70), we then have

$$E - e\sin E = M \equiv n\,(t - \tau) \qquad \ldots\ldots(71).$$

This is Kepler's equation. It is a relation between the eccentric anomaly $E$ and the mean anomaly $M$. If $M$ and $e$ are known, it is then possible to determine the corresponding value of $E$. It is to be noted, from the manner in which (71) has been derived, that both $E$ and $M$ are supposed to be expressed in circular measure.

**69.** *Solution of Kepler's equation.*

The general method of solution depends on deriving an approximate value of $E$, nearly satisfying the equation, either by inspection or by special tables or by one of the many graphical processes devised at different times. Let this approximate value be denoted by $E_0$ and let the true value be $E_0 + \Delta E_0$. Then, rigorously,

$$(E_0 + \Delta E_0) - e\sin(E_0 + \Delta E_0) = M,$$

or

$$E_0 + \Delta E_0 - e\sin E_0\cos\Delta E_0 - e\cos E_0\sin\Delta E_0 = M.$$

Since $\Delta E_0$ is supposed to be small, we can write as an approximation: $\cos\Delta E_0 = 1$, $\sin\Delta E_0 = \Delta E_0$, so that

$$(E_0 - e\sin E_0) + \Delta E_0\,(1 - e\cos E_0) = M.$$

As $E_0$ and $e$ are known we can compute a quantity $M_0$ given by

$$M_0 = E_0 - e\sin E_0 \qquad \ldots\ldots(72),$$

so that we then obtain

$$\Delta E_0 = \frac{M - M_0}{1 - e\cos E_0} \qquad \ldots\ldots(73),$$

from which $\Delta E_0$ can be calculated. Then $(E_0 + \Delta E_0)$ is a more accurate value of $E$. The process can then be repeated, if necessary, with $(E_0 + \Delta E_0)$ as a new approximation to $E$.

If the eccentricity is small, say less than 0·1, we obtain by inspection of Kepler's equation an approximate value of the eccentric anomaly. For, neglecting the term $e\sin E$, we obtain for the first approximation $E_0$ the simple result

$$E_0 = M,$$

and by the application of (73) a more accurate value of the eccentric anomaly can be derived.

When the eccentricity is large, it is not so easy to obtain an approximate value of the eccentric anomaly satisfying Kepler's equation. In these circumstances the use of special tables, such as Bauschinger's* or Astrand's†, or of a graphical construction, greatly facilitates the computations. In Bauschinger's tables, the values of $E$ are tabulated for different values of the eccentricity $e$ and the mean anomaly $M$; then, by inspection or by an easy interpolation, a very good approximate value $E_0$ can be obtained. The application of (73) follows as already indicated.

*Example.* Required to calculate the eccentric anomaly of Mars 200 days after perihelion passage, given: $e = 0{\cdot}09334$, $T = 1{\cdot}8809$ years.

The mean angular motion $n$ is defined by $n = 2\pi/T$ or $360/1{\cdot}8809$ degrees per year or $1886{\cdot}52$ seconds of arc per mean solar day. Hence

$$M = 200 \times 1886''{\cdot}52 = 377304'' = 104^\circ\ 48'\ 24''.$$

If we neglect the term $e \sin E$ in Kepler's equation we can take, as a rough value of the eccentric anomaly, $E_0 = 105^\circ$ (the value of $M$ to the nearest degree).

Compute now the value of $M_0$ given by (72), namely

$$M_0 = E_0 - e \sin E_0.$$

We first notice that, in this theoretical formula, $M_0$ and $E_0$ are necessarily expressed in circular measure. If we now suppose them expressed in seconds of arc, we have

$$M_0 \sin 1'' = E_0 \sin 1'' - e \sin E_0,$$

or

$$M_0 = E_0 - e \sin E_0 \operatorname{cosec} 1''.$$

$$\log e = \bar{2}{\cdot}97007$$

$$\log \sin E_0$$

$$(\equiv \log \sin 105^\circ) = \bar{1}{\cdot}98494$$

$$\log \operatorname{cosec} 1'' = 5{\cdot}31443$$

$$4{\cdot}26944 \quad \text{which is } \log 18597.$$

Hence $M_0 = E_0 - 18597'' = 105^\circ - 5^\circ\ 9'\ 57''$,

or $M_0 = 99^\circ\ 50'\ 3''$.

* J. Bauschinger, *Tafeln zur theoretischen Astronomie* (Leipzig, 1901).

† J. J. Astrand, *Hülfstafeln* (Leipzig, 1890).

Then the formula (73) for $\Delta E_0$ becomes

$$\Delta E_0 = \frac{104^\circ\,48'\,24'' - 99^\circ\,50'\,3''}{1 - e\cos 105^\circ}.$$

But $e\cos 105^\circ = -0{\cdot}09334 \sin 15^\circ = -0{\cdot}0242.$

Hence $\Delta E_0 = \dfrac{4^\circ\,58'\,21''}{1{\cdot}0242} = \dfrac{17901}{1{\cdot}0242}$ seconds of arc

$= 17478''$

$= 4^\circ\,51'\,18''.$

Hence $E_0 + \Delta E_0 = 109^\circ\,51'\,18'',$

which is a more accurate value of the eccentric anomaly satisfying Kepler's equation. We ought to repeat the process again, taking now $109^\circ\,51'\,18''$ as an approximation to the value of $E$, and to continue until the required accuracy is obtained. We leave the step thus indicated to the student.

From Bauschinger's tables we extract the following:

| | $e = 0{\cdot}0$ | $e = 0{\cdot}1$ |
|---|---|---|
| $M = 104^\circ$ | $E = 104^\circ$ | $E = 109^\circ{\cdot}40$ |
| $M = 105^\circ$ | $E = 105^\circ$ | $E = 110^\circ{\cdot}37$ |

Hence, by a rough interpolation for $M = 104^\circ\,48'$ and $e = 0{\cdot}093$, we derive $E_0 = 109^\circ{\cdot}8$, which is close to the result of our first computation. The tables thus save, in this instance, the computation of our approximate value of $E$. The tables are never needed, however, if the interative calculation has been programmed as the process will converge ultimately in any case.

**70.** *Summary of the formulae of elliptic motion.*

We now collect the more important formulae of elliptic motion which have been derived in the previous pages:

$$\left.\begin{aligned}
&\text{(i)} && r = \frac{a(1-e^2)}{1+e\cos v}\\
&\text{(ii)} && n^2a^3 = \mu \equiv G(M+m)\\
&\text{(iii)} && h^2 = \mu a(1-e^2)\\
&\text{(iv)} && E - e\sin E = M \equiv n(t-\tau)\\
&\text{(v)} && r = a(1-e\cos E)\\
&\text{(vi)} && \tan\frac{v}{2} = \left(\frac{1+e}{1-e}\right)^{\frac{1}{2}}\tan\frac{E}{2}
\end{aligned}\right\} \quad \ldots\ldots(74).$$

In these formulae $a$, $e$ and $\tau$ are *elements of the elliptic orbit.*

To obtain the position of the planet in its orbit at time $t$, the elements $a$, $e$, $\tau$ and the period being given, the procedure is as follows:

($a$) Calculate the eccentric anomaly $E$ by means of (iv), in the manner outlined in section 69, or otherwise.

($b$) Calculate the radius vector $r$ by means of (v).

($c$) Calculate* the true anomaly $v$ by means of (vi).

($d$) As a check, calculate the radius vector $r$ by means of (i), using the value of the true anomaly $v$ found in ($c$).

**71.** *The eccentric anomaly expressed as a series in terms of* e *and the mean anomaly.*

We now express Kepler's equation in a different form, in which we shall derive $E$ as a series in terms of the eccentricity $e$ and the mean anomaly $M$. We have Kepler's equation

$$E = M + e \sin E \qquad \ldots\ldots(75).$$

We shall regard $e$ as a small fraction and the first approximation to $E$—denoted by $E_1$—will evidently be given by neglecting $e \sin E$, so that

$$E_1 = M.$$

A more accurate value of $E$—the second approximation denoted by $E_2$—will clearly be given by writing $E_1$ (or $M$) on the right of (75). Thus

$$E_2 = M + e \sin M \qquad \ldots\ldots(76).$$

In the same way, denoting the third approximation to $E$ by $E_3$, we write

$$E_3 = M + e \sin E_2,$$

which by (76) becomes

$$E_3 = M + e \sin [M + e \sin M],$$

or $$E_3 = M + e \sin M \cos [e \sin M] + e \cos M \sin [e \sin M].$$

But, since $e$ is small, we can write this last equation

$$E_3 = M + e \sin M + e \cos M \,.\, e \sin M \qquad \ldots\ldots(77),$$

which is correct up to terms in $e^2$. (77) is equivalent to

$$E_3 = M + e \sin M + \tfrac{1}{2} e^2 \sin 2M \qquad \ldots\ldots(78).$$

We can proceed in this way as far as we like. A further approximation will give

$$E = M + \left(e - \frac{e^3}{8}\right) \sin M + \tfrac{1}{2} e^2 \sin 2M + \tfrac{3}{8} e^3 \sin 3M \ldots(79).$$

* In the method to be described in section 77 the quantities $E$ and $r$ are alone calculated.

When terms containing $e^4$ and higher powers can be neglected, formula (79) enables us to calculate the value of $E$ when $e$ and $M$ are given.

**72.** *The true anomaly expressed as a series in terms of* e *and the eccentric anomaly.*

We begin with the formula

$$\tan\frac{v}{2} = \left(\frac{1+e}{1-e}\right)^{\frac{1}{2}} \tan\frac{E}{2} \qquad \text{.....(80).}$$

Let us define an angle $\phi$, lying between 0 and $\frac{\pi}{2}$, by

$$\sin\phi = e \qquad \text{.....(81).}$$

Then we can write (80) in the form

$$\tan\frac{v}{2} = \frac{1+\tan\frac{\phi}{2}}{1-\tan\frac{\phi}{2}} \tan\frac{E}{2},$$

or, putting $\tan\frac{\phi}{2} = x$,

$$\tan\frac{v}{2} = \frac{1+x}{1-x}\tan\frac{E}{2} \qquad \text{.....(82).}$$

Since $2\cos\theta = e^{i\theta} + e^{-i\theta}$ and $2i\sin\theta = e^{i\theta} - e^{-i\theta}$, where $i$ is defined by $i^2 = -1$, we can write

$$\tan\frac{v}{2} = \frac{e^{iv/2} - e^{-iv/2}}{i\,(e^{iv/2} + e^{-iv/2})} = \frac{e^{iv}-1}{i\,(e^{iv}+1)},$$

with a similar formula for $\tan\frac{E}{2}$. Hence (82) can be written

$$\frac{e^{iv}-1}{e^{iv}+1} = \frac{1+x}{1-x}\cdot\frac{e^{iE}-1}{e^{iE}+1},$$

from which

$$e^{iv} = \frac{e^{iE}-x}{1-xe^{iE}},$$

or

$$e^{iv} = e^{iE}(1-xe^{-iE})/(1-xe^{iE}),$$

from which, by taking logarithms of both sides,

$$iv = iE + \log(1-xe^{-iE}) - \log(1-xe^{iE}).$$

Now $x = \tan\frac{\phi}{2}$ and $\phi = \sin^{-1}(e)$; hence $x < 1$ numerically, since

$e < 1$. Applying the formula for the logarithmic series we derive, after some reduction,

$$v = E + 2\left(x \sin E + \frac{x^2}{2} \sin 2E + \frac{x^3}{3} \sin 3E + \ldots\right) \ldots(83).$$

Now

$$x \equiv \tan \frac{\phi}{2} = \frac{\sin^2 \frac{\phi}{2}}{\sin \frac{\phi}{2} \cos \frac{\phi}{2}} = \frac{1 - \cos \phi}{e} = \frac{1 - (1 - e^2)^{\frac{1}{2}}}{e},$$

or

$$x = \tfrac{1}{2} e + \tfrac{1}{8} e^3 + \ldots \qquad \ldots\ldots(84).$$

Hence from (83) and (84) we obtain **finally**

$$v = E + (e + \tfrac{1}{4} e^3) \sin E + \tfrac{1}{4} e^2 \sin 2E + \tfrac{1}{12} e^3 \sin 3E \ \ldots(85),$$

correct to the third power of $e$.

**73.** *The equation of the centre.*

By means of (79) and (85) we can express $v$ as a series in $e$ and $M$. We shall keep only terms up to $e^3$. Now in (85), $E$ occurs in the form $\sin E$, $\sin 2E$ and $\sin 3E$, and we first find from (79) the values of these quantities expressed in terms of $e$ and $M$. Thus, by (79),

$$\sin E = \sin\left[M + \left(e - \frac{e^3}{8}\right) \sin M + \tfrac{1}{2} e^2 \sin 2M + \tfrac{3}{8} e^3 \sin 3M\right] \qquad \ldots\ldots(86),$$

but, as in (85) $\sin E$ is multiplied by the factor $e$, we need only keep terms in (86) up to $e^2$, so that

$$\sin E = \sin [M + e \sin M + \tfrac{1}{2} e^2 \sin 2M],$$

which can be expanded in the form

$$\begin{aligned}
\sin E &= \sin M \cos [e \sin M + \tfrac{1}{2} e^2 \sin 2M] \\
&\qquad + \cos M \sin [e \sin M + \tfrac{1}{2} e^2 \sin 2M] \\
&= \sin M \left[1 - \frac{e^2}{2} \sin^2 M\right] + \cos M \, [e \sin M + \tfrac{1}{2} e^2 \sin 2M] \\
&= \sin M + \tfrac{1}{2} e \sin 2M + \tfrac{1}{2} e^2 (\sin 2M \cos M - \sin^2 M) \\
&= (1 - \tfrac{1}{8} e^2) \sin M + \tfrac{1}{2} e \sin 2M + \tfrac{3}{8} e^2 \sin 3M.
\end{aligned}$$

Similarly, keeping only the necessary terms, we derive

$$\sin 2E = \sin 2M + e (\sin 3M - \sin M),$$

$$\sin 3E = \sin 3M.$$

Hence from (79) and the expressions for $\sin E$, $\sin 2E$ and $\sin 3E$ just found, the formula (85) becomes

$$v - M = (2e - \tfrac{1}{4}e^3)\sin M + \tfrac{5}{4}e^2 \sin 2M + \tfrac{13}{12}e^3 \sin 3M \quad \ldots\ldots(87).$$

This last formula is known as the *equation of the centre*. Its importance lies in the fact that the true anomaly $v$ is expressed directly in terms of the eccentricity $e$ and the mean anomaly $M$. When $e$ and $M$ are given, $v$ can thus be calculated. We shall make use of the equation of the centre in a later chapter.

**74.** *The orbit in space.*

We shall take the plane of the ecliptic as the fundamental plane. In general, the orbital plane of any planet, other than the earth, will be inclined at some particular angle to the ecliptic. Suppose a sphere drawn with the sun as centre (Fig. 49).

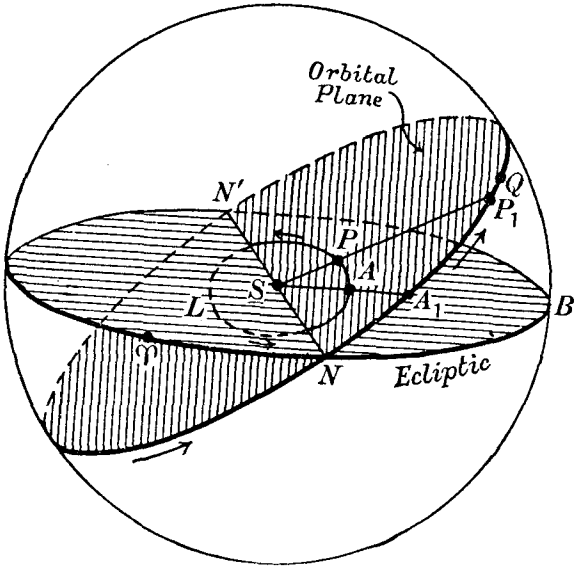



Fig. 49.

As the planet's orbital plane (shown shaded in the figure) passes through the sun, it will therefore cut the sphere in a great circle $NQN'$. This great circle intersects the ecliptic in two points $N$ and $N'$, called the *nodes*. The orbital ellipse $APL$ is drawn in the

figure, $A$ being perihelion; $SA$ produced cuts the great circle $NQN'$ in $A_1$ and, if $P$ is the position of the planet in its orbit at any time $t$, the radius vector $SP$ when produced will cut the great circle $NQN'$ in $P_1$. Thus, since $A\hat{S}P$ is the true anomaly $v$, the angle $A_1SP_1$ or the great circle arc $A_1P_1$ is also $v$. In the figure, $v$ increases in the direction of the arrow between $A_1$ and $P_1$. $N$ is then called the *ascending node* and $N'$ the *descending node.*

The angle $P_1NB$ (denoted by $i$) defines the inclination of the orbital plane to the ecliptic; $i$ is simply called the *inclination.* The arc $NA_1$, measured from the ascending node to the point $A_1$, is called the argument of perihelion and is denoted by $\omega$.

Let $\Upsilon$ be the position of the vernal equinox. The arc $\Upsilon N$ is the *longitude of the ascending node,* which we denote by $\theta$. The sum of the arcs $\Upsilon N$ and $NA_1$ is called the *longitude of perihelion*, denoted by $\varpi$. Hence

$$\varpi = \theta + \omega \qquad \text{.....(88).}$$

It is to be noted that it is only the part $\theta$ of the longitude of perihelion that is measured along the ecliptic.

$\theta$, $i$, $\varpi$ are three elements of the orbit. The longitude of the node ($\theta$) determines the points $N$ and $N'$ on the celestial sphere at which the plane of the orbit intersects the ecliptic; the element $i$ specifies the angle at which the orbital plane is inclined to the ecliptic; the longitude of perihelion ($\varpi$) determines the direction of perihelion with reference to the ecliptic and the vernal equinox.

As the three elements $a$, $e$ and $\tau$ considered in section 70 refer to the orbital ellipse only, we now see that the complete specification of a planetary orbit requires six elements; they are $a$, $e$, $\tau$, $\theta$, $i$ and $\varpi$.

In Fig. 49 the sum of the arcs $\Upsilon N$ and $NP_1$ is called the *true longitude of the planet* in its orbit. Denote it by $L$. Then

$$L = \theta + \omega + v,$$

or, by (88),

$$L = \varpi + v \qquad \text{.....(89).}$$

Now consider a radius vector coinciding with $SA$ at time $\tau$ and moving in the plane of the orbit with mean angular velocity $n$. At time $t$, the radius vector has described the angle $n\,(t - \tau)$—the mean anomaly $M$—and we shall suppose that it then inter-

sects the sphere in $Q$. Thus the arc $A_1Q$ is $M$. Let $l$ denote the sum of the arcs $\Upsilon N$ and $NQ$. Then

$$l = \theta + \omega + n(t - \tau),$$

or
$$l = \varpi + n(t - \tau) \qquad \ldots\ldots(90).$$

$l$ is called the *mean longitude* of the planet.

The formula (90) is generally written in the form

$$l = nt + \epsilon \qquad \ldots\ldots(91),$$

where $\epsilon$ is given by
$$\epsilon = \varpi - n\tau \qquad \ldots\ldots(92).$$

From (91) it is seen that $\epsilon$ is the mean longitude when $t = 0$. This instant is called the "epoch" and $\epsilon$ is thus the *mean longitude at the epoch.*

From (90) we have

$$M \equiv n(t - \tau) = l - \varpi,$$

so that, by (91),
$$M = nt + \epsilon - \varpi.$$

Hence Kepler's equation can be written

$$E - e \sin E = nt + \epsilon - \varpi \qquad \ldots\ldots(93).$$

Since $\epsilon$ is defined by (92) in terms of the two elements $\varpi$ and $\tau$, it may be regarded as one of the six elements of the orbit in place of $\tau$.

As an illustration, the elements of the orbit of Mars for the epoch 1929 January 0, Greenwich Mean Noon, are given below:

| | |
|---|---|
| $a = 1{\cdot}52369$ astronomical units | $\theta = 49^\circ\ 0'\ 36''{\cdot}0$ |
| $e = 0{\cdot}09334$ | $i = 1^\circ\ 51'\ 0''{\cdot}5$ |
| $\epsilon = 84^\circ\ 42'\ 33''{\cdot}9$ | $\varpi = 334^\circ\ 45'\ 7''{\cdot}3$ |

**75.** *The heliocentric ecliptic rectangular co-ordinates of a planet.*

In Fig. 50 let $S\Upsilon$, $SB$, $SK$ be rectangular axes through the sun, forming a right-handed system, $\Upsilon$ being the vernal equinox, $B$ the point on the ecliptic 90° from $\Upsilon$ and $K$ the pole of the ecliptic. Let $P$ be the position of the planet in its orbit at time $t$ so that $SP = r$. Let $P_1$ be the point of intersection of $SP$ with the celestial sphere. Denote by $(x_1, y_1, z_1)$ the rectangular co-ordinates of $P$ with respect to the axes $S\Upsilon$, $SB$ and $SK$. Then

$$\frac{x_1}{r} = \cos PS\Upsilon = \cos P_1S\Upsilon,$$

or, writing the arc $P_1\Upsilon$ for $P_1\hat{S}\Upsilon$, we have

$$x_1 = r\cos P_1\Upsilon \quad \text{.....(94).}$$

Similarly,

$$y_1 = r\cos P_1 B \quad \text{.....(95),}$$

$$z_1 = r\cos P_1 K \quad \text{.....(96).}$$

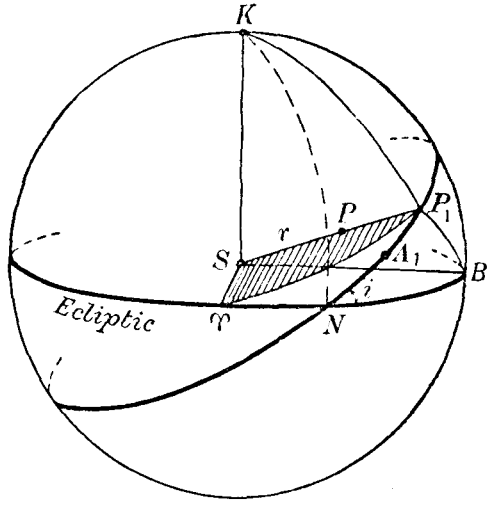



Fig. 50.

Consider now the spherical triangle $P_1\Upsilon N$ in which $\Upsilon N = \theta$, $NP_1 = \omega + v$ and $P_1\hat{N}\Upsilon = 180° - i$. Then, by the cosine formula **A**,

$$\cos P_1\Upsilon = \cos\theta\cos(\omega + v) - \sin\theta\sin(\omega + v)\cos i.$$

Hence

$$x_1 = r[\cos\theta\cos(\omega + v) - \sin\theta\sin(\omega + v)\cos i] \quad \text{...(97).}$$

Similarly, from the triangle $P_1NB$ in which $NB = 90° - \theta$ and $P_1\hat{N}B = i$,

$$y_1 = r[\sin\theta\cos(\omega + v) + \cos\theta\sin(\omega + v)\cos i] \quad \text{...(98).}$$

In the triangle $KP_1N$, $KN = 90°$, $K\hat{N}P_1 = 90° - i$; hence by applying formula **A** we derive

$$z_1 = r\sin(\omega + v)\sin i \quad \text{.....(99).}$$

Thus from the formulae (97), (98) and (99) the heliocentric co-ordinates $(x_1, y_1, z_1)$ can be calculated when $r$ and $v$ and the elements $\theta$, $i$ and $\omega$ are given.

It is convenient in practice to calculate auxiliary angles $a$, $A$, $b_1$ and $B_1$ defined by

$$\left.\begin{aligned} \sin a \sin A &= \cos\theta; \quad \sin a \cos A = -\sin\theta\cos i \\ \sin b_1 \sin B_1 &= \sin\theta; \quad \sin b_1 \cos B_1 = \cos\theta\cos i \end{aligned}\right\} \quad (100),$$

in which $\theta$ and $i$ are supposed known. Then

$$\left.\begin{aligned} x_1 &= r \sin a \sin(A + \omega + v) \\ y_1 &= r \sin b_1 \sin(B_1 + \omega + v) \\ z_1 &= r \sin(\omega + v) \sin i \end{aligned}\right\} \quad \ldots\ldots(101).$$

**76.** *The heliocentric equatorial co-ordinates of a planet.*

In Fig. 51 let $\Upsilon C$ be the equator on the celestial sphere. Consider the right-handed system of equatorial rectangular axes $S\Upsilon$, $SC$ and $SQ$, where $C$ is the point on the equator 90° from $\Upsilon$

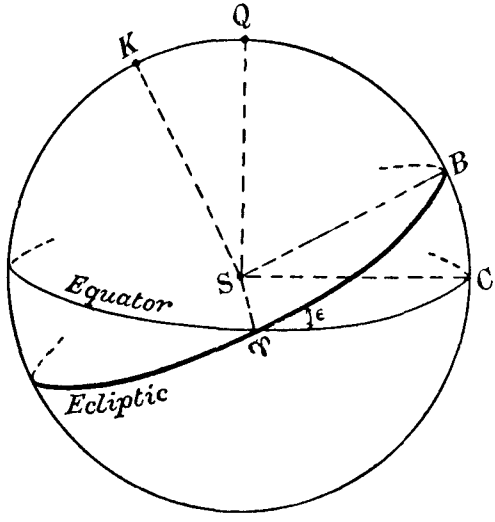



Fig. 51.

and $Q$ is the north pole of the equator. Let $(x, y, z)$ be the co-ordinates of the planet referred to these axes. Now $B\hat{\Upsilon}C = KQ = \epsilon$, the obliquity of the ecliptic. The axes $SC$ and $SQ$ can be obtained from $SB$ and $SK$ by rotating the latter

about $ST$ through the angle $\epsilon$. Hence we obtain the following relations between $(x, y, z)$ and $(x_1, y_1, z_1)$:

$$\left.\begin{aligned} x &= x_1 \\ y &= y_1 \cos\epsilon - z_1 \sin\epsilon \\ z &= y_1 \sin\epsilon + z_1 \cos\epsilon \end{aligned}\right\} \quad \text{.....(102).}$$

Inserting the values of $x_1$, $y_1$ and $z_1$ given by (97), (98) and (99) we derive the expressions for $x$, $y$ and $z$ in terms of $r$ and $v$, the elements $\theta$, $i$ and $\omega$, and $\epsilon$.

Define auxiliary angles $b$, $B$, $c$ and $C$ by the following:

$$\left.\begin{aligned} &\sin b \sin B = \sin\theta\cos\epsilon;\ \sin b\cos B = \cos\theta\cos i\cos\epsilon - \sin i\sin\epsilon \\ &\sin c\sin C = \sin\theta\sin\epsilon;\ \sin c\cos C = \cos\theta\cos i\sin\epsilon + \sin i\cos\epsilon \end{aligned}\right\} \quad \text{.....(103).}$$

Then we have

$$\left.\begin{aligned} x &= r\sin a\sin(A+\omega+v) \\ y &= r\sin b\sin(B+\omega+v) \\ z &= r\sin c\sin(C+\omega+v) \end{aligned}\right\} \quad \text{.....(104).}$$

The convenience of these formulae is seen when the rectangular co-ordinates are required for several positions of the planet. For when the auxiliary quantities $\sin a$, $\sin b$, $\sin c$, $A$, $B$ and $C$ have been computed by means of (100) and (103) from the known values of the elements $\theta$, $i$ and $\omega$, and of $\epsilon$ they can be used in (104) for determining the several values of $(x, y, z)$. For each position of the planet the radius vector $r$ and the true anomaly $v$ are supposed to be calculated in advance by the methods already described.

An alternative method of deriving the co-ordinates $(x, y, z)$ is described in the following section.

**77.** *The heliocentric equatorial co-ordinates of a planet (alternative method).*

The heliocentric equatorial co-ordinates can be put in another form* which is now favoured by computers. The first equation of (104) can be written

$$x = r\cos v\,[\sin a\sin(A+\omega)] + r\sin v\,[\sin a\cos(A+\omega)].$$

Put $P_x = \sin a\sin(A+\omega)$ and $Q_x = \sin a\cos(A+\omega)$.

* *Journal of the Brit. Astron. Association*, vol. XXXII, p. 231 (C. E. Adams).

Then $P_x = \sin a \sin A \cos \omega + \sin a \cos A \sin \omega$,

which by (100) becomes

$$P_x = \cos \theta \cos \omega - \sin \theta \sin \omega \cos i.$$

Similarly, $Q_x = -\cos \theta \sin \omega - \sin \theta \cos \omega \cos i.$

Now by (62), $r \cos v = a (\cos E - e)$, and by (61),

$$r \sin v = b \sin E,$$

where $E$ is the eccentric anomaly and $b$ is the semi-minor axis of the orbit. We then have, adding the formulae for $y$ and $z$ which can be derived in a similar manner,

$$\left.\begin{aligned} x &= aP_x \cos E + bQ_x \sin E - aeP_x \\ y &= aP_y \cos E + bQ_y \sin E - aeP_y \\ z &= aP_z \cos E + bQ_z \sin E - aeP_z \end{aligned}\right\} \quad \ldots\ldots(104\,a),$$

where

$$\left.\begin{aligned} P_x &= \cos \theta \cos \omega - \sin \theta \sin \omega \cos i \\ Q_x &= -\cos \theta \sin \omega - \sin \theta \cos \omega \cos i \\ P_y &= (\sin \theta \cos \omega + \cos \theta \sin \omega \cos i) \cos \epsilon - \sin \omega \sin i \sin \epsilon \\ Q_y &= (-\sin \theta \sin \omega + \cos \theta \cos \omega \cos i) \cos \epsilon - \cos \omega \sin i \sin \epsilon \\ P_z &= \sin \omega \sin i \cos \epsilon + (\sin \theta \cos \omega + \cos \theta \sin \omega \cos i) \sin \epsilon \\ Q_z &= \cos \omega \sin i \cos \epsilon - (\sin \theta \sin \omega - \cos \theta \cos \omega \cos i) \sin \epsilon \end{aligned}\right\}$$

$\ldots\ldots(104\,b).$

For each position of the planet, the eccentric anomaly $E$ is calculated from the elements, supposed known. The quantities $P_x$, etc., are also found from the elements. The co-ordinates $(x, y, z)$ can then be easily calculated from (104 $a$) for as many positions of the planet as are desired.

**78.** *The heliocentric rectangular co-ordinates of the earth.*

In the case of the earth, the equations (104) assume a simple form as the inclination of the earth's orbit to the ecliptic is zero. Writing $(x', y', z')$ for the heliocentric equatorial co-ordinates of the earth at any time $t$, we obtain easily the following formulae:

$$\left.\begin{aligned} x' &= r' \cos (\theta' + \omega' + v') \\ y' &= r' \sin (\theta' + \omega' + v') \cos \epsilon \\ z' &= r' \sin (\theta' + \omega' + v') \sin \epsilon \end{aligned}\right\} \quad \ldots\ldots(105),$$

in which dashes refer to the respective quantities associated with the earth's orbit. But $(\theta' + \omega')$ is the longitude of perihelion for the earth's orbit; denote it by $\varpi'$. Then

$$\left.\begin{aligned} x' &= r' \cos(\varpi' + v') \\ y' &= r' \sin(\varpi' + v') \cos\epsilon \\ z' &= r' \sin(\varpi' + v') \sin\epsilon \end{aligned}\right\} \quad \ldots\ldots(106).$$

The values of $r'$ and $v'$ can be calculated for any time $t$ from the elements of the earth's orbit; hence $x'$, $y'$, $z'$ can be obtained.

An alternative method is based on the formulae (104 *a*) and (104 *b*); in the latter, the corresponding values of $P_x$, etc., are obtained by putting the inclination $i$ equal to zero. As before, the values of the eccentric anomaly for the earth are calculated and the co-ordinates derived by means of (104 *a*).

**79.** *The planet's geocentric right ascension and declination.*

Consider now rectangular axes drawn through the earth (Fig. 52) and parallel to the equatorial axes through the sun illustrated in Fig. 51. Let $X$, $Y$, $Z$ denote the co-ordinates of the sun with reference to the axes drawn through $E$. Then it is clear that

$$X = -x', \quad Y = -y', \quad Z = -z' \quad \ldots\ldots(107).$$

$X$, $Y$ and $Z$ are the *geocentric* equatorial rectangular coordinates of the sun and are calculable by means of (106) as already indicated. Their values are tabulated in the almanacs for every day throughout the year.

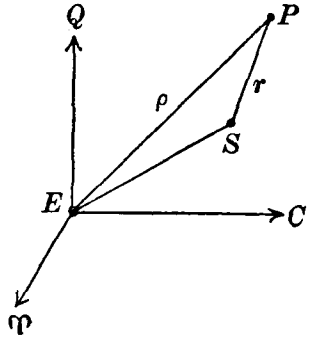


Fig. 52.

Let $(\xi, \eta, \zeta)$ denote the co-ordinates at time $t$ of the planet $P$ with respect to $E$ as origin (Fig. 52). Then since $(x, y, z)$ are the co-ordinates of $P$ with respect to $S$ as origin, we have

$$\xi = X + x; \quad \eta = Y + y; \quad \zeta = Z + z \quad \ldots\ldots(108).$$

Consider in Fig. 53 the celestial sphere centred at $E$ (the earth's centre). The straight line joining $E$ and the planet cuts the sphere in $R$. Denote the distance $EP$ by $\rho$. Then $\rho$ is the

*geocentric distance* of the planet at time $t$. Draw the meridian $QRT$ intersecting the equator $\Upsilon C$ in $T$. Denote by $\alpha$, $\delta$ the geocentric right ascension and declination of the planet so that $\Upsilon T = \alpha$, $TR = \delta$. Then

$$\xi/\rho = \cos P\hat{E}\Upsilon = \cos R\hat{E}\Upsilon,$$

or $$\xi = \rho \cos R\Upsilon.$$

Similarly, $$\eta = \rho \cos RC,$$

$$\zeta = \rho \cos RQ.$$

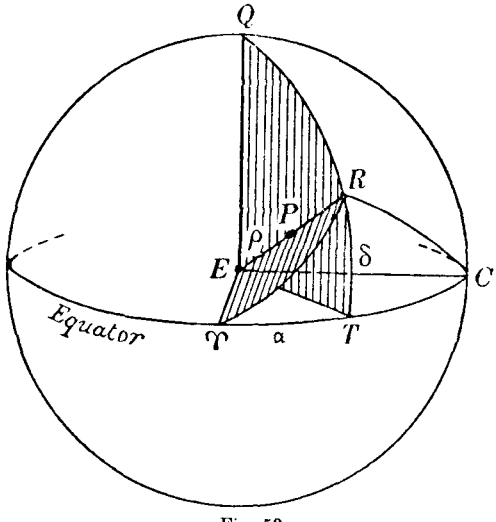



Fig. 53.

But formula **A** gives, for the triangle $R\Upsilon T$ (right-angled at $T$),

$$\cos R\Upsilon = \cos\alpha \cos\delta.$$

Hence $$\xi = \rho \cos\alpha \cos\delta.$$

Similarly, $$\eta = \rho \sin\alpha \cos\delta$$

$$\zeta = \rho \sin\delta$$ .....(109).

Hence from (104), (108) and (109) we obtain

$$\rho \cos\alpha \cos\delta = X + r \sin a \sin(A + \omega + v) \quad .....(110),$$

$$\rho \sin\alpha \cos\delta = Y + r \sin b \sin(B + \omega + v) \quad .....(111),$$

$$\rho \sin\delta = Z + r \sin c \sin(C + \omega + v) \quad .....(112).$$

These equations in the form stated were first given by Gauss.

Given the elements of the planet's orbit, the right-hand sides of (110)–(112) can be calculated for any time $t$, as we have already explained ($X$, $Y$ and $Z$ are given in the *Astronomical Ephemeris*). Thus dividing (111) by (110) we obtain

$$\tan \alpha = \frac{Y + r \sin b \sin (B + \omega + v)}{X + r \sin a \sin (A + \omega + v)},$$

from which $\alpha$ can be derived. Similarly, by dividing (112) by (111), we obtain an expression for $\tan \delta \operatorname{cosec} \alpha$, from which $\delta$ is easily calculated. Thus the geocentric right ascension and declination of the planet can be derived for any time $t$. The observed position will then be the geocentric position as calculated, modified by the correction for refraction and certain other corrections to be discussed in later chapters.

We have thus indicated how the position of a planet in the sky at any time can be derived from a knowledge of its orbit.

The converse problem—*the determination of the elements of a planetary orbit* from an adequate number of observations—is one of much greater difficulty. A vast amount of astronomical literature is devoted to this problem alone and the subject is outside the scope of this book. The reader is referred, for the general principles of the methods of determining orbits from observation, to *Dynamical Astronomy* by H. C. Plummer (Cambridge, 1918) or to *Bahnbestimmung der Planeten und Kometen* by G. Stracke (Berlin, 1929).

**80.** *The orbital and synodic periods of a planet.*

We have assumed throughout this chapter that the orbital period of a planet is known and we now indicate how this period can be obtained from observation. We shall take the simple case when the planet's orbital plane is assumed to coincide with the plane of the ecliptic and the orbits of the earth and planet are assumed to be circular (Fig. 54). In this figure, the planet considered is taken to be an *outer* planet, for which the heliocentric distance is greater than the heliocentric distance of the earth. At some moment, let the sun, the earth and the planet be collinear as at $S$, $E_1$ and $P_1$ respectively. The earth and planet are then said to be in heliocentric conjunction. As viewed from the earth at $E_1$ (we neglect the dimensions of the earth in comparison with the dimensions of the orbits concerned), the planet's

position $P_1$ will be diametrically opposite in the sky to the position of the sun; the planet is then said to be in *opposition*. If heliocentric conjunction occurred at a particular place—say, Greenwich—at apparent midnight (when the sun's hour angle is $12^h$) the planet would then be on the meridian of the place; this instant could be determined accurately from observation of the planet with the meridian circle. This particular assumption, however, is unlikely to be realised in practice and so observations must be made on a few nights before and after the expected heliocentric conjunction. On each night the difference between the right ascensions of the planet and the sun, at the moment of transit of the planet, can be determined; on some nights this difference will be a little less than $12^h$ and on other nights a little more than $12^h$. By interpolation the instant at which the difference is precisely $12^h$ can be derived. This gives the time of heliocentric conjunction. Let this time be denoted by $t_1$.

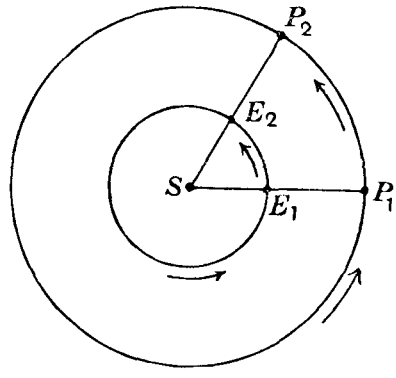


Fig. 54.

Let $t_2$ be the time at which the earth and planet are next in heliocentric conjunction and let their respective positions be $E_2$ and $P_2$. Since the planet is at a greater distance from the sun than the earth, by Kepler's third law its mean angular velocity is less than that of the earth. Hence at the end of the interval $(t_2 - t_1)$ the earth has described $360° + E_2\hat{S}E_1$ or $2\pi + \phi$, where $\phi = E_2\hat{S}E_1$, and the planet has described the angle $P_2SP_1$ or $\phi$. Let $T_1$, $T$ denote the orbital periods of the earth and planet. Then their mean angular motions are $2\pi/T_1$ and $2\pi/T$ respectively. Hence

$$2\pi + \phi = \frac{2\pi}{T_1}(t_2 - t_1),$$

and

$$\phi = \frac{2\pi}{T}(t_2 - t_1).$$

Hence, eliminating $\phi$,

$$(t_2 - t_1)\left(\frac{1}{T_1} - \frac{1}{T}\right) = 1 \qquad \text{.....(113).}$$

The interval between two successive heliocentric conjunctions in longitude is called the *synodic period* of the planet, which we denote by $S$. Hence $S = t_2 - t_1$ and therefore, by (113),

$$\frac{1}{S} = \frac{1}{T_1} - \frac{1}{T}.$$

Now, for the earth, $T_1$ is one year. Also $S$ is obtained from observations and if it is expressed in terms of the year as unit we derive the formula for the planet's orbital period $T$, in the same unit, as

$$\frac{1}{T} = 1 - \frac{1}{S},$$

or

$$T = \frac{S}{S-1} \qquad \text{.....(114).}$$

For an *inner* planet, the period $T'$ is given by

$$T' = \frac{S'}{1+S'}$$

where $S'$ is the corresponding synodic period.

For the outer planets, the synodic periods are evidently greater than one year. For Mercury and Venus the synodic periods, found in this way, are 116 and 584 days.

Owing to the assumptions made in this section with reference to the orbits of the earth and the planet under consideration, the orbital periods derived in this way are approximate; accurate values can only be obtained from a more thorough study of the planetary orbits.

**81.** *The earth's orbit.*

The eccentricity of the earth's orbit is 0·016739 and the longitude of perihelion is 101° 45′ 8″·1; these quantities refer to the beginning of 1931.

The *sidereal year* is defined to be the interval between two successive returns of the earth to the same point among the stars as viewed from the sun. The sidereal year is 365·2564 mean solar days.

**82.** *The sun's apparent orbit.*

For many of our purposes, it will be convenient to consider the sun's apparent orbit, that is, the orbit it appears to describe

relative to the earth. The eccentricity is, of course, the same as that given in the previous section.

The point of the orbit at which the sun is nearest the earth is *perigee* and the point most remote is *apogee*. As the direction of perihelion (as viewed from the sun) is exactly opposite to the direction of perigee (as viewed from the earth), it follows that

$$\varpi = \varpi_1 + 180^\circ \qquad \text{.....(115),}$$

where $\varpi$ denotes the longitude of perigee and $\varpi_1$ denotes the longitude of perihelion.

It is clear that the sidereal year is the interval between two returns of the sun (moving in the apparent orbit) to the same point among the stars, as viewed from the earth.

The *tropical year* is defined to be the interval between two successive passages of the sun through the vernal equinox. As we shall see in a later chapter, the vernal equinox is not a fixed point on the ecliptic, as we have hitherto assumed it to be, and consequently the tropical year differs somewhat in length from the sidereal year. The tropical year is 365·2422 mean solar days. When the term "year" is used without any qualifying adjective it is the tropical year that is referred to.

Owing to the gravitational attraction of the planets on the earth, the elements of the earth's orbit are not quite constant. In particular, the longitude of perihelion (or, by (115), the longitude of perigee) undergoes small changes. The interval between two successive passages of the earth, in its orbit, through perihelion—or the interval between two successive passages of the sun, in the apparent orbit, through perigee—is called the *anomalistic year*, which is 365·2596 mean solar days.

### 83. *The moon's orbit.*

The formulae which we have derived for the motion of a planet around the sun are applicable to the motion of the moon around the earth. The moon's orbit in space is defined with reference to the ecliptic as in the case of a planet and the appropriate figure, corresponding to Fig. 49, would be identical in character with the latter, with this exception, that the centre of the celestial sphere is now $E$ (the earth) instead of $S$ (the sun). The longitude of the moon's node, the longitude of *perigee* (the point of the moon's orbit nearest the earth—the furthest point is *apogee*) and the

inclination of the orbital plane are defined in the same way as in section 74.

So far as the gravitational attraction of the earth alone is concerned, the moon's orbit would be an ellipse; but, owing to the gravitational influence of the sun and—to a lesser degree—of the planets, the elements of the orbit undergo considerable changes —*perturbations*—which have to be taken into account in many problems with which the moon is associated and, in particular, in the theory of precession and nutation, to be discussed in Chapter x.

The *synodical month* is the interval between two successive "new moons". *New moon* occurs when the geocentric longitudes of the sun and moon are the same. The average value of the synodical month is 29·5306 mean solar days.

The *sidereal month* is defined to be the interval given by the moon's complete circuit of the stars as seen from the earth; its mean value is 27·3217 mean solar days.

Owing to the perturbations, the direction of perigee is altering and the interval required by the moon to move in its path around the earth from perigee to perigee is called the *anomalistic month*; its value is 27·5546 mean solar days.

Again, owing to perturbations, the moon's ascending node has a backward movement along the ecliptic, the longitude of the node decreasing at the rate of nearly 20° per annum (the period required by the moon's node to make a circuit of the ecliptic is 18·6 years). The *nodical month* is defined to be the interval between two successive passages of the moon through the ascending node; its value is 27·2122 mean solar days.

## EXERCISES

[Symbols used:

$a$ = semi-major axis of orbit, $E$ = eccentric anomaly,
$e$ = eccentricity, $M$ = mean anomaly.]
$v$ = true anomaly,

**1.** If the semi-major axes of the orbits of Mercury and Jupiter are 0·387 and 5·203 astronomical units and Jupiter's orbital period is 11·862 years, show that Mercury's orbital period is 0·2406 year.

**2.** Neglecting the mass of the first satellite of Jupiter, calculate the mass of this planet in terms of that of the earth from the following data:

Period of first satellite: $1^d\ 18^h\ 28^m$.
Mean distance of first satellite from Jupiter's centre: 267,000 miles.
Radius of the earth: 3960 miles.
Acceleration of gravity at earth's surface: 32·2 feet per second per second.
[*Lond.* 1926.]

**3.** The orbital period of Jupiter is 4333 mean solar days and Jupiter's mass is 1/1048 times the sun's mass. Show that the period of a small body, of negligible mass, moving in an elliptic orbit round the sun with the same major axis as that of Jupiter, is $4335\frac{1}{16}$ days. [*Lond.* 1930.]

**4.** If $T$ is the orbital period of a planet, show that a small increase $\Delta a$ in the semi-major axis $a$ will produce an increase $\dfrac{3T\Delta a}{2a}$ in the period.

**5.** If $V_1$ and $V_2$ are the linear velocities of a planet at perihelion and aphelion respectively, prove that $(1-e)\,V_1 = (1+e)\,V_2$.

**6.** If $e = \sin\phi$, prove that

$$\tan\frac{v}{2} = \tan(45^\circ + \tfrac{1}{2}\phi)\tan\tfrac{1}{2}E.$$

**7.** If $e = \sin\phi$, show that, when powers of $e$ above the second are neglected, the value of $E$ satisfying Kepler's equation is given by

$$\tan E = \sec\phi\tan 2\chi,$$

where $$\tan\chi = \tan(45^\circ + \tfrac{1}{2}\phi)\tan\tfrac{1}{2}M,$$

and $$\tan E = \sin M/(\cos M - e).$$

**8.** Prove that

$$\cos v = \frac{\cos E - e}{1 - e\cos E},$$

$$\sin v = \frac{(1-e^2)^{\frac{1}{2}}\sin E}{1 - e\cos E}.$$

**9.** Prove that, if the fourth and higher powers of $e$ are neglected,

$$E = M + \frac{e\sin M}{1 - e\cos M} - \frac{1}{2}\left(\frac{e\sin M}{1 - e\cos M}\right)^3$$

is a solution of Kepler's equation. [*Ball.*]

**10.** The relevant elements of an orbit are as follows: $e = 0{\cdot}961733$, $T$ (period) = 76·085 years; time of perihelion passage, 1910 May 24. If $E = 101^\circ{\cdot}3$ is an approximate solution of Kepler's equation when $M = 47^\circ{\cdot}3$ and $e = 0{\cdot}96$, show that the value of $E$ for 1900 May 24 is $101^\circ\ 20'\ 33''{\cdot}1$. [*Ball.*]

**11.** A comet describes a hyperbolic orbit around the sun; prove that the velocity $V$ is given by

$$V^2 = \mu\left(\frac{2}{r} + \frac{1}{a}\right).$$

If its minimum heliocentric distance is $k$ astronomical units and its maximum linear velocity $l$ times the earth's velocity, show that the angle between the asymptotes of the hyperbola is

$$2\sin^{-1}\left(\frac{1}{l^2k - 1}\right).$$ [*Edin.* 1921.]

12. If $\psi$ is the angle between the direction of a planet's motion and the direction perpendicular to the radius vector, prove that

$$\tan\psi = \frac{e\sin E}{(1-e^2)^{\frac{1}{2}}}.$$

13. A planet moves in an orbit inclined at a small angle $i$ to the ecliptic; show that, if its declination is a maximum, either the motion in latitude vanishes or the longitude is approximately $90° + i\cot\epsilon\sin\theta$, where $\theta$ is the longitude of the ascending node and $\epsilon$ is the obliquity of the ecliptic.

14. The mean distance of Venus from the sun is 0·72 of that of the earth. Determine the greatest altitude at which Venus, supposed to have a circular orbit in the plane of the ecliptic, can be visible after sunset in a given latitude, and the time of year at which this may occur. [*M.T.*]

15. If $(\lambda_1, \beta_1)$, $(\lambda_2, \beta_2)$ and $(\lambda_3, \beta_3)$ are the heliocentric longitudes and latitudes of a planet at three different points of its orbit, prove that

$$\tan\beta_1\sin(\lambda_2-\lambda_3) + \tan\beta_2\sin(\lambda_3-\lambda_1) + \tan\beta_3\sin(\lambda_1-\lambda_2) = 0.$$

16. Prove that the equation of the centre is given in terms of the true anomaly $v$ by the expression

$$\sum_{p=1}^{\infty}(-)^{p-1}\frac{2\lambda^p\{(p+1)-(p-1)\lambda^2\}}{p(1+\lambda^2)}\sin pv,$$

where $\lambda = e/(1+\sqrt{1-e^2})$, $e$ being the eccentricity of the orbit.

Show that the maximum value of the equation of the centre occurs approximately when $v = \frac{1}{2}\pi + \sin^{-1}(3e/4)$, $e$ being small. [*Lond.* 1930.]

17. The perihelion distance of a parabolic comet is $a$ astronomical units ($a < 1$). Assuming that the earth's orbit is circular and that the comet moves in the ecliptic, show that, if $t$ (measured in years) is the interval during which the comet is within the earth's orbit,

$$t = \frac{1}{3\pi}(1+2a)(2-2a)^{\frac{1}{2}}.$$

18. (*Euler's Theorem.*) If $r$ and $r_1$ are the radii vectores of two points $C$ and $C_1$ in a parabolic orbit and if $k$ is the distance $CC_1$, prove that the time in the orbit between $C$ and $C_1$ is

$$\frac{T_0}{12\pi}\left\{\left(\frac{r+r_1+k}{a}\right)^{\frac{3}{2}} - \left(\frac{r+r_1-k}{a}\right)^{\frac{3}{2}}\right\},$$

where $T_0$ is the length of the sidereal year and $a$ is the semi-major axis of the earth's orbit.

19. (*Lambert's Theorem.*) If $r$ and $r_1$ are the radii vectores of two points $C$ and $C_1$ in an elliptic orbit, $k$ the distance $CC_1$, $t$ the time required by the planet to move from $C$ to $C_1$ and $T$ the orbital period, prove that

$$\frac{2\pi t}{T} = \eta - \sin\eta - (\eta_1 - \sin\eta_1),$$

where $$\sin\tfrac{1}{2}\eta = \frac{1}{2}\left(\frac{r+r_1+k}{a}\right)^{\frac{1}{2}};\quad \sin\tfrac{1}{2}\eta_1 = \frac{1}{2}\left(\frac{r+r_1-k}{a}\right)^{\frac{1}{2}}.$$

## CHAPTER VI

# TIME

**84.** *Sidereal time.*

In Chapter II we briefly discussed the subject of time and now it is necessary to consider the problem in greater detail. Sidereal time at any instant at a given place is the hour angle of the vernal equinox. Now in Chapter II we regarded the ecliptic and celestial equator as fixed great circles on the celestial sphere, so that the vernal equinox was also regarded as a fixed point whose direction, for convenience, we might visualise as that of a particular star. But owing to the phenomena of precession and nutation the celestial equator can no longer be regarded as a fixed great circle and, consequently, the vernal equinox must be treated as a point on the celestial sphere moving slowly, according to well-ascertained laws, with reference to the background of the stars.

Precession and nutation will be fully considered in Chapter X but for our present purpose we shall assume certain results. In our discussion of sidereal time it will be sufficient to regard the ecliptic as a fixed great circle. Owing to precession, the north celestial pole $P$ (defined by the direction of the earth's axis) describes a small circle around the pole $K$ of the ecliptic in a period of about 26,000 years (Fig. 55). At present the north celestial pole $P$ is a little under 1° from the second magnitude star $\alpha$ Ursae Minoris (Polaris or the pole-star), but their relative positions are altering from day to day and from year to

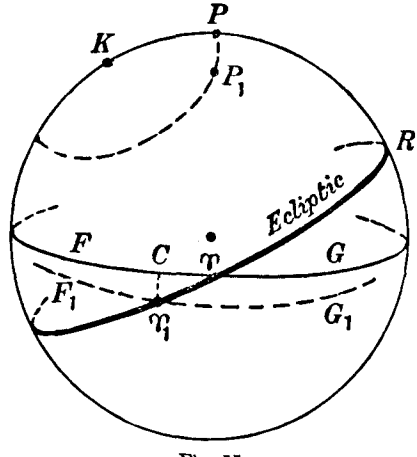


Fig. 55.

year. Two thousand years ago the pole $P$ was 12° from Polaris and 12,000 years hence it will be within a few degrees of the first magnitude star Vega. It is the direction of the earth's axis that is altering continuously with reference to the background of the stars.

Now consider Fig. 55, in which $P$ is the north celestial pole, say, at the beginning of 1900 (denoted by 1900·0) and $P_1$ its position one year later. $PP_1$ is the arc of a small circle of which $K$ is the pole. $FTG$ is the celestial equator corresponding to the position $P$, and $F_1\Upsilon_1G_1$ is the equator one year later corresponding to the position $P_1$ of the pole. $\Upsilon$ is the vernal equinox for 1900·0 and $\Upsilon_1$ the vernal equinox for 1901·0. $\Upsilon$ and $\Upsilon_1$ are called the *mean equinoxes* at the dates in question and the corresponding celestial equators are called the *mean equators*. We assume that owing to precession the north celestial pole moves uniformly along the small circle arc $PP_1$ and that the mean equinox moves uniformly backwards along the ecliptic from $\Upsilon$ to $\Upsilon_1$. It is found that the motion of $\Upsilon$ along the ecliptic is at the rate of 50″·3 per annum.

Let us return to our definition of sidereal time. Suppose for simplicity that there is a star in the direction of $\Upsilon$ which we shall assume for the moment to be fixed in direction. The sidereal time would then be defined as the hour angle of $\Upsilon$ or the hour angle of the star, and the period of rotation of the earth is then simply the interval between two successive transits of this star (or of $\Upsilon$) across the meridian of any particular observatory. But when we define sidereal time in relation to the moving equinox, we can no longer regard the earth's rotational period to be the interval between two successive transits of the equinox. In Fig. 55 let $C\Upsilon_1$ be a great circle arc drawn through $\Upsilon_1$ perpendicularly to the equator $FTG$. Then the equinox at any given date is separating, in right ascension, from the equinox $\Upsilon$ for 1900·0 at the annual rate measured by $\Upsilon C$. But from the small triangle $\Upsilon C\Upsilon_1$,

$$\Upsilon C = \Upsilon\Upsilon_1 \cos\epsilon,$$

where $\epsilon$ is the obliquity of the ecliptic. Hence

$$\Upsilon C = 50''\cdot3 \cos\epsilon,$$

and inserting the value of $\epsilon$ (23° 27′) we easily find that, in time measure, the mean equinox is separating, in right ascension, from

$\Upsilon$ at the rate of $0^s{\cdot}008$ per sidereal day. Now the direction of motion of the equinox is westwards in the sky—opposite to that in which right ascension increases—and it follows that the interval between two successive transits of the moving equinox over any meridian is $0^s{\cdot}008$ less than the interval given by a fixed equinox or star. The first interval is a *sidereal day* (defined with reference to the moving equinox) and the second interval is the rotational period of the earth.

Owing to nutation, the true equator at any instant is slightly different from the mean equator at that instant. Consequently the true equinox is displaced slightly along the ecliptic relative to the mean equinox; these small displacements are periodic in character, the period being about 18 years. The difference in right ascension between the true equinox at any instant and the mean equinox at that instant is evidently periodic in character also and may amount numerically to $1^s{\cdot}2$.

We define *mean sidereal time* to be associated with the moving mean equinox (precession only being involved) and *apparent sidereal time* to be associated with the true equinox. As the motion of the true equinox along the ecliptic can be regarded as compounded of (*a*) the uniform motion due to precession ($50''{\cdot}3$ per annum), and (*b*) the small oscillatory motion, with respect to the mean equinox, due to nutation, it is evident that the interval between two successive transits of the true equinox over a meridian will differ from the interval between two successive transits of the mean equinox by a small amount which is periodic in character. But the differences from day to day are so small that in practice, generally, the sidereal day is taken to mean the interval between two successive transits of the mean equinox. As we have seen, the adopted sidereal day is $0^s{\cdot}008$ shorter than the earth's rotational period. Sidereal clocks are regulated according to mean sidereal time.

When a star's position is observed at any instant, that position is referred to the true equator and equinox at that instant. For example, when the transit of a star is observed, the star's right ascension is the apparent sidereal time of transit; or supposing that the right ascension is known (referred to the true equator) we obtain the apparent sidereal time of the star's transit. This, however, will not be the time shown by the sidereal clock, which,

as we have said, is regulated (we assume, correctly) to keep mean sidereal time, the difference being the small effect due to nutation. The magnitude of this quantity, which has a principal period of 18 years, can be derived from other considerations and it is tabulated for every day in the almanacs under the heading "equation of equinoxes" in the sense:

Apparent S.T. = Mean S.T. + equation of equinoxes ...(1).

**85.** *Ephemeris and Universal Time.*

Sidereal time was defined as the hour angle of the vernal equinox, a definite direction in space. The passage of sidereal time is, therefore, completely determined by the rotation of the earth, and as this is subject to irregular and unpredictable variations, the passage of sidereal time is not uniform. The same is true of universal time, which was defined in section 28 in terms of the hour angle of the mean sun, a fictitious body moving at a constant rate round the celestial equator. Notice, however, if Universal Time is to be precisely related to Sidereal Time, then this rate must be constant with respect to the rotation of the earth. It is, not, therefore, a strictly uniform rate. When the mean sun was first introduced it was not realised that the rotation of the earth was variable. No distinction was therefore, made between Universal Time, defined by the rotation of the earth, and Ephemeris Time (see Appendix E) which is uniform and is defined by the gravitational dynamics of the solar system, independently of the earth's rotation. If, therefore, we are to retain the concept of the mean sun, we must define it in two separate ways for Universal and Ephemeris Time.

We shall begin by ignoring precession and nutation, thus assuming for the present that the equinox $\Upsilon$ and the equator $FYG$ are fixed (Fig. 56). Let $A$ be the direction of the sun as seen from the earth when the sun is nearest the earth. Since the sun appears to describe an ellipse relative to the earth in the course of a year, $A$ is the direction of perigee. Let $n$ denote the mean angular velocity of the sun in its apparent orbit around the earth. When the sun's position on the celestial sphere is at $A$, suppose a fictitious body to start from $A$ and to move along the ecliptic with the mean angular velocity $n$. When the sun reaches the

point $S$ on the ecliptic we shall suppose the fictitious body to be at $M_1$. If $t$ and $\tau$ are the times corresponding to the position of the sun at $S$ and at $A$ respectively, the arc $AM_1$ is the mean anomaly $M$ which is given by

$$M = n\,(t - \tau) \quad \ldots(2).$$

When the fictitious body is at $\Upsilon$ (at time $\tau_1$) suppose a second body, which we call the *fictitious mean sun*, to start from $\Upsilon$ and to move along the equator with the sun's mean angular velocity $n$. When the first fictitious body is at $M_1$, the second is at $M$, so that $\Upsilon M_1 = \Upsilon M$. But $\Upsilon M_1$ is the sun's mean longitude $l$ and $\Upsilon M$ is the right ascension of the fictitious mean sun; hence

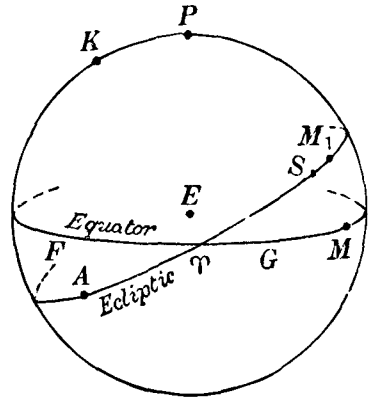


Fig. 56.

$$\text{R.A.F.M.S.} = l = n(t - \tau_1) \quad \ldots\ldots(3).$$

It is evident that according to this equation the R.A.F.M.S. increases at a uniform rate.

When the movements of the equator and equinox due to precession are taken into account, the fictitious mean sun is defined to travel along the mean equator in such a way that its mean right ascension is always equal to the sun's mean longitude. This right ascension is, therefore, independent of the earth's rotation, and the fictitious mean sun is a suitable reference point for the definition of Ephemeris Time. The hour angle of the fictitious mean sun will, however, be affected by irregularities in the earth's rotation, since it depends on the instantaneous sidereal direction of the observer's meridian. So an alternative meridian is defined, called the Ephemeris Meridian, which corresponds with the sidereal direction that the Greenwich meridian would have if the earth were rotating strictly uniformly. The hour angle measured from the ephemeris meridian, instead of the observer's meridian, is called the Ephemeris Hour Angle (E.H.A.). Ephemeris Time is defined in terms of the Ephemeris Hour Angle of the fictitious mean sun by

$$\text{E.T.} = 12^{\text{h}} + \text{E.H.A.F.M.S.} \quad \ldots\ldots(4).$$

A slightly different reference point, which is simply called the mean sun, is required to define Universal Time. The mean sun also moves round the mean equator but at a rate that is directly proportional at each instant to the earth's angular velocity. At a certain epoch in the past the mean sun is defined to be coincident with the fictitious mean sun, and the ephemeris meridian is defined to be coincident with the Greenwich meridian. At some later time, if the earth's angular velocity is decreasing, the ephemeris meridian will be displaced slightly east of the Greenwich meridian, and the right ascension of the mean sun will be less than that of the fictitious mean sun by a proportional, but much smaller, amount. A precise formal definition of the mean sun is given in Appendix E, and Universal Time is defined in terms of the Greenwich H.A.M.S., as in section 28, by the equation

$$\text{U.T.} = 12^{\text{h}} + \text{Greenwich H.A.M.S.} \qquad \ldots\ldots(5).$$

The difference between ephemeris and universal time, about $45^{\text{s}}$ in 1975, is denoted by $\Delta T$, that is

$$\Delta T = \text{E.T.} - \text{U.T.} \qquad \ldots\ldots(6).$$

The value of $\Delta T$ cannot, however, be calculated in advance since the earth's rotational rate is not wholly predictable.

**86.** *The sidereal year and the tropical year.*

The time required by the sun to make a complete circuit of the ecliptic is called a *sidereal year*. Thus, it is the interval between its passage through any fixed point on the ecliptic and its next passage through the same point.

The *tropical year* is the average interval between two consecutive passages of the sun through the vernal equinox (which must now be regarded as in motion owing to precession). In Fig. 55, p. 136, we shall now suppose that $\Upsilon$ is the mean equinox when the sun's R.A. and declination are both zero—the sun is then in the direction of $\Upsilon$. As the sun moves along the ecliptic in the direction $\Upsilon R$, the mean equinox moves slowly in the opposite direction. Let $\Upsilon_1$ denote the position of the mean equinox when it and the sun are again coincident. Then the tropical year is the time taken by the sun to describe 360° less $\Upsilon_1\Upsilon$. From observations it is found that

The tropical year = 365·2422 ephemeris days.

The relation between the tropical and sidereal years is evidently given by

$$\text{Sid. year} : \text{trop. year} = 360° : (360° - 50''\cdot3) \quad ...(7)$$

(the precessional motion of the equinox, i.e. $\Upsilon\Upsilon_1$, is $50''\cdot3$ per *tropical* year). It is then found that

The sidereal year = 365·2564 ephemeris days.

For strict accuracy we use ephemeris days; but, to the precision to which the two years are given above, we may equally well regard them as given in mean solar days of universal time.

**87.** *Relation between universal and mean sidereal time.*

We have defined a tropical year as the average interval between two successive passages of the sun through the moving mean equinox. Ignoring the difference between U.T. and E.T., this interval is the same as that between two successive passages of the mean sun through the mean equinox. So in the course of a tropical year the R.A.M.S. increases from 0° to 360°, that is the increase in the R.A.M.S. is at the rate of $360° \div 365\cdot2422$ or $59'\ 8''\cdot33$ per mean solar day. Let $t_1$ be the mean sidereal time when the hour angle of the mean sun at a given place is $H_1$ and let $R_1$ denote the corresponding value of the R.A.M.S. Then

$$t_1 = H_1 + R_1 \qquad .....(8).$$

Let $t_2$ be the mean sidereal time one mean solar day later. The hour angle of the mean sun has increased by 360° and the R.A.M.S. by $59'\ 8''\cdot33$, or in time measure by $24^h$ and $3^m\ 56^s\cdot556$ respectively. Hence

$$t_2 = (H_1 + 24^h) + (R_1 + 3^m\ 56^s\cdot556) \qquad .....(9),$$

so that, by (8), $\qquad t_2 - t_1 = 24^h\ 3^m\ 56^s\cdot556.$

But $t_2 - t_1$ is the interval of sidereal time corresponding to $24^h$ of universal time. Hence

$$24^h \text{ U.T.} = 24^h\ 3^m\ 56^s\cdot556 \text{ mean sidereal time} \quad ...(10).$$

It is easily calculated from (10) that

$$24^h \text{ mean sidereal time} = (24^h - 3^m\ 55^s\cdot910) \text{ U.T.} \quad ...(11).$$

The relation (10) may also be derived from the following consideration. At a particular instant the mean sun and the mean equinox are coincident and after a tropical year they are together

again. In this interval, the earth has rotated about its axis 365·2422 times with respect to the mean sun and once more with respect to the mean equinox. Hence

365·2422 mean solar days = 366·2422 mean sid. days

from which .....(12),

$$24^h \text{ U.T.} = \left(1 + \frac{1}{365{\cdot}2422}\right) 24^h \text{ mean sidereal time} \quad \text{.....(13).}$$

This last relation is evidently the same as (10).

To facilitate the conversion of any interval of U.T. into its equivalent, expressed in mean sidereal time, and *vice versa*, the following tables are given; the entries are easily derived from (10) and (11).

Table I. *Conversion of mean solar to mean sidereal time*

| | | | | | |
|---|---|---|---|---|---|
| $24^h$ | mean solar time | $\equiv$ | $(24^h$ | $+ 3^m\ 56^s{\cdot}556)$ | mean sid. time |
| $1^h$ | „ „ | $\equiv$ | $(\ 1^h$ | $+\ 9^s{\cdot}8565)$ | „ |
| $1^m$ | „ „ | $\equiv$ | $(\ 1^m$ | $+\ 0^s{\cdot}1643)$ | „ |
| $1^s$ | „ „ | $\equiv$ | $(\ 1^s$ | $+\ 0^s{\cdot}0027)$ | „ |

Table II. *Conversion of mean sidereal to mean solar time*

| | | | | | | |
|---|---|---|---|---|---|---|
| $24^h$ | mean sid. time | $\equiv$ | $(24^h$ | $- 3^m\ 55^s{\cdot}910)$ | mean solar time | |
| $1^h$ | | $\equiv$ | $(\ 1^h$ | $-\ 9^s{\cdot}8296)$ | „ | „ |
| $1^m$ | | $\equiv$ | $(\ 1^m$ | $-\ 0^s{\cdot}1638)$ | „ | „ |
| $1^s$ | | $\equiv$ | $(\ 1^s$ | $-\ 0^s{\cdot}0027)$ | „ | „ |

In the almanacs there are more extensive tables which facilitate the problem of time-conversion.

*Example* 1. To find the Greenwich mean sidereal time on 1931 February 24 at $8^h\ 47^m\ 38^s{\cdot}52$ U.T.

From the almanacs, the Greenwich sidereal time on February 24 at $0^h$ U.T. (midnight) is $10^h\ 11^m\ 37^s{\cdot}67$. The mean solar interval concerned is $8^h\ 47^m\ 38^s{\cdot}52$ and by means of Table I this can be expressed in mean sidereal time (taking the hours, minutes, seconds separately) as follows:

| Mean solar time | | Mean sidereal time | | |
|---|---|---|---|---|
| $8^h$ | $\equiv$ | $8^h + 8 \times 9^s{\cdot}8565$ or $8^h$ | + | $1^m\ 18^s{\cdot}85$ |
| $47^m$ | $\equiv$ | $47^m + 47 \times 0^s{\cdot}1643$ or $47^m$ | + | $7^s{\cdot}72$ |
| $38^s$ | $\equiv$ | $38^s + 38 \times 0^s{\cdot}0027$ or $38^s$ | + | $0^s{\cdot}10$ |
| $0^s{\cdot}52$ | $\equiv$ | $0^s{\cdot}52$ | + | $0^s{\cdot}00$ |

The sum of the quantities on the right is

$$8^h\ 47^m\ 38^s{\cdot}52 + 1^m\ 26^s{\cdot}67 \text{ or } 8^h\ 49^m\ 5^s{\cdot}19.$$

Hence the interval of $8^h\ 47^m\ 38^s{\cdot}52$ mean solar time is equivalent to $8^h\ 49^m\ 5^s{\cdot}19$ mean sidereal time.

But at $0^h$ U.T. the mean sidereal time is $10^h\ 11^m\ 37^s{\cdot}67$. Hence at $8^h\ 47^m\ 38^s{\cdot}52$ U.T. the mean sidereal time is $19^h\ 0^m\ 42^s{\cdot}86$.

*Example* 2. To find the U.T. on 1931 April 5 when the Greenwich mean sidereal time is $18^h\ 31^m\ 52^s{\cdot}38$.

From the almanac, the mean sid. time at $0^h$ U.T. on April 5 is $12^h\ 49^m\ 19^s{\cdot}83$, which subtracted from $18^h\ 31^m\ 52^s{\cdot}38$ gives the sidereal interval concerned. This interval is thus $5^h\ 42^m\ 32^s{\cdot}55$ mean sidereal time. We use Table II as follows:

| Mean sidereal time | | Mean solar time |
|---|---|---|
| $5^h$ | $\equiv$ | $5^h - 5 \times 9^s{\cdot}8296 = 5^h - 49^s{\cdot}15$ |
| $42^m$ | $\equiv$ | $42^m - 42 \times 0^s{\cdot}1638 = 42^m - 6^s{\cdot}88$ |
| $32^s$ | $\equiv$ | $32^s - 32 \times 0^s{\cdot}0027 = 32^s - 0^s{\cdot}09$ |
| $0^s{\cdot}55$ | | $= 0^s{\cdot}55 - 0^s{\cdot}00$ |

The sum of the quantities on the right is

$$5^h\ 42^m\ 32^s{\cdot}55 - 56^s{\cdot}12 \quad \text{or} \quad 5^h\ 41^m\ 36^s{\cdot}43.$$

Hence when the Greenwich mean sidereal time is as stated in the problem, the U.T. is $5^h\ 41^m\ 36^s{\cdot}43$.

The computations can be considerably curtailed by means of the special tables in the almanacs already mentioned.

**88.** *The calendar.*

We have already mentioned that the tropical year is the unit on which civil reckoning is based. For obvious reasons, the *civil year* contains an integral number of mean solar days and, as we have seen, the tropical year is equal to 365·2422 mean solar days.

In the Julian calendar, introduced by Julius Caesar, the tropical year was taken to be $365\frac{1}{4}$ days; three years out of every four were each given 365 days while the fourth—called a *leap-year*—was given 366 days. The leap-year was chosen to be that one which was divisible by 4 and the extra day was added to February. Thus, according to the Julian rule, the years 1928, 1932, 1936, etc., are all leap-years in each February of which there are 29 days. If the assumed length of the tropical year had been accurate, then there would have been an exact accordance between cycles of four tropical years and cycles of four civil years.

In 1582, Pope Gregory introduced the calendar now in use; it was not, however, till 1752 that it was adopted in England. Now this Gregorian calendar is designed to introduce the necessary correction, based on the accurate value of the tropical year, to the Julian system. According to the Julian system, the years 1700, 1800, 1900 and 2000 would all be leap-years; in the Gregorian calendar only 2000 is defined to be a leap-year, the rule being that, when a year ends in two "noughts", it is not a leap-year unless it is divisible by 400. In a cycle of 400 years, there are 100 leap-years according to the Julian calendar and 3 less, or 97, in the Gregorian calendar. Hence, according to the Gregorian calendar,

$$400 \text{ civil years} = (400 \times 365 + 97) \text{ mean solar days},$$

from which the average civil year is 365·2425 mean solar days. This is so near the value of the tropical year that no appreciable discrepancy can arise for many centuries.

There is one small matter relating to the civil year that we can conveniently consider here. The first day in the year is, according to civil usage, denoted January 1 and an event that happens, say, at 6 a.m. at Greenwich on January 1 is said to occur on January 1, $6^h\ 0^m$ U.T. or on Jan. $1^d{\cdot}25$. The event thus occurs 1·25 days after an epoch, which is denoted astronomically by Jan. $0^d{\cdot}0$ which, in effect, is the midnight that ushers in the day December 31. Thus the instant given by 1931 December 31, $18^h\ 0^m$ U.T. can be written as 1932 Jan. $0^d{\cdot}75$. A similar procedure is applicable to the other months.

**89.** *The Besselian year.*

We have defined the length of the tropical year but not the instant, according to civil reckoning, at which it is assumed to begin. It is the general astronomical practice to define the beginning of the tropical year (or solar year, as it is sometimes called) as the instant when the R.A. of the fictitious mean sun —or the sun's mean longitude—is exactly $18^h\ 40^m$ or 280°. This instant falls near the beginning of the civil year. The year defined in this way is usually called the Besselian year, after the German astronomer Bessel who first introduced it into astronomical practice. Thus the beginning of the Besselian year in 1975 is 1975 January $0^d{\cdot}978$ Ephemeris Time, that is 1974

December 31, $23^h\ 28^m$ E.T. To the accuracy quoted here the difference between ephemeris and universal time is almost negligible, since in 1975 $\Delta T = 45^s{\cdot}6$, which is less than one thousandth of a day. The beginning of the next Besselian year is clearly obtained by adding 365·2422 days. Hence in 1976, the beginning of the Besselian year is 1976 Jan. $1^d{\cdot}221$ E.T., the slight apparent discrepancy is due to rounding.

*It is the general practice to denote the beginning of any Besselian year by the notation* 1975·0, 1976·0, 1977·0, etc.

The year defined in this way is used in calculations and observations relating to the heavenly bodies. For example, if the R.A. and declination of a star are observed at a particular instant, the co-ordinates refer to the actual or true equinox and equator at that instant. By means of certain principles which we shall elaborate in Chapter X, the star's position can be deduced with respect to the mean equinox and equator at the beginning of the Besselian year. Thus for example, the positions of stars observed at different dates during 1975 can all be referred to the mean equinox and equator for 1975·0 and by a further process to some standard equinox such as that for 1900·0 or 1950·0.

**90.** *The Julian Date.*

In certain observations (such as observations of variable stars) it is found convenient to express the instant of observation as so many days and fraction of a day after a definite fundamental epoch. The epoch chosen is Greenwich mean noon of January 1, 4713 B.C., and for any given date the number of days which have elapsed since this epoch defines the *Julian Date* (J.D.) of the date in question. For example, for 1975 January 1, $12^h$ U.T., the Julian Date is denoted by J.D. 2442414; the time of an observation made say on 1975 Jan. 3, $18^h$ U.T. (i.e. 6 hours after Greenwich mean noon on Jan. 3) is denoted by J.D. 2442416·25. It is essential to note that the Julian Day begins at Greenwich mean noon. In the almanacs the Julian Date is given for every day in the year, and there are also tables which enable the astronomer to find the Julian Date for any day in any year.

**91.** *The equation of time.*

As in Chapter II (p. 43) the equation of time ($E$) is defined by

$$E = \text{H.A.}\ \odot - \text{H.A.M.S.} \qquad \ldots\ldots(14),$$

in which H.A. ⊙ denotes the hour angle of the sun and H.A.M.S. the hour angle of the mean sun at a given time and place. We have also

Sid. time = H.A. ⊙ + R.A. ⊙ = H.A.M.S. + R.A.M.S. .....(15),

and $E$ = R.A.M.S. − R.A. ⊙ .....(16).

The R.A.M.S. is known in advance for any universal time, while R.A. ⊙ is similarly known for any emphemeris time, but the exact relationship between the two times cannot be predicted. Hence the equation of time cannot be computed in advance and so it is no longer given in the *Astronomical Ephemeris*, though it is still used in the *Nautical Almanac*. If, however, we replace the mean sun with the fictitious mean sun and write instead

$E'$ = R.A.F.M.S. − R.A. ⊙ .....(17)

then $E'$, which may be called the equation of ephemeris time, differs from $E$ by only $0{\cdot}0027\Delta T$ and can be computed in advance.

In Fig. 57 the celestial sphere is drawn with the earth as centre. Relative to the earth, the sun will appear to describe an elliptic orbit lying in the plane of the ecliptic. The principles and formulae of Chapter v apply to the sun's apparent orbit around the earth.

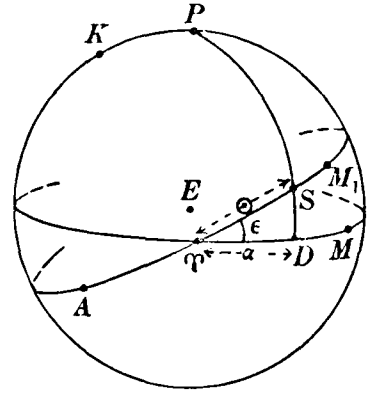


Fig. 57.

Let ⊙ denote the longitude of the sun at a given instant, let $\alpha$ be the sun's right ascension and let $l$ be the sun's mean longitude. Then, by (3),

$$E' = (\odot - \alpha) - (\odot - l) \quad \text{.....(18).}$$

In Fig. 57 let $A$ denote perigee. Then the arc $AS$ is the sun's true anomaly $v$. Let $M_1$ be the position of a fictitious body coinciding with the true sun at perigee and moving in the ecliptic with the sun's mean angular motion $n$; then the arc $AM_1$ is the mean anomaly $M$ and the arc $\Upsilon M_1$ is $l$. Thus,

$$\odot - l = \Upsilon S - \Upsilon M_1$$
$$= AS - AM_1,$$

that is $\odot - l = v - M$ .....(19).

Hence $E' = -(\alpha - \odot) - (v - M)$ .....(20).

The quantity $E'$ will thus consist of two parts: (1) the quantity $(\alpha - \odot)$ called the *reduction to the equator*, and (2) the quantity $(v - M)$ which is the *equation of the centre* discussed in section 73.

We first express the quantity $(\odot - \alpha)$ in the form of a series. From the right-angled triangle $\Upsilon SD$ ($PSD$ is the meridian through $S$) in which

$$\Upsilon S = \odot, \quad \Upsilon D = \alpha, \quad S\hat{\Upsilon}D = \epsilon \text{ and } S\hat{D}\Upsilon = 90°,$$

we have by the four-parts formula **D**

$$\cos\alpha \cos\epsilon = \sin\alpha \cot\odot,$$

or

$$\tan\alpha = \cos\epsilon \tan\odot,$$

which can be written

$$\tan\alpha = \frac{1 - \tan^2\frac{\epsilon}{2}}{1 + \tan^2\frac{\epsilon}{2}} \tan\odot \qquad \ldots\ldots(21).$$

Since $\epsilon$ is about $23\frac{1}{2}°$, $\tan^2\frac{\epsilon}{2}$ is approximately $1/25$. Write $y$ for $\tan^2\frac{\epsilon}{2}$; then

$$\tan\alpha = \frac{1-y}{1+y}\tan\odot \qquad \ldots\ldots(22),$$

in which we regard $y$ as a small quantity.

(22) is essentially of the same form as formula (82) of Chapter v, p. 118, which is given in the form of a series by (83), p. 119. Hence writing $y$ for $-x$ in (83) and $\alpha$ for $\frac{v}{2}$ and $\odot$ for $\frac{E}{2}$, we have

$$\odot - \alpha = y\sin 2\odot - \tfrac{1}{2}y^2\sin 4\odot + \tfrac{1}{3}y^3\sin 6\odot \ldots\ldots(23).$$

In this formula, $\odot$ and $\alpha$ are expressed in circular measure. Expressing the difference $(\odot - \alpha)$ in seconds of time we obtain

$$\odot - \alpha = \text{cosec } 1^s\,[y\sin 2\odot - \tfrac{1}{2}y^2\sin 4\odot \ldots].$$

Now cosec $1^s = 206265/15$ and $y\left(\equiv \tan^2\frac{\epsilon}{2}\right) = 0{\cdot}0430468$, using the value $23° 26' 33''$ for the obliquity $\epsilon$ in 1975. We then obtain

$$\odot - \alpha = 591^s{\cdot}94\sin 2\odot - 12^s{\cdot}74\sin 4\odot + 0^s{\cdot}37\sin 6\odot \qquad \ldots\ldots(24).$$

Formula (24) gives the part of the equation of ephemeris time, depending on the obliquity of the ecliptic, expressed as a series involving the sun's true longitude $\odot$.

The other part of the equation of ephemeris time depends on the eccentricity and is the equation of the centre. Quoting formula (87) of Chapter v we write, keeping only terms up to $e^2$,

$$v - M \equiv \odot - l = 2e \sin M + \tfrac{5}{4}e^2 \sin 2M \quad \ldots\ldots(25).$$

Expressing $(v - M)$ in seconds of time and using the value of $e$ for 1975·0, namely 0·016720, we find

$$v - M = 459^s{\cdot}83 \sin M + 4^s{\cdot}81 \sin 2M \quad \ldots\ldots(26).$$

Now the right-hand side of (23) is expressed in terms of $\odot$, the sun's longitude, which by (25) is expressed in terms of the mean anomaly $M$ by

$$\odot = l + 2e \sin M + \tfrac{5}{4}e^2 \sin 2M \quad \ldots\ldots(27).$$

We substitute the value of $\odot$ given by (27) on the right-hand side of (23). As we have pointed out, $y$ is about 1/25 and $e$ is about 1/60; regarding $y$ and $e$ as small quantities of the same order of smallness and keeping only terms up to the second order in the value of $\odot - \alpha$, we can write with the accuracy indicated

$$\begin{aligned} \sin 2\odot &= \sin (2l + 4e \sin M) \\ &= \sin 2l + 4e \sin M \cos 2l \end{aligned}$$

(we require the development of $\sin 2\odot$ up to the first order only, since in (23) $\sin 2\odot$ is multiplied by $y$). Similarly we have, with the limitations imposed,

$$\sin 4\odot = \sin 4l.$$

Hence (23) becomes—up to the second order—

$$\odot - \alpha = y \sin 2l + 4ey \sin M \cos 2l - \tfrac{1}{2}y^2 \sin 4l \quad \ldots(28).$$

Combining (28) with (25) we have

$$\begin{aligned} E' = y \sin 2l - 2e \sin M + 4ey \sin M \cos 2l \\ - \tfrac{1}{2}y^2 \sin 4l - \tfrac{5}{4}e^2 \sin 2M \quad \ldots(29). \end{aligned}$$

Inserting the numerical values of $y$ and $e$, we obtain

$$\begin{aligned} E' = 591^s{\cdot}9 \sin 2l - 459^s{\cdot}8 \sin M + 39^s{\cdot}6 \sin M \cos 2l \\ - 12^s{\cdot}7 \sin 4l - 4^s{\cdot}8 \sin 2M \quad \ldots(30). \end{aligned}$$

Now in Fig. 57, the mean anomaly $M$ is the arc $AM_1$ which is equal to $A\Upsilon + \Upsilon M_1$, so that

$$M = A\Upsilon + l.$$

Also, if $\varpi$ denotes the longitude of perigee, we have $\varpi = 360° - A\Upsilon$. Hence

$$M = 360° + (l - \varpi) \quad \ldots\ldots(31).$$

In the formula (30) we can thus write $(l - \varpi)$ for $M$ on the right-hand side. For 1975·0 the value of $\varpi$ is 282°·5099. After some reduction, we then obtain

$$E' = -103^s{\cdot}9 \sin l - 429^s{\cdot}6 \cos l + 596^s{\cdot}3 \sin 2l - 2^s{\cdot}0 \cos 2l + 4^s{\cdot}3 \sin 3l + 19^s{\cdot}3 \cos 3l - 12^s{\cdot}7 \cos 4l \quad \ldots(32).$$

Up to the order of approximation adopted, this formula gives the equation of time in terms of the sun's mean longitude.

The quantity tabulated in the *Astronomical Ephemeris* is not $E'$ but the sun's ephemeris transit. This is the ephemeris time at the instant of the sun's transit across the ephemeris meridian, and this time is easily seen to be $12^h$ minus the equation of ephemeris time at this instant.

92. *The seasons.*

Consider the apparent path of the sun during the year (Fig. 58). The earth $E$ is the centre of the celestial sphere and relative to $E$ the sun appears to describe the great circle $C\Upsilon AB$ on the sphere —the ecliptic—which intersects the celestial equator in the two points $\Upsilon$ and $B$, the vernal and autumnal equinoxes (sometimes called the First Point of Aries and the First Point of Libra respectively). $P$ is the north pole of the equator. The sun's declination increases from 0° at $\Upsilon$ (about March 21) to a maximum of 23° 27′ at $A$ (about June 21). The point $A$ is called the *summer solstice*. Similarly, the point $C$ where the sun (about December 21) has its greatest southerly declination (− 23° 27′) is called the *winter solstice*.

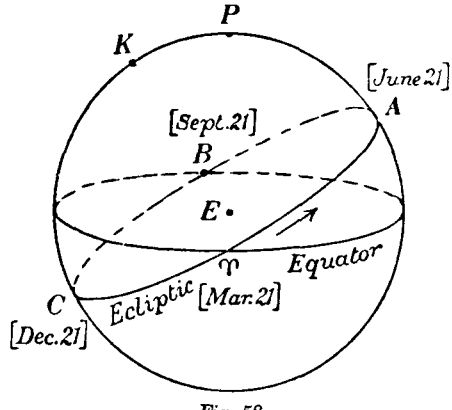


Fig. 58.

Astronomically, the four parts of the year, or seasons, during which the sun is successively in the quadrants $\Upsilon A$, $AB$, $BC$ and $C\Upsilon$ are called, *spring*, *summer*, *autumn* and *winter* respectively. As these terms are used rather loosely in ordinary speech, we shall consider them only in their astronomical significance.

The general characteristics of the seasons at any place depend on the relative amounts of heat received from the sun from day to day. The two astronomical factors are (*a*) the interval during which the sun is above the horizon on any day, and (*b*) the general sequence of altitudes which the sun attains during this interval. It is convenient to use the sun's altitude or zenith distance at apparent noon as a criterion of (*b*).

Refraction being neglected, the number of hours during which the sun (strictly, the sun's centre) is above the horizon is given by $2H$, where $H$ is the angle between $0^h$ and $12^h$ given by

$$\cos H = -\tan\phi \tan\delta \qquad \text{.....(33),}$$

where $\phi$ is the latitude of the place; this is formula (24), p. 47.

If the declination $\delta$ is positive, then in northern latitudes $H$ lies between $6^h$ and $12^h$ and therefore the sun is above the horizon for more than 12 hours out of the 24. This occurs between March 21 (approximately), when the sun is at $T$, and September 21 (approximately), when the sun is at $B$. Thus as far as (*a*) is concerned, the days increase in warmth from March 21 to June 21 corresponding to the increase in $\delta$ from $0°$ to $+ 23° 27'$; from the latter date, a decrease ensues. If $\delta$ is negative, then in northern latitudes the hour angle of setting is by (33) less than $6^h$, and consequently the sun is above the horizon for less than $12^h$; this refers to the two seasons autumn and winter. The minimum number of hours of daylight occurs when the sun attains its greatest southerly declination, that is at the winter solstice (December 21). It is easily seen that the number of hours of daylight increases from December 21 to June 21 and decreases from June 21 to December 21.

The sun's zenith distance $z$ at apparent noon on any day is given by

$$z = \phi - \delta \qquad \text{.....(34).}$$

It is clear that at any place whose north latitude lies between $0°$ and $23° 27'$, the sun is in the zenith (or practically so) twice during spring and summer, that is, between March 21 and September 21. For latitude $23° 27'$ N, the sun is in the zenith only on June 21. This northern parallel of latitude ($23° 27'$ N) is called the *Tropic of Cancer*. The corresponding south parallel ($23° 27'$ S) is the *Tropic of Capricorn*, and for places between this parallel and the equator, the sun is in the zenith twice during

the interval between September 21 and March 21. The zone of the earth's surface bounded by the Tropics of Cancer and Capricorn is the *Torrid Zone*. The parallels of latitude 66° 33′ N and 66° 33′ S are called the *Arctic Circle* and *Antarctic Circle* respectively. The zone between the parallels of 23° 27′ N and 66° 33′ N is called the *North Temperate Zone*, and the corresponding zone in the southern hemisphere is the *South Temperate Zone*.

From (33), it is seen that when $\phi = 66° 33' \text{ N}$ and $\delta = + 23° 27'$, the hour angle of setting is $12^h$; that is to say, at the Arctic Circle on June 21 the sun is above the horizon during the whole 24 hours. In latitude 70°, for example, the sun is above the horizon continuously—by (33)—during the period for which the sun's declination is between + 20° and + 23° 27′, that is to say (referring to the almanac for 1931), between May 22 and July 24. At the north pole the sun is above the horizon continuously between March 21 and September 21 and below the horizon for the remainder of the year.

Summarising for the northern hemisphere, we have these characteristics of the seasons. During spring, the number of hours of daylight increases from 12 hours at March 21 until June 21 when the maximum is reached; this maximum can be 24 hours on particular days for places on or north of the Arctic Circle; during spring, the sun's altitude at noon increases, for places north of the Tropic of Cancer, and reaches a maximum on June 21; for places between the equator and the parallel of 23° 27′ N, it reaches a maximum of 90° on some date between March 21 and June 21.

During summer, the number of hours of daylight decreases from the maximum on June 21 to 12 hours on September 21.

During autumn, the number of hours of daylight decreases from 12 hours on September 21 to a minimum on December 21, and during this period the sun's altitude at noon progressively decreases.

During winter, the number of hours of daylight increases from the minimum on December 21 to 12 hours on March 21, and during this season the sun's altitude at noon progressively increases.

The relation between the seasons and the sun's declination in

the southern hemisphere can readily be inferred from the preceding discussion.

The respective lengths (in mean solar days) of the seasons can be obtained by considering the equation of the centre which, for our present purpose, we write simply in the form—from (25)—

$$l = \odot - 2e \sin M$$

or, using (31),

$$l = \odot - 2e \sin (l - \varpi) \qquad \text{.....(35).}$$

As $l$ is approximately equal to $\odot$, we can replace $l$ on the right of (35) by $\odot$ and then

$$l = \odot - 2e \sin (\odot - \varpi) \qquad \text{.....(36).}$$

Let $l_1, l_2, l_3$ and $l_4$ denote the values of the sun's mean longitude at the beginning of spring, summer, autumn and winter respectively. At the beginning of spring $\odot = 0$, therefore

$$l_1 = 2e \sin \varpi \qquad \text{.....(37).}$$

At the beginning of summer, $\odot = 90°$ or $\frac{\pi}{2}$ in radian measure; hence

$$l_2 = \frac{\pi}{2} - 2e \cos \varpi \qquad \text{.....(38).}$$

Similarly

$$l_3 = \pi - 2e \sin \varpi \qquad \text{.....(39),}$$

$$l_4 = \frac{3\pi}{2} + 2e \cos \varpi \qquad \text{.....(40).}$$

Let $t_1, t_2, t_3$ and $t_4$ denote the instants when the mean longitude of the sun has the values $l_1, l_2, l_3$ and $l_4$. Then

$$l_2 - l_1 = n (t_2 - t_1) \qquad \text{.....(41),}$$

where $n$ is the mean angular motion. If $t_2 - t_1$ is expressed in mean solar days, $n = 2\pi/T$, where $T = 365{\cdot}2422$ mean solar days. Hence by (41)

$$t_2 - t_1 = \frac{T}{2\pi} (l_2 - l_1) \quad \text{mean solar days} \qquad \text{.....(42).}$$

But $(t_2 - t_1)$ is the number of mean solar days in spring, and if we denote this interval by $I_1$ we have from (37), (38) and (42),

$$I_1 = \frac{T}{2\pi} \left\{ \frac{\pi}{2} - 2e (\sin \varpi + \cos \varpi) \right\},$$

or

$$I_1 = 91{\cdot}31 - \frac{eT}{\pi} (\sin \varpi + \cos \varpi) \qquad \text{.....(43).}$$

Similarly if $I_2$, $I_3$ and $I_4$ denote the number of mean solar days in summer, autumn and winter respectively, we obtain

$$I_2 = 91{\cdot}31 - \frac{eT}{\pi}(\sin\varpi - \cos\varpi) \qquad \ldots\ldots(44),$$

$$I_3 = 91{\cdot}31 + \frac{eT}{\pi}(\sin\varpi + \cos\varpi) \qquad \ldots\ldots(45),$$

$$I_4 = 91{\cdot}31 + \frac{eT}{\pi}(\sin\varpi - \cos\varpi) \qquad \ldots\ldots(46).$$

Putting $T = 365{\cdot}2422$, $e = 0{\cdot}01672$, $\varpi = 282°{\cdot}510$, we find that, for the northern hemisphere,

| | | | | | |
|---|---|---|---|---|---|
| Spring | contains | 92 | days | 19 | hours |
| Summer | ,, | 93 | ,, | 15 | ,, |
| Autumn | ,, | 89 | ,, | 20 | ,, |
| Winter | ,, | 89 | ,, | 0 | ,, |

**93.** *Time of the sun's transit over any meridian.*

We shall conclude this chapter by considering two problems involving time. An example of the first problem is the following: to calculate, to the nearest second, the U.T. and standard time of the transit of the sun's centre on 1975 January 4, at the Dominion Observatory, Victoria, B.C. where the longitude is $8^h\ 13^m\ 40^s{\cdot}2$ W.

From the 1975 almanac, we have the following:

On Jan. 4, E.T. of the sun's emphemeris transit is $12^h\ 04^m\ 45^s{\cdot}5$.
On Jan. 5, ,, ,, ,, ,, $12^h\ 05^m\ 12^s{\cdot}6$.

These two times are not exactly the times for which the Greenwich hour angle of the sun is zero, since they are times of transit over the ephemeris, not the Greenwich, meridian. It can be shown (see exercise 23 at the end of this chapter) that the ephemeris meridian is displaced east of the Greenwich meridian by an amount $1{\cdot}0027\ \Delta T$, and so it follows that

$$\text{E.H.A. } \odot = \text{G.H.A. } \odot + 1{\cdot}0027\Delta T \qquad \ldots\ldots(47).$$

If now we neglect any change in the equation of time during the short interval between the two transits, we readily deduce

U.T. (Greenwich transit) = E.T. (eph. transit) + $0{\cdot}0027\ \Delta T$.

The small correction $0{\cdot}0027\ \Delta T$, which for 1975·0 is $0^s{\cdot}12$, is

normally neglected, and the entries in the almanac are then treated as if they were simply the universal times of transit at Greenwich. When the sun's centre is on the meridian of Victoria, the sun's Greenwich hour angle is $8^h\ 13^m\ 40^s{\cdot}2$, or approximately $8^h{\cdot}23$. From the two times quoted above we see that it requires $27^s{\cdot}1$ more than 24 hours for the sun's hour angle to increase from $0^h$ to $24^h$. The additional time relevant to Victoria is $(8{\cdot}23/24) \times 27^s{\cdot}1$ or $9^s{\cdot}3$, and so we find the U.T. of solar transit at Victoria is $T$, where

$$T = 12^h\ 04^m\ 45^s{\cdot}5 + 8^h\ 13^m\ 40^s{\cdot}2 + 9^s{\cdot}3.$$

Thus the U.T. of transit at Victoria is $20^h\ 18^m\ 35^s$.

The standard time kept at Victoria is "Pacific time" corresponding to the meridian of $8^h$ west. Hence the standard time of the sun's transit at Victoria on January 4 is $12^h\ 18^m\ 35^s$ (Pacific time).

**94.** *Time of the moon's transit over any meridian.*

This is the second problem and, as an example, we shall find the U.T. and standard time of the moon's transit at Victoria on the night of May 24/25 1975.

From the *Astronomical Ephemeris* information is obtained about both the upper and the lower ephemeris transit of the moon. As in the previous section we may regard these entries as the universal times of transit over the Greenwich meridian. Thus it is found that on May 24 the moon's transit occurs at $23^h{\cdot}7174$, and that its lower culmination occurs $12^h{\cdot}4717$ later. During this interval the Greenwich hour angle of the moon will increase from $0^h$ to $12^h$. Clearly transit at Victoria will occur at an interval of $(12{\cdot}4717/12) \times 8^h\ 13^m\ 40^s{\cdot}2$ after the transit at Greenwich. This interval can be written as

$$\left(1 + \frac{0{\cdot}4717}{12}\right) \times 8^h{\cdot}2278 \quad \text{or} \quad 8^h{\cdot}5512.$$

Hence when the moon transits at Victoria, the corresponding U.T. is $(23^h{\cdot}7174 + 8^h{\cdot}5512)$ May 24, or May 25 $8^h{\cdot}2686$ U.T.

A more accurate result can now be obtained as follows. Taking $\Delta T$ as $45^s{\cdot}6$ (or $0^h{\cdot}0127$), the E.T. of this transit at Victoria is found to be May 25 $8^h{\cdot}2813$. In the *Astronomical Ephemeris* the moon's apparent right ascension is tabulated at

hourly intervals of ephemeris time. For the computed time of transit it is found to be $16^h\ 11^m\ 41^s$. Since the Greenwich hour angle is then $8^h\ 13^m\ 40^s$, the apparent sidereal time at Greenwich is

$$\text{G.S.T.} = 16^h\ 11^m\ 41^s + 8^h\ 13^m\ 40^s \quad \text{or} \quad 0^h\ 25^m\ 21^s.$$

This Greenwich sidereal time is readily converted to U.T. by using the Universal and Sidereal Times table in the almanac. We then find the U.T. of transit at Victoria to be $8^h\ 16^m\ 09^s$. The standard time of transit is May 25 $0^h\ 16^m\ 09^s$ (Pacific time).

## EXERCISES

[Symbols used:

$\phi$ = latitude of observer, $\epsilon$ = obliquity of the ecliptic.]

$e$ = eccentricity of earth's orbit,

**1.** For a place within the tropics, prove that the hour angle $H$ of the sun $(\alpha, \delta)$ when the ecliptic is vertical is given by

$$H = \sin^{-1}(\sin\alpha \cot\delta \tan\phi) - \alpha.$$

**2.** A star $(\alpha, \delta)$ rises at the same time as the sun at a place in north latitude $\phi$ when the sun's right ascension is $\alpha_1$. Prove that

$$\alpha_1 - \sin^{-1}(\sin\alpha_1 \tan\phi \tan\epsilon) = \alpha - \sin^{-1}(\tan\delta \tan\phi).$$

**3.** Prove that at a place on the Arctic Circle the daily displacement of the point of sunset is equal to the sun's change in longitude during the same interval. [*M.T.*]

**4.** Neglecting the eccentricity of the earth's orbit, prove that at a place within the Arctic Circle the sun will be above the horizon for

$$\frac{365\frac{1}{4}}{\pi} \cos^{-1}\left(\frac{\cos\phi}{\sin\epsilon}\right) \text{ days.}$$

**5.** If $S$ is the sun's semi-diameter in minutes of arc, show that at a solstice the time taken by the sun's disc to cross the prime vertical at a place in latitude $\phi$ ( $> \epsilon$) is, in minutes,

$$\frac{2S}{15(\sin^2\phi - \sin^2\epsilon)^{\frac{1}{2}}}.$$

**6.** Observations of equal altitude of $\zeta$ Persei are taken with a theodolite on 1931 January 3, at U.T. $18^h\ 45^m\ 8^s$ and $22^h\ 44^m\ 19^s$. Calculate the longitude of the observer. [*Lond.* 1925.]

**7.** The longitude of Columbia University, New York, is $4^h\ 55^m\ 50^s$ west of Greenwich. The sidereal time at mean noon at Greenwich on a certain day is $17^h\ 23^m\ 8^s$. Show that on the same day when the sidereal time at Columbia University is $20^h\ 8^m\ 4^s$ the hour angle of the mean sun at the same place is $2^h\ 43^m\ 41^s$.

**8.** If $\delta$ is the sun's declination, $S$ its semi-diameter in minutes of arc, $I$ the interval in minutes between the disappearance of its lower and of its upper limb at sunset for a place in latitude $\phi$, prove that $\phi$ is given approximately by

$$\sin^2\phi = \cos^2\delta - \frac{4}{225}\frac{S^2}{I^2}.$$ [*Lond.* 1930.]

**9.** Two stars, of equal R.A. and of declinations $\delta$ and $\delta'$, are observed when their altitudes are the same; the sidereal interval $h$ between the observations is noted. Prove that $H$, the hour angle of the first star at the moment of observation, is given by

$$\cos(\psi + H) = \frac{2\sin\psi\tan\phi}{\cos\delta'\sin h}\cdot\sin\tfrac{1}{2}(\delta - \delta')\cos\tfrac{1}{2}(\delta + \delta'),$$

where

$$\cot\psi = \frac{\cos\delta'\cos h - \cos\delta}{\cos\delta'\sin h}.$$

**10.** The right ascension and declination of the sun near the summer solstice are $(90° - 2\beta)$ and $\delta$ respectively. Prove that $(\epsilon - \delta)$ expressed in seconds of arc is given by

$$\epsilon - \delta = \operatorname{cosec} 1''\,[\tan^2\beta\sin 2\epsilon - \tfrac{1}{2}\tan^4\beta\sin 4\epsilon + \ldots].$$

**11.** Let $T$ and $T'$ be the times shown by a sidereal clock when a star is at the same altitude, first on one side of the meridian and then on the other, and let $\alpha$ be the right ascension of the star. Show that $\alpha - \frac{1}{2}(T + T')$ is the correction to be applied to clock time to obtain true sidereal time.

**12.** If the sidereal clock times when the sun arrives at equal altitudes on each side of the meridian are $T$ and $T'$, and if the change of declination $\delta$ of the sun in the interval is $d\delta$, and the right ascension of the sun at culmination is $\alpha$, show that the correction to be applied to clock time to obtain the true sidereal time is

$$\alpha - \tfrac{1}{2}(T + T') - \tfrac{1}{2}\left(\frac{\tan\delta}{\tan\frac{1}{2}(T' - T)} - \frac{\tan\phi}{\sin\frac{1}{2}(T' - T)}\right)d\delta.$$

**13.** From the side of a church running due east a tall buttress projects southwards forming a rectangular corner in which the afternoon sun casts a triangle of shadow on the ground. Show that in winter the total amount of shade during the day (ground-area integrated with respect to time) is proportional to

$$\frac{1}{\sin\phi}\log\frac{\sin(-\delta)}{\cos\phi\sin(\phi - \delta)},$$

where $\delta$ is the declination of the sun. [*M.T.* 1925.]

**14.** If the sun's right ascension increases by $\Delta\alpha$ while its longitude increases by a small amount $\Delta L$, show that

$$\Delta\alpha = \Delta L\cos\epsilon\sec^2\delta,$$

where $\epsilon$ is the obliquity of the ecliptic and $\delta$ is the sun's declination.

Hence find the approximate dates when that part of the equation of time which is known as the reduction to the equator is a maximum.

[*Lond.* 1929.]

**15.** If $\odot$ is the longitude of the sun and $\alpha$ its right ascension, show that the greatest value of $(\alpha - \odot)$ occurs when

$$\tan \odot = (\sec \epsilon)^{\frac{1}{2}} \quad \text{and} \quad \tan \alpha = (\cos \epsilon)^{\frac{1}{2}}.$$

**16.** Prove that, if the eccentricity of the earth's orbit were zero, the equation of ephemeris time in minutes would be

$$\frac{720}{\pi} \tan^{-1} \frac{(1 - \cos \epsilon) \tan \odot}{1 + \cos \epsilon \tan^2 \odot},$$

where $\epsilon$ is the obliquity of the ecliptic and $\odot$ the sun's longitude.

[*M.T.* 1912.]

**17.** Show that the part of the equation of time due to the orbital eccentricity is stationary in value when the sun's true anomaly is

$$\cos^{-1}\left[\frac{(1 - e^2)^{\frac{3}{4}} - 1}{e}\right],$$

and that the part due to the obliquity is stationary in value when the sun's longitude is

$$\tan^{-1} \{(\sec \epsilon)^{\frac{1}{2}}\}.$$

**18.** Show that when the equation of time is a maximum or minimum the sun's longitude $\odot$ is given by

$$(1 - e^2)^{\frac{3}{2}} (\cos^2 \odot + \cos^2 \epsilon \sin^2 \odot) = \cos \epsilon \{1 + e \cos (\odot - \varpi)\}^2,$$

where $\varpi$ is the longitude of perigee.

**19.** Show that in latitude 45° the difference between the times from sunrise to apparent noon and from apparent noon to sunset is

$$\frac{D}{365\frac{1}{4}} \tan \delta \sec \delta (\sec 2\delta)^{\frac{1}{2}} \cot (360° \, T/365\tfrac{1}{4}),$$

where $D$ is the length of the day, $\delta$ the sun's declination, and $T$ the number of days since the vernal equinox, the earth's orbit being supposed circular.

[*Ball.*]

**20.** Assuming the sun's semi-diameter at mean distance to be 961″, show that the time $t$, expressed in sidereal seconds, required for the semi-diameter to cross the meridian is expressed by the relation

$$\frac{961}{r \cos \delta} = 15 \left(1 - \frac{\cos \epsilon}{r^2 (\tau + 1) \cos^2 \delta}\right) t,$$

where $\tau$ is the length of the year in mean solar days and $r$ the sun's distance in astronomical units. [*Coll. Exam.*]

**21.** Prove that, for a ship steaming $V$ nautical miles per hour on a course S $\theta$° W, the interval $I$ between the passage of the sun over the meridian of the ship and the moment when the sun's altitude is a maximum, is given by

$$I = 15^{s} \cdot 3 \, (V \cos \theta + \Delta\delta) (\tan \phi - \tan \delta) \Big/ \left(1 - \frac{V \sin \theta \sec \phi}{900}\right)^2,$$

where $\phi$ is the ship's latitude at apparent noon and $\Delta\delta$ is the northerly change (in minutes of arc) of the sun's declination per hour at apparent noon.

[*Lond.* 1926.]

**22.** Given that:

| | | |
|---|---|---|
| the tropical year | = 365·2422 | mean solar days |
| the sidereal year | = 365·2564 | „ „ |
| the anomalistic year | = 365·2596 | „ „ |

determine the amount and sign of annual precession and the motion of perihelion.

**23.** Define the ephemeris meridian and show that consistent definitions may be made of the ephemeris hour angle of a star, of the ephemeris longitude of a place and of the ephemeris sidereal time. Show that the tables in the *Astronomical Ephemeris* relating mean solar and mean sidereal time may be used without modification to relate ephemeris time and (mean) ephemeris sidereal time.

Prove that for any celestial object $X$

$$\text{E.H.A.}X. = \text{G.H.A.}X. + 1{\cdot}0027\Delta T,$$

where $\Delta T$ is, as usual, defined as the difference between the ephemeris and the universal time.

**24.** Prove that the right ascension of the fictitious mean sun exceeds that of the mean sun by $0{\cdot}0027\Delta T$.

CHAPTER VII

# PLANETARY PHENOMENA AND HELIOGRAPHIC CO-ORDINATES

**95.** *The geocentric motion of a planet.*

We shall assume for the present that the orbits of the earth and a planet $P$ are circular and in the plane of the ecliptic. In Fig. 59 let $E$ and $P$ denote the positions of the earth and planet at any given moment. The plane of the paper represents the plane of the ecliptic. Let $S\Upsilon$ denote the direction of the vernal equinox from the sun $S$ and $E\Upsilon$ its direction from the earth $E$. The

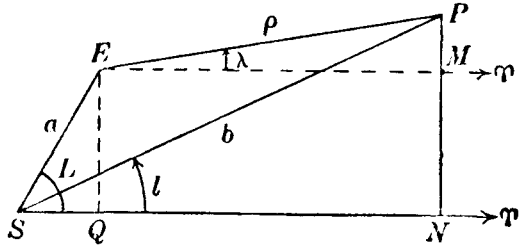



Fig. 59.

angles $EST$ and $PST$ are accordingly the heliocentric longitudes of the earth and planet (denoted by $L$ and $l$ respectively) and $P\hat{E}\Upsilon$ is the geocentric longitude of the planet (denoted by $\lambda$). Let $a$ and $b$ be the radii of the orbits so that $SE = a$ and $SP = b$; also let the planet's geocentric distance $EP$ be denoted by $\rho$. Our object is to express the changes in the geocentric longitude $\lambda$ in terms of the heliocentric longitudes $L$ and $l$. Draw perpendiculars $PMN$, $EQ$ to $S\Upsilon$. Then we have at once

$$\rho \sin\lambda = b \sin l - a \sin L \qquad \ldots\ldots(1),$$

and

$$\rho \cos\lambda = b \cos l - a \cos L \qquad \ldots\ldots(2).$$

In these equations $\rho$, $\lambda$, $l$ and $L$ vary with the time. Differentiating, we obtain

$$\rho \cos\lambda \frac{d\lambda}{dt} + \sin\lambda \frac{d\rho}{dt} = b \cos l \frac{dl}{dt} - a \cos L \frac{dL}{dt} \qquad \ldots\ldots(3),$$

$$\rho \sin\lambda \frac{d\lambda}{dt} - \cos\lambda \frac{d\rho}{dt} = b \sin l \frac{dl}{dt} - a \sin L \frac{dL}{dt} \qquad \ldots\ldots(4).$$

Multiply (3) by $\rho \cos \lambda$ and (4) by $\rho \sin \lambda$ and add; we obtain

$$\rho^2 \frac{d\lambda}{dt} = b\rho \cos (l - \lambda) \frac{dl}{dt} - a\rho \cos (L - \lambda) \frac{dL}{dt} \quad \ldots\ldots(5).$$

Multiply (1) by $\sin l$ and (2) by $\cos l$ and add. Then

$$\rho \cos (l - \lambda) = b - a \cos (L - l) \quad \ldots\ldots(6).$$

Multiply (1) by $\sin L$ and (2) by $\cos L$ and add. Then

$$\rho \cos (L - \lambda) = b \cos (L - l) - a \quad \ldots\ldots(7).$$

Using (6) and (7) we obtain from (5),

$$\rho^2 \frac{d\lambda}{dt} = \{b^2 - ab \cos (L - l)\} \frac{dl}{dt} + \{a^2 - ab \cos (L - l)\} \frac{dL}{dt} \quad \ldots\ldots(8).$$

Now, since the planet's orbit is supposed to be circular, $\frac{dl}{dt}$ is the mean angular motion $n$ which is related to the radius $b$ by

$$n^2 b^3 = G (M + m),$$

where $M$, $m$ are the masses of the sun and planet and $G$ is the constant of gravitation. We can neglect $m$ in comparison with $M$ in this problem and writing $G.M = \mu$, we obtain

$$n \equiv \frac{dl}{dt} = \mu^{\frac{1}{2}} b^{-\frac{3}{2}} \quad \ldots\ldots(9).$$

Similarly,

$$\frac{dL}{dt} = \mu^{\frac{1}{2}} a^{-\frac{3}{2}} \quad \ldots\ldots(10).$$

Insert these values of $\frac{dl}{dt}$ and $\frac{dL}{dt}$ in (8) and there results,

$$\rho^2 \frac{d\lambda}{dt} = \mu^{\frac{1}{2}} \{(b^{\frac{1}{2}} + a^{\frac{1}{2}}) - (ab^{-\frac{1}{2}} + ba^{-\frac{1}{2}}) \cos (L - l)\} \quad \ldots\ldots(11).$$

Also, from the triangle $PES$, we obtain

$$\rho^2 = a^2 + b^2 - 2ab \cos (L - l) \quad \ldots\ldots(12).$$

Thus the value of $\frac{d\lambda}{dt}$ given by (11) can be expressed completely in terms of $L$ and $l$.

The formula (11) can be written as

$$\rho^2 \frac{d\lambda}{dt} = \mu^{\frac{1}{2}} (ab^{-\frac{1}{2}} + ba^{-\frac{1}{2}}) \left\{ \frac{(b^{\frac{1}{2}} + a^{\frac{1}{2}}) a^{\frac{1}{2}} b^{\frac{1}{2}}}{b^{\frac{1}{2}} + a^{\frac{1}{2}}} - \cos (L - l) \right\} \quad \ldots\ldots(13).$$

Now $(b^{\frac{1}{2}} + a^{\frac{1}{2}})\, a^{\frac{1}{2}} b^{\frac{1}{2}}$ is less than $(b^{\frac{3}{2}} + a^{\frac{3}{2}})$ if $a^{\frac{1}{2}} b^{\frac{1}{2}} < b + a - a^{\frac{1}{2}} b^{\frac{1}{2}}$, that is, if $(a^{\frac{1}{2}} - b^{\frac{1}{2}})^2 > 0$, which is evidently always true. Hence it is possible to define an angle $\alpha$ between 0° and 90° such that

$$\cos \alpha = \frac{(b^{\frac{1}{2}} + a^{\frac{1}{2}})\, a^{\frac{1}{2}} b^{\frac{1}{2}}}{b^{\frac{3}{2}} + a^{\frac{3}{2}}} \qquad \ldots\ldots(14).$$

The angle $\alpha$ can be calculated from the known values of $a$ and $b$. Hence, from (13),

$$\rho^2 \frac{d\lambda}{dt} = A\,[\cos \alpha - \cos (L - l)] \qquad \ldots\ldots(15),$$

where $A$ is evidently a positive quantity. It is clear from (15) that $\frac{d\lambda}{dt}$ can be positive or negative or zero according to the values taken by $(L - l)$.

When $\frac{d\lambda}{dt}$ is positive, that is when the geocentric longitude of the planet is increasing, the geocentric motion is said to be *direct*; when $\frac{d\lambda}{dt}$ is negative, that is when the geocentric longitude is decreasing, the geocentric motion is said to be *retrograde*; when $\frac{d\lambda}{dt} = 0$, the planet is said to be *stationary*.

For a stationary point, (15) gives $\cos (L - l) = \cos \alpha$, whence $L - l = \alpha$ or $360° - \alpha$. Hence when $(L - l)$ is within the range defined by $(360° - \alpha) \to 0° \to \alpha$, $\cos (L - l) > \cos \alpha$ and the geocentric motion is accordingly retrograde. When $(L - l)$ is outside the range indicated, the geocentric motion is direct.

The interval of time during which the planet's geocentric longitude $\lambda$ increases from 0° to 360° is the synodic period (p. 131) which we denote by $S$ days. Hence during each synodic period the geocentric motion is retrograde for $\frac{\alpha . S}{180}$ days and direct for $\frac{180 - \alpha}{180} . S$ days, $\alpha$ being expressed in degrees.

For Jupiter, $b = 5{\cdot}20a$ and from (14) it is easily found that $\alpha = 54\frac{1}{2}°$. As the synodic period of Jupiter is 399 days, it follows that the motion of the planet is retrograde for 121 days and direct for 278 days.

In Fig. 60, the geocentric motion of Jupiter against the back-

ground of the stars is shown between 1931 January 15 and 1931 May 15. Between January 15 and March 7 the motion is retrograde and the right ascension of the planet is decreasing between these dates. On March 7 the planet is stationary, after which the

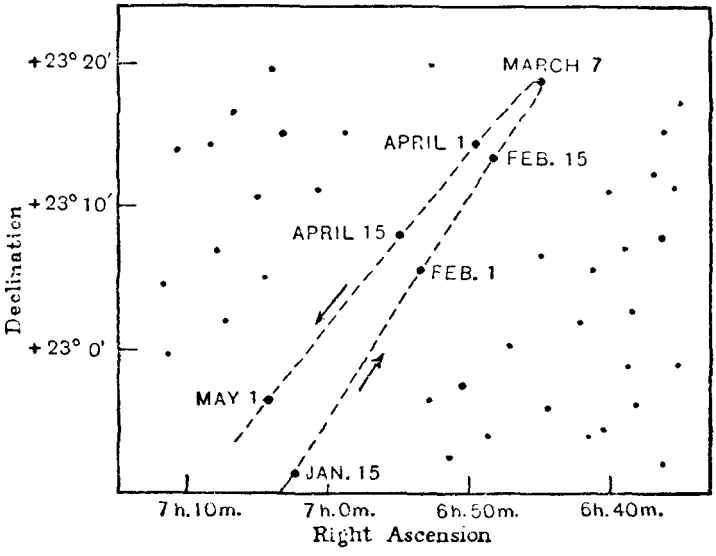



Fig. 60.

motion is direct and the right ascension is increasing. The separation of the two parts of the path is due to the effect of the inclination of the planet's orbit to the ecliptic. The effect of the inclination on the determination of the stationary points will be discussed later in section 97.

**93.** *The heliocentric distance of a planet, when stationary, in terms of its elongation.*

In Fig. 59, let $P$ now denote the planet's position at a stationary point so that $E\hat{S}P = \alpha$, defined by (14). Let $E$ denote the angle $SEP$ which is called the *elongation* of the planet from the sun, as viewed from the earth. We shall suppose that the right ascension and declination of the planet at the stationary point are observed; the co-ordinates of the sun being taken from the almanac, the calculation of $E$ can then be easily effected.

From the triangle $ESP$, we have

$$a \sin E = b \sin (E + \alpha),$$

from which
$$\tan E = \frac{b \sin \alpha}{a - b \cos \alpha} \qquad \text{.....(16).}$$

We consider the case when $b$ is greater than $a$.

From (14), since by definition $0 < \alpha < 90°$, we have

$$\sin \alpha = + \frac{(b - a)^{\frac{1}{2}} (b + a)^{\frac{1}{2}}}{b^{\frac{1}{2}} + a^{\frac{1}{2}}}.$$

Hence (16) becomes, after a little reduction,

$$\tan E = - \frac{b}{(ab + a^2)^{\frac{1}{2}}} \qquad \text{.....(17),}$$

from which it is evident that in the case concerned the value of $E$ lies between 90° and 180°.

As $a$ is the earth's heliocentric distance and $E$ is supposed known, this last formula enables the value of $b$, the planet's heliocentric distance, to be calculated. We must remember, of course, that the orbital eccentricities and the inclination to the ecliptic of the planet's orbital plane have been neglected in deriving (17) and consequently, in the general case, this formula can be expected to give only an approximate value of the planet's heliocentric distance. The minor planet Pallas was stationary soon after its discovery and its approximate heliocentric distance was first derived by means of the relation (17), the assumption of a circular orbit being made.

When $b$ is less than $a$, the procedure is similar.

**97.** *The stationary points, the inclination being taken into account.*

We consider in this section the circumstances in which a planet is stationary in longitude when its orbital plane is inclined at an angle $i$ to the plane of the ecliptic. We shall assume that the orbits of the earth and the planet are circular and of radii $a$ and $b$ respectively.

In Fig. 61, let $NP$ define the plane of the planet's orbit about the sun $S$, the plane of the ecliptic being defined by $NJ$; the angle $PNJ = i$. Through $K$, the pole of the ecliptic, draw a great circle $KPJ$. Let $NP$ be denoted by $\psi$, $NJ$ by $l$ and $JP$ by $\beta$. The projection of the planet's radius vector $b$ on the plane of the

ecliptic is $b \cos \beta$ which we shall denote by $B$. From the triangle $PJN$ (in which $P\hat{J}N = 90°$), we have

$$\sin \psi \cos i = \cos \beta \sin l \quad \text{.....(18),}$$

$$\cos \psi = \cos \beta \cos l \quad \text{.....(19).}$$

Consider now Fig. 62 in which $SQ$ is the projection of the planet's radius vector on the plane of the ecliptic. Let $SN$ be the direction of a node and let $\phi$ denote the angular distance of

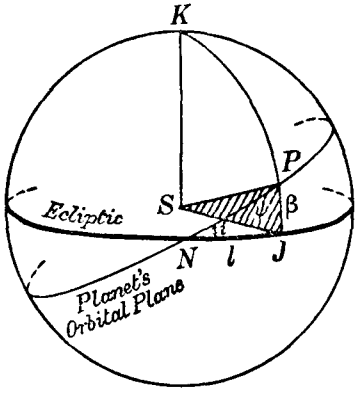



Fig. 61.

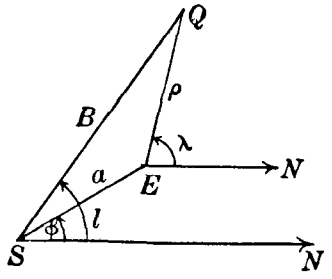



Fig. 62.

the earth $E$ from the node. Let $EQ$ be denoted by $\rho$ and $N\hat{E}Q$ by $\lambda$ ($EN$ gives the geocentric direction of the node). We have also $N\hat{S}Q = l$ and $SQ = B$. As in section 95 we have

$$\rho \sin \lambda = B \sin l - a \sin \phi,$$

$$\rho \cos \lambda = B \cos l - a \cos \phi.$$

Now $B = b \cos \beta$; hence, using (18) and (19), we have

$$\left.\begin{aligned} \rho \sin \lambda &= b \cos i \sin \psi - a \sin \phi \\ \rho \cos \lambda &= b \cos \psi - a \cos \phi \end{aligned}\right\} \quad \text{.....(20),}$$

and differentiating,

$$\rho \cos \lambda \frac{d\lambda}{dt} + \frac{d\rho}{dt} \sin \lambda = b \cos i \cos \psi \frac{d\psi}{dt} - a \cos \phi \frac{d\phi}{dt},$$

$$\rho \sin \lambda \frac{d\lambda}{dt} - \frac{d\rho}{dt} \cos \lambda = b \sin \psi \frac{d\psi}{dt} - a \sin \phi \frac{d\phi}{dt}.$$

Multiplying these equations, in order, by $\rho \cos\lambda$ and $\rho \sin\lambda$, adding and making use of (20), we obtain

$$\rho^2 \frac{d\lambda}{dt} = \left(b\cos i \cos\psi \frac{d\psi}{dt} - a\cos\phi \frac{d\phi}{dt}\right)(b\cos\psi - a\cos\phi)$$

$$+ \left(b\sin\psi \frac{d\psi}{dt} - a\sin\phi \frac{d\phi}{dt}\right)(b\cos i \sin\psi - a\sin\phi)$$

$$= b^2 \cos i \frac{d\psi}{dt} + a^2 \frac{d\phi}{dt} - ab\frac{d\psi}{dt}[\cos\psi\cos\phi\cos i + \sin\psi\sin\phi]$$

$$- ab\frac{d\phi}{dt}[\cos\psi\cos\phi + \sin\psi\sin\phi\cos i].$$

If the planet is stationary in longitude we must have $\frac{d\lambda}{dt} = 0$. Also by Kepler's law

$$\frac{d\psi}{dt} = \mu^{\frac{1}{2}} b^{-\frac{3}{2}} \text{ and } \frac{d\phi}{dt} = \mu^{\frac{1}{2}} a^{-\frac{3}{2}}.$$

We thus have for stationary points

$$b^{\frac{1}{2}}\cos i + a^{\frac{1}{2}} = ab^{-\frac{1}{2}}[\cos\psi\cos\phi\cos i + \sin\psi\sin\phi]$$
$$+ a^{-\frac{1}{2}}b[\cos\psi\cos\phi + \sin\psi\sin\phi\cos i] \ldots(21).$$

Writing $\cos i = \cos^2\frac{i}{2} - \sin^2\frac{i}{2}$, we finally obtain, after multiplying throughout by $a^{\frac{1}{2}}b^{\frac{1}{2}}$,

$$a^{\frac{1}{2}}b^{\frac{1}{2}}(a^{\frac{1}{2}} + b^{\frac{1}{2}}\cos i) = (a^{\frac{3}{2}} + b^{\frac{3}{2}})\cos(\psi - \phi)\cos^2\frac{i}{2}$$

$$- (a^{\frac{3}{2}} - b^{\frac{3}{2}})\cos(\psi + \phi)\sin^2\frac{i}{2} \ldots(22).$$

This is the relation sought. It is to be remembered that $\psi$ and $\phi$ denote the angular distances of the planet and of the earth, measured in the plane of their respective orbits, from the line of nodes.

**98.** *The phases of the planets and the moon.*

We shall assume the heavenly bodies concerned to be spherical. In Fig. 63, let $P$ be the centre of a planet (or the moon) and let the straight lines joining $P$ to the earth and the sun cut the planet's surface at $E$ and $S$. The hemisphere illuminated by the sun will be bounded by the great circle $BCKD$, of which $S$ is the pole. The hemisphere towards the earth is bounded by the great

circle $ACLD$ of which $E$ is the pole. Hence the only part of the planet visible from the earth is the area consisting of the spherical triangles $ABC$ and $ABD$. The plane $ACLD$ is the plane

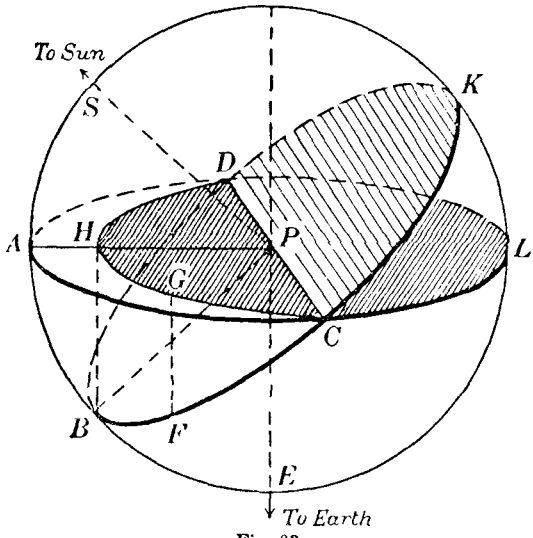


Fig. 63.

of the planet's *disc* as viewed from the earth. The great circle arc $CAD$ is the visible *limit* of the planet. Any point on $CB$ such as $F$ will be seen on the disc at the point $G$, $FG$ being parallel to the line of sight $EP$. The projection of all points on the semicircle $CBD$ thus leads to a curve $CHD$ in the plane of the great circle $ACLD$, and this curve is an ellipse of which $CD$ is the major axis; the semi-minor axis is $HP$, where $H$ is the projection of $B$ on the plane of the disc. The planet is seen as shown in Fig. 64. The visible area of the planet's disc is the area bounded by the semicircle $CAD$ and the semi-circumference $CHD$ of the ellipse. If $r$ denotes the linear radius of the planet (or moon) this visible area $A$ is given by $A = \frac{1}{2}\pi r^2 - \frac{1}{2}\pi r . PH$.

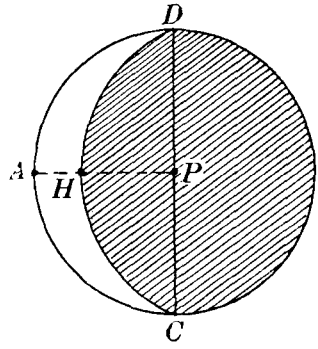


Fig. 64.

Let $d$ denote the angle $SPE$; then $d$ is the elongation of the earth from the sun as seen from $P$. Now $PH = PB \cos A\hat{P}B$, and since $S\hat{P}B = 90°$, it follows that the angle $APB = 180° - d$. Hence

$$A = \frac{\pi r^2}{2}(1 + \cos d) \qquad \text{.....(23).}$$

The *phase* is measured by the fraction of the diameter, perpendicular to the line of cusps, lying in the visible portion of the disc; the phase is thus $AH/2r$ or $(1 + \cos d)/2$. From (23) it is seen that the phase is also represented by the fraction of the area of the disc illuminated.

We now consider in greater detail the phases of the moon. In Fig. 65, let $M$ denote the moon and $E$ the earth. $MS$ is the direction of the sun from $M$; $S\hat{M}E$ is the angle $d$. The moon's phase is then given by (23). Now the moon's distance from the earth is very small compared with the sun's distance from the earth, and in Fig. 65 we may assume $ES$ parallel to $MS$ without any serious loss of accuracy. Hence, denoting $S\hat{E}M$ by $E$, we have $E = 180° - d$ or $d = 180° - E$. This angle $E$ is the elongation of the moon from the sun as viewed from the earth.

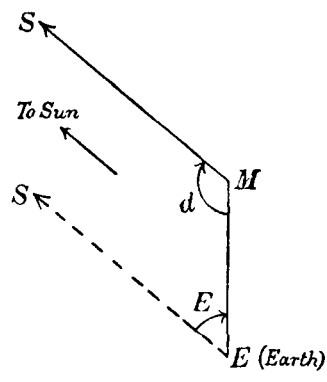


Fig. 65.

At new moon $E = 0$, for the moon is then directly, or almost directly, between the earth and the sun; thus $d = 180°$ and the phase is zero. At *quadrature*, the elongation is 90° and thus $d = 90°$, so that the phase is $\frac{1}{2}$; the moon then presents one-half of its illuminated surface towards the earth. When the moon is in opposition, the elongation is 180°, so that $d = 0$; hence the phase is unity, that is, the complete disc of the moon is visible; the moon is then "full". After full moon, the phases are repeated in reverse order until the next new moon occurs. When more than one-half of the disc is visible, the moon is said to be *gibbous*.

**99.** *The brightness of the planets.*

The amount of light which reaches the earth from a planet depends (i) on the phase, and (ii) on the geocentric distance $\rho$ of the planet, varying inversely as $\rho^2$. Thus if $B$ denotes the apparent brightness, we can write

$$B = \frac{c\,(1 + \cos d)}{\rho^2} \qquad \ldots\ldots(24),$$

where $c$ is a constant depending on the surface illumination and reflective power. But from Fig. 66

$$\cos d = \frac{\rho^2 + b^2 - a^2}{2\rho b} \qquad \ldots\ldots(25),$$

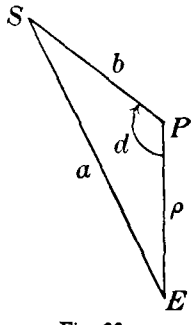


Fig. 66.

where $a$ and $b$ are the heliocentric distances of the earth and planet. Hence

$$B = c\,[\rho^2 + 2\rho b + b^2 - a^2]/2b\rho^3 \qquad \ldots\ldots(26).$$

When $B$ is a maximum, the orbits being assumed circular, we must have $\frac{dB}{d\rho} = 0$

or $\qquad \rho^2 + 4b\rho + 3\,(b^2 - a^2) = 0,$

from which $\qquad \rho = (b^2 + 3a^2)^{\frac{1}{2}} - 2b \qquad \ldots\ldots(27).$

For the planet Venus, we have $b = 0{\cdot}723a$ and for maximum brightness (27) gives $\rho = 0{\cdot}430$. From (25), the corresponding value of $d$ is $117°{\cdot}9$. The elongation from the sun—the angle $SEP$—is then found to be $39°{\cdot}7$.

It should be remarked that in the case of the planet Mercury, for which the orbital eccentricity is large (about 0·2), the symbol $b$ in (26) represents the radius vector in the orbit. The brightness of Mercury is thus dependent on the planet's position in its orbit as well as on its geocentric distance.

**100.** *Heliographic co-ordinates.*

In studying sun-spots and the solar phenomena associated with them, it is of importance to know their co-ordinates on the solar surface; these co-ordinates are defined in much the same way as that in which points on the surface of the earth are expressed in terms of longitude and latitude. In Fig. 67, let the sphere represent the solar globe. The great circle $\Upsilon EN$, of which $K$ is the pole, is the intersection of the plane of the ecliptic with

the sphere. The straight line from the sun's centre $C$, drawn in the direction of the vernal equinox, cuts the sphere in $T$. Since the earth is in the ecliptic, the straight line joining $C$ to the earth cuts the solar surface in $E$. Any plane perpendicular to $EC$ (which is the line of sight) we can regard as the plane of the sun's disc; for convenience we shall take the shaded plane defined by the radii $CK$ and $CT$ to be the plane of the disc ($E$ is the pole of the great circle $UKT$). Any point $X$ on the solar surface will be seen on the disc at $X_1$, such that $XX_1$ is perpendicular to the plane of the disc and therefore parallel to $EC$. The

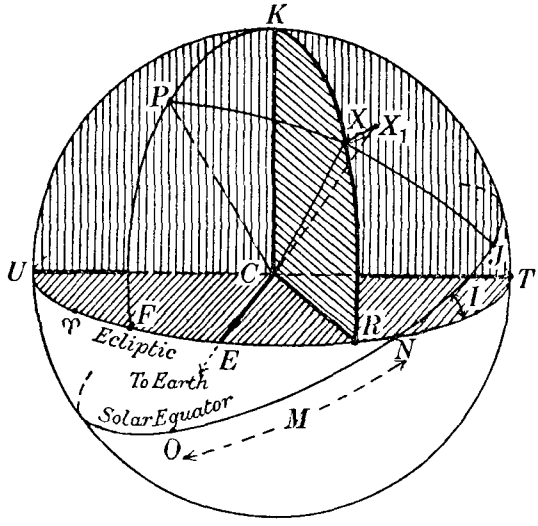



Fig. 67.

position of $X_1$ can be ascertained, from visual or photographic observations, with reference to certain rectangular axes in the plane of the disc. The problem is to deduce the solar or heliographic longitude and latitude of $X$ from the observed position $X_1$.

The sun rotates about an axis, the northern extremity of which is the point $P$ in Fig. 67, and the plane perpendicular to this axis cuts the sun's surface in the great circle $ONJ$, which is called the *solar equator*. The inclination $JNT$ of the solar equator to the ecliptic is usually denoted by $I$. The point $N$ is the ascending node of the solar equator on the ecliptic and its longi-

tude $\Upsilon N$ (measured along the ecliptic from $\Upsilon$) is denoted by $\Omega$. The value of $\Omega$ is found to be given by

$$\Omega = 73^\circ\ 40' + 50''{\cdot}25\ (t - 1850{\cdot}0),$$

where $t$ is expressed in years, so that for 1975·0 the value of $\Omega$ is $75^\circ\ 24'{\cdot}7$.

The *heliographic co-ordinates* of a point $X$ on the sun's surface are defined with reference to the solar equator. As the sun is a gaseous body, there is no definitely recognisable point on the solar equator from which to measure longitudes. A reference point $O$ is chosen thus. At Greenwich mean noon on 1854 Jan. 1, the node $N$ defined a particular point on the solar surface. Owing to the solar rotation this point is carried round the equator and at any subsequent time we suppose that it is at $O$; assuming that the equatorial rotation occurs in a period of 25·38 days, the position of $O$ with reference to $N$ can be calculated for the instant concerned; in particular the arc $ON$ which we denote by $M$ can be found. The rotation occurs in the direction $\vec{ON}$. In Fig. 67, let $PXJ$ be the solar meridian through $X$. Then $OJ$ is the *heliographic longitude* of $X$ (denoted by $L$) and the arc $JX$ is the *heliographic latitude* of $X$ (denoted by $B$).

**101.** *The heliographic co-ordinates of the centre of the disc.*

Referring to Fig. 67, we see that the point $E$ on the solar surface will be seen as the centre of the disc. Let the heliographic co-ordinates of $E$ be $L_0$ and $B_0$. Then in the spherical triangle $PEN$ we have:

$$PN = 90^\circ,\ P\hat{N}E = 90^\circ - I,\ PE = 90^\circ - B_0,\ \text{and}\ E\hat{P}N = M - L_0.$$

We require still the arc $EN$ expressed in terms of known quantities. Now $\Upsilon E$ is the angle between the direction of the vernal equinox and the direction of the earth as seen from the sun's centre $C$, so that $\Upsilon E$ is the earth's heliocentric longitude. If $\odot$ denotes the geocentric longitude of the sun, we have

$$\Upsilon E = \odot + 180^\circ \qquad \text{.....(28).}$$

Also since $\Upsilon N = \Omega$, we see that

$$EN = \Omega - \odot - 180^\circ \qquad \text{.....(29).}$$

From the triangle $PEN$ we derive the following formulae, using **D** and **A** in succession,

$$\tan (L_0 - M) = \tan (\odot - \Omega) \cos I \qquad \ldots\ldots(30),$$

$$\sin B_0 = \sin (\odot - \Omega) \sin I \qquad \ldots\ldots(31).$$

Assuming that $M$ has been calculated for the time of observation, these formulae enable the heliographic co-ordinates $L_0$ and $B_0$ of the centre of the disc to be found; $L_0$ and $B_0$ are tabulated in the almanacs for every day during the year.

**102.** *Position angle of the sun's axis of rotation.*

Consider rectangular axes $CE$, $CT$ and $CK$ (Fig. 67). Let $KXR$ be a great circle through $K$ (the pole of the ecliptic) meeting the ecliptic in $R$. The radius of the sphere being taken as unity, the co-ordinates of $X$ referred to these axes are (cos $ER$ cos $RX$, sin $ER$ cos $RX$, sin $RX$). Now $X_1$ is the projection of $X$ on the plane $KCT$—the plane of the disc—and the co-ordinates of $X_1$ referred to $CT$ and $CK$ as axes are therefore (sin $ER$ cos $RX$, sin $RX$) or, since $KR = 90°$, (sin $ER$ sin $KX$, cos $KX$). Let $\psi$ denote the angle between $CX_1$ and $CK$. Then we have

$$\tan \psi = \sin ER \tan KX \qquad \ldots\ldots(32).$$

Suppose that $X$ now represents the point on the solar surface intersected by the radius drawn parallel to the *earth's axis of rotation*. The projection $CX_1$ on the disc defines the *north direction*, from which the position angle of any point on the disc is measured in practice. Now $KX$ is the angle between the pole of the ecliptic and the pole of the earth's equator; hence $KX = \epsilon$, where $\epsilon$ is the obliquity of the ecliptic. Also the vernal equinox is 90° from both $K$ and $X$. Hence it is 90° from the great circle $KXR$; hence $\Upsilon R = 90°$. But by (28), $\Upsilon E = 180° + \odot$; hence $ER = 90° - (180° + \odot)$ or

$$ER = 270° - \odot \qquad \ldots\ldots(33).$$

Denoting the angle $X_1CK$ by $x$, we have from (32), writing $x$ for $\psi$,

$$\tan x = - \cos \odot \tan \epsilon \qquad \ldots\ldots(34).$$

Again, the projection of $CP$ on the plane $KCT$ will make an angle, say, $y$ with $CK$. Hence, by (32),

$$\tan y = \sin EF \tan KP \qquad \ldots\ldots(35),$$

where $F$ is the point on the ecliptic cut by the great circle $KP$. Now since $K$ is the pole of $\Upsilon RT$ and $P$ is the pole of $ONJ$, it follows that $N$ is the pole of $KPF$; hence $FN = 90°$. But $\Upsilon N = \Omega$ and therefore $\Upsilon F = \Omega - 90°$. Also, by (28), $\Upsilon E = 180° + \odot$; hence $EF = 180° + \odot - (\Omega - 90°)$ or

$$EF = 270° + (\odot - \Omega) \qquad \text{.....(36).}$$

Also $KP$ is the inclination $I$. Hence (35) becomes, using (36),

$$\tan y = -\cos(\odot - \Omega)\tan I \qquad \text{.....(37).}$$

For convenience, we shall assume that we are dealing with a photograph of the solar disc. Let $CN$ (Fig. 68) be the radius in the plane of the disc corresponding to the projection of the solar radius parallel to the earth's axis. This radius $CN$ can be inferred from the diurnal motion of the sun across the sky; if the telescope is at rest, the image of the sun will move along a parallel of declination, that is, perpendicular to $CN$; hence by suitable means the radius $CN$ can be drawn on the photograph.

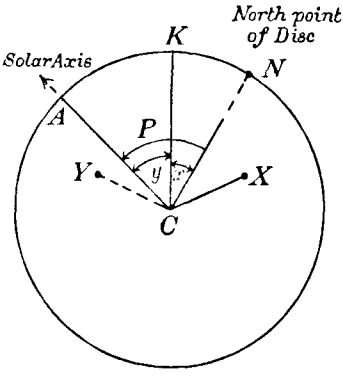


Fig. 68.

The *position angle* of any marking such as $Y$ on the disc is measured from $CN$ eastwards, that is, towards the left as we view the sun. Thus the position angle of $Y$ is the angle $NCY$ and of $X$ it is $(360° - N\hat{C}X)$. Now if $CA$ is the projection on the disc of the sun's axis $NCK = x$ and $KCA = y$,

where $x$ and $y$ are given by (34) and (37). Thus the position angle $P$ of the sun's axis $CA$ is given by

$$P = x + y \qquad \text{.....(38).}$$

Thus $P$ can be calculated; it is tabulated in the almanacs.

If $\chi$ denotes the angle $ACX$ (Fig. 68), $X$ being any marking on the disc, and $\theta$ is the position angle of $X$ we have, since $NCX = 360° - \theta$, that $\chi = P + 360° - \theta$ or

$$\chi = P - \theta \qquad \text{.....(39).}$$

As $\theta$ and $P$ may be supposed known, the former by means of the photograph and the latter by means of the almanac, the angle $\chi$ can be found.

**103.** *The heliographic co-ordinates of a sun-spot.*

Let $R$ and $d$ denote in linear measure the radius of the sun and the distance of the earth from the sun respectively. If $S$ is the angular semi-diameter of the sun—this can be found for the day of observation from the almanac—we have $\sin S = R/d$ or, since $S$ is about 16′, we can write with sufficient accuracy

$$S = \frac{R}{d} \operatorname{cosec} 1' \qquad \text{.....(40),}$$

in which $S$ is expressed in minutes of arc. Let $r_0$ and $r$ denote the measures of the radius of the disc and of the distance of the spot $X$ from the centre $C$ of the disc (Fig. 68); $r_0$ and $r$ can be measured in any convenient unit on the photograph. If $\rho_1$ (in minutes of arc) is the angle between the direction of the sun's centre and the direction of the spot, both viewed from the earth, we have, with sufficient accuracy

$$\frac{\rho_1}{S} = \frac{r}{r_0} \qquad \text{.....(41),}$$

from which $\rho_1$ can be determined.

In Fig. 69, let $S$ be the position of the spot on the solar surface

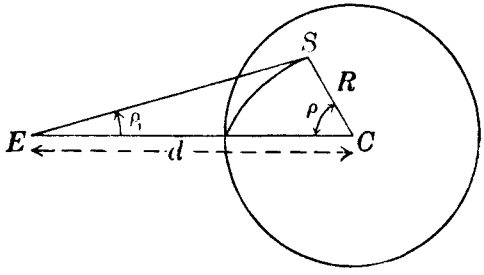



Fig. 69.

and let $S\hat{C}E$ be denoted by $\rho$, $E$ being the earth. Then $ESC = 180° - (\rho + \rho_1)$. But from the plane triangle $ESC$,

$$\sin ESC = \frac{d}{R} \sin \rho_1$$

or, since $\rho_1$ is small (less than 16′) and is expressed in minutes of arc,

$$\sin(\rho + \rho_1) = \frac{d}{R}\rho_1 \sin 1'$$

or, using (40),

$$\sin(\rho + \rho_1) = \frac{\rho_1}{S} \qquad \ldots\ldots(42).$$

Since $\rho_1$ can be found by means of (41), this last formula (42) enables $\rho$ to be calculated.

Consider now Fig. 70 in which $S$ is the spot on the solar surface, $CE$ gives the direction of the earth from the sun's centre $C$, and $P$ is the north pole of the sun's axis. The great circle arc $ES$ is thus the angle $\rho$ given by (42). Denote the heliographic longitude and latitude of $S$ by $L$ and $B$; the corresponding co-ordinates of $E$ are $L_0$ and $B_0$. In the spherical triangle $PES$ we have: $PS = 90° - B$, $PE = 90° - B_0$, $ES = \rho$ and $E\hat{P}S = L - L_0$. Also, since the point $E$ is seen as the centre $C$ of the disc and since the great circles $EP$ and $ES$ project into the straight lines $CA$ and $CX$ of Fig. 68, the angle $PES$ is equal to the angle $ACX$ or $\chi$. Hence, by (39), $P\hat{E}S = P - \theta$, where $\theta$ is the position angle of the spot on the disc. Hence from the triangle $PES$ we have the formulae, using **A** and **B**,

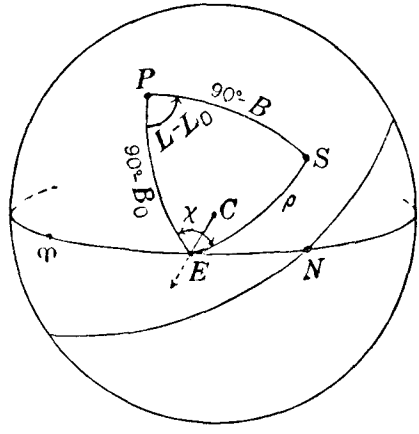


Fig. 70.

$$\sin B = \sin B_0 \cos\rho + \cos B_0 \sin\rho \cos(P - \theta) \quad \ldots(43),$$

$$\sin(L - L_0) = \sin\rho \sin(P - \theta) \sec B \qquad \ldots\ldots(44).$$

Since $B_0$, $\rho$, $P$, $\theta$ and $L_0$ are all supposed now to be known, the formulae (43) and (44) determine the heliographic latitude and longitude of the spot.

## EXERCISES

**1.** Two planets $P_1$ and $P_2$ revolve in circular orbits at distances $b_1$, $b_2$ from the sun. Prove that when they appear stationary to one another,

$$\frac{b_1}{b_2} = \tfrac{1}{2} \tan \frac{\theta}{2} \tan \theta,$$

where $\tan \theta = 2 \cot E$, $E$ being the elongation of $P_2$ as seen from $P_1$.

**2.** If $\theta$ is the angle subtended at the earth by the sun and a stationary point of a planet's orbit, and $\phi$ is the maximum elongation of the planet, prove that

$$2 \cot \theta = \sec \tfrac{1}{2}\phi + \operatorname{cosec} \tfrac{1}{2}\phi.$$

**3.** Show that, if the earth and a planet be supposed to describe coplanar, circular orbits, and the difference in longitude of the sun and the planet be $\theta$, the rate of change of $\theta$ is, numerically,

$$\frac{2\pi}{S}\left(1 - \frac{a}{\rho}\cos\theta\right),$$

where $S$ is the synodic period of the planet, $a$ the radius of the earth's orbit, and $\rho$ the distance of the planet from the earth at the moment.

[*Coll. Exam.*]

**4.** If the line joining two planets to one another subtends an angle of 60° at the sun when the planets appear to each other to be stationary, show that $a^2 + b^2 = 7ab$, where $a$, $b$ are the distances of the planets from the sun.

[*M.T.*]

**5.** If $u$ and $v$ are the velocities of two planets in circular and coplanar orbits, show that the period of direct motion is to the period of retrograde motion as $(180° - \theta) : \theta$, where $\cos\theta = uv/(u^2 - uv + v^2)$. [*Coll. Exam.*]

**6.** If $a$ and $b$ are the radii of the orbits (assumed to be circular and coplanar) of the earth $E$ and a superior planet $P$, and $u$ and $v$ are their respective linear velocities, prove that the square of the velocity of $P$ relative to $E$ at a stationary point is

$$\frac{(u^2 - v^2)(bu - av)}{bu + av}.$$

**7.** One planet whose mean distance from the sun is $a$ appears to have a phase $E$ to another planet whose mean distance from the sun is $b$, and the latter appears to have a phase $V$. Prove that, if the inclinations of the orbits to each other and their eccentricities be neglected,

$$b^2 V (1 - V) = a^2 E (1 - E).$$

If the distance of Venus from the sun is 0·72 astronomical units, find what part of the earth's surface appears illuminated as seen from Venus.

**8.** The heliocentric distance of an inferior planet $P$, moving in a circular orbit in the ecliptic, is $b$ astronomical units; the orbit of the earth ($E$) is also assumed circular. If the heliocentric co-ordinates of $P$ and $E$ are $(b\cos f\theta, b\sin f\theta)$ and

(cos $\theta$, sin $\theta$) respectively, $\theta$ being measured from inferior conjunction, show that for a stationary point

$$\cos(f-1)\theta = \frac{1+fb^2}{b(1+f)}.$$

**9.** Prove that when a planet whose orbit does not coincide with the ecliptic is absolutely stationary as seen from the earth, its direction of motion and that of the earth must intersect on the line of nodes, and that its projection on the plane of the ecliptic is also stationary. [*M.T.*]

**10.** Neglecting the inclination of the plane of a planet's orbit to the ecliptic, prove that, in latitude $\phi$ ($> \epsilon$), the altitude of an inferior planet at sunrise or sunset never exceeds

$$\sin^{-1}(\sin E \cos\overline{\phi - \epsilon}),$$

where $E$ is the maximum elongation of the planet; and that this most favourable case can only occur when maximum elongation coincides with an equinox.

How is this result modified if the observer is in the tropical zone?

[*Lond.* 1925.]

**11.** If the orbit of an outer planet is an ellipse of eccentricity $e$ and semi-axis $a$, the inclination to the ecliptic being zero, and if it is in opposition at perihelion, show that its motion will appear direct if $a$, in astronomical units, is less than $(1+e)/(1-e)$. [*Ball.*]

**12.** Show that the phase of a superior planet, as seen from the earth, is least when the earth appears half illuminated to the planet, but that the apparent brightness of the planet is a maximum at opposition and a minimum at conjunction. [*Coll. Exam.*]

**13.** Assuming that Venus and the earth describe circular orbits in the ecliptic, show that Venus will appear brightest at elongation $\theta$ given by

$$\cos\theta = \tfrac{2}{3}\{(3+a^2)^{\frac{1}{2}} - a\},$$

where $a$ is the heliocentric distance of Venus in astronomical units.

[*Lond.* 1926.]

**14.** Show that the "correction for defective illumination" of Mars (i.e. the apparent width in arc of the unilluminated crescent) is a maximum when Mars is at a geocentric distance $(b^2-a^2)/b$, the orbits of the earth and Mars being treated as circles of radii $a$ and $b$. Show that when the disc of the moon is seen at night by earthshine there is a narrow crescent not illuminated by either sunlight or earthshine and that the width of this crescent does not exceed $0''{\cdot}5$. (Lunar parallax = $57'$; semi-diameter = $15'$.) [*M.T.* 1926.]

**15.** Assuming the diameters of the earth and Venus to be negligible, show that $\psi$, the heliocentric elongation of Venus from the earth at the beginning or end of a transit, is given by the equation

$$b^2r^2\cos^2\psi - 2brR^2\cos\psi + R^2(b^2+r^2) - b^2r^2 = 0,$$

where $R$ is the sun's radius, and $b$, $r$ the distances of Venus and the earth from the sun's centre. [*Ball.*]

CHAPTER VIII

# ABERRATION

**104.** *The law of aberration.*

The phenomenon of aberration was discovered in 1728 by Bradley, later Astronomer Royal, as a result of a series of meridian observations of the second magnitude star $\gamma$ Draconis. In 1675, the Danish astronomer Roemer had established that light travelled with a finite velocity and Bradley's observations were interpreted by recognising that a star's position in the sky could be displaced by an amount depending on the ratio of the earth's orbital velocity to the velocity of light, and on the position of the particular star concerned. The earth's average orbital speed is about $18\frac{1}{2}$ miles per second and the velocity of light is 186, 280 miles per second; thus the ratio just mentioned is small but not negligible.

In Fig. 71, let $C$ represent the centre of a telescope's object-glass and $E$ the eye-piece at the moment when a ray from a star $X$ arrives at $C$. Let $EF$ be parallel, at this moment, to the direction in which the earth is moving around the sun. Let $\tau$ be the time required by the ray to pass through the telescope; in this interval the earth has moved through a distance $EE_1$ or $V\tau$, where $V$ is the earth's velocity. Denote by $c$ the velocity of light. Then when the ray from the star reaches the eye-piece, the latter is at $E_1$ and we have $CE_1 = c\tau$. If the earth had no velocity, the direction in which the telescope would be pointed is along $E_1C$, which we describe as the true direction of the star. Actually, owing to the earth's motion, the telescope has to be pointed in the direction $EC$. Complete the parallelogram $EE_1C_1C$. Then $E_1C_1$ is parallel to $EC$ and thus $E_1C_1$ is the apparent direction of the star at the moment of observation. Let $\theta$, $\theta_1$ denote $C\hat{E}_1F$,

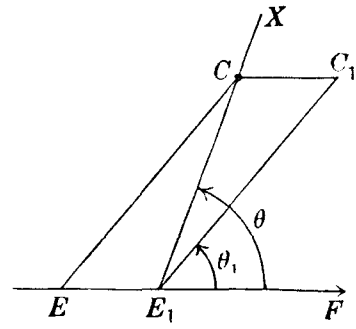


Fig. 71.

$C_1\hat{E}_1F$ respectively. Then the angular displacement $\theta - \theta_1$ is said to be due to aberration. From Fig. 71 it is seen that aberration displaces the star's true direction towards the direction $EF$ of the earth's motion and in the plane $XE_1F$.

In the triangle $CE_1C_1$,

$$\frac{\sin CE_1C_1}{\sin CC_1E_1} = \frac{CC_1}{CE_1}.$$

But $CC_1 = EE_1 = V\tau$ and $CE_1 = c\tau$. Hence we derive

$$\sin(\theta - \theta_1) = \frac{V}{c}\sin\theta_1.$$

Now $V/c$ is small and consequently $\theta - \theta_1$ is small. We can then write with all needful accuracy ($\theta - \theta_1$ being expressed in seconds of arc)

$$\theta - \theta_1 = \frac{V}{c}\sin\theta_1 \operatorname{cosec} 1''$$

or
$$\theta - \theta_1 = \kappa \sin\theta_1 \qquad \ldots\ldots(1),$$

in which
$$\kappa = \frac{V}{c}\operatorname{cosec} 1'' \qquad \ldots\ldots(2).$$

$\kappa$ is defined to be the constant of aberration; its value which can be calculated from the values of $V$ and $c$ is about $20''{\cdot}5$. The definition and value of $\kappa$ will be considered later in greater detail. It is clear that as $\theta - \theta_1$ cannot exceed $20''{\cdot}5$ we can write (1) with sufficient accuracy in the form

$$\theta - \theta_1 = \kappa \sin\theta \qquad \ldots\ldots(3).$$

The law of aberration is contained in formula (1) or (3).

**105.** *Annual aberration in ecliptic longitude and latitude.*

For the present we shall neglect the eccentricity of the earth's orbit and assume simply that the orbit is a circle with the sun at the centre, and that the orbital velocity $V$ is constant. If $E$ (Fig. 72) is the position of the earth at any time, its direction of motion is along $EF$ at right angles to $SE$. Let $E\Upsilon$ or $S\Upsilon_1$ be the direction of the vernal equinox; then $\Upsilon\hat{E}S$ is the geocentric longitude of the sun (denoted by $\odot$). The geocentric longitude of $F$ is thus $(\odot - 90°)$. Accordingly the point $F$ of the ecliptic, towards which the earth's motion is directed, is 90° behind the position of the sun on the ecliptic. With reference to the earth as centre of the celestial sphere, the point $F$ evidently makes a

complete circuit of the ecliptic in a year; the aberration in this instance is called *annual aberration*, and the displacements to which it gives rise are repeated in yearly cycles.

In Fig. 73, $F$ defines the direction of the earth's motion and $\Upsilon F = \odot - 90°$. Let $X$ be the true position of a star (it defines the direction in which the star is viewed from the sun). Then as the aberrational displacement takes place in the plane $XEF$, the star will appear to be displaced along the great circle $XF$ and

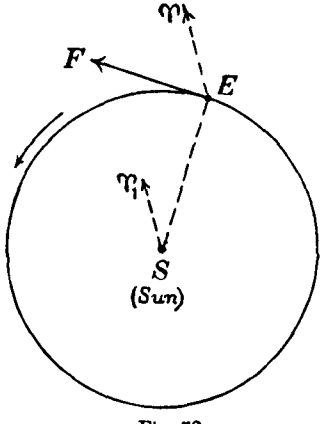



Fig. 72.

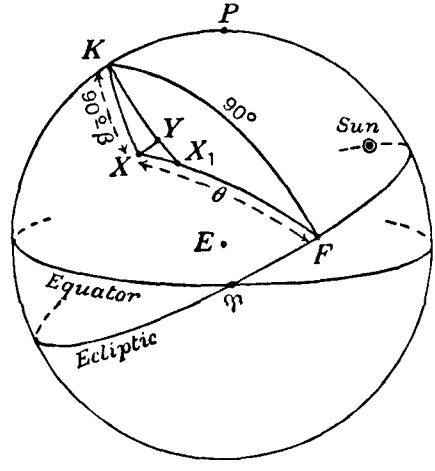



Fig. 73.

will be seen at $X_1$. With the notation of the previous section, $XF = \theta$, $X_1F = \theta_1$ and therefore $XX_1$ is given by (3), so that

$$XX_1 = \kappa \sin \theta \qquad \text{.....(4).}$$

Join $KX$ and $KX_1$ by great circles, $K$ being the pole of the ecliptic, and draw the small circle $XY$ parallel to the ecliptic. If $\lambda$, $\beta$ are the longitude and latitude of $X$ and $\lambda_1$, $\beta_1$ are the corresponding co-ordinates of $X_1$, then $\lambda_1 - \lambda = X\hat{K}Y$ and, since $XY = X\hat{K}Y \sin KX$, we have $XY = (\lambda_1 - \lambda)\cos\beta$. Also $\beta - \beta_1 = X_1Y$. Write

$$\Delta\lambda = \lambda_1 - \lambda \text{ and } \Delta\beta = \beta_1 - \beta.$$

Then $$XY = \Delta\lambda \cos\beta \text{ and } X_1Y = -\Delta\beta \qquad \text{.....(5).}$$

In the infinitesimal plane triangle $XX_1Y$, let $\phi$ denote $Y\hat{X}X_1$. Then

$$XY = XX_1 \cos\phi \text{ and } X_1Y = XX_1 \sin\phi.$$

Hence from (4) and (5)

$$\Delta\lambda = \kappa \sin\theta \cos\phi \sec\beta \qquad \ldots\ldots(6),$$

$$\Delta\beta = -\kappa \sin\theta \sin\phi \qquad \ldots\ldots(7).$$

Now in the spherical triangle $KXF$, $KX = 90° - \beta$, $KF = 90°$, $XF = \theta$, $K\hat{X}F = 90° + \phi$ and $X\hat{K}F$ is the difference of longitude of $F$ and $X$, so that $X\hat{K}F = (\odot - 90°) - \lambda$.

By the sine-formula **B**,

$$\sin XF \sin KXF = \sin KF \sin XKF,$$

and consequently $\sin\theta \cos\phi = -\cos(\odot - \lambda) \qquad \ldots\ldots(8).$

By formula **C**,

$$\sin XF \cos KXF = \cos KF \sin KX - \sin KF \cos KX \cos XKF,$$

and thus $\sin\theta \sin\phi = \sin\beta \sin(\odot - \lambda) \qquad \ldots\ldots(9).$

From (6)...(9), we obtain the formulae

$$\Delta\lambda = -\kappa \sec\beta \cos(\odot - \lambda) \qquad \ldots\ldots(10),$$

$$\Delta\beta = -\kappa \sin\beta \sin(\odot - \lambda) \qquad \ldots\ldots(11).$$

These formulae—(10) and (11)—give the displacements in longitude and latitude due to annual aberration.

**106.** *The aberrational ellipse.*

Referring to Fig. 73 we see that the aberrational displacement from $X$ to $X_1$ is equivalent to the two displacements from $X$ to $Y$ and from $Y$ to $X_1$. Denote these by $x$ and $y$ respectively. Then

$$x = \Delta\lambda \cos\beta = -\kappa \cos(\odot - \lambda) \qquad \ldots\ldots(12),$$

$$y = -\Delta\beta \quad = \kappa \sin\beta \sin(\odot - \lambda) \qquad \ldots\ldots(13),$$

from which, by eliminating $(\odot - \lambda)$, we obtain

$$\frac{x^2}{\kappa^2} + \frac{y^2}{\kappa^2 \sin^2\beta} = 1 \qquad \ldots\ldots(14).$$

This is the equation of an ellipse, known as the aberrational ellipse. During the course of a year the star appears to describe this curve on the celestial sphere, the ellipse centre being the star's true position. The semi-major axis is $\kappa$, parallel to the ecliptic, and is therefore constant for all stars; the semi-minor axis is $\kappa \sin\beta$, perpendicular to the semi-major axis. The aberrational displacement is greatest when $x = \pm\kappa$, that is—from (12)—when $\odot - \lambda = 0°$ or $180°$. Assuming that we know the value

of $\lambda$ for a particular star, we can thus obtain the two values of $\odot$ (the geocentric longitude of the sun) corresponding to maximum displacement and so we can derive from the *Astronomical Ephemeris* the date when this occurs. Thus, if $\lambda = 0°$, the aberrational displacement is a maximum when $\odot = 0°$ or $180°$, that is, at the spring or autumn equinox.

For a star on the ecliptic ($\beta = 0°$), the aberrational ellipse degenerates into a straight line (an element of the ecliptic) and for a star at the pole of the ecliptic, it becomes a circle.

**107.** *Aberration in right ascension and declination.*

As before, let $X$ be the true position of the star and $X_1$ its apparent position (Fig. 74). Let $\alpha$, $\delta$ be the right ascension and

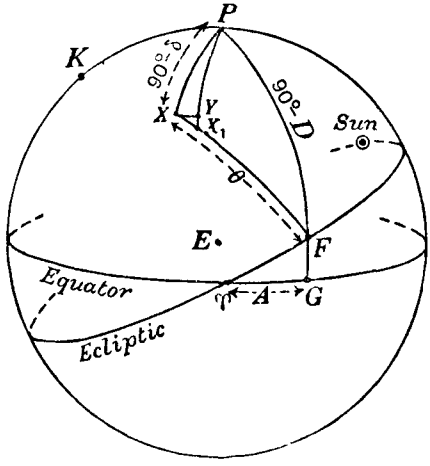



Fig. 74.

declination of $X$, and $\alpha_1$, $\delta_1$ the corresponding co-ordinates of $X_1$. Draw a small circle $XY$ parallel to the equator. Then

$$X\hat{P}Y = \alpha_1 - \alpha \quad \text{and} \quad XY = X\hat{P}Y \sin PX.$$

Let $\Delta\alpha = \alpha_1 - \alpha$ and $\Delta\delta = \delta_1 - \delta$.

Then $\Delta\alpha = XY \operatorname{cosec} PX = XY \sec \delta$.

From the figure, $X_1Y = -\Delta\delta$. Denote $Y\hat{X}X_1$ by $\psi$. Then

$$XY = XX_1 \cos\psi \quad \text{and} \quad X_1Y = XX_1 \sin\psi.$$

Hence by (3), since $XX_1 \equiv \theta - \theta_1$ in the notation of section 104,

$$\Delta\alpha = \kappa \sin\theta \cos\psi \sec\delta \quad \text{.....(15),}$$

$$\Delta\delta = -\kappa \sin\theta \sin\psi \quad \text{.....(16).}$$

Let $A$, $D$ denote the right ascension and declination of $F$ (the point on the ecliptic towards which the earth's motion is directed). In the spherical triangle $PXF$, we have: $PX = 90° - \delta$, $PF = 90° - D$, $X\hat{P}F = A - \alpha$, $XF = \theta$ and $P\hat{X}F = 90° + \psi$.

By formula **B**,

$$\sin XF \sin PXF = \sin PF \sin XPF,$$

from which $\sin\theta \cos\psi = \cos D \sin(A - \alpha)$ .....(17).

By formula **C**,

$$\sin XF \cos PXF = \cos PF \sin PX - \sin PF \cos PX \cos XPF,$$

from which

$$-\sin\theta \sin\psi = \sin D \cos\delta - \cos D \sin\delta \cos(A - \alpha) \quad \text{...(18).}$$

We thus derive from (15)...(18),

$$\Delta\alpha = \kappa \sec\delta \cos D \sin(A - \alpha) \quad \text{.....(19),}$$

$$\Delta\delta = \kappa \sin D \cos\delta - \kappa \cos D \sin\delta \cos(A - \alpha) \quad \text{.....(20).}$$

Now consider the triangle $\Upsilon FG$ in which $PFG$ is the meridian through $F$. We have: $\Upsilon F = \odot - 90°$, $F\hat{\Upsilon}G = \epsilon$ (the obliquity of the ecliptic), $\Upsilon G = A$, $FG = D$ and $F\hat{G}\Upsilon = 90°$. By formulae **A**, **B** and **C**, we derive

$$\left.\begin{aligned} \sin\odot &= \cos A \cos D \\ -\cos\odot \sin\epsilon &= \sin D \\ -\cos\odot \cos\epsilon &= \sin A \cos D \end{aligned}\right\} \quad \text{.....(21).}$$

Now from (19) we have

$$\Delta\alpha = \kappa \sec\delta \,[\cos\alpha . \sin A \cos D - \sin\alpha . \cos A \cos D],$$

so that, using (21), we obtain

$$\Delta\alpha \equiv \alpha_1 - \alpha = -\kappa \sec\delta \,[\cos\alpha \cos\odot \cos\epsilon + \sin\alpha \sin\odot] \quad \text{.....(22).}$$

Similarly we obtain from (20) and (21),

$$\Delta\delta \equiv \delta_1 - \delta = -\kappa \cos\odot \cos\epsilon (\tan\epsilon \cos\delta - \sin\alpha \sin\delta) - \kappa \cos\alpha \sin\delta \sin\odot \quad \text{...(23).}$$

As $\kappa$ is expressed in seconds of arc, the formula for $\Delta\alpha$ or $(\alpha_1 - \alpha)$ gives this quantity also in seconds of arc. It is of course

usual to express right ascension in time-measure and we must then suppose the right of (22) to be divided by 15.

Let

$$\left.\begin{array}{ll} C = -\kappa \cos\epsilon \cos\odot, & D = -\kappa \sin\odot \\ c = \frac{1}{15}\cos\alpha \sec\delta, & c' = \tan\epsilon\cos\delta - \sin\alpha\sin\delta \\ d = \frac{1}{15}\sin\alpha\sec\delta, & d' = \cos\alpha\sin\delta \end{array}\right\} \quad \ldots\ldots(24)$$

Then we have from (22), (23) and (24),

$$\alpha_1 = \alpha + Cc + Dd \quad \ldots\ldots(25),$$

$$\delta_1 = \delta + Cc' + Dd' \quad \ldots\ldots(26).$$

In these equations, $C$ and $D$ depend on the sun's longitude and consequently on the time of the year to which they refer; the values of $C$ and $D$, which are known as *Besselian Day Numbers*, vary rapidly and so are tabulated in the almanacs for every day of the year. The quantities $c$, $c'$, $d$, $d'$ depend only on the co-ordinates of the star and the obliquity of the ecliptic, and can therefore be calculated once for all.

It is to be remarked that (25) and (26) give only the effect of aberration on the star's co-ordinates.

Another method of simplifying the computations is as follows. Let

$$h\cos H = -\kappa\sin\odot\,; \quad h\sin H = -\kappa\cos\epsilon\cos\odot\,;$$

$$i = -\kappa\sin\epsilon\cos\odot\,.$$

Then when $\alpha_1$ and $\alpha$ are expressed in time-measure, equations (22) and (23) become

$$\alpha_1 = \alpha + \tfrac{1}{15}h\sin(H+\alpha)\sec\delta \quad \ldots\ldots(27),$$

and

$$\delta_1 = \delta + i\cos\delta + h\cos(H+\alpha)\sin\delta \quad \ldots\ldots(28).$$

The values of $H$, $h$ and $i$ (the *Independent Day Numbers*) are also tabulated in the almanacs for every day of the year. In most cases they are simpler to use than the Besselian Day Numbers.

**103.** *The elliptic motion of the earth and aberration.*

So far, we have assumed that the earth moves with constant speed in a circular orbit. We consider now the problem in its general aspect. It has been shown in section 66 that the velocity along $ET$ (the tangent at $E$) in an elliptic orbit (Fig. 75) is equivalent to a constant velocity $h/p$ along $EF$ perpendicular to the radius vector $SE$, together with a constant velocity $eh/p$

along $Ef$ at right angles to the major axis $AB$. In the notation of Chapter v, we have $h^2 = n^2a^3p$ and $p = a(1-e^2)$; hence, if we write $n = 2\pi/T$, where $T$ is the period in the orbit, we have

$$\frac{h}{p} = \frac{2\pi a}{T(1-e^2)^{\frac{1}{2}}} \qquad \text{.....(29).}$$

The total displacement, in any co-ordinate, of a star's position due to aberration can be regarded as made up of two displacements, the first due to the constant velocity $h/p$ at right angles to the radius vector, and the second due to the constant velocity $eh/p$ perpendicular to the major axis. We consider these in turn.

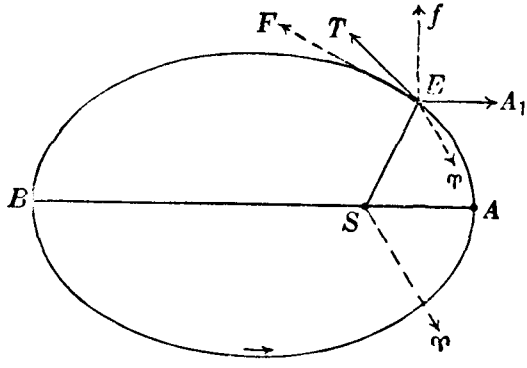



Fig. 75.

Since $EF$ is perpendicular to $SE$, the geocentric longitude of $F$ is the geocentric true longitude of the sun less 90° or $(\odot - 90°)$. We thus have the same geometrical conditions as are represented in Fig. 73. We consider the corresponding displacements in longitude and latitude. If these are denoted by $\Delta\lambda_1$ and $\Delta\beta_1$, they are given directly by (10) and (11), so that

$$\Delta\lambda_1 = -\kappa \sec\beta \cos(\odot - \lambda) \qquad \text{.....(30),}$$

$$\Delta\beta_1 = -\kappa \sin\beta \sin(\odot - \lambda) \qquad \text{.....(31),}$$

in which $\kappa$ (in seconds of arc) is defined after the manner of formula (2) by

$$\kappa = \frac{h}{p} \cdot \frac{\operatorname{cosec} 1''}{c},$$

or, using (29), by

$$\kappa = \frac{2\pi a \operatorname{cosec} 1''}{cT(1-e^2)^{\frac{1}{2}}} \qquad \text{.....(32).}$$

This is the precise definition of the *aberration constant* $\kappa$.

We now consider the aberrational displacements $\Delta\lambda_2$, $\Delta\beta_2$ arising from the constant velocity $eh/p$ perpendicular to the major axis. In Fig. 75 let $S\Upsilon$ be the direction of the vernal equinox; then $\Upsilon\hat{S}A$ is $\varpi$, the longitude of perihelion. Draw $EA_1$ parallel to $SA$; then the geocentric longitude of $A_1$ is $\varpi$. Also $Ef$ is perpendicular to $EA_1$; therefore the geocentric longitude of $f$ is $\varpi + 90°$. In Fig. 73, where $\Upsilon F$ is $\odot - 90°$, we derive the aberrational displacements $\Delta\lambda_1$, $\Delta\beta_1$ corresponding to the velocity $h/p$. It is now seen from Fig. 76, in which $\Upsilon f$ is $\varpi + 90°$, that the formulae for $\Delta\lambda_2$, $\Delta\beta_2$ corresponding to the constant velocity $eh/p$ can be written down from (30) and (31) by substituting $\varpi + 90°$ for $\odot - 90°$, that is, $180° + \varpi$ for $\odot$, and by writing $eh/p$ for $h/p$ or $e\kappa$ for $\kappa$. We thus derive

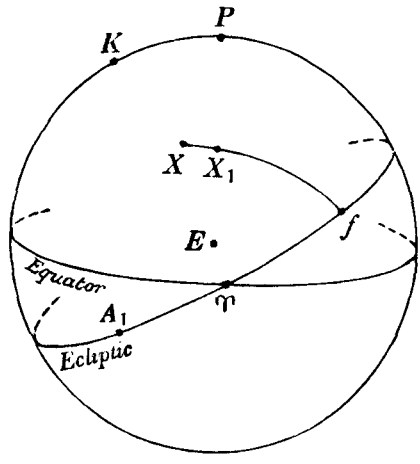


Fig. 76.

$$\Delta\lambda_2 = + e\kappa \sec\beta \cos(\varpi - \lambda) \quad \text{.....(33),}$$

$$\Delta\beta_2 = + e\kappa \sin\beta \sin(\varpi - \lambda) \quad \text{.....(34).}$$

Hence the total aberrational displacements resulting from the motion of the earth in its elliptic orbit are given by

$$\Delta\lambda \equiv \Delta\lambda_1 + \Delta\lambda_2 = - \kappa \sec\beta \cos(\odot - \lambda) + e\kappa \sec\beta \cos(\varpi - \lambda) \quad \text{.....(35),}$$

$$\Delta\beta \equiv \Delta\beta_1 + \Delta\beta_2 = - \kappa \sin\beta \sin(\odot - \lambda) + e\kappa \sin\beta \sin(\varpi - \lambda) \quad \text{.....(36).}$$

Note that the expressions for $\Delta\lambda_2$ and $\Delta\beta_2$, which are called the *E-terms*, are independent of the sun's longitude and so are the same for any given star throughout the year. As the value of the eccentricity $e$ is about 1/60, the value of $e\kappa$ is about $0''{\cdot}3$. Although $\varpi$, $e$, $\lambda$, $\beta$ are slowly varying quantities, the changes in the $E$-terms are so minute that for several hundred years at least they can be regarded strictly as constants.

Since $\Delta\lambda = \lambda_1 - \lambda$, we have

$$\lambda_1 = \lambda + \Delta\lambda_1 + \Delta\lambda_2,$$

with a similar equation in $\beta$. Instead of applying the value of $\Delta\lambda_2$ in this equation, it is more convenient to regard $(\lambda + \Delta\lambda_2)$ as the true longitude of the star; similarly for the true latitude.

In the same way we can derive the aberrational effects, taking into account the ellipticity of the earth's orbit, on the equatorial co-ordinates $\alpha$ and $\delta$. Again we suppose that the $E$-terms, depending on the velocity $eh/p$ perpendicular to the major axis, are incorporated in the true co-ordinates $\alpha$ and $\delta$.

Thus the effective formulae for giving the aberrational displacements are, in ecliptic and equatorial co-ordinates,

$$\lambda_1 = \lambda - \kappa \sec\beta \cos(\odot - \lambda) \quad \text{.....(37)},$$

$$\beta_1 = \beta - \kappa \sin\beta \sin(\odot - \lambda) \quad \text{.....(38)},$$

and

$$\alpha_1 = \alpha + Cc + Dd \quad \text{.....(39)},$$

$$\delta_1 = \delta + Cc' + Dd' \quad \text{.....(40)},$$

where $C$, $c$, $c'$, $D$, $d$ and $d'$ are defined by (24).

**109.** *The measurement of the constant of aberration.*

The accurate determination of $\kappa$ is a practical problem that appears at first sight to be comparatively simple and straightforward. In order to reduce the effects of refraction which, even under the best conditions, may not be known with the certainty essential in this investigation, only stars culminating very near to the zenith are selected for observation. Let $\delta$ be the declination of a star culminating a little south of the zenith on a certain date. For simplicity we shall neglect the effects of precession and nutation, which will be discussed in a later chapter. The apparent declination $\delta_1$ which will include the aberration displacement will be given by (23), which we write in the form

$$\delta_1 = \delta + \kappa x \quad \text{.....(41)},$$

where $x$ is the coefficient of $\kappa$ on the right of (23).

In Fig. 77 let $ZX'$ be the observed zenith distance (denoted by $z$) of a star on the meridian and let $XX'$ represent the refraction $r$. Then $ZX = z + r$. Now $PX = 90° - \delta_1$ and $PZ = 90° - \phi$, where $\phi$ is the latitude, so that

$$ZX = \phi - \delta_1 = \phi - \delta - \kappa x.$$

Hence $$z + r = \phi - \delta - \kappa x$$

or $$\phi = \delta + z + r + \kappa x \qquad \ldots\ldots(42).$$

Suppose that another star culminates at $Y'$, a little north of the zenith, and let $Z$ denote its observed zenith distance and $R$ the refraction. Then $ZY = Z + R$. Also if $D$ is its true declination and $D_1$ its apparent declination (as affected by aberration), then by (23)

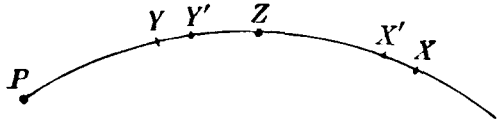



Fig. 77.

$$D_1 = D + \kappa X,$$

where $X$ is the value for this star of the coefficient of $\kappa$ on the right of (23). Now $PY = 90° - D_1$ and $PZ = 90° - \phi$. Hence $ZY = D_1 - \phi$, and we have

$$Z + R = D + \kappa X - \phi$$

or $$\phi = D - Z - R + \kappa X \qquad \ldots\ldots(43).$$

Hence from (42) and (43),

$$2\phi = (\delta + D) + (z - Z) + (r - R) + \kappa (x + X) \quad \ldots(44).$$

In this equation we assume $(\delta + D)$ to be known and $(z - Z)$ is the difference of the measured zenith distances. As the stars culminate close to the zenith, the telescope can remain unaltered in position for the observations of the two stars, the interval between the transits—if the stars have been carefully chosen—being at most a few minutes. Although the meridian circle has been used in this problem it has been superseded by the photographic zenith telescope which records, on a photographic plate, the trails of the stars as they pass across the meridian. The measurement of the distance between the trails of the two stars we are considering gives the value of $(z - Z)$ very accurately. The difference $(r - R)$ of the refractions may be assumed known with the necessary accuracy. The quantity $(x + X)$ can be easily calculated.

About six months later, the observations are repeated. We have, as before, a subscript referring to the new observations,

$$2\phi = (\delta + D) + (z_1 - Z_1) + (r_1 - R_1) + \kappa (x_1 + X_1) \quad \ldots(45).$$

For simplicity we shall assume that the interval is such that the sun's longitude has increased by 180° exactly. By referring to (23) we see that the values of $x_1$ and $X_1$ will now be equal to $-x$ and $-X$ respectively, so that we can write (45) as

$$2\phi = (\delta + D) + (z_1 - Z_1) + (r_1 - R_1) - \kappa(x + X) \quad \ldots(46).$$

Subtracting (44) and (46) we obtain

$$2\kappa(x + X) + (z - Z) - (z_1 - Z_1) + (r - R) - (r_1 - R_1) = 0 \quad \ldots\ldots(47),$$

from which $\kappa$ can be derived.

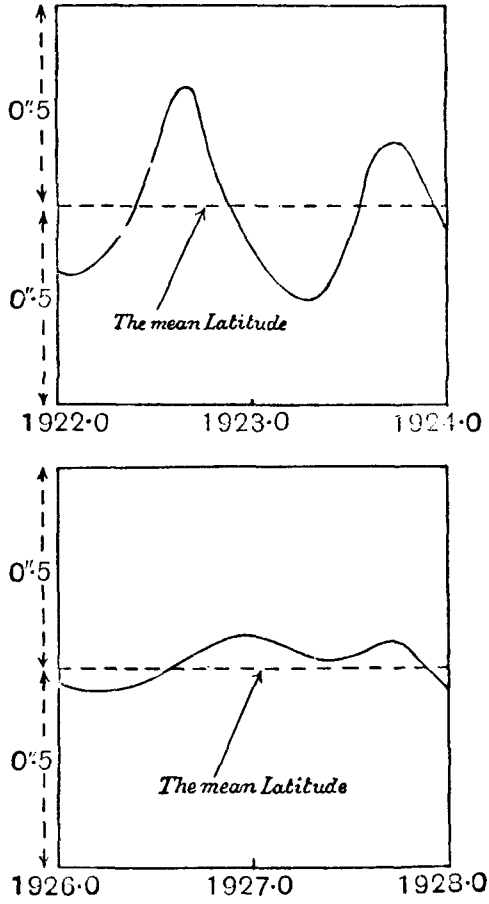



Fig. 78.

This procedure—of eliminating the latitude in (44) and (46) —is, however, unjustified when the object of the investigation is the *precise* determination of $\kappa$. It was discovered by Küstner that the earth's axis of rotation is not quite fixed relatively to the crust, and as the latitude $\phi$ is defined with reference to the axis of rotation, the value of $\phi$ undergoes minute changes of the order of a few tenths of a second of arc. Thus the values of $\phi$ in (44) and (46) must not be presumed to be identical. The determination of the aberrational constant is consequently bound up intricately with the problem of the *variation of latitude*. We shall not prolong the discussion further, but merely recognise that although the measurement of the aberrational constant is straightforward in principle, in practice it is not so, and the constant of aberration is more precisely determined from the theoretical considerations of the next section. Fig. 78 shows the changes in the latitude of Greenwich during 1922–3 and 1926–7.

**110.** *The theoretical value of the aberration constant.*

From (32), $\kappa$ is given by

$$\kappa = \frac{2\pi a \operatorname{cosec} 1''}{cT(1-e^2)^{\frac{1}{2}}} \qquad \ldots\ldots(48),$$

in which $a$ is the semi-major axis of the earth's orbit, $c$ is the velocity of light, $T$ is the sidereal year expressed in mean solar seconds and $e$ is the eccentricity of the earth's orbit. The accepted values of these quantities are:

$$a = 149{,}600{,}000 \text{ kilometres},$$
$$c = 299{,}792{\cdot}5 \text{ kilometres per second},$$
$$T = 31{,}558{,}150 \text{ seconds},$$
$$e = 0{\cdot}01672,$$

from which, by calculation, $\kappa$ is found to be $20''{\cdot}496$. The values of the astronomical quantities quoted above are those adopted by the International Astronomical Union in 1964 (*Trans. IAU*, vol. XIIB, p. 594, 1966). These values are generally used in the *Astronomical Ephemeris*, although the apparent co-ordinates of the sun and the inner planets are based on an older value of $\kappa$, namely $20''{\cdot}47$. The formulae necessary for conversion to the IAU system of constants are, however, provided in the Explanation that follows the main text.

**111.** *Diurnal aberration.*

In addition to the yearly motion of the observer around the sun, there is also the daily motion due to the rotation of the earth about its axis. Let $\rho$ denote the radius of the earth and $\phi$ the latitude of an observatory, which describes in the course of a sidereal day, due to rotation alone, the circumference of the small parallel of latitude whose radius is $\rho \cos \phi$. As there are 86,164 mean solar seconds in a sidereal day the speed concerned is $\frac{2\pi\rho \cos \phi}{86164}$ kilometres per second, where we suppose $\rho$ to be expressed in kilometres. Taking $\rho$ to be 6378 kilometres, the speed is easily found to be $0{\cdot}465 \cos \phi$ kilometres per second. Define $k$ (in seconds of arc) by

$$k = \frac{0{\cdot}465}{c} \cos \phi \operatorname{cosec} 1'' \qquad \ldots\ldots(49),$$

$c$ being the velocity of light in kilometres per second. Then for an observatory in latitude $\phi$, $k$ is the *constant of diurnal aberration*. Inserting the value of $c$ which we have stated previously, we find $k = 0''{\cdot}32 \cos \phi$.

Now the direction of motion is perpendicular to the plane of the observer's meridian and it is eastwards. In Fig. 79, let $X$ be

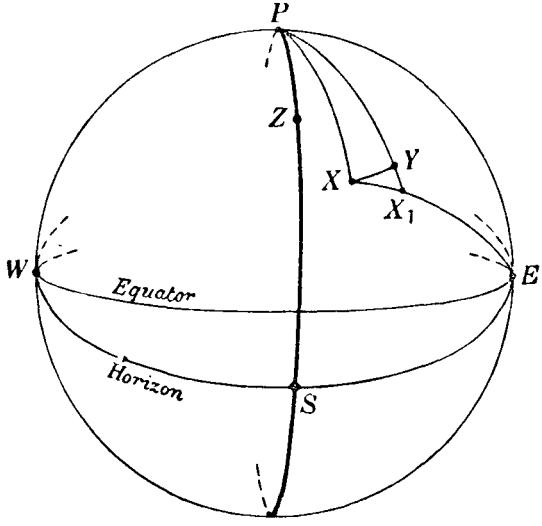



Fig. 79.

the true position of a star at a given moment on the celestial sphere. Owing to diurnal aberration it will be displaced to $X_1$ towards the east point $E$. Denote $XE$ by $\theta$ and $X_1E$ by $\theta_1$. Then by the law of aberration [formula (3)] we have

$$XX_1 \equiv \theta - \theta_1 = k \sin \theta \qquad \ldots\ldots(50).$$

Draw a parallel of declination $XY$. Since hour angle is measured westward from the observer's meridian, the hour angle of $X_1$ is less than that of $X$, so that if $H$ and $H_1$ are the respective hour angles, $H - H_1 = X\hat{P}Y$. Let $\Delta H = H_1 - H$. Now

$$XY = X\hat{P}Y \sin PX = X\hat{P}Y \cos \delta,$$

where $\delta$ is the true declination of the star.

Hence $$XY = -\Delta H \cos \delta.$$

Denote by $\delta_1$ the declination of $X_1$. Then $X_1Y = \delta - \delta_1 = -\Delta\delta$. If $Y\hat{X}X_1 = \psi$, we have

$$XY = XX_1 \cos \psi; \quad X_1Y = XX_1 \sin \psi,$$

that is, using (50),

$$-\Delta H \cos \delta = k \sin \theta \cos \psi; \quad -\Delta\delta = k \sin \theta \sin \psi \ldots(51).$$

In the spherical triangle $PXE$, $PX = 90° - \delta$, $PE = 90°$, $XE = \theta$, $P\hat{X}E = 90° + \psi$ and $X\hat{P}E = H - 270°$. By formulae **B** and **C** we have

$$\sin XE \sin PXE = \sin XPE \sin PE,$$

and

$$\sin XE \cos PXE = \cos PE \sin PX - \sin PE \cos PX \cos XPE.$$

These become, on inserting the values of $PX$, etc.,

$$\sin \theta \cos \psi = \cos H,$$

and $$\sin \theta \sin \psi = -\sin \delta \sin H.$$

Hence from (51) and putting $k = 0''{\cdot}32 \cos \phi$, we obtain

$$\Delta H \equiv H_1 - H = -0''{\cdot}32 \cos \phi \cos H \sec \delta \ldots\ldots(52),$$

and $$\Delta\delta \equiv \delta_1 - \delta = 0''{\cdot}32 \cos \phi \sin H \sin \delta \qquad \ldots\ldots(53).$$

These equations—(52) and (53)—give the effect of diurnal aberration on the hour angle and declination of a star.

When the star is on the meridian it is seen from (53) that the effect on declination is zero.

The displacements due to diurnal aberration are so small that they are generally neglected. An exception occurs in the obser-

vation of stars with the meridian circle. It is seen from Fig. 79, or from formula (52), that diurnal aberration displaces the star eastwards of its true position; the star will therefore transit later than it would if the diurnal aberration were ineffectual. From (52) the transit is delayed by

$$\frac{0{\cdot}32}{15}\cos\phi\sec\delta \quad \text{(seconds of time)}$$

(the hour angle $H$ at the time of transit is $0^h$). For a fixed observatory, this expression is of the form $C\sec\delta$, which is precisely the form due to the effect of collimation error on the time of transit of a star. If $t$ is the time of transit actually observed, the true time (considering only the effect of diurnal aberration) is

$$t - 0^s{\cdot}021\cos\phi\sec\delta \qquad \text{.....(54).}$$

The second term in (54) is the correction mentioned in section 46 and inserted in formula (9) of Chapter IV.

112. *The correction for light-time.*

Let $t$ be the instant at which the sun or moon or a planet or comet is observed. Since light has a finite velocity, the moment of observation is—let us say—$\tau$ seconds after the instant at which the ray, which finally enters the observer's eye, has left the particular object concerned. Let us suppose the object to be a planet. The position observed at time $t$ is affected by annual aberration and, after applying the general formulae for aberration derived in previous sections, we deduce the planet's position as it would be found if the earth were at rest. But this position is not the position of the planet at the time $t$ of the observation; it is the position corresponding to the instant when the ray left the planet, that is to say, it is the position at time $t - \tau$. This correction for the light-time and the annual aberration correction are usually applied together as they have very similar origins. The former arises from the planet's velocity and the latter from the earth's, and only the relative velocity of the two bodies is significant. When the effects are combined, their combination is called *planetary precession*. The resulting correction is very simple. We merely add to the observed right ascension and declination quantities $\tau\Delta\alpha$ and $\tau\Delta\delta$, where $\Delta\alpha$ and $\Delta\delta$ are the respective co-ordinates' rates of change per second.

## EXERCISES

**1.** Find the positions of stars which (i) are unaffected by aberration, (ii) are affected only in latitude.

**2.** The angular distance between two stars which have the same latitude $\beta$ is $\theta$ and the mean of their longitudes is $\lambda$; show that the increase in $\theta$ due to aberration is $2\kappa \tan \frac{1}{2}\theta \sin(\lambda - \odot)(\cos^2\beta - \sin^2\frac{1}{2}\theta)^{\frac{1}{2}}$. [*M.T.*]

**3.** Prove that the angular distance between two stars whose ecliptic co-ordinates are $(\lambda, \beta)$ and $(\lambda_1, \beta_1)$ is unaltered when the sun's longitude $\odot$ is given by $\cos\beta \sin(\odot - \lambda) + \cos\beta_1 \sin(\odot - \lambda_1) = 0$. [*Ball.*]

**4.** Show that, in latitude $\phi$, a star of declination $\delta$ will, owing to diurnal aberration, appear to move in an ellipse whose semi-axes are $m\cos\phi$ and $m\cos\phi\sin\delta$, where $m$ is the ratio of the earth's circumference to the distance described by light in one day. [*Coll. Exam.*]

**5.** Prove that, when the aberration in declination has its greatest numerical value, the arcs on the celestial sphere joining the star to the sun and to the pole of the equator are at right angles. [*Ball.*]

**6.** If $\kappa$ is the constant of aberration, $c$ the velocity of light, $G$ the constant of gravitation, $M$ the mass of the sun, and $l$ the semi-latus rectum of the earth's orbit, show that $GM = l\kappa^2c^2$. [*M.T.* 1922.]

**7.** Show that at any place and at any time there is a position of a star such that the aberrational effect is equal and opposite to that of refraction.

Show also that at midnight on the shortest day the zenith distance of this star is given by an equation of the form

$$\sin^2 z + \lambda \sin z = 1,$$

the correction for refraction being assumed proportional to the tangent of the zenith distance, and the earth's orbit to be circular. [*M.T.* 1900.]

**8.** If two planets revolve about the sun in circular orbits of radii $a$ and $b$, show that the aberration of one as seen from the other is greater at opposition than at conjunction by an amount equal to

$$\frac{\sqrt{a} - \sqrt{b}}{\sqrt{a} + \sqrt{b}}.$$ [*M.T.*]

**9.** If $u$ and $v$ are the equatorial components of the earth's velocity in the direction of $\Upsilon$ and perpendicular to this direction respectively, show that the Besselian Day Numbers may be defined as

$$C = v/c - e\kappa\cos\varpi\cos\epsilon,$$
$$D = -u/c - e\kappa\sin\varpi.$$

## CHAPTER IX

# PARALLAX

### 113. *Introductory.*

The method of determining the distance of a heavenly body is, in principle, essentially similar to that adopted in land surveys. If a surveyor wishes to determine the distance of an object $A$ from a point $B$, he first measures a base-line $BC$ and with a suitable instrument, such as a theodolite, he observes the angles $ABC$ and $ACB$. The solution of the plane triangle gives him the distance of the object $A$ from $B$. As regards the astronomical survey of the solar system, we shall at first suppose the necessary base-line to be defined by the straight line joining two widely separated points on the earth's surface. It is thus necessary to consider more particularly the dimensions and figure of the earth. The maximum base-line obtainable in this way is not quite 8000 miles, corresponding to the earth's diameter; as this is wholly inadequate in the investigation of the distances of even the nearest stars, a different choice of base-line has to be made, as we shall see later, in dealing with stellar distances.

### 114. *The geoid.*

The approximate figure of the earth is called the *geoid*, and for astronomical purposes it is sufficient to consider it a spheroid

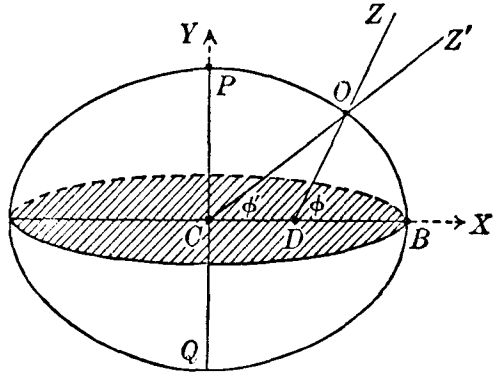



Fig. 80.

of revolution, its minor axis being coincident with the diameter joining the north and south poles. A terrestrial meridian, such as $POBQ$ in Fig. 80, is then an ellipse, the major axis $CB$ being in the plane of the equator. If $a$ denotes $CB$, the earth's equator is a circle of radius $a$. Denote the minor axis $PC$ by $b$ and the eccentricity of a meridian section by $e$. Then

$$b^2 = a^2(1 - e^2) \qquad \ldots\ldots(1).$$

Referring the ellipse $POBQ$ to axes $CX$ and $CY$, we have its equation in the well-known form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \ldots\ldots(2).$$

**115.** *Astronomical and geocentric latitude.*

Consider an observer at $O$ (Fig. 80). If we neglect any local gravitational irregularities, in the neighbourhood of $O$, which affect slightly the direction given by a plumb-line, the direction of the observer's zenith is defined to be perpendicular to the surface of the geoid at $O$; that is to say, it is in the direction $DOZ$, this line being the normal at $O$ to the ellipse $POBQ$. $Z$ is said to be the *astronomical zenith* of the observer; it is from this zenith that zenith distances are measured, for example, with the meridian circle. The angle which $DOZ$ makes with the equatorial radius $CB$ is the astronomical latitude of the observer; in Fig. 80, it is the angle $ODB$, denoted by $\phi$. Join $C$ and $O$ and produce $CO$ to $Z'$; then $Z'$ is said to be the observer's geocentric zenith and the angle $OCB$, denoted by $\phi'$, is called the *geocentric latitude* of $O$. The angle $ZOZ'$ between the directions of the astronomical and geocentric zeniths $Z$ and $Z'$ is called the *angle of the vertical*, usually denoted by $v$. We now investigate the relationship between $v$ and $\phi$.

If $x$, $y$ are the rectangular co-ordinates of $O$ with respect to the axes $CX$ and $CY$, we have $y/x = \tan\phi'$. Also $\tan\phi$ is the slope of the normal $DOZ$, and it is given by a well-known formula

$$\tan\phi = \frac{y}{x}\frac{a^2}{b^2}.$$

Hence

$$\frac{y}{x} = \tan\phi' = \frac{b^2}{a^2}\tan\phi \qquad \ldots\ldots(3).$$

On substituting in (2) the value of $y$, given by (3) in terms of $x$ and $\phi$, we obtain

$$\frac{x^2}{a^2} + x^2 \frac{b^2}{a^4} \tan^2 \phi = 1,$$

from which
$$x^2 = \frac{a^4 \cos^2 \phi}{a^2 \cos^2 \phi + b^2 \sin^2 \phi} \qquad \ldots\ldots(4),$$

or, replacing $b^2$ by $a^2(1 - e^2)$,

$$x^2 = \frac{a^2 \cos^2 \phi}{1 - e^2 \sin^2 \phi} \qquad \ldots\ldots(5).$$

Similarly
$$y^2 = \frac{a^2 (1 - e^2)^2 \sin^2 \phi}{1 - e^2 \sin^2 \phi} \qquad \ldots\ldots(6).$$

Let $\rho$ denote the radius vector $CO$; then

$$x = \rho \cos \phi'; \quad y = \rho \sin \phi'.$$

Hence, by (5) and (6),

$$x = \rho \cos \phi' = \frac{a \cos \phi}{(1 - e^2 \sin^2 \phi)^{\frac{1}{2}}} \qquad \ldots\ldots(7),$$

$$y = \rho \sin \phi' = \frac{a (1 - e^2) \sin \phi}{(1 - e^2 \sin^2 \phi)^{\frac{1}{2}}} \qquad \ldots\ldots(8),$$

from which
$$\rho^2 = \frac{a^2 [1 - (2e^2 - e^4) \sin^2 \phi]}{1 - e^2 \sin^2 \phi} \qquad \ldots\ldots(9).$$

Equations (7) and (8) are written in short as $x = aC \cos \phi$, $y = aS \sin \phi$, where $C$ and $S$ are clearly functions of $\phi$ only. They are tabulated in the *Astronomical Ephemeris*.

Multiply (7) by $\sin \phi$ and (8) by $\cos \phi$ and subtract; then, since $\phi - \phi' = v$, we obtain

$$\rho \sin v = \frac{a e^2 \sin \phi \cos \phi}{(1 - e^2 \sin^2 \phi)^{\frac{1}{2}}} \qquad \ldots\ldots(10).$$

Again, multiply (7) by $\cos \phi$ and (8) by $\sin \phi$ and add. Then

$$\rho \cos v = a (1 - e^2 \sin^2 \phi)^{\frac{1}{2}} \qquad \ldots\ldots(11).$$

Hence, from (10) and (11),

$$\tan v = \frac{e^2 \sin 2\phi}{2 (1 - e^2 \sin^2 \phi)}$$

$$= \frac{e^2 \sin 2\phi}{2 - e^2 + e^2 \cos 2\phi} \qquad \ldots\ldots(12).$$

Denoting $e^2/(2-e^2)$ by $m$, formula (12) becomes

$$\tan v = \frac{m \sin 2\phi}{1 + m \cos 2\phi} \qquad \text{.....(13).}$$

Now $e$ is about 0·08, so that $m$ is about 0·003. We can, in consequence, obtain a rapidly converging series for $v$ in the following manner. From (13) we obtain, $i$ denoting the square root of $-1$,

$$\frac{1 + i \tan v}{1 - i \tan v} = \frac{1 + m(\cos 2\phi + i \sin 2\phi)}{1 + m(\cos 2\phi - i \sin 2\phi)},$$

or
$$e^{2iv} = \frac{1 + me^{2i\phi}}{1 + me^{-2i\phi}},$$

whence $2iv = \log(1 + me^{2i\phi}) - \log(1 + me^{-2i\phi})$.

Using the logarithmic expansion, we have

$$2iv = m(e^{2i\phi} - e^{-2i\phi}) - \frac{m^2}{2}(e^{4i\phi} - e^{-4i\phi}) + \ldots,$$

from which
$$v = m \sin 2\phi - \frac{m^2}{2} \sin 4\phi + \ldots.$$

Expressing $v$ in seconds of arc, we obtain

$$\phi - \phi' \equiv v = m \frac{\sin 2\phi}{\sin 1''} - \frac{1}{2} \frac{m^2 \sin 4\phi}{\sin 1''} + \frac{1}{3} \frac{m^3 \sin 6\phi}{\sin 1''} - \ldots$$

.....(14).

When $\phi$ is given and $e$ (or $m$) is known, this formula enables the calculation of the angle of the vertical, and hence of the geocentric latitude $\phi'$, to be made simply and easily.

**116.** *The figure of the earth.*

In geodesy it is usual to deal with the *flattening* or ellipticity (denoted by $f$) of the terrestrial meridian; it is related to the eccentricity by

$$f = \frac{a-b}{a} = 1 - (1 - e^2)^{\frac{1}{2}} \qquad \text{.....(15).}$$

The dimensions of the geoid, adopted by the IAU in 1964, are:

$$a = 6378{\cdot}160 \text{ km},$$
$$f = 1/298{\cdot}25.$$

Then
$$b = 6356{\cdot}7747 \text{ km},$$
$$e^2 = 0{\cdot}00669454$$

and
$$m = 0{\cdot}00335851.$$

Hence
$$\frac{m}{\sin 1''} = 692''{\cdot}74 \text{ and } \frac{m^2}{2 \sin 1''} = 1''{\cdot}16.$$

Keeping only the first two terms in (14), we derive the formula for the angle of the vertical as

$$v \equiv \phi - \phi' = 692''{\cdot}74 \sin 2\phi - 1''{\cdot}16 \sin 4\phi \quad \ldots(16).$$

By means of this formula we can calculate the geocentric latitude $\phi'$ when the astronomical latitude $\phi$ is given. For example, the value of $\phi$ for Greenwich is 51° 28′ 38″·2, and

$$\phi - \phi' = 675''{\cdot}09 + 0''{\cdot}51,$$

whence

$$\phi' = 51° \ 17' \ 22''{\cdot}6.$$

**117.** *Geocentric parallax.*

In Fig. 81, let $O$ be the observer, $Z'$ his geocentric zenith and $M$ the centre of the moon or sun or planet. Let $z'$ and $z$ denote the angles $Z'OM$ and $Z'CM$ and $p$ the angle $OMC$. Let $\rho$ denote

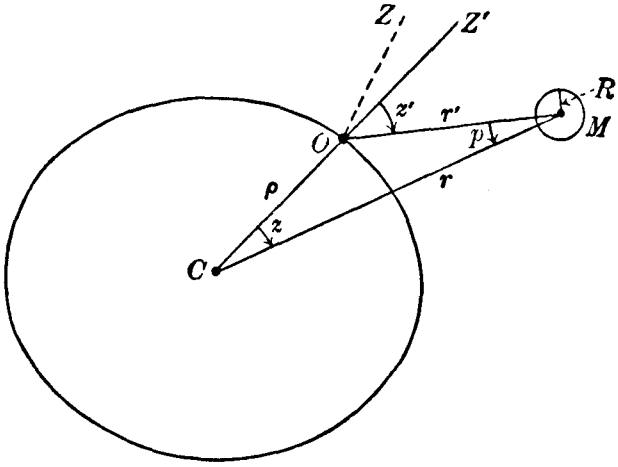



Fig. 81

the distance of $O$ from the earth's centre $C$ and $r$ the geocentric distance $CM$ of the body $M$. Then

$$z' = z + p \quad \ldots\ldots(17),$$

$$\sin p = \frac{\rho}{r} \sin z' \quad \ldots\ldots(18).$$

The angle $p$ is called the *parallax* of the body $M$; it is the angle between the direction of the body as seen from $O$ and the direction from the earth's centre regarded as the standard view-

point. If $p$ can be determined from observations, the distance $r$ of the body can then be derived from (18), in which $\rho$ may be presumed to be known and $z'$ derived from zenith observations. The greater the value of $r$, the smaller is the angle of parallax $p$. As $z'$ is greater than $z$, the effect of parallax is to increase the zenith distance of the body as observed at $O$, and this displacement takes place in the plane defined by $CZ'$ and $CM$.

When $z'$ is 90°, the parallax $p'$ is given by $\sin p' = \rho/r$. In this case, $p'$ is called the *horizontal parallax* for the observer at $O$. If the observer is on the equator and $M$ is on the horizon, its parallax $P$ will then be given by

$$\sin P = \frac{a}{r} \qquad \text{.....(19).}$$

$P$ is called the *equatorial horizontal parallax.*

Owing to the effect of elliptic motion, the geocentric distance $r$ of the moon or sun will vary during the orbital period, and if $r_0$ denotes the mean geocentric distance, we obtain from (19) the *mean equatorial horizontal parallax* $P_0$ given by

$$\sin P_0 = \frac{a}{r_0} \qquad \text{.....(20).}$$

Writing (18) in the form

$$\sin p = \frac{\rho}{a} \cdot \frac{a}{r_0} \cdot \frac{r_0}{r} \sin z',$$

we have

$$\sin p = \left(\frac{\rho}{a}\right)\left(\frac{r_0}{r}\right) \sin P_0 \sin z' \qquad \text{.....(21),}$$

in which it will be sufficient to regard $\rho/a$ and $r_0/r$ as known, the former by formula (9) and the latter from the circumstances of the orbital motion.

**118.** *The parallax of the moon.*

We shall now indicate how the parallax of the moon can be determined from observation. We suppose that there are two observatories $O$ and $O_1$ (Fig. 82) on the same terrestrial meridian, one in high northern latitude and the other in high southern latitude. This choice is made to ensure that the base-line determined by $O$ and $O_1$ is as large as possible. Let us assume that the meridian zenith distances of the moon's centre are measured on the same day with meridian circles at $O$ and $O_1$. These zenith

distances are of course measured with reference to the astronomical zeniths $Z$ and $Z_1$ at $O$ and $O_1$ respectively (Fig. 82). After the corrections for refraction have been made, we shall denote these results by $\zeta$ and $\zeta_1$. Thus $Z\hat{O}M = \zeta$ and $Z_1\hat{O}_1M = \zeta_1$. If $v$ and $v_1$ denote the angles of the vertical at $O$ and $O_1$ and $z'$ and $z_1'$ the angles $Z'OM$ and $Z_1'O_1M$ ($Z'$ and $Z_1'$ are the geocentric zeniths at $O$ and $O_1$), we have

$$z' = \zeta - v \text{ and } z_1' = \zeta_1 - v_1.$$

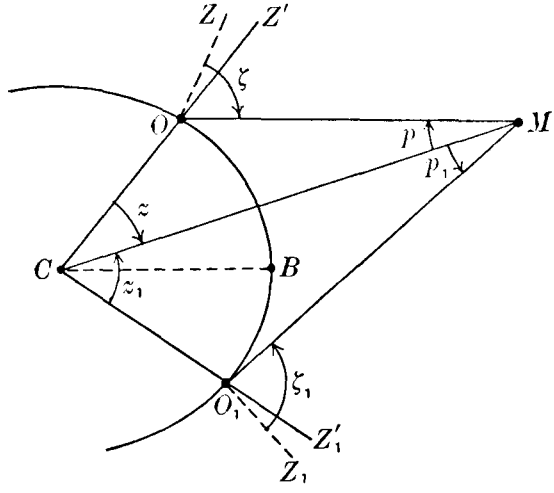



Fig. 82.

Denoting the angles of parallax for $O$ and $O_1$ by $p$ and $p_1$ and the angles $OCM$ and $O_1CM$ by $z$ and $z_1$, we have

$$p = z' - z \text{ and } p_1 = z_1' - z_1,$$

and therefore

$$p + p_1 = \zeta + \zeta_1 - (v + v_1) - (z + z_1) \quad \text{.....(22).}$$

But $z + z_1 = O\hat{C}O_1 = O\hat{C}B + O_1\hat{C}B$, where $CB$ is the equatorial radius in the meridian of $O$ and $O_1$. But $OCB$ and $O_1CB$ are respectively the geocentric latitudes of $O$ and $O_1$, which we denote by $\phi'$ and $\phi_1'$. Hence

$$z + z_1 = \phi' + \phi_1'.$$

Also if $\phi$ and $\phi_1$ are the astronomical latitudes of $O$ and $O_1$, $\phi = \phi' + v$ and $\phi_1 = \phi_1' + v_1$. Hence (22) becomes

$$p + p_1 = \zeta + \zeta_1 - \phi - \phi_1 \quad \text{.....(23).}$$

The quantities on the right are now all known and we shall write

$$p + p_1 = \theta \qquad \text{.....(24).}$$

Now

$$\sin p = \frac{\rho}{r} \sin (\zeta - v) \qquad \text{.....(25),}$$

and

$$\sin p_1 = \frac{\rho_1}{r} \sin (\zeta_1 - v_1) \qquad \text{.....(26).}$$

By means of (24), the equation (26) can be written

$$\sin \theta \cos p = \cos \theta \sin p + \frac{\rho_1}{r} \sin (\zeta_1 - v_1),$$

or, using (25),

$$\sin \theta \cos p = \frac{\rho}{r} \cos \theta \sin (\zeta - v) + \frac{\rho_1}{r} \sin (\zeta_1 - v_1) \quad (27).$$

Dividing (25) by (27), we eliminate $r$ and obtain

$$\tan p = \frac{\rho \sin \theta \sin (\zeta - v)}{\rho \cos \theta \sin (\zeta - v) + \rho_1 \sin (\zeta_1 - v_1)} \qquad \text{...(28).}$$

All the quantities on the right of (28) are known and this equation thus determines $p$.

We now write (25) in the form

$$\sin p = \left(\frac{\rho}{a}\right) \left(\frac{r_0}{r}\right) \sin P_0 \sin (\zeta - v) \qquad \text{.....(29),}$$

in which $p$, $\rho/a$, $r_0/r$, $\zeta - v$ are all presumed to be known. We thus determine from (29) the mean equatorial horizontal parallax $P_0$ of the moon.

It has been assumed that the observations at $O$ and $O_1$ refer to the moon's centre. In practice, however, it is the zenith distance of the moon's upper or lower limb, or of a small crater or mountain peak that is measured. The adjustments to the procedure outlined above are, however, slight and the details need not be more fully described here. Also it has been assumed that $O$ and $O_1$ are on the same meridian and clearly it is unlikely that two fixed observatories can be found to satisfy this assumption exactly. The Royal Observatories at Greenwich and the Cape of Good Hope have collaborated in the past in the investigation of the moon's parallax, and as they differ in longitude by $1^h\ 13^m\ 55^s$, the effect of the change in the moon's declination during the interval between the meridian transits at these two observatories has to be taken into account.

The value of the moon's mean equatorial horizontal parallax adopted by the IAU and used in the *Astronomical Ephemeris* is 3422″·608, and this corresponds to a mean distance of 384400 km. As a method of determining the moon's distance with the highest accuracy, the above considerations are no longer relevant, as this can now be determined from the measured time lag of radar and laser echoes from the lunar surface. The first radar echoes were detected from the moon in 1946, and accurate radar distances became available about ten years later. The laser echoes have been made possible by the reflectors set up on the surface of the moon by the Apollo astronauts.

**119.** *Semi-diameter.*

In Fig. 81, let the linear radius of the moon be $R$ miles and let $r$, $r'$ denote the distances, in the same unit, of the moon's centre from $C$ and $O$. Let $S$ denote the angle subtended by the moon's radius at $C$ and $s$ that subtended at $O$. Then we have $\sin S = R/r$ and $\sin s = R/r'$. The first can be written

$$\sin S = \frac{R}{a} \cdot \frac{a}{r},$$

where $a$ is the earth's equatorial radius. Since $a/r = \sin P$, we have

$$\sin S = \frac{R}{a} \sin P \qquad \text{.....(30).}$$

If $S_0$ denotes the angle subtended at $C$ by the moon's radius when the moon is at its mean distance $r_0$, we have

$$\sin S_0 = \frac{R}{a} \sin P_0 \qquad \text{.....(31);}$$

$S_0$ is called the mean semi-diameter of the moon.

Also
$$\sin s = \frac{R}{r'} = \frac{R}{r} \cdot \frac{\sin z'}{\sin z},$$

or
$$\sin s = \sin S \frac{\sin z'}{\sin z} \qquad \text{.....(32).}$$

$s$ can be measured, and as $z'$ and $z$ can be assumed to be known, equation (32) determines $S$. From (30) and (31) $\dfrac{\sin S}{\sin S_0} = \dfrac{\sin P}{\sin P_0}$ and assuming $P_0$ and $P$ known we determine $S_0$. By (30) or (31) the moon's radius $R$ is derived; it is found to be 1738 kms or

1080 miles. Thus the moon's radius is about one-quarter that of the earth. The value of $S_0$ is 15′ 32″·6. The semi-diameters of the sun and planets are defined in a similar manner.

**120.** *Parallax in right ascension and declination.*

We now investigate the effect of parallax on the right ascension and declination of the moon with respect to an observer at $O$. We shall develop rigorous formulae which are only necessary in the case of the moon and artificial satellites, for which $P$ can be very large indeed. For the sun and other bodies in the solar system the parallax is a small angle, and consequently the general formulae can be greatly modified and simplified in this case.

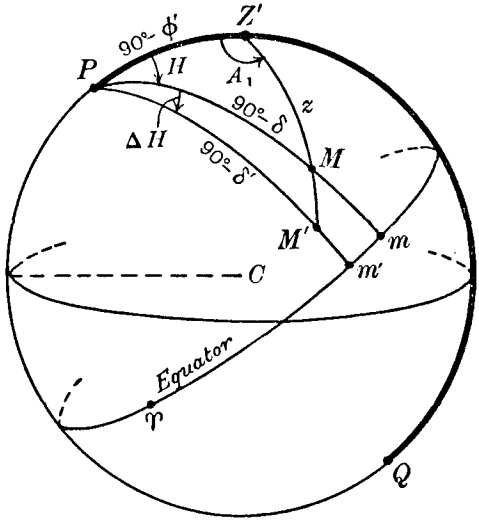



Fig. 83.

Consider, in Fig. 83, the celestial sphere centred at $C$ (the earth's centre); $P$ is the north pole, $Z'$ is the observer's geocentric zenith and $PZ' = 90° - \phi'$. Let $M$ be the position of the body on the celestial sphere as viewed from $C$; then $Z'M = z$. Produce the great circle arc $Z'M$ to $M'$ so that $Z'M' = z'$, $z'$ being the zenith distance of $M$ with respect to the observer at $O$. Then, since $z' = z + p$, we have $MM' = p$, the angle of parallax corresponding to the observation made at $O$. Let

$\alpha$, $\delta$ be the right ascension and declination of the point $M$ on the celestial sphere and $\alpha'$, $\delta'$ the corresponding co-ordinates of $M'$. Then $PM = 90° - \delta$; $PM' = 90° - \delta'$. Let the hour angles of $M$ and $M'$ be $H$ and $H'$ respectively. We shall write $H' - H = \Delta H$. Now, from the figure, $H' - H = m'm$, and if $\Upsilon$ is the vernal equinox $\alpha' = \Upsilon m'$ and $\alpha = \Upsilon m$ so that, if we write $\alpha' - \alpha = \Delta\alpha$, then $\Delta\alpha = -m'm$ and therefore $\Delta H = -\Delta\alpha$. Let $A_1$ denote the angle $PZ'M$.

We shall now investigate the problem of expressing the co-ordinates $\alpha'$ (or $H'$) and $\delta'$ in terms of $\alpha$ (or $H$) and $\delta$.

From the triangle $PZ'M$ we have, using formula **D**,

$$\sin\phi' \cos A_1 = \cos\phi' \cot z - \sin A_1 \cot H \quad \ldots\ldots(33),$$

and from the triangle $PZ'M'$,

$$\sin\phi' \cos A_1 = \cos\phi' \cot z' - \sin A_1 \cot H' \ldots\ldots(34).$$

Subtracting (33) from (34), we obtain

$$\cos\phi' (\cot z - \cot z') = \sin A_1 (\cot H - \cot H'),$$

or, since $z' - z = p$, and $H' - H = \Delta H$,

$$\frac{\cos\phi' \sin p}{\sin z \sin z'} = \frac{\sin A_1 \sin \Delta H}{\sin H \sin H'}.$$

But by (18), $\sin p = \frac{\rho}{r} \sin z'$ and, by the sine-formula **B**,

$$\sin A_1 \sin z = \sin H \cos\delta.$$

Hence
$$\frac{\rho}{r} \cos\phi' = \frac{\cos\delta \sin \Delta H}{\sin (H + \Delta H)},$$

from which we obtain

$$\tan \Delta H \equiv -\tan \Delta\alpha = \frac{\rho}{r} \frac{\sin H \cos\phi'}{\cos\delta - \frac{\rho}{r} \cos H \cos\phi'} \quad \ldots\ldots(35).$$

This is a rigorous formula which enables us to calculate $\Delta H$ or $\Delta\alpha$ (the effect of parallax on right ascension) when $\rho/r$, $\phi'$ and the geocentric co-ordinates $H$ and $\delta$ are known. We then obtain $H'$ or $\alpha'$. It is only when $\rho/r$ is not a small quantity that this exact formula must be used.

We now find the general formula to give $\Delta\delta$, the effect of parallax on declination ($\Delta\delta = \delta' - \delta$).

By the cosine-formula **A**, from the triangles $PZ'M$ and $PZ'M'$,

$$\sin\delta = \sin\phi'\cos z + \cos\phi'\sin z\cos A_1,$$
$$\sin\delta' = \sin\phi'\cos z' + \cos\phi'\sin z'\cos A_1,$$

whence, eliminating $\cos A_1$, we obtain

$$\sin\delta\sin z' - \sin\delta'\sin z = \sin\phi'\sin p$$
$$= \frac{\rho}{r}\sin\phi'\sin z' \text{ by (18)},$$

which can be written

$$\sin\delta'\frac{\sin z}{\sin z'} = \sin\delta - \frac{\rho}{r}\sin\phi' \qquad \text{.....(36).}$$

Again, by **C**, we have

$$\cos\delta\cos H = \cos\phi'\cos z - \sin\phi'\sin z\cos A_1,$$
$$\cos\delta'\cos H' = \cos\phi'\cos z' - \sin\phi'\sin z'\cos A_1,$$

and, by eliminating $\cos A_1$ from these equations and using (18),

$$\cos\delta\cos H\sin z' - \cos\delta'\cos H'\sin z = \frac{\rho}{r}\cos\phi'\sin z',$$

whence $$\cos\delta'\cos H'\frac{\sin z}{\sin z'} = \cos\delta\cos H - \frac{\rho}{r}\cos\phi' \text{.....(37).}$$

Dividing (36) by (37), we have

$$\frac{\tan\delta'}{\cos H'} = \frac{\sin\delta - \frac{\rho}{r}\sin\phi'}{\cos\delta\cos H - \frac{\rho}{r}\cos\phi'}.$$

Writing $\delta' = \delta + \Delta\delta$ and $T = \tan\Delta\delta$, we obtain

$$\frac{\tan\delta + T}{1 - T\tan\delta} = \frac{\cos H'\left(\sin\delta - \frac{\rho}{r}\sin\phi'\right)}{\cos\delta\cos H - \frac{\rho}{r}\cos\phi'} \qquad \text{.....(38).}$$

$\cos H'$ occurs on the right of (38), and since $H' = H + \Delta H$ and $\Delta H$ is given by (35), we can then calculate $T$ from (38). We thus obtain $\Delta\delta$ or $\delta' - \delta$, and hence $\delta'$ is deduced.

121. *Parallax in zenith distance and azimuth.*

We now consider briefly the effect of parallax on the zenith distance and azimuth of the moon or the sun when the body concerned is not on the meridian. In Fig. 84, let $Z$ and $Z'$ be the astronomical and geocentric zeniths of an observer at $O$, the

centre of the celestial sphere being the earth's centre $C$. As before, let $M$ and $M'$ be the positions of the body on the celestial sphere as viewed from $C$ and $O$ respectively; $Z'M = z$, $Z'M' = z' = z + p$. Now observations of zenith distance and azimuth are made with reference to the astronomical zenith $Z$ and so we can regard $ZM'$ and $PZM'$ as the observed quantities. Denote these by $\zeta$ and $A$ respectively. Now $ZZ'$ is the angle of the vertical $v$, which we assume to be known, and $Z'\hat{Z}M' = 180° - A$. Formula **A** then enables us to express $z'$ in terms of $v$, $A$ and $\zeta$. Also $\sin p = (\rho/r) \sin z'$, from which $p$ and then $z\ (\equiv z' - p)$ may be derived if $\rho/r$ is known. Thus $z$ can be expressed in terms of $v$, $\zeta$, $A$ and $\rho/r$.

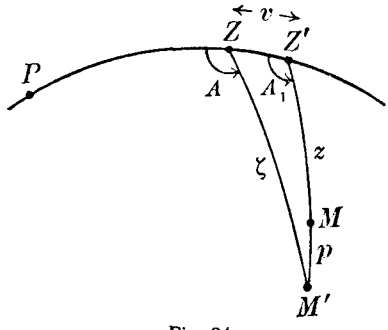


Fig. 84.

Let $A_1$ denote the geocentric azimuth $PZ'M$. By formula **D**, $A_1$ can be expressed in terms of $v$, $\zeta$ and $A$.

It is clear that if $v$, $z$, $A_1$, $\rho/r$ are known, $\zeta$ and $A$ can be derived by an analogous process.

The actual formulae giving $(z - \zeta)$ and $(A_1 - A)$, which can be derived from the procedure outlined above, are not of very great practical importance, and it is sufficient here to draw attention to the principles involved.

### 122. *The solar parallax.*

The method which we have described for the determination of the moon's parallax cannot be applied with sufficient accuracy to the direct measurement of the sun's parallax. The principle of the traditional method of deriving the sun's distance is based on the measurement of the distance of an exterior planet, such as Mars or Eros, and on the application of Kepler's third law. If $a$ (semi-major axis), $T$ (orbital period) and $m$ (mass) refer to the earth then, by formula (18), p. 102, $M$ denoting the sun's mass,

$$\frac{4\pi^2 a^3}{T^2} = G(M + m),$$

and if $a_1$, $T_1$, $m_1$ refer to Mars or Eros,

$$\frac{4\pi^2 a_1^3}{T_1^2} = G(M + m_1).$$

We can safely neglect $m$ and $m_1$ in comparison with $M$ and consequently we have $a^3 : a_1^3 = T^2 : T_1^2$.

Now the periods of orbital revolution are known very accurately; consequently the ratio $a : a_1$ can be determined very accurately. This can be expressed more generally by the statement that the relative dimensions of the planetary orbits are known with high precision. If the distance (say, in miles) of a planet from the earth can be accurately measured at a particular time, the scale of the planetary system can then be deduced; in particular, the sun's distance from the earth can then be found in miles or in terms of the earth's equatorial radius as the unit.

Consider the earth's orbit and the highly elliptical orbit of the minor planet Eros (Fig. 85). If the earth and Eros are simultaneously at $E$ and $M$ respectively (the planet is then in opposition), it is clear that conditions are favourable for measuring the distance $EM$, for then the planets are at their minimum distance apart. The opposition shown by the configuration $SE_1M_1$ is evidently the most unsatisfactory. At the most favourable opposition, when Eros makes its closest approach to the earth, the distance between the two bodies is about 14 million miles, a very much smaller distance to be measured than the distance of the earth from the sun, which is about 93 million miles. A favourable opportunity of measuring the distance of Eros occurred in 1900–1 and in 1930–1 the circumstances were again favourable.

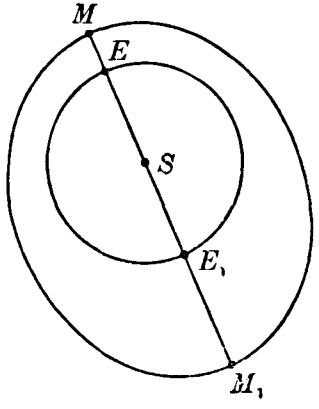


Fig. 85.

In section 120, we obtained general formulae for the effect of parallax on the co-ordinates of a heavenly body. It was only necessary to use such rigorous formulae in connection with the moon because, as we have seen, the mean horizontal parallax

is about 57′. For a planet such as Eros the value of the horizontal parallax may be no more than about 30″; it follows that for such an object $\rho/r$ is a very small quantity (it is of the order of 0·0002), and therefore we can safely neglect squares and higher powers of $\rho/r$ in the formulae mentioned. In this way, we can reduce (35), for example, to the form

$$\Delta\alpha = -\frac{\rho}{r}\sin H\cos\phi'\sec\delta.$$

We shall now derive this formula for $\Delta\alpha$ and the corresponding one for $\Delta\delta$ by another method. In Fig. 86 let $M$ be the position of the planet on the celestial sphere as viewed from the earth's centre $C$ and $M'$ the position as seen by an observer at $O$ whose geocentric zenith is $Z'$. Let $p$ denote the displacement $MM'$ due to parallax. Let $\alpha$, $\delta$ and $\alpha'$, $\delta'$ be the equatorial co-ordinates of $M$ and $M'$ respectively, and $H$, $H'$ the corresponding hour angles. Let $ML$ be a small circle arc parallel to the equator. Denote $P\hat{M}'Z'$ by $\eta$. Then as $p$ is very small we have, from the sensibly plane triangle $LMM'$,

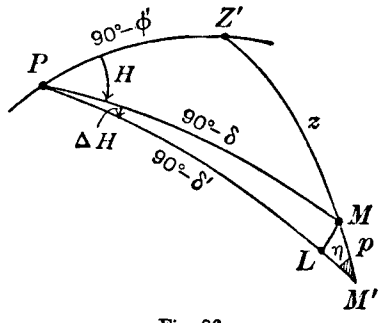


Fig. 86.

$$LM = p\sin\eta \quad\text{and}\quad LM' = p\cos\eta.$$

Now from (18) we have, writing $p$ for $\sin p$,

$$p = \frac{\rho}{r}\sin z' \qquad \text{.....(39).}$$

Put $\Delta H = H' - H$ and $\Delta\delta = \delta' - \delta$.

We have then

$$LM = (H' - H)\cos\delta = \Delta H\cos\delta.$$

Hence $$\Delta H\cos\delta = \frac{\rho}{r}\sin\eta\sin z'.$$

But by **B**, $$\sin\eta\sin z' = \cos\phi'\sin H'.$$

Hence $$\Delta H = \frac{\rho}{r}\sin H'\cos\phi'\sec\delta \qquad \text{.....(40).}$$

We can write $H$ for $H'$ on the right of (40) without sensible loss of accuracy and, since $\alpha' - \alpha \equiv \Delta\alpha = -(H' - H) = -\Delta H$, we obtain

$$\alpha' - \alpha \equiv \Delta\alpha = -\frac{\rho}{r}\cos\phi'\sec\delta\sin H \qquad \text{.....(41).}$$

Again, $M'L = \delta - \delta' = -\Delta\delta$, and therefore

$$\Delta\delta = -p\cos\eta,$$

so that by (39), $$\Delta\delta = -\frac{\rho}{r}\sin z'\cos\eta,$$

and, by applying formula **C**, we obtain

$$\Delta\delta = -\frac{\rho}{r}(\sin\phi'\cos\delta' - \cos\phi'\sin\delta'\cos H').$$

Replacing $\delta'$, $H'$ by $\delta$ and $H$ on the right of this equation we have

$$\delta' - \delta \equiv \Delta\delta = -\frac{\rho}{r}(\sin\phi'\cos\delta - \cos\phi'\sin\delta\cos H) \quad \ldots\ldots(42),$$

which gives the displacement in declination due to parallax.

As in the case of the moon, the observation of the planet's zenith distance at two widely separated observatories on the same (or nearly the same) meridian will lead to the determination of $\rho/r$. Thus the distance $r$ of the planet can be found at the time of observation and, the scale of the planetary system being now known, the mean distance of the earth from the sun can be deduced. The base-line in this instance is the straight line joining the two observatories.

**123.** *The solar parallax (diurnal method).*

We now describe an alternative method called the diurnal method. The principle is as follows. The planet is observed several hours before meridian passage and again several hours after meridian passage, at the *same* observatory. In the interval between observations the observer's position, with reference to the earth's centre and the planet, has been changed owing to the rotation of the earth, and the distance between the two positions constitutes essentially the base-line appropriate for the measurement of the planet's parallax.

Consider now a star whose position in the sky is near that of the planet. The parallax observations consist, in effect, of measuring the difference between the apparent right ascension $\alpha'$ of the planet and the right ascension $\alpha_0$ of the star, or the corresponding difference in declination. For simplicity we shall consider the former only. This difference in right ascension can be measured by means of the meridian circle or filar micrometer

or, more accurately, by making a photograph of the region of the sky in which the planet is situated. The distance between the image of the planet and the image of a neighbouring star can be accurately determined and, by the principles which we shall describe in Chapter XII, the difference $\alpha' - \alpha_0$ can be derived with great precision.

As the planet is in or near opposition when a parallax investigation is undertaken, the planet will be on or near the meridian at midnight. Our two sets of observations are then made a few hours before midnight and a few hours after midnight. We shall refer to these as the evening and morning observations. Consider first an evening observation when the difference $m_1$ in right ascension of the planet and of the star is measured. We have

$$m_1 = \alpha' - \alpha_0.$$

Now the displacement due to parallax is $\alpha' - \alpha$, that is, $(\alpha' - \alpha_0) + (\alpha_0 - \alpha)$. Hence by (41),

$$m_1 + \alpha_0 - \alpha = -\frac{\rho}{r} \cos \phi' \sec \delta \sin H,$$

or if we denote the values of $\alpha$, $r$, $\delta$, $H$ at the time of the evening observation by $\alpha_1$, $r_1$, $\delta_1$, $H_1$ we have

$$m_1 + \alpha_0 - \alpha_1 = -\frac{\rho}{r_1} \cos \phi' \sec \delta_1 \sin H_1 \quad \ldots\ldots(43).$$

During the interval between the evening and morning observations the true right ascension, declination and distance of the planet from the earth's centre are altering; at the time of the morning observation let these quantities be $\alpha_2$, $\delta_2$, $r_2$ and let the planet's hour angle be $H_2$. Then we shall have an equation similar to (43); it is

$$m_2 + \alpha_0 - \alpha_2 = -\frac{\rho}{r_2} \cos \phi' \sec \delta_2 \sin H_2 \quad \ldots\ldots(44).$$

Subtracting (43) from (44) we eliminate the star's right ascension $\alpha_0$ and obtain

$$m_2 - m_1 = (\alpha_2 - \alpha_1) - \rho \cos \phi' \left\{ \frac{\sec \delta_2 \sin H_2}{r_2} - \frac{\sec \delta_1 \sin H_1}{r_1} \right\};$$

$(\alpha_2 - \alpha_1)$ is the increase in the planet's geocentric right ascension between the observations due to the relative orbital motions of the earth and planet. From a series of observations, for example, meridian observations, this quantity may be deduced. Also we

may write without loss of accuracy the mean values $\delta$ and $r$ of $\delta_1$, $\delta_2$ and $r_1$, $r_2$ respectively. Hence

$$m_2 - m_1 = (\alpha_2 - \alpha_1) - \frac{\rho}{r} \cos \phi' \sec \delta \, (\sin H_2 - \sin H_1) \quad \ldots\ldots(45).$$

As $(m_2 - m_1)$ is given by the measures and $(\alpha_2 - \alpha_1)$, $\rho$, $\phi'$, $\delta$, $H_1$, $H_2$ are all supposed known, we can determine $r$ by (45). Thus the scale of the planetary system is found and the value of the solar parallax deduced.

The methods that we have considered in this and the previous section require a favourable opposition of the minor planet Eros to be applied most effectively. A particularly favourable opposition occurred in 1930–31, and the determination of the astronomical unit based on the observations made at that time was generally accepted for many years as the most accurate determination available. Since 1961, however, greater accuracy has been achieved by using radar measurements. The underlying principle of the method is the same. By using dynamical theory, in particular Kepler's third law, the relative dimensions of all the planetary orbits are known. The absolute scale of the planetary system is set by the measurement of a single planetary distance. The distance that can be most accurately measured by radar is the distance of Venus at inferior conjunction. This was first attempted in 1959 when very encouraging preliminary results were obtained. At the next inferior conjunction in 1961 a number of independent determinations of the solar parallax were made by this radar method. These taken together have provided a very substantial improvement in the accuracy to which the astronomical unit is now known. The value of the solar parallax adopted by the IAU in 1964 is 8″·794.

**124.** *The solar parallax (other methods).*

(*a*) *Transit of Venus.* For a description of this method, which is now entirely of historical interest, see Ball's *Spherical Astronomy*, p. 312.

(*b*) *Gravitational methods.* These depend on comparing certain observed perturbations in the motion of the moon with the theoretical expressions derived from celestial mechanics, which involve the semi-major axis $a$ of the earth's orbit.

(*c*) *The constant of aberration.* We have seen in Chapter VIII that the constant of aberration $\kappa$ is given by

$$\kappa = \frac{2\pi a \operatorname{cosec} 1''}{cT\,(1-e^2)^{\frac{1}{2}}}.$$

Denote the mean equatorial parallax of the sun by $P_0$ and the earth's equatorial radius by $\rho_0$. Then $\sin P_0 = \rho_0/a$ or, expressing $P_0$ in seconds of arc,

$$P_0 = \frac{\rho_0}{a} \operatorname{cosec} 1''.$$

Hence, combining these two formulae, we obtain

$$\kappa P_0 = \frac{2\pi\rho_0 \operatorname{cosec}^2 1''}{cT\,(1-e^2)^{\frac{1}{2}}} \qquad \ldots\ldots(46).$$

Here $\rho_0 = 6378{\cdot}16\,\text{km}$, $c = 299{,}792\,\text{km per second}$, $T = 31{,}558{,}150$ seconds and $e = 0{\cdot}01672$. We thus obtain

$$\kappa P_0 = 180{\cdot}24 \qquad \ldots\ldots(47),$$

$\kappa$ and $P_0$ being both expressed in seconds of arc. The constant of aberration can be determined from observations as we have shown in section 109, and therefore the solar parallax $P_0$ can be derived from (47). However, as we noted in that section, a precise determination of $\kappa$ is extremely difficult to achieve, and so the method considered here has not yielded the most accurate results. It is decidedly preferable to determine the astronomical unit by the methods of the previous section and then derive $\kappa$ by the theoretical considerations of section 110.

(*d*) *Spectroscopic method.* The basis of this method depends on one of the achievements of spectroscopic astronomy, namely, the measurement* of the velocity of approach, or of recession, of any star to or from the earth. This velocity (called the *radial velocity*) is obtained directly in linear units, for example, as so many kilometres per second. Consider, for simplicity, a star whose position is in the ecliptic and assume for a moment that the earth's orbit is a circle and that its orbital velocity is constant. At some definite time during the year the earth's orbital velocity is directed towards the star. In general, the star will be moving relatively to the sun which we may consider to be at rest. Denote the component of the star's velocity, directed towards the sun,

* For an account of the spectroscopic principles and methods, see *Astronomical Spectroscopy* by A. D. Thackery (Eyre and Spottiswoode, 1961).

by $V$. The spectroscopic method determines the star's velocity relative to and directed towards the earth. This is $V$ plus the earth's orbital velocity. Six months later, the earth's velocity is directed away from the star and the spectroscopist then measures $V$ minus the earth's orbital velocity. By subtraction $V$ is eliminated and the earth's orbital velocity is determined; it is usually expressed in kilometres per second. The period of revolution in the earth's orbit being known, the circumference in kilometres is then deduced, from which the radius of the earth's orbit is finally found in kilometres. Thus the astronomical unit of distance is determined and from it the solar parallax. We now consider the general problem.

In Fig. 87, let $\lambda$ and $\beta$ be the longitude and latitude of a star $X$. The earth's orbital velocity, as we have seen on p. 185, can be resolved into two parts: (i), $h/p$ directed to a point $F$ on the ecliptic, 90° behind the sun in longitude (the longitude of $F$ is thus $\odot - 90°$, where $\odot$ is the sun's true longitude), and (ii), $eh/p$ directed to a point $f$ on the ecliptic, 90° ahead of the point $A$, which is such that $\vec{EA}$ is parallel to the direction of perihelion as viewed from the sun. Consider first the velocity $h/p$ and its component resolved in the direction of the star $X$. This component is $\frac{h}{p}\cos XF$. Now

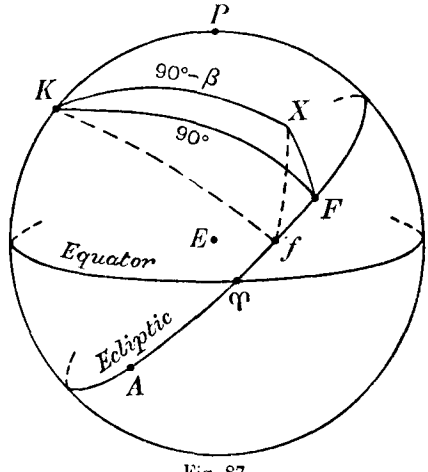


Fig. 87.

$$F\hat{K}X = \lambda - (\odot - 90°), \quad KF = 90° \quad \text{and} \quad KX = 90° - \beta.$$

Hence by the cosine-formula **A**,

$$\cos XF = \cos\beta \sin(\odot - \lambda),$$

and consequently the component of the velocity $h/p$ in the direction of the star is

$$\frac{h}{p}\cos\beta \sin(\odot - \lambda).$$

Consider now the component $eh/p$ of the earth's velocity directed towards the point $f$. The component of $eh/p$ along $EX$ is $\frac{eh}{p}\cos Xf$. Now the longitude of perihelion is $\varpi$ and therefore the longitude of $f$ is $\varpi + 90°$. Hence the angle $fKX$ is $\lambda - (\varpi + 90°)$. Using the cosine-formula we find that

$$\cos Xf = -\cos\beta\sin(\varpi - \lambda),$$

and the corresponding component of radial velocity is

$$-\frac{eh}{p}\cos\beta\sin(\varpi - \lambda).$$

For any given star this quantity is constant.

If the star is moving towards the sun (which we may regard as at rest) with velocity $V$, the velocity of approach of the star to the earth is

$$V + \frac{h}{p}\cos\beta\sin(\odot - \lambda) - \frac{eh}{p}\cos\beta\sin(\varpi - \lambda).$$

This is the radial velocity, which we shall denote by $-R$ (the convention is that a velocity of recession is taken to be positive).

Consider now two radial velocity determinations made several months apart. We shall then have

$$\left.\begin{aligned} -R_1 &= V + \frac{h}{p}\cos\beta\{\sin(\odot_1 - \lambda) - e\sin(\varpi - \lambda)\} \\ -R_2 &= V + \frac{h}{p}\cos\beta\{\sin(\odot_2 - \lambda) - e\sin(\varpi - \lambda)\} \end{aligned}\right\} \ldots(48),$$

in which $\odot_1$ and $\odot_2$ are the sun's true longitudes at the two dates concerned. Then by subtraction

$$R_2 - R_1 = \frac{h}{p}\cos\beta\{\sin(\odot_1 - \lambda) - \sin(\odot_2 - \lambda)\} \ldots(49).$$

$R_2 - R_1$ is numerically greatest when $\beta = 0$ and when $\odot_1 - \lambda = -(\odot_2 - \lambda) = 90°$ or $270°$. It follows that the best conditions for carrying out an investigation of this kind are obtained when (i) the star observed is in or near the ecliptic, (ii) the two sets of observations are made at an interval of six months, (iii) the dates are chosen so that the difference between the longitudes of the sun and star on either date is $90°$ or $270°$.

Now we have

$$\frac{h}{p} = \frac{2\pi a}{T(1-e^2)^{\frac{1}{2}}},$$

in which $a$ is the semi-major axis of the earth's orbit. Hence from (49),

$$R_2 - R_1 = \frac{2\pi a}{T(1-e^2)^{\frac{1}{2}}} \cos\beta \{\sin(\odot_1 - \lambda) - \sin(\odot_2 - \lambda)\} \quad \ldots\ldots(50).$$

In this equation, $R_2 - R_1$ is given by the spectroscopic observations in kilometres per second, $T$ is the orbital period in seconds, $e$, $\odot_1$, $\odot_2$ and $\lambda$ are known. Hence $a$ is determined in kilometres and thus the solar parallax is found.

In the past this method gave valuable results, though they were less accurate than those based on the parallax of Eros, and so this method has also been superseded by the radar method.

In deriving the value of $a$ in (50) from the radial velocity measures, we omitted one consideration. Unless the star is on the meridian at the time of observation, the linear velocity of the observatory due to the diurnal motion will have a component in the direction of the star, and this component will be included in the observed radial velocity. It is clear that this effect must first be removed before formula (50) can be applied. Consider the observer $O$ on the earth's surface (Fig. 88). Due to the earth's rotation he is moving along $OT$ (tangential to the parallel of latitude through $O$) with the linear velocity $\frac{2\pi\rho}{\tau}\cos\phi$ km per second, where $\rho$ is the radius of the earth in kilometres and $\tau$ is the length of the sidereal day in mean seconds ($\tau$ is 86164). Let $X$ denote the point on the earth's surface at which the star is vertically overhead at the time of observation. Let $E$ be the east point of the horizon. Then $CE$ is parallel to $OT$. Hence the component of the observer's velocity along $CX$ is

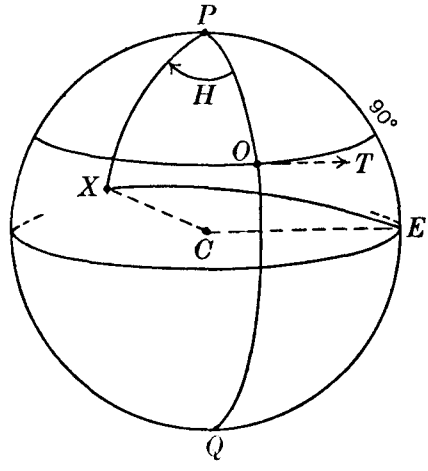


Fig. 88.

$$\frac{2\pi\rho}{\tau}\cos\phi\cos XE.$$

But if $H$ is the hour angle of the star, $X\hat{P}O = H$ and $X\hat{P}E = H + 90°$. Hence by the cosine-formula **A**,

$$\cos XE = -\cos\delta \sin H.$$

Let $R'$ denote the radial velocity of the star (positive, when it is a velocity of recession) due to the diurnal motion: then

$$R' = +\frac{2\pi\rho}{\tau}\cos\phi\cos\delta\sin H \qquad \ldots\ldots(51).$$

It is seen that $R'$ is zero when $H$ is $0^h$ or $12^h$. The maximum value of $R'$ is 0·47 cos $\phi$ km per second. The value of $R'$ just found must be subtracted from the observed radial velocity so as to eliminate the effect of the diurnal motion.

In formula (50), $R_1$ and $R_2$ are the radial velocities so corrected.

125. *Stellar parallax.*

We now describe the fundamental method of determining stellar distances. Fig. 89 shows the earth's orbit, which we can regard as circular; $S$ is the sun and $a$ is the orbital radius. Let $X$ denote the position of a star at a distance $d$ from the sun, and we shall here assume simply that the star is stationary with respect to the sun. On a given date let the earth be at $E$ and six months later at $E_1$. The distance $EE_1$ constitutes the base-line—it is about 300 million kilometres—by means of which the star's distance from the sun can eventually be determined.

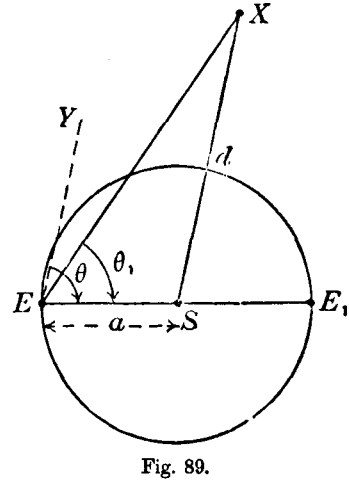


Fig. 89.

Let $EY$ be parallel to $SX$ and denote the angles $SEY$, $SEX$ by $\theta, \theta_1$ respectively. From the triangle $SXE$, in which $S\hat{X}E = \theta - \theta_1$ we have

$$\sin(\theta - \theta_1) = \frac{a}{d}\sin\theta_1 \qquad \ldots\ldots(52).$$

Define $\Pi$ by

$$\sin\Pi = \frac{a}{d} \qquad \ldots\ldots(53).$$

Then $\Pi$ is called the star's parallax (it is sometimes referred to as the *annual parallax*). It is the angle subtended at the star by the radius of the earth's orbit. For the nearest star, $\Pi$ is $0''{\cdot}76$ and it is evident that we can write (52) with the help of (53) in the simple and sufficiently accurate form

$$\theta - \theta_1 = \Pi \sin \theta \qquad \ldots\ldots(54),$$

in which $(\theta - \theta_1)$ and $\Pi$ are expressed in the same unit and $\theta$ is written for $\theta_1$.

In Fig. 89, $SX$ is the direction of the star as seen from the sun and $EX$ is the direction as seen from the earth on the date in question; we refer to these as the heliocentric and geocentric directions respectively. If the star were at an infinite distance it would be seen from the earth in the direction $EY$, which has been drawn parallel to $SX$. The angle $\theta$ is the angle between the heliocentric direction $EY$ and the direction $ES$ of the sun from the earth. Fig. 89 shows that the geocentric direction $EX$ is displaced from the heliocentric direction $EY$ *towards* the direction $ES$ of the sun, and that this displacement takes place in the plane $ESX$.

Consider now Fig. 90, which represents the celestial sphere centred at $E$. Let $X$ denote the position of the star corresponding to its heliocentric direction. $S$ is the sun's position on the

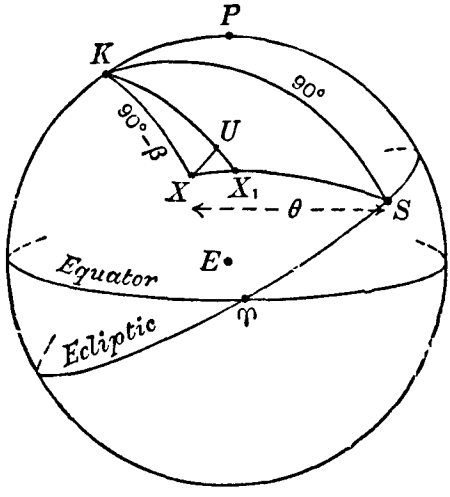



Fig. 90.

ecliptic at the date concerned. If $X_1$ is the star's position on the celestial sphere corresponding to its geocentric direction, then $X_1$ is in the plane of $X$, $E$ and $S$ and therefore $X_1$ is on the great circle arc $XS$. Moreover $X_1$ is between $X$ and $S$. We shall refer to $X$ and $X_1$ as the heliocentric and geocentric positions of the star respectively.

126. *Effect of parallax on the star's longitude and latitude.*

Let $\lambda$ and $\beta$ denote the star's heliocentric longitude and latitude (these quantities refer to the position $X$), and let $\lambda_1$ and $\beta_1$ denote the geocentric longitude and latitude (these refer to $X_1$). Draw the small circle arc $UX$ parallel to the ecliptic. Denote $\lambda_1 - \lambda$ by $\Delta\lambda$ and $\beta_1 - \beta$ by $\Delta\beta$. Denote the angle $UXX_1$ by $\phi$. Then in the infinitesimal plane triangle $UXX_1$,

$$UX \equiv \Delta\lambda \cos\beta = XX_1 \cos\phi,$$

and

$$UX_1 \equiv -\Delta\beta = XX_1 \sin\phi.$$

Now with the notation of Fig. 89, $XX_1 = \theta - \theta_1$ and $XS = \theta$. Hence using (54),

$$\Delta\lambda \cos\beta = \Pi \sin\theta \cos\phi \qquad \text{.....(55),}$$

and

$$\Delta\beta = -\Pi \sin\theta \sin\phi \qquad \text{.....(56).}$$

In the triangle $KXS$, $X\hat{K}S = \odot - \lambda$, where $\odot$ is the sun's true longitude $\Upsilon S$; also $KS = 90°$, $KX = 90° - \beta$ and $K\hat{X}S = 90° + \phi$. Hence by the sine-formula **B**,

$$\sin\theta \cos\phi = \sin(\odot - \lambda),$$

and by **C**,

$$\sin\theta \sin\phi = +\sin\beta \cos(\odot - \lambda).$$

The formulae (55) and (56) now become

$$\Delta\lambda \cos\beta = \Pi \sin(\odot - \lambda) \qquad \text{.....(57),}$$

$$\Delta\beta = -\Pi \sin\beta \cos(\odot - \lambda) \qquad \text{.....(58).}$$

These equations give the displacements of the star's position, due to parallax, in longitude and latitude.

127. *The parallactic ellipse.*

Let $x$ denote the displacement $UX$ parallel to the ecliptic and $y$ the displacement $UX_1$ in latitude. Then by (57) and (58),

$$x = \Pi \sin(\odot - \lambda),$$

$$y = \Pi \sin\beta \cos(\odot - \lambda),$$

whence, eliminating $\odot$, we derive

$$\frac{x^2}{\Pi^2} + \frac{y^2}{\Pi^2 \sin^2 \beta} = 1 \qquad \ldots\ldots(59),$$

the equation of an ellipse, known as the *parallactic ellipse*. This curve is the locus of the geocentric position $X_1$ throughout the year. The major axis is parallel to the ecliptic and the minor axis perpendicular to the ecliptic. Whatever the star observed, the semi-major axis is independent of the star's latitude and is equal to $\Pi$. For a star on the ecliptic the semi-minor axis vanishes and the parallactic displacement takes place wholly in the ecliptic.

**128.** *The effect of stellar parallax on right ascension and declination.*

In Fig. 91 let $A$, $D$ be the right ascension and declination of the sun $S$. Let $\alpha$, $\delta$ be the right ascension and declination of $X$ and $\alpha_1$, $\delta_1$ the corresponding quantities for $X_1$. Denote $\alpha_1 - \alpha$

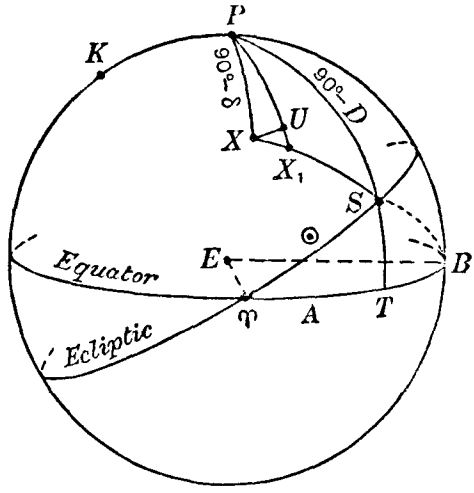



Fig. 91.

by $\Delta\alpha$ and $\delta_1 - \delta$ by $\Delta\delta$. Let $UX$ be the parallel of declination through $X$. Denote $U\hat{X}X_1$ by $\psi$. Then in the small plane triangle $UXX_1$,

$$UX \equiv \Delta\alpha \cos\delta = XX_1 \cos\psi,$$
$$UX_1 \equiv -\Delta\delta \quad = XX_1 \sin\psi.$$

As before $XS = \theta$ and $XX_1 = \theta - \theta_1$. Thus, using (54),

$$\Delta\alpha \cos\delta = \Pi \sin\theta \cos\psi \quad \ldots\ldots(60),$$

$$\Delta\delta = -\Pi \sin\theta \sin\psi \quad \ldots\ldots(61).$$

Now in the spherical triangle $PXS$, $PX = 90° - \delta$, $PS = 90° - D$, $X\hat{P}S = A - \alpha$, $XS = \theta$ and $P\hat{X}S = 90° + \psi$.

By formulae **B** and **C**,

$$\sin\theta \cos\psi = \cos D \sin(A - \alpha) \quad \ldots(62),$$

$$-\sin\theta \sin\psi = \sin D \cos\delta - \cos D \sin\delta \cos(A - \alpha) \quad \ldots(63).$$

Now from the triangle $\Upsilon ST$, in which $\Upsilon S = \odot$, $S\hat{\Upsilon}T = \epsilon$ (the obliquity of the ecliptic), $\Upsilon T = A$, $\Upsilon\hat{T}S = 90°$ and $TS = D$, we have, by **A**, **B** and **C** successively,

$$\left.\begin{aligned} \cos\odot &= \cos D \cos A \\ \sin\odot \sin\epsilon &= \sin D \\ \sin\odot \cos\epsilon &= \cos D \sin A \end{aligned}\right\} \quad \ldots\ldots(64).$$

Hence from (60), (62) and (64) we have

$$\Delta\alpha \cos\delta = \Pi(\cos\alpha \cos\epsilon \sin\odot - \sin\alpha \cos\odot) \quad \ldots(65).$$

Similarly, from (61), (63) and (64) we obtain

$$\Delta\delta = \Pi(\cos\delta \sin\epsilon \sin\odot - \cos\alpha \sin\delta \cos\odot - \sin\alpha \sin\delta \cos\epsilon \sin\odot) \quad \ldots\ldots(66).$$

Formulae (65) and (66) may be modified as follows. Consider rectangular equatorial axes $E\Upsilon$, $EB$ and $EP$ in Fig. 91, where $\Upsilon B$ is 90°. Let $X$, $Y$ and $Z$ denote the co-ordinates of the sun, expressed in terms of the radius of the earth's orbit (which we have assumed to be circular) as the unit of distance. Then

$$X = \cos\Upsilon S, \quad Y = \cos BS, \quad Z = \cos PS,$$

or $$X = \cos\odot, \quad Y = \sin\odot \cos\epsilon, \quad Z = \sin\odot \sin\epsilon.$$

Making the substitutions in (65) and (66) we have

$$\Delta\alpha \cos\delta = \Pi(Y \cos\alpha - X \sin\alpha) \quad \ldots\ldots(65\,a),$$

and $$\Delta\delta = \Pi(Z \cos\delta - X \cos\alpha \sin\delta - Y \sin\alpha \sin\delta) \quad \ldots\ldots(66\,a).$$

The values of $X$, $Y$ and $Z$ are tabulated in the almanacs for every day in the year and thus the coefficients of $\Pi$ in (65 $a$) and (66 $a$) can be readily calculated.

The formulae (65) and (66) or (65 $a$) and (66 $a$) give the displacements, due to parallax, in right ascension and declination.

129. *The measurement of stellar parallax.*

We shall consider only the application of formula (65). We write it in the form

$$\Delta\alpha \cos\delta \equiv (\alpha_1 - \alpha)\cos\delta = F\Pi \quad \ldots\ldots(67),$$

where $$F = \cos\alpha\cos\epsilon\sin\odot - \sin\alpha\cos\odot \quad \ldots\ldots(68),$$

or $$F = Y\cos\alpha - X\sin\alpha \quad \ldots\ldots(68\,a).$$

$F$ is called the *parallax factor* in right ascension and its value for any given star varies with the sun's true longitude. As we have seen, its value can be readily calculated.

Suppose there is a faint star $B$, presumably at a very great distance, near the star $A$ whose distance is to be measured. Let $\alpha_0$ be the right ascension of $B$. Suppose on a given date the difference between the geocentric right ascension $\alpha_1$ of $A$ and the right ascension $\alpha_0$ of $B$ is measured. Let $m_1 = \alpha_1 - \alpha_0$. Then $m_1 = (\alpha_1 - \alpha) + (\alpha - \alpha_0)$, where $\alpha$ is the heliocentric right ascension of $A$. If $B$ be regarded as at an infinite distance, $\alpha_0$ will be simply the heliocentric right ascension of $B$ and the quantity $(\alpha - \alpha_0)$ will remain constant throughout the year. Using (67), we have

$$m_1 \equiv (\alpha_1 - \alpha) + (\alpha - \alpha_0) = F_1\Pi\sec\delta + (\alpha - \alpha_0) \quad \ldots(69),$$

where $F_1$ is the parallax factor at the date in question. Suppose the observation is repeated some months later. We shall then have

$$m_2 = F_2\Pi\sec\delta + (\alpha - \alpha_0) \quad \ldots\ldots(70),$$

and by eliminating $(\alpha - \alpha_0)$ between (69) and (70) we obtain

$$m_2 - m_1 = (F_2 - F_1)\,\Pi\sec\delta$$

and hence

$$\Pi = \frac{(m_2 - m_1)\cos\delta}{F_2 - F_1} \quad \ldots\ldots(71).$$

If there is an error $e$ in the measured quantity $(m_2 - m_1)$, the parallax $\Pi$ will be determined with an error $e\cos\delta/(F_2 - F_1)$. The error in $\Pi$ will then be least when $F_2 - F_1$ is greatest. This condition leads us to the consideration of the most favourable circumstances under which parallax determinations can be carried out.

Let $$\cos\alpha\cos\epsilon = g\cos h \quad \ldots\ldots(72),$$

and $$\sin\alpha = g\sin h \quad \ldots\ldots(73).$$

Then we can write $F$ from (68) in the form

$$F = g\sin(\odot - h) \quad \ldots\ldots(74).$$